MATHEMATICS

APPLICATION OF DERIVATIVE

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APPLICATION OF DERIVATIVE

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TANGENT AND NORMAL AT A POINT

1. If a variable tangent to the curve $x^2y = c^3$ makes intercepts a, b on x and y axis respectively, then the value of a^2b is :

यदि वक्र $x^2y = c^3$ की चर स्पर्श रेखा, x- अक्ष और y-अक्ष से क्रमशः a एवं b अन्तःखण्ड काटती है तब a^2b का मान है :

(A) $27\,c^3$ (B) $\frac{4}{27}c^3$ (C) $\frac{27}{4}c^3$ (D) $\frac{4}{9}c^3$

Ans. C

Sol. $x^2y = c^3$

$$x^2\frac{dy}{dx} + 2xy = 0 \Rightarrow \frac{dy}{dx} = -\frac{2y}{x}$$

equation of tangent at (x,y)

$$Y - y = -\frac{2y}{x}(X - x)$$

$$Y = 0,\ \text{gives}\ ,\ X = \frac{3x}{2} = a$$

and $X = 0$, gives , $Y = 3y = b$

Now $a^2b = \frac{9x^2}{4}.3y = \frac{27}{4}x^2y = \frac{27}{4}c^3 \Rightarrow$ (C)

2. At the point $P(a, a^n)$ on the graph of $y = x^n\ (n \in N)$ in the first quadrant a normal is drawn. The normal intersects the y-axis at the point (0, b). If $\lim_{a\to 0} b = \frac{1}{2}$, then n equals :

प्रथम चतुर्थांश में वक्र $y = x^n\ (n \in N)$ पर बिन्दु $P(a, a^n)$ पर अभिलम्ब, y-अक्ष को बिन्दु (0, b) पर प्रतिच्छेद करता है। यदि $\lim_{a\to 0} b = \frac{1}{2}$ तब n बराबर :

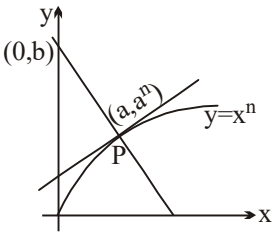
(A) 1 (B) 3 (C) 2 (D) 4

Ans. C

Sol. $y = x^n$

$$\frac{dy}{dx} = n\,x^{n-1} = na^{n-1}$$

$$\text{slope of normal} = -\frac{1}{na^{n-1}}$$



equation of normal $y - a^n = -\frac{1}{na^{n-1}}(x - a)$

put $x = 0$ to get y-intercept

$y = a^n + \frac{1}{na^{n-2}}$; Hence $b = a^n + \frac{1}{na^{n-2}}$

$$\lim_{a\to 0} b = \begin{cases} 0 & \text{if } n < 2 \\ \frac{1}{2} & \text{if } n = 2 \\ \infty & \text{if } n > 2 \end{cases} \quad \Rightarrow \quad \text{(A)}$$

3. The normal to the curve $x = a(1+\cos\theta)$, $y = a\sin\theta$ at $P(\theta)$ always passes through the fixed point :

वक्र $x = a(1+\cos\theta)$, $y = a\sin\theta$ के बिन्दु $P(\theta)$ पर अभिलम्ब सदैव अचर बिन्दु से होकर जाता है, वह बिन्दु है :

(A) (a, 0) (B) (0, a) (C) (0,0) (D) (a, a)

Ans. A

Sol. $x = a(1+\cos\theta)$, $y = a\sin\theta$

$\frac{dx}{d\theta} = -a\sin\theta$; $\frac{dy}{d\theta} = a\cos\theta$

$\left(\frac{dy}{dx}\right) = -\frac{\cos\theta}{\sin\theta}$ slope of normal $= -\left(\frac{dx}{dy}\right) = \frac{\sin\theta}{\cos\theta}$

$y - a\sin\theta = \frac{\sin\theta}{\cos\theta}(x - a - a\cos\theta)$

$y\cos\theta - a\sin\theta\cos\theta = x(\sin\theta) - a\sin\theta(1+\cos\theta)$

$x\sin\theta - y\cos\theta = a\sin\theta(1+\cos\theta - \cos\theta)$

clearly passes through (a, 0)

4. The normal to the curve, $x^2 + 2xy - 3y^2 = 0$ at (1, 1) : **[JEE(Main)2015]**

(A) does not meet the curve again (B) meets the curve again in the second quadrant

(C) meets the curve again in the third quadrant (D) meets the curve again in the fourth quadrant

वक्र $x^2 + 2xy - 3y^2 = 0$ के बिन्दु (1, 1) पर अभिलम्ब : **[JEE(Main)2015]**

(A) वक्र को दोबारा नहीं मिलता (B) वक्र को दोबारा द्वितीय चतुर्थांश मे मिलता है

(C) वक्र को दोबारा तृतीय चतुर्थांश में मिलता है (D) वक्र को दोबारा चतुर्थ चतुर्थांश में मिलता है

Ans. D

Sol. $(x + 3y)(x - y) = 0$

Equation of normal at (1, 1) $x + y = 2$
will meet the curve again at $(3, -1)$

5. Consider $f(x) = \tan^{-1}\left(\sqrt{\frac{1+\sin x}{1-\sin x}}\right), x \in \left(0, \frac{\pi}{2}\right)$. A normal to $y = f(x)$ at $x = \frac{\pi}{6}$ also passes through the point :

[JEE(Main)2016]

$f(x) = \tan^{-1}\left(\sqrt{\frac{1+\sin x}{1-\sin x}}\right), x \in \left(0, \frac{\pi}{2}\right)$ पर विचार कीजिए। $y = f(x)$ के बिन्दु $x = \frac{\pi}{6}$ पर खींचा गया अभिलम्ब निम्न बिन्दु से भी होकर जाता है :

[JEE(Main)2016]

(A) $\left(\frac{\pi}{6}, 0\right)$ (B) $\left(\frac{\pi}{4}, 0\right)$ (C) $(0, 0)$ (D) $\left(0, \frac{2\pi}{3}\right)$

Ans. D

Sol. $f(x) = \tan^{-1}\left(\frac{1+\sin x}{\cos x}\right)$

$= \tan^{-1}\left(\tan\left(\frac{\pi}{4} + \frac{x}{2}\right)\right)$

$f(x) = \frac{\pi}{4} + \frac{x}{2}$

$f'(x) = \frac{1}{2}$

Eq. of normal $y - \frac{\pi}{3} = -2\left(x - \frac{\pi}{6}\right)$

$2x + y = \frac{2\pi}{3}$

6. The normal to the curve $y(x-2)(x-3) = x + 6$ at the point where the curve intersects the y-axis passes through the point : **[JEE(Main)2017]**

वक्र $y(x-2)(x-3) = x + 6$ के उस बिन्दु पर, जहाँ वक्र y-अक्ष को काटती है, खींचा गया अभिलंब निम्न में से किस बिन्दु से होकर जाता है : **[JEE(Main)2017]**

(A) $\left(\frac{1}{2}, -\frac{1}{3}\right)$ (B) $\left(\frac{1}{2}, \frac{1}{3}\right)$ (C) $\left(-\frac{1}{2}, -\frac{1}{2}\right)$ (D) $\left(\frac{1}{2}, \frac{1}{2}\right)$

Ans. D

Sol. Point (0, 1)

$y(x-2)(x-3) = x+6$

$\left(\frac{dy}{dx}\right)(x-2)(x-3) + (2x-5)y = 1$

Slope of tangent at (0, 1) = 1

Equation of norm.

$y - 1 = -1(x-y)$

$x + y = 1$

ULTIMATE CONCEPT

7. The point(s) at each of which the tangents to the curve $y = x^3 - 3x^2 - 7x + 6$ cut off on the positive semi axis OX a line segment half that on the negative semi axis OY then the co-ordinates the point(s) is/are given by :

(A) $(-1, 9)$ (B) $(3, -15)$ (C) $(1, -3)$ (D) None

बिन्दु जहाँ वक्र $y = x^3 - 3x^2 - 7x + 6$ पर बनायी गई स्पर्श रेखायें धनात्मक x-अक्ष को, ऋणात्मक y-अक्ष की तुलना में आधा रेखाखण्ड काटती है :

(A) $(-1, 9)$ (B) $(3, -15)$ (C) $(1, -3)$ (D) कोई नहीं

Ans. B

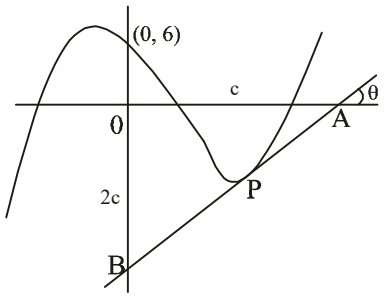
Sol. If OA = a ; OB = 2a $\Rightarrow \tan\theta = 2$

slope of the tangent is 2

$\Rightarrow \left(\frac{dy}{dx}\right)_{x_1 y_1} = 2 \Rightarrow 3x_1^2 - 6x_1 - 7 = 2 \Rightarrow 3x_1^2 - 2x_1 - 3 = 0$

$\Rightarrow x_1 = 3$ or -1 (rejected)

$\Rightarrow (3, -15) \Rightarrow$ B



8. Let C be the curve $y = x^3$ (where x takes all real values). The tangent at A meets the curve again at B. If the gradient at B is K times the gradient at A then K is equal to :

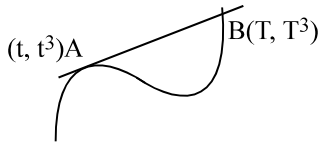
माना C वक्र $y = x^3$ हैं। A पर स्पर्श रेखा वक्र को पुनः बिन्दु B पर मिलती है। यदि B पर प्रवणता, A पर प्रवणता की K गुना है तब K बराबर :

(A) 4 (B) 2 (C) -2 (D) $\frac{1}{4}$

Ans. A

Sol. $\frac{dy}{dx} = 3x^2 = 3t^2$ at 'A'

$\therefore \quad 3t^2 = \dfrac{T^3 - t^3}{T - t} = T^2 + Tt + t^2$



$T^2 + Tt - 2t^2 = 0$

$(T - t)(T + 2t) = 0 \quad \Rightarrow \quad T = t \quad \text{or} \quad T = -2t \quad (T = t \text{ is not possible})$

$\dfrac{m_B}{m_A} = \dfrac{3T^2}{3t^2} = \dfrac{4t^2}{t^2}$ (using $T = -2t$)

$m_B = 4$

9. A circle with centre at (15, –3) is tangent to $y = \dfrac{x^2}{3}$ at a point in the first quadrant. The radius of the circle is equal to :

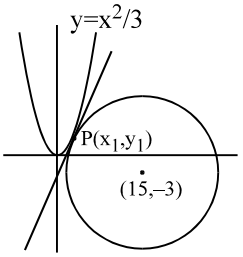
केन्द्र (15, –3) वाला वृत्त वक्र $y = \dfrac{x^2}{3}$ को प्रथम चतुर्थांश में किसी बिन्दु पर स्पर्श करता है तब वृत्त की त्रिज्या होगी :

(A) $5\sqrt{6}$ (B) $8\sqrt{3}$ (C) $9\sqrt{2}$ (D) $6\sqrt{5}$

Ans. D

Sol. $y = \dfrac{x^2}{3} \Rightarrow \left.\dfrac{dy}{dx}\right|_P = \dfrac{2x_1}{3}$

$\therefore \quad m_N = -\dfrac{3}{2x_1} = \dfrac{y_1 + 3}{x_1 - 15} \Rightarrow 2x_1y_1 + 6x_1 = -3x_1 + 45$



Put $y_1 = \dfrac{x_1^2}{3}$, we get $2x_1 \cdot \dfrac{x_1^2}{3} + 9x_1 - 45 = 0$; $2x_1^3 + 27x_1 - 135 = 0$

$2x_1^2(x_1 - 3) + 6x_1(x_1 - 3) + 45(x_1 - 3) = 0$

$(x_1 - 3)(2x_1^2 + 6x_1 + 45) = 0 \quad \Rightarrow \quad x_1 = 3$

10. The equation of the tangent to the curve $y = x + \dfrac{4}{x^2}$, that is parallel to x-axis, is : **[AIEEE-2010]**

वक्र $y = x + \dfrac{4}{x^2}$ की स्पर्श रेखा, जो x-अक्ष के समांतर है, का समीकरण है : **[AIEEE-2010]**

(A) $y = 0$ (B) $y = 1$ (C) $y = 2$ (D) $y = 3$

Ans. D

Sol. $y = x + \frac{4}{x^2}$

$\frac{dy}{dx} = 1 - \frac{8}{x^3}$

Equation of tangent is parallel to x-axis

$\therefore \frac{dy}{dx} = 0$

$\Rightarrow 1 - \frac{8}{x^3} = 0 \Rightarrow x^3 = 8 \Rightarrow x = 2$

At, $x = 2$, $y = 2 + \frac{4}{4} = 3 \Rightarrow y_1 = 3$

$\therefore$ point is (2, 3)

equation of tangent is :

$y - y_1 = 0(x - x_1)$

$y = 3$

MINIMUM DISTANCE BETWEEN TWO CURVES

11. The coordinates of the point of the parabola $y^2 = 8x$, which is at minimum distance from the circle $x^2 + (y + 6)^2 = 1$ are :

(A) (2 , – 4) (B) (18 , –12) (C) (2 , 4) (D) None of these

परवलय $y^2 = 8x$ पर उस बिन्दु के निर्देशांक जो वृत्त $x^2 + (y + 6)^2 = 1$ से न्यूनतम दूरी पर हैं :

(A) (2 , – 4) (B) (18 , –12) (C) (2 , 4) (D) इनमें से कोई नहीं

Ans. A

Sol. Here, the minimum distance occurs along the common noral.

Any normal to

$y^2 = 8x$ at $(2t^2\ 4t)$ is

$tx + y = 4t + 2t^3$

It passes through the centre of circle (0, 6)

$6 = 4t + 2t^3$

$\Rightarrow 2t^3 + 4t - 6 = 0$

$\Rightarrow t^3 + 2t - 3 = 0$

$\Rightarrow (t - 1)\ (t^2 + t + 3 \neq 0)\ (\because\ D < 0)$

At $t = 1$, required point is (2, 4)

12. The shortest distance between the line $y - x = 1$ and the curve $x = y^2$ is : **[AIEEE-2009]**

रेखा $y - x = 1$ और वक्र $x = y^2$ के बीच की लधुतम दूरी है : **[AIEEE-2009]**

(A) $\frac{3\sqrt{2}}{5}$ (B) $\frac{\sqrt{3}}{4}$ (C) $\frac{3\sqrt{2}}{8}$ (D) $\frac{2\sqrt{3}}{8}$

Ans. C

Sol. Point (t^2, t) is on the parabola $x = y^2$

Its distance from $y - x = 1$

$$d(t) = \frac{t^2 - t + 1}{\sqrt{2}}$$

$$d'(t) = \frac{1}{\sqrt{2}}[2t-1] = 0$$

$$t = \frac{1}{2}$$

$$d''(t) = \frac{2}{\sqrt{2}} > 0$$

d(t) is min at $t = \frac{1}{2}$

Its value

$$d\left(\frac{1}{2}\right) = \frac{1}{\sqrt{2}}\left(\frac{1}{4} - \frac{1}{2} + 1\right)$$

$$d\left(\frac{1}{2}\right) = \frac{3\sqrt{2}}{8}$$

13. The lines tangent to the curves $y^3 - x^2y + 5y - 2x = 0$ and $x^4 - x^3y^2 + 5x + 2y = 0$ at the origin intersect at an angle θ equal to :

वक्रों $y^3 - x^2y + 5y - 2x = 0$ तथा $x^4 - x^3y^2 + 5x + 2y = 0$ की मूल बिन्दु पर स्पर्श रेखाएँ θ कोण पर प्रतिच्छेद करती है तब θ बराबर :

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{2}$

Ans. D

Sol. Given curve is

$y^3 - x^2y + 5y - 2x = 0$

Slope of tangent to curve at (0, 0) = $m_1 = \frac{2}{5}$

Given curve II as

$x^4 - x^3y^2 + 5x + 2y = 0$

Slope of tangent to curve at (0, 0) = $m_2 = -\frac{5}{2}$

Since $m_1.m_2 = -1$

Hence, angle between tangent to the curves is $\frac{\pi}{2}$

14. The tangent to the curve $y = e^x$ drawn at the point (c, e^c) intersects the line joining the points $(c - 1, e^{c-1})$ and $(c + 1, e^{c+1})$:

(A) on the left of $x = c$ (B) on the right of $x = c$

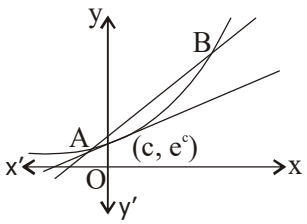
(C) at no point (D) at all points

वक्र $y = e^x$ के बिन्दु (c, e^c) पर खीचीं गई स्पर्श रेखा बिन्दुओं $(c - 1, e^{c-1})$ तथा $(c + 1, e^{c+1})$ को मिलाने वाली रेखा को कहाँ प्रतिच्छेद करती है :

(A) $x = c$ के बाई ओर (B) $x = c$ के दाई ओर (C) कही भी नहीं (D) सभी बिन्दुओं पर

Ans. A

Sol. Slope of the line joining the points $(c - 1, e^{c-1})$ and $(c + 1, e^{c+1})$ is equal to $\frac{e^{c+1} - e^{c-1}}{2} > e^c$



tangent to the curve $y = e^x$ will intersecct the given line to the left of the line $x = c$.

ORTHOGONALITY OF TWO CURVES

15. If curve $y = 1 - ax^2$ and $y = x^2$ intersect orthogonally then the value of a is :

यदि वक्र $y = 1 - ax^2$ और $y = x^2$ परस्पर लम्बवत् प्रतिच्छेद करते हैं, तो a का मान है :

(A) 1/2 (B) 1/3 (C) 2 (D) 3

Ans. B

Sol. $y = 1 - ax^2 \Rightarrow \frac{dy}{dx} = -2ax$

$y = x^2 \Rightarrow \frac{dy}{dx} = 2x$

two curve intersect orthogonally if

$(-2ax)(2x) = -1$

$\Rightarrow 4ax2 = 1$ i

$\Rightarrow x^2 = \frac{1}{1+a}$.....ii

from i and ii

$\frac{4a}{1+a} = 1$

$\Rightarrow a = \frac{1}{3}$

16. If the curves $y^2 = 6x$, $9x^2 + by^2 = 16$ intersect each other at right angles, then the value of b is : **[JEE(Main)2018]**

यदि वक्र $y^2 = 6x$, $9x^2 + by^2 = 16$ समकोण पर प्रतिच्छेद करते हैं, तो b का मान है : **[JEE(Main)2018]**

(A) 4 (B) $\frac{9}{2}$ (C) 6 (D) $\frac{7}{2}$

Ans. B

Sol. $y^2 = 6x$

$2yy' = 6$

$y' = \frac{3}{y_1}$

$9x^2 + by^2 = 16$

$18x + 2byy' = 0$

$y' = \frac{-9x_1}{by_1}$

$m_1 m_2 = -1$

$27x_1 = by_1^2$ ..... (i)

$6x_1 = y_1^2$ ..... (ii)

(i) ÷ (ii)

$b = \frac{9}{2}$

CASE OF TWO CURVES TOUCHING EACH OTHER

17. Number of positive integral values of 'a' for which the curve $y = a^x$ intersects the line $y = x$ is :

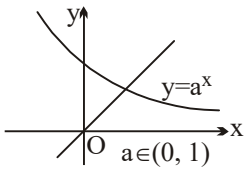
(A) 0 (B) 1 (C) 2 (D) More than 2

'a' के धनात्मक पूर्णांक मानों की संख्या जब कि वक्र $y = a^x$ रेखा $y = x$ को प्रतिच्छेद करता है :

(A) 0 (B) 1 (C) 2 (D) 2 से अधिक

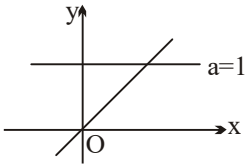
Ans. B

Sol. for $0 < a \le 1$ the line always cuts $y = a^x$



for $a > 1$ say $a = e$

consider $f(x) = e^x - x$



$f'(x) = e^x - 1$

$f'(x) > 0$ for $x > 0$ and $f'(x) < 0$ for $x < 0$

$\therefore$ f(x) is incresing ($\uparrow$) for $x > 0$

and decreasing ($\downarrow$)for $x < 0$

$y = e^x$ always lies above $y = x$ i.e. $e^x - x \ge 1$ for $a > 1$

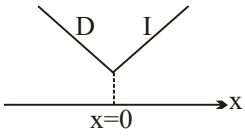
hence never cuts = $a = (0, 1] \Rightarrow$ (B)

$f(0)$ is the minimum

$f(x)|_{min} = 1$

$f(x) \ge 1$

$e^x - x \ge 1$



18. The value of k for which the equation $e^{2x} = k\sqrt{x}$ has exactly one solution, is :

k का वह मान जिसके लिए समीकरण $e^{2x} = k\sqrt{x}$ का केवल एक हल होगा :

(A) $\frac{1}{\sqrt{e}}$ (B) $\sqrt{e}$ (C) $2\sqrt{e}$ (D) $\frac{2}{\sqrt{e}}$

Ans. C

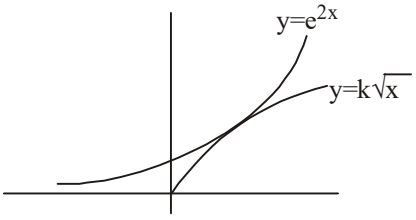
Sol. k can not be negative

$y = e^{2x}$

$\left.\frac{dy}{dx}\right|_P = 2e^{2x_1}$

$\therefore \quad \left.\frac{dy}{dx}\right|_P = \frac{k}{2\sqrt{x_1}}$

$2e^{2x_1} = \frac{k}{2\sqrt{x_1}}$ (but $e^{2x_1} = k\sqrt{x_1}$)



$$2k\sqrt{x_1} = \frac{k}{2\sqrt{x_1}};\ x_1 = \frac{1}{4}\ ;\quad \therefore \qquad e^{\frac{1}{2}} = \frac{k}{2}\ ;\ k = 2\sqrt{e}$$

19. The radius of a circle, having minimum area, which touches the curve $y = 4 - x^2$ and the lines, $y = |x|$ is :

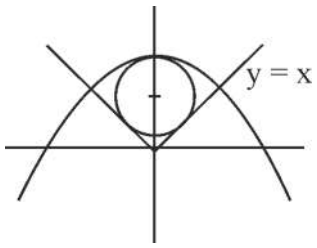
[JEE(Main)2017]

न्यूनतम क्षेत्रफल वाले ऐसे वृत, जो वक्र $y = 4 - x^2$ तथा रेखाओं $y = |x|$ को स्पर्श करता है, की त्रिज्या है :

(A) $4\left(\sqrt{2}-1\right)$ (B) $4\left(\sqrt{2}+1\right)$ (C) $2\left(\sqrt{2}+1\right)$ (D) $2\left(\sqrt{2}-1\right)$

Ans. A

Sol.



Center will be $(0, 4 - r)$

Distance of $(0, 4 - r)$ from $L : x - y = 0$

radius = r

$$\left|\frac{4-r}{\sqrt{2}}\right| = r$$

$$4 - r = \sqrt{2}r$$

$$r = 4\left(\sqrt{2}-1\right)$$

20. At any point of a curve $\sqrt{\frac{\text{sub normal}}{\text{sub tangent}}}$ is equal to :

(A) the abscissa of that point (B) the ordinate of that point

(C) slope of the tangent at that point (D) slope of the normal at that point

वक्र के किसी बिन्दु पर $\sqrt{\frac{v/kqyEc}{v/kuUe'kkUrj}}$ बराबर है :

(A) उस बिन्दु के भुज (B) उस बिन्दु की कोटि

(C) उस बिन्दु पर स्पर्श रेखा की प्रवणता (D) उस बिन्दु पर अभिलम्ब की प्रवणता

Ans. C

Sol. $\sqrt{\dfrac{y\dfrac{dy}{dx}}{\dfrac{y}{\dfrac{dy}{dx}}}} = \sqrt{\left(\dfrac{dy}{dx}\right)^2} = \left|\dfrac{dy}{dx}\right|$

RATE MEASURE

21. Water is poured at the rate of $2m^3$/sec. into a cone of semi-vertical angle $45°$. The rate at which periphery of water surface changes when height of the water in the cone is 2 meter, is :

एक शंकु जिसका अर्ध शीर्ष कोण $45°$ है, में जल $2m^3$/sec की नियत दर से भरा जा रहा है। जब शंकु में जल की सतह की ऊँचाई 2 m है तब जल की सतह की परिधी के बदलने की दर है :

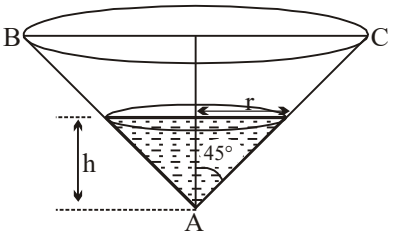
(A) 1 m/sec. (B) 2 m/sec. (C) 3m/sec. (D) 4 m/sec.

Ans. A

Sol. We have

$$\frac{dV}{dt} = 2 \Rightarrow \frac{d}{dt}\left(\frac{1}{3}\pi r^3\right) = 2 \text{ [Here r = h, as } \theta = 45° \text{]}$$

$$\Rightarrow \pi r^2 \frac{dr}{dt} = 2 \Rightarrow \frac{dr}{dt} = \frac{2}{\pi r^2} \quad ....(1)$$



Now, perimeter = $2\pi r = p$ (let)

$$\Rightarrow \frac{d}{dt}(2\pi r) = 2\pi\left(\frac{2}{\pi r^2}\right) = \frac{4}{r^2} \quad ....(2) \quad \text{(Using equation (1))}$$

When h = 2 meters $\Rightarrow$ r = 2 meters

Hence $\dfrac{dp}{dt} = \dfrac{4}{4} = 1$ m/sec.

22. A point is moving along the curve $y^3 = 27x$. The interval in which the abscissa changes at slower rate than ordinate, is :

वक्र $y^3 = 27x$ पर एक चर बिन्दु किसी अन्तराल में इसप्रकार से गमन करता है कि उसके भुज में परिवर्तन की दर कोटि में परिवर्तन की दर से कम है, तब वह अन्तराल होगा :

(A) $(-3, 3)$ (B) $(-\infty, \infty)$ (C) $(-1, 1)$ (D) $(-\infty, -3) \cup (3, \infty)$

Ans. C

Sol. $y^3 = 27x \Rightarrow 3y^2\dfrac{dy}{dt} = 27\dfrac{dx}{dt}$

But $\frac{|dx/dt|}{|dy/dt|} < 1 \Rightarrow \frac{y^2}{9} < 1 \Rightarrow -3 < y < 3$ for $y \in (-3,3)$, $x \in (-1,1) \Rightarrow$ (C)

23. A spherical iron ball 10 cm in radius is coated with a layer of ice of uniform thickness that melts at a rate of 50 cm^3/ min. When the thickness of ice is 5 cm, then the rate of which the thickness of ice decreases, is :

(A) $\frac{1}{36\pi}$ cm/min. (B) $\frac{1}{18\pi}$ cm/min. (C) $\frac{1}{54\pi}$ cm/min. (D) $\frac{5}{6\pi}$ cm/min.

एक गोलाकार लोहे का गेंद, जिसकी त्रिज्या 10 सेमी है, बर्फ की एक समान मोटाई की परत से ढका है, जो 50 सेमी^{3}/मिनट की दर से पिघल रही है। जब बर्फ की परत की मोटाई 5 सेमी है, वह दर जिससे बर्फ की परत की मोटाई कम हो रही है, है :

(A) $\frac{1}{36\pi}$ सेमी/मि. (B) $\frac{1}{18\pi}$ सेमी/मि. (C) $\frac{1}{54\pi}$ सेमी/मि. (D) $\frac{5}{6\pi}$ सेमी/मि.

Ans. B

Sol. $V = 4\frac{\pi}{3}(10+r)^3$, $0 \le r \le 15$ $\because$ $\frac{dV}{dt} = -50.$

$4\pi (10 + r)^2 \frac{dr}{dt} = -50 \Rightarrow \frac{dr}{dt} = \frac{-1}{18\pi}$ (where $r = 5$)

MONOTONOCITY

DETERMINE MONOTONOCITY

24. The function $f(x) = \tan^{-1}\left(\frac{1-x^2}{1+x^2}\right)$ is :

(A) increasing in its domain (B) decreasing in its domain
(C) decreasing in $(-\infty, 0)$ and increasing in $(0, \infty)$ (D) increasing in $(-\infty, 0)$ and decreasing in $(0, \infty)$

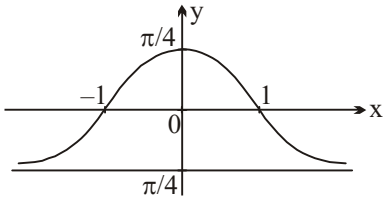
फलन $f(x) = \tan^{-1}\left(\frac{1-x^2}{1+x^2}\right)$ है :

(A) अपने प्रान्त में वर्धमान (B) अपने प्रान्त में ह्यसमान
(C) $(-\infty, 0)$ में ह्यसमान, एवं $(0, \infty)$ में वर्धमान (D) $(-\infty, 0)$ में वर्धमान और $(0, \infty)$ में ह्यसमान

Ans. D

Sol. put $x^2 = \tan\theta$ to get

$f(x) = \frac{\pi}{4} - \tan^{-1}(x^2)$



$\Rightarrow \quad f'(x) = -\dfrac{2x}{1+x^4}$ which is greater than zero for $x < 0$ and less than zero for $x > 0$

$\Rightarrow$ (D)

25. If $f(x) = xe^{x(1-x)}$, then f(x) is :

(A) increasing on $\left(-\dfrac{1}{2}, 1\right)$ (B) decreasing on $\left[-\dfrac{1}{2}, 1\right]$

(C) increasing on R (C) decreasing on R

यदि $f(x) = xe^{x(1-x)}$, तो f(x) एक :

(A) $\left(-\dfrac{1}{2}, 1\right)$ में वर्द्धमान फलन है (B) $\left[-\dfrac{1}{2}, 1\right]$ में ह्रासमान फलन है

(C) R में वर्द्धमान है (C) R में ह्रासमान है

Ans. A

Sol. $f(x) = xe^{x(x-1)}$

$\Rightarrow f'(x) = e^{x(1-x)} + x^2(-1)e^{x(1-x)} + x(1-x)e^{x(1-x)}$

$= -e^{x(1-x)}\left(2x^2 - x - 1\right) = e^{x(1-x)}(x-1)\left(x + \dfrac{1}{2}\right)$

$f'(x) = 0$ at $x = -\dfrac{1}{2}, 1$ and $f'(x) > 0$ when $-\dfrac{1}{2} < x < 1$

$\Rightarrow$ f(x) is increasing function on $\left[-\dfrac{1}{2}, 1\right]$

26. A function is matched below against an interval whre it is supposed to be increasing. Which of the followig pairs is incorrectly matched?

interval	**function**
(A) $(-\infty, \infty)$	$x^3 - 3x^2 + 3x + 3$
(B) $[2, \infty)$	$2x^3 - 3x^2 - 12x + 6$
(C) $\left(-\infty, \dfrac{1}{3}\right]$	$3x^2 - 2x + 1$
(D) $(-\infty, -4]$	$x^3 + 6x^2 + 6$

नीचे दिये गए फलनों को उनके सामने दिये गये अन्तरालों से मिलान किया गया है, जिनमें फलन वर्धमान हैं। निम्न युग्मों में से कौन से युग्म का मिलान ठीक नहीं है?

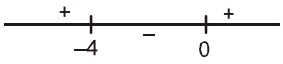
अन्तराल	**फलन**
(A) $(-\infty, \infty)$	$x^3 - 3x^2 + 3x + 3$
(B) $[2, \infty)$	$2x^3 - 3x^2 - 12x + 6$
(C) $\left(-\infty, \frac{1}{3}\right]$	$3x^2 - 2x + 1$
(D) $(-\infty, -4]$	$x^3 + 6x^2 + 6$

Ans. C

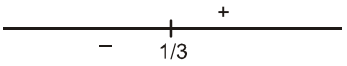
Sol. Let $f_1(x) = x^3 + 6x^2 + 6 \Rightarrow f_1'(x) = 3x(x+4)$

Increasing in $(-\infty, -4]$

Let $f_2(x) = 3x^2 - 2x + 1 \Rightarrow f_2'(x) = 6x - 2$

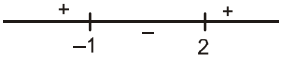


Not increasing in $\left(-\infty, \frac{1}{3}\right]$



Let $f_3(x) = 2x^3 - 3x^2 - 12x + 6 \Rightarrow f_3'(x) = 6x^2 - 6x - 12$

$= 6(x+1)(x-1)$



Increasing in $[2, \infty)$

Let $f_4(x) = x^3 - 3x^2 + 3x + 3 \Rightarrow f'(x) = 3(x-1)^2 \geq 0$

Increasing in $(-\infty, \infty)$

27. The function $f(x) = \tan^{-1}(\sin x + \cos x)$ is an increasing function in :

फलन $f(x) = \tan^{-1}(\sin x + \cos x)$, जिस अन्तराल में वर्धमान फलन है वह है :

(A) $(\pi/4, \pi/2)$ (B) $(-\pi/2, \pi/4)$ (C) $(0, \pi/2)$ (D) $(-\pi/2, \pi/2)$

Ans. B

Sol. $f(x) = \tan^{-1}(\sin x + \cos x)$

$$f'(x) = \frac{1 \times (\cos x - \sin x)}{1 + (\sin x + \cos x)^2} > 0$$

$\cos x - \sin x > 0 \qquad \cos x > \sin x$

$\sin x < \cos x \qquad \tan x < 1$

$x < \frac{\pi}{4} \qquad \boxed{x \in \left(-\frac{\pi}{2}, \frac{\pi}{4}\right)}$

ALWAYS INCREASING/ ALWAYS DECREASING BASED QUESTIONS

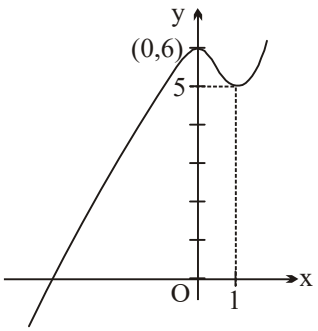
28. The function $f : [a, \infty) \to R$ where R denotes the range corresponding to the given domain, with rule $f(x) = 2x^3 - 3x^2 + 6$, will have an inverse provided :

यदि फलन $f : [a, \infty) \to R$ जहाँ R दिये गये प्रान्त के सापेक्ष परिसर को दर्शाता है इस प्रकार है कि $f(x) = 2x^3 - 3x^2 + 6$ तब f(x) प्रतिलोमीय होगा यदि :

(A) $a \geq 1$ (B) $a \geq 0$ (C) $a \leq 0$ (D) $a \leq 1$

Ans. A

Sol. $f(x) = 2x^3 - 3x^2 + 6$



$f'(x) = 6x^2 - 6x = 6(x^2 - x) = 0$ gives $x = 0$ or $x = 1$

for inverse to exist function must be one one onto

hence Domain is $[1, \infty)$

Hence $a \geq 1$

29. Let $f(x) = x - \frac{1}{x}$ then which one of the following statement is true :

(A) Function is invertible if defined from $R - \{0\} \to R$.

(B) $f(x_1) > f(x_2)$, $\forall$ $x_1 > x_2$ and $x_1, x_2 \neq 0$.

(C) Graph of the function has exactly one asymptote.

(D) Function is one-one in every continuous interval [a, b] defined on one side of origin.

माना $f(x) = x - \frac{1}{x}$ तब निम्न में से कौनसा कथन सत्य है :

(A) फलन प्रतिलोमीय है यदि f, $R - \{0\} \to R$ पर परिभाषित हो

(B) $f(x_1) > f(x_2)$, $\forall$ $x_1 > x_2$ और $x_1, x_2 \neq 0$.

(C) वक्र की केवल एक अनन्तस्पर्शी रेखा है

(D) मूल बिन्दु के एक और परिभाषित प्रत्येक सतत् अन्तराल [a, b] में फलन एकैकी है।

Ans. D

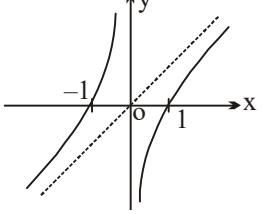
Sol. note that $x = 0$ and $y = x$ are the two asymptote $\Rightarrow$ (C) is wrong

A is wrong as f is many one

$f(1) = f(-1) = 0$

$f'(x) = 1 + \frac{1}{x^2} > 0 \quad \Rightarrow \quad$ f is increasing

$x \to 0^+; \quad f \to -\infty \quad x \to \infty, \ f \to \infty$



$x \to 0^-$; $f \to +\infty$ $x \to -\infty,\ f \to -\infty$

30. If the function $f(x) = 2x^2 - kx + 5$ is increasing in $[1, 2]$, then 'k' lies in the interval :

यदि $f(x) = 2x^2 - kx + 5$ अन्तराल $[1, 2]$ में वर्धमान है तब 'k' :

(A) $(-\infty, 4)$ (B) $(4, \infty)$ (C) $(-\infty, 8]$ (D) $(8, \infty)$

Ans. A

Sol. Given, $f(x) = 2x^2 - kx + 5$ Now $f'(x) > 0$ for $1 \le x \le 2$

$f'(x) = 4x - k$, $f''(x) = 4 \Rightarrow f''(x) > 0$ for all $x \in [1, 2]$

$\Rightarrow$ $f'(x)$ is an increasing function in $[1, 2] \Rightarrow f'(x) > 0$

$\therefore$ $f'(1)$ is the least value of $f'(x)$ in $[1, 2] \Rightarrow f'(1) > 0 \Rightarrow 4 - k > 0$

$\Rightarrow$ $k < 4 \Rightarrow k \in (-\infty, 4)$

31. Consider the following behaviours of function in (–1, 1)

I. Increasing II. Decreasing III. Continuity IV. Derivability

Which one of the following four functions exhibits atleast three of the four mentioned above? (where [ ] denotes greatest integer function)

(–1, 1) में माना फलन की निम्न प्रकृति पर विचार कीजिये

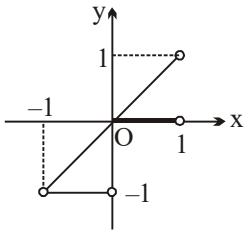
I. वर्धमान II. ह्रासमान III. सांतत्यता IV. अवकलनीयता

तब निम्न में से कौनसा फलन उपरोक्त चार में से कम से कम तीन गुणधर्म रखता है? (जहाँ [ ] महत्तम पूर्णांक फलन है)

(A) max $\{x, x^3\}$ (B) max $\{x, x^2\}$ (C) max $\{x, |x|\}$ (D) max $\{x, [x]\}$

Ans. D

Sol. max. $\{x, [x]\}$ in (–1, 1) is x as $[x] \le x$



32. Consider the following statements S and R

S : Both sin x & cos x are decreasing functions in the interval $(\pi/2, \pi)$.

R : If a differentiable function decreases in an interval (a, b), then its derivative also decreases in (a, b).

Which of the following is true?

(A) both S and R are wrong

(B) both S and R are correct, but R is not the correct explanation for S

(C) S is correct and R is the correct explanation for S

(D) S is correct and R is wrong

निम्नलिखित कथनों S तथा R के लिये

S : sin x तथा cos x दोनों फलन अन्तराल $(\pi/2, \pi)$ में ह्रासमान है।

R : यदि कोई फलन अन्तराल (a, b) में ह्रासमान है तो उसका अवकलज भी अन्तराल (a, b) ह्रासमान होगा

तो निम्न में से कौनसा सत्य है?

(A) S तथा R दोनों असत्य है

(B) S तथा R दोनों सत्य है परन्तु R कथन S की सही व्याख्या नही करता

(C) S सत्य है तथा R भी सत्य है एवं कथन S की सही व्याख्या है

(D) S सत्य है परन्तु R असत्य है

Ans. D

Sol. Since, it is true that sin x and cos x is decreasing in the interval $\left(\frac{\pi}{20}, \pi\right)$.

Now, let $f(x) = \sin x$, $x \in \left(\pi, \frac{3\pi}{3}\right)$, f(x) decrease from 0 to – 1 but $f'(x) = \cos\ x$ increases from –1 to 0.

CRITICAL AND STATIONARY POINTS

33. Number of critical points of the function, $f(x) = \frac{2}{3}\sqrt{x^3} - \frac{x}{2} + \int_1^x \left(\frac{1}{2} + \frac{1}{2}\cos 2t - \sqrt{t}\right) dt$ which lie in the interval $[-2\pi, 2\pi]$ is :

फलन $f(x) = \frac{2}{3}\sqrt{x^3} - \frac{x}{2} + \int_1^x \left(\frac{1}{2} + \frac{1}{2}\cos 2t - \sqrt{t}\right) dt$ के अन्तराल $[-2\pi, 2\pi]$ में क्रांतिक बिन्दुओं की संख्या है :

(A) 2 (B) 4 (C) 6 (D) 8

Ans. B

Sol. note that f is defined for $x > 0$

$f'(x) = \frac{1}{2}\cos 2x = 0 \Rightarrow x = n\pi \pm \frac{\pi}{4} \Rightarrow$ B]

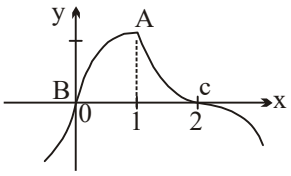
34. Let $f(x) = \begin{cases} x^{3/5} & \text{if } x \le 1 \\ -(x-2)^3 & \text{if } x > 1 \end{cases}$, then the number of critical points on the graph of the function is :

यदि $f(x) = \begin{cases} x^{3/5} & \text{if } x \le 1 \\ -(x-2)^3 & \text{if } x > 1 \end{cases}$, तब फलन f(x) के क्रान्तिक बिन्दुओं की संख्या है :

(A) 1 (B) 2 (C) 3 (D) 4

Ans. C

Sol. A, B, C are the 3 critical points of $y = f(x)$



$f''(x) = 0$ for $x = 2$ and fails to exists at $x = 0$

35. Let a function f be defined as $f(x) = \begin{cases} \frac{|x-1|}{x^2+1} & \text{if} \quad x > -1 \\ x^2 & \text{if} \quad x \le -1 \end{cases}$, Then the number of critical point(s) on the graph of this function is/are :

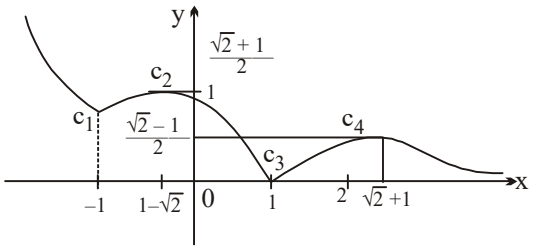
माना $f(x) = \begin{cases} \frac{|x-1|}{x^2+1} & \text{if} \quad x > -1 \\ x^2 & \text{if} \quad x \le -1 \end{cases}$, तब फलन के आरेख में क्रांतिक बिन्दुओं की संख्या है :

(A) 4 (B) 3 (C) 2 (D) 1

Ans. A

Sol. $f(x) = \begin{cases} \frac{x-1}{1+x^2} & \text{if} \quad x \ge 1 \\ \frac{1-x}{1+x^2} & \text{if} \quad -1 < x < 1 \\ x^2 & \text{if} \quad x \le -1 \end{cases}$

$f'(x) = 0$ gives $x = \sqrt{2} + 1$ or $1 - \sqrt{2}$. The function has a continuity at $x = -1$. As $x \to \infty$, $f(x) \to 0$.
The graph is as shown .
Note that critical points are those where $dy/dx = 0$ if it exists or dy/dx is non existent .



The points $x = -1, 1 - \sqrt{2}$, 1 and $1 + \sqrt{2}$ are the four critical points on the graph of this function $\Rightarrow$ A

36. Suppose that f is a polynomial of degree 3 and that $f''(x) \neq 0$ at any of the stationary point. Then :

(A) f has exactly one stationary point (B) f must have no stationary point
(C) f must have exactly 2 stationary points (D) f has either 0 or 2 stationary points

यदि f , तीन घात का बहुपदीय फलन है और $f''(x) \neq 0$ किसी भी स्थिर बिन्दु (stationary point) के लिए तब :

(A) f का केवल एक स्थिर बिन्दु होगा (B) f का कोई स्थिर बिन्दु नही होगा

(C) f के दो स्थिर बिन्दु होगें (D) f के 0 या 2 स्थिर बिन्दु होगें

Ans. D

Sol. The derivative of a degree 3 polynomial is a quadratic. This must have either 0, 1 or 2 roots. If this has precisely one root, then this must be repeated. Hence we have $f'(x) = m(x-\alpha)^2$, where α is repeated root and $m \in R$. So our original function f has a critical point at $x = \alpha$.

Also $f''(x) = 2m(x-\alpha)$, in which case $f''(\alpha) = 0$. But we are told that the 2nd derivative is non zero at critical point. Hence there must be either 0 or 2 critical point.

37. The set of all values of 'a' for which the function, $f(x) = (a^2-3a+2)\left(\cos^2\frac{x}{4} - \sin^2\frac{x}{4}\right) + (a-1)x + \sin 1$ does not possess critical points is :

'a' के मानों का समुच्चय जिसके लिए $f(x) = (a^2-3a+2)\left(\cos^2\frac{x}{4} - \sin^2\frac{x}{4}\right) + (a-1)x + \sin 1$ के कोई क्रांतिक बिन्दु नहीं है :

(A) $[1, \infty)$ (B) $(0, 1) \cup (1, 4)$ (C) $(-2, 4)$ (D) $(1, 3) \cup (3, 5)$

Ans. B

Sol. $f(x) = \left(a^2 = -3a+2\right)\left(\cos^2\frac{x}{4} - \sin^2\frac{x}{4}\right) + (a-1)x + \sin 1$

$$\Rightarrow f(x) = (a-1)(a-2)\cos\frac{x}{2} + (a-1)x + \sin 1$$

$$\Rightarrow f'(x) = -\frac{1}{2}(a-1)(a-2)\sin\frac{x}{2} + (a-1)$$

$$\Rightarrow f'(x) = (a-1)\left[1 - \frac{(a-2)}{2}\sin\frac{x}{2}\right]$$

If f(x) does not possess critical points, then $f'(x) = 0$ for any $x \in R$

$$\Rightarrow (a-1)\left[1 - \frac{(a-2)}{2}\sin\frac{x}{2}\right] = 0 \text{ for any } x \in R$$

$$\Rightarrow a = 1 \text{ and } 1 - \left(\frac{a-2}{2}\right)\sin\frac{x}{2} = 0$$

must not have any solution in R.

$\Rightarrow a = 1$ and $\sin\frac{x}{2} = \frac{2}{a-2}$ is not solvable in R.

$\Rightarrow a = 1$ and $\left|\frac{2}{a-2}\right| > 1$ [For $a = 2$, $f(x) = x + \sin 1 \ \therefore f'(x) = 1 = 0$]

$\Rightarrow a = 1$ and $|a - 2| < 2 \Rightarrow a = 1$ and $-2 < a - 2 < 2$

$\Rightarrow a = 1$ and $0 < a < 4 \Rightarrow a \in (0, 1) \cup (1, 4)$.

PROVING INEQUALITIES

38. For all $x \in (0, 1)$:

सभी $x \in (0, 1)$ के लिये :

(A) $e^x < 1 + x$ (B) $\log_e(1 + x) < x$ (C) $\sin x > x$ (D) $\log_e x > x$

Ans. B

Sol. $y = e^x \Rightarrow \frac{dy}{dx} = e^{e^x}$

Then equation of the tangent at $x = 0$ is $y - 1 = 1(x - 0)$ or $y = x + 1$

Graph of $y = e^x$ always lies above the graph of $y = 1 + x$,

Hence, $e^x > 1 + x \Rightarrow x > \log_e(1 + x)$. Hence, b is true.

39. The true set of real values of x for which the function, $f(x) = x\,\ell n\,x - x + 1$ is positive is :

(A) $(1, \infty)$ (B) $(1/e, \infty)$ (C) $[e, \infty)$ (D) $(0, 1)$ and $(1, \infty)$

x के सभी मानों के समुच्चय जिसके लिये फलन $f(x) = x\,\ell n\,x - x + 1$ का मान धनात्मक है :

(A) $(1, \infty)$ (B) $(1/e, \infty)$ (C) $[e, \infty)$ (D) $(0, 1)$ तथा $(1, \infty)$

Ans. D

Sol. We have $f(x) = \log_e x - x + 1, x > 0$

$\therefore f(x) = 1 + \log_e x - 1 = \log_e x$

$\therefore f(X) < 0$ if $0 < x < 1$

$f(X) > 0$ if $x > 1$

also $\lim\limits_{x \to 0^+} (x \log_x x - x + 1) = 1$

So in $(0, 1)$, $f(x)$ decreases from 1 to 0 and in $(1, \infty)$ increase from $0 \to \infty$

Hence f(X) > 0 f or $x \in (0, 1) \cup (1, \infty)$

40. The set of all x for which $\ell n (1 + x) \le x$ is equal to :

(A) $x > 0$ (B) $x > -1$ (C) $-1 < x < 0$ (D) null set

$\ell n (1 + x) \le x$ के लिए x के समस्त मानों का समुच्चय है :

(A) $x > 0$ (B) $x > -1$ (C) $-1 < x < 0$ (D) रिक्त समुच्चय

Ans. B

Sol. $\ell n(1 + x) - x \le 0$

Let $f(x) = \ell n (1 + x) - x$, {Domain is $(-1, \infty)$}

$f'(x) = \dfrac{1}{x+1} - 1 = \dfrac{-x}{x+1}$ $\quad \underset{-1}{+} \quad \underset{0}{-}$

$\Rightarrow f(x) \le f(0) \; \forall \; x \in (-1, \infty) \Rightarrow f(x) \le 0 \; \forall \; x \in (-1, \infty)$.

ROLLE'S THERORM

41. The equation $\sin x + x \cos x = 0$ has at least one root in :

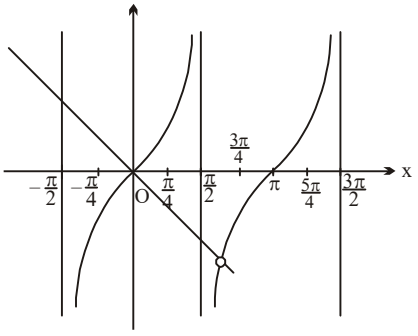
समीकरण $\sin x + x \cos x = 0$ का कम से कम एक मूल निम्न अन्तराल में है :

(A) $\left(-\dfrac{\pi}{2}, 0\right)$ (B) $(0, \pi)$ (C) $\left(\pi, \dfrac{3\pi}{2}\right)$ (D) $\left(0, \dfrac{\pi}{2}\right)$

Ans. B

Sol. Let $f(x) = \sin x + x \cos x$

consider $g(x) = \int_0^x (\sin t + t \cos t)\, dt = t \sin t \Big]_0^x = x \sin x$



$g(x) = x \sin x$ which is differentiable

now $g(0) = 0$ and $g(\pi) = 0$, using Rolles Theorem

hence $\exists$ atleast one $x \in (0, \pi)$ where $g'(x) = 0$

i.e. $x \cos x + \sin x = 0$ for atleast one $x \in (0, \pi)$

Alter: Draw graph of $y = \tan x$ & $y = -x$ Ans. $\Rightarrow$ (B)

42. Let $f(x) = (x-4)(x-5)(x-6)(x-7)$ then :

(A) $f'(x) = 0$ has four roots

(B) Three roots of $f'(x) = 0$ lie in $(4, 5) \cup (5, 6) \cup (6, 7)$

(C) The equation $f'(x) = 0$ has only one real root

(D) Three roots of $f'(x) = 0$ lie in $(3, 4) \cup (4, 5) \cup (5, 6)$

यदि $f(x) = (x-4)(x-5)(x-6)(x-7)$ हो, तो :

(A) $f'(x) = 0$ के चार मूल हैं

(B) $f'(x) = 0$ के तीन मूल $(4, 5) \cup (5, 6) \cup (6, 7)$ में स्थित हैं

(C) समीकरण $f'(x) = 0$ का केवल एक वास्तविक मूल है

(D) $f'(x) = 0$ के लिए तीन मूल $(3, 4) \cup (4, 5) \cup (5, 6)$ में स्थित हैं

Ans. B

Sol. $f(4) = f(5) = f(6) = f(7) = 0$

By Rolle's theorem on interval

$[4, 5], [5, 6], [6, 7]$ we have

$f'(x) = 0$ for at least once in each intervals $(4, 5), (5, 6), (6, 7)$.

43. Let a, b, c, d are non-zero real numbers such that $6a + 4b + 3c + 3d = 0$, then the equation $ax^3 + bx^2 + cx + d = 0$ has :

(A) atleast one root in $[-2, 0]$ (B) atleast one root in $[0, 2]$

(C) atleast two roots in $[-2, 2]$ (D) no root in $[-2, 2]$

यदि a, b, c, d अशून्य वास्तविक संख्याऐं इस प्रकार है कि $6a + 4b + 3c + 3d = 0$, तब समीकरण $ax^3 + bx^2 + cx + d = 0$ के

(A) अन्तराल $[-2, 0]$ में कम से कम एक मूल है (B) अन्तराल $[0, 2]$ में कम से कम एक मूल है

(C) अन्तराल $[-2, 2]$ में कम से कम दो मूल है (D) $[-2, 2]$ में कोई मूल नहीं है

Ans. B

Sol. We have $ax^3 + bx^2 + cx + d = 0$

Let $f(x) = \frac{ax^4}{4} + \frac{bx^3}{3} + \frac{cx^2}{2} + dx + e$

$\therefore$ $f(0) = e$

$$f(2) = 4a + \frac{8b}{3} + 2c + 2d + e = \frac{(12a + 8b + 6c + 6d)}{3} + e = \frac{2}{3}(6a + 4b + 3c + 3d) + e = 0 + e$$

$\Rightarrow$ $f(2) = e$

$\therefore$ By Rolle's theorem, there exist atleast one value of $x \in (0, 2)$ such that $f'(x) = 0$

$\Rightarrow$ The equation $ax^3 + bx^2 + cx + d = 0$ has atleast one real root in $[0, 2]$

44. If the equation $a_n x^n + a_{n-1} x^{n-1} + ... + a_1 x = 0$; $a_1 \neq 0$, $n \geq 2$, has a positive root $x = \alpha$, then the equation $na_n x^{n-1} + (n-1) a_{n-1} x^{n-2} + ... + a_1 = 0$ has a positive root, which is -

(A) greater than α (B) smaller than α

(C) greater than or equal to α (D) equal to α

यदि समीकरण $a_n x^n + a_{n-1} x^{n-1} + ... + a_1 x = 0$; $a_1 \neq 0$, $n \geq 2$ का एक धनात्मक मूल $x = \alpha$ है, तो समीकरण $na_n x^{n-1} + (n-1) a_{n-1} x^{n-2} + ... + a_1 = 0$ का एक धनात्मक मूल होगा जो कि :

(A) α से बड़ा है (B) α से छोटा है

(C) α से बड़ा है या बराबर है (D) α के बराबर है

Ans. B

Sol. Let $f(x) = a_0 X^n + a_1 X^{n-1} + ... + a_n X$

Now for $x = 0$

$f(0) = 0$

And as a is one root

$f(a) = 0$

Hence f'(x) will have one root in (0, a)

As $f'(x) = na_n x^{n-1} + (n-1) a_{n-1} x^{n-2} + ... + a_1 = 0$

Then root of $na_n x^{n-1} + (n-1) a_{n-1} x^{n-2} + ... + a_1 = 0$ is less than a

45. If f and g are differentiable functions in [0, 1] satisfying f(0) = 2 =g(1), g(0) = 0 and f(1) = 6, then for some $c \in]0, 1[$ **[JEE(Main)2014]**

यदि f तथा g, [0, 1] में अवकलनीय फलन है जो f(0) = 2 =g(1), g(0) = 0 और f(1) = 6 को संतुष्ट करते है, तो किसी $c \in]0, 1[$ के लिए : **[JEE(Main)2014]**

(A) f'(c) = g'(c) (B) f'(c) = 2g'(c) (C) 2f'(c) = g'(c) (D) 2f'(c) = 3g'(c)

Ans. B

Sol. Consider f(x) – 2g(x) = h(x)

Then, h(x) is continuous and differentiable in [0, 1]

Also h(0) = 2 & h(1) = 2

Hence h(x) satisfies conditions of Rolles Theorem in (0, 1)

Thus, There exist a 'c' such that h'(c) = 0 where $c \in (0, 1)$

$\Rightarrow$ f'(c) = 2g'(c)

LMVT

46. Let $f(x) = x + \sqrt{x}$ on [1, 4]. The mean value theorem says that there must be some number 'c' between 1 and 4 so that f'(c) is equal to the average slope of f(x) on [1, 4]. the number 'c' must be :

यदि $f(x) = x + \sqrt{x}$ अन्तराल [1, 4] में मध्यमान प्रमेय द्वारा कम से कम एक 'c' (1, 4) में इस प्रकार से होगा की f'(c) अन्तराल [1, 4] में f(x) की औसत प्रवणता के बराबर है तो 'c' होना चाहिये :

(A) $\frac{5}{2}$ (B) $\frac{9}{4}$ (C) $\frac{11}{4}$ (D) 3

Ans. B

Sol. Hint: $f'(c) = \frac{f(4) - f(1)}{4 - 1}$

$1 + \frac{1}{2\sqrt{c}} = \frac{6-2}{3} = \frac{4}{3} \Rightarrow \frac{1}{2\sqrt{c}} = \frac{1}{3} \Rightarrow \sqrt{c} = \frac{3}{2} \Rightarrow c = \frac{9}{4}$

47. Let f be differentiable for all x. If f(1) = – 2 and f' (x) ≥ 2 for x ∈ [1, 6], then :

(A) f(6) ≥ 8 (B) f(6) < 8 (C) f(6) < 5 (D) f(6) = 5

माना f सभी x के लिए अवकलनीय है। यदि f(1) = – 2 तथा x ∈ [1, 6] के लिए f' (x) ≥ 2, तो :

(A) f(6) ≥ 8 (B) f(6) < 8 (C) f(6) < 5 (D) f(6) = 5

Ans. A

Sol. By LMVT

$\frac{f(6) - f(1)}{6 - 1} = f'(x), \quad x \in (1, 6)$

and (तथा) $f'(x) \geq 2 \Rightarrow \frac{f(6) - f(1)}{6 - 1} \geq 2$

$\Rightarrow \frac{f(6) + 2}{5} \geq 2 \Rightarrow f(6) \geq 8.$

48. A value of C for which the conclusion of Mean Value Theorem holds for the function $f(x) = \log_e x$ on the interval [1, 3] is :

C का वह मान जिसके लिये अन्तराल [1, 3] में फलन $f(x) = \log_e x$ के लिये माध्यमान प्रमेय सत्य है :

(A) $2\log_3 e$ (B) $\frac{1}{2}\log_e 3$ (C) $\log_3 e$ (D) $\log_e 3$

Ans. A

Sol. $\frac{f(3) - f(1)}{3 - 1} = f'(c), \ 1 < c < 3$

$\frac{\log_e^3 - 0}{2} = \frac{1}{c}$

$$c = \frac{2}{\log_e^3} = 2\log_3 e$$

MAXIMA & MINIMA

MAXIMA AND MINIMA AT A POINT

49. Difference between the greatest and the least values of the function $f(x) = x(\ln x - 2)$ on $[1, e^2]$ is :

फलन $f(x) = x(\ln x - 2)$ के अन्तराल $[1, e^2]$ में अधिकतम और न्यूनतम मानों का अन्तर है :

(A) 2 (B) e (C) e^2 (D) 1

Ans. B

Sol. $y = x(\ln x - 2)$

$$y' = x\left(\frac{1}{x}\right) + (\ln x - 2) = \ln x - 1$$

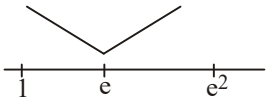
$$\frac{dy}{dx} = \ln x - 1 = 0 \quad \Rightarrow \quad x = e$$

now $f(1) = -2$

$f(e) = -e$ (least)

$f(e^2) = 0$ (greatest)

$\therefore$ difference $= 0 - (-e) = e$



50. Which one of the following statements does not hold good for the function $f(x) = \cos^{-1}(2x^2 - 1)$?

(A) f is not differentiable at $x = 0$ (B) f is monotonic

(C) f is even (D) f has an extremum

फलन $f(x) = \cos^{-1}(2x^2 - 1)$ के लिए निम्न में से कौनसा सत्य नहीं हैं :

(A) f, $x = 0$ पर अवकलनीय नहीं है (B) f एकदिष्ट है

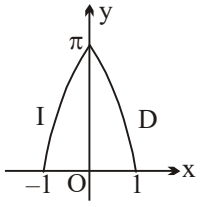
(C) f समफलन है (D) f का एक चरम मान है

Ans. B

Sol. $\cos^{-1}(2x^2 - 1) = \begin{cases} 2\cos^{-1} x & \text{for} \quad x \in [0,1] \\ 2\pi - 2\cos^{-1} x & \text{for} \quad x \in [-1,0] \end{cases}$

f is increasing in $(-1, 0)$ and decreasing in $(0, 1)$

$\Rightarrow$ non monotonic $\Rightarrow$ (B) is wrong



51. The minimum value of $f(x, y) = x^2 - 4x + y^2 + 6y$ when x and y are subjected to the restrictions $0 \le x \le 1$ and $0 \le y \le 1$, is :

$f(x, y) = x^2 - 4x + y^2 + 6y$ का न्यूनतम मान जहाँ $0 \le x \le 1$ तथा $0 \le y \le 1$ है

(A) -1 (B) -2 (C) -3 (D) -5

Ans. C

Sol. We have $f(x, y) = x^2 + y^2 - 4x + 6y$

Let $(x, y) = (\cos\theta, \sin\theta)$, then $\theta \in [0, \pi/2]$ and

$f(x, y) = f(\theta) = \cos^2\theta + \sin^2\theta - 4\cos\theta + 6\sin\theta$

$f'(\theta) = 6\cos\theta + 4\sin\theta > 0 \ \forall \ \theta \in [0, \pi/2]$

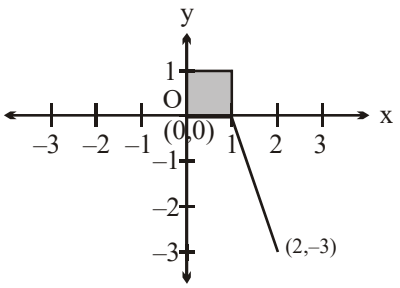
$\therefore$ $f'(\theta)$ is strictly increasing in $[0, \pi/2]$

$\therefore$ $f(\theta)_{min} = f(0) = 1 - 4 + 0 = -3$ Ans.]

Alternatively: $f(x, y) = (x-2)^2 + (y+3)^2 - 13$.

So, to minimise f over the unit square is the same as minimising the distance from a point in this square to the point of coordinates $(2, -3)$.

The closest point in the square to $(2, -3)$ is then $(1, 0)$.



So $f(1, 0) = -3$ must be the minimum.

52. Let $f(x) = ax^2 - b|x|$, where a and b are constants. Then at $x = 0$, $f(x)$ has :

(A) a maxima whenever $a > 0, b > 0$ (B) a maxima whenever $a > 0, b < 0$

(C) minima whenever $a > 0, b > 0$ (D) neither a maxima nor minima whenever $a > 0, b < 0$

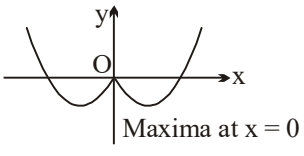
माना $f(x) = ax^2 - b|x|$, जहाँ a तथा b अचर है तब $x = 0$ पर फलन $f(x)$:

(A) एक उच्चिष्ठ मान देगा जबकि $a > 0, b > 0$ (B) एक उच्चिष्ठ मान देता जबकि $a > 0, b < 0$

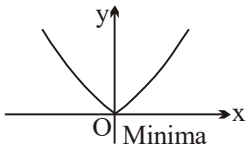
(C) एक निम्निष्ठ मान देगा जबकि $a > 0, b > 0$ (D) न तो उच्चिष्ठ न ही निम्निष्ठ देगा जबकि $a > 0, b < 0$

Ans. A

Sol. for $a > 0, b > 0$



for $a > 0, b < 0$



53. If a function f is continuous for all x and if f has a relative maximum at $(-1, 4)$ and a relative minimum at $(3, -2)$, then which of the following statements must be true?

(A) The graph of f has a point of inflection somewhere between $x = -1$ and $x = 3$.

(B) $f'(-1) = 0$.

(C) The graph of f has a horizontal tangent line at $x = 3$.

(D) The graph of f intersect both co-ordinate axes.

यदि फलन f सभी x के लिये सतत् है तथा f का स्थानीय उच्चिष्ठ $(-1, 4)$ पर है तथा स्थानीय निम्नष्ठ $(3, -2)$ पर है तो निम्न में से कौनसे कथन सत्य होना चाहिये?

(A) f का आरेख पर $x = -1$ एवं $x = 3$ के मध्य कही पर एक नति परिवर्तन बिन्दु होगा

(B) $f'(-1) = 0$

(C) f का आरेख में $x = 3$ पर स्पर्श रेखा क्षैतिज होगी

(D) f का आरेख दोनों निर्देशांक अक्षो को प्रतिच्छेद करता है

Ans. D

Sol. Because f is continuous for all x, the intermediate value theorem implies that the graph of f must intersect the x-axis. The graph must also intersect the y-axis since f is defined for all x, in particular, at $x = 0$. As f need not be differentiable. Hence A, B and C need not be correct.

54. If the function $f(x) = 2x^3 - 9ax^2 + 12a^2x + 1$, where $a > 0$, attains its maximum and minimum at p and q respectively such that $p^2 = q$, then a equals :

यदि फलन $f(x) = 2x^3 - 9ax^2 + 12a^2x + 1$, (जहाँ $a > 0$) के बिन्दु p तथा q पर क्रमशः उच्चिष्ठ एवं निम्निष्ठ हों तथा $p^2 = q$ हो, तो a का मान होगा :

(A) 1/2 (B) 3 (C) 1 (D) 2

Ans. D

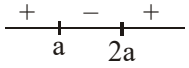
Sol. $f(x) = 2x^3 - 9ax^2 + 12a^2x + 1, \quad a > 0$

$f'(x) = 6x^2 - 18ax + 12a^2$

$= 6(x^2 - 3ax + 2a^2)$

$= 6(x - a)(x - 2a)$

$p = a, \quad q = 2a$



$\Rightarrow a^2 = 2a$

$\Rightarrow a = 0$ (rejected) or $a = 2$

$a = 2$

55. If p and q are positive real numbers such that $p^2 + q^2 = 1$, then the maximum value of $(p + q)$ is :

यदि p तथा q इस प्रकार से धनात्मक वास्तविक संख्या है कि $p^2 + q^2 = 1$, तब $(p + q)$ का अधिकतम मान होगा :

(A) 2 (B) $\frac{1}{2}$ (C) $\frac{1}{\sqrt{2}}$ (D) $\sqrt{2}$

Ans. D

Sol. Using A.M. ≥ G.M.

$$\frac{p^2+q^2}{2} \geq p.q$$

$$\Rightarrow \quad pq \leq \frac{1}{2}$$

$$\Rightarrow \quad (p+q)^2 = p^2 + q^2 + 2pq$$

$$\Rightarrow \quad (p+q) \leq \sqrt{2}$$

56. Suppose the cubic $x^3 - px + q$ has three distinct real roots where $p > 0$ and $q > 0$. Then which one of the following holds? **[AIEEE-2008]**

(A) The cubic has minima at $-\sqrt{\frac{p}{3}}$ and maxima at $\sqrt{\frac{p}{3}}$

(B) The cubic has manima at both $\sqrt{\frac{p}{3}}$ and $-\sqrt{\frac{p}{3}}$

(C) The cubic has maxima at both $\sqrt{\frac{p}{3}}$ an $-\sqrt{\frac{p}{3}}$

(D) The cubic has minima at $\sqrt{\frac{p}{3}}$ and maxima at $-\sqrt{\frac{p}{3}}$

माना त्रिघात बहुपद $x^3 - px + q$ के तीन विभिन्न वास्तविक मूल हैं जबकि $p > 0$ तथा $q > 0$ हैं। तो निम्न में से कौनसा एक सत्य है? **[AIEEE-2008]**

(A) त्रिपद का न्यूनतम मान $-\sqrt{\frac{p}{3}}$ पर है तथा अधिकतम मान $\sqrt{\frac{p}{3}}$ पर है

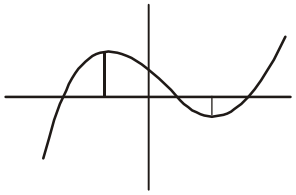
(B) त्रिपद का दोनों $\sqrt{\frac{p}{3}}$ तथा $-\sqrt{\frac{p}{3}}$ पर न्यूनतम मान है

(C) त्रिपद का दोनों $\sqrt{\frac{p}{3}}$ तथा $-\sqrt{\frac{p}{3}}$ पर अधिकतम मान है

(D) त्रिपद का $\sqrt{\frac{p}{3}}$ पर न्यूनतम मान तथा $-\sqrt{\frac{p}{3}}$ पर अधिकतम मान है

Ans. D

Sol. Graph of $y = x^3 - px + q$ cuts x-axis at three distinct pints



$\frac{dy}{dx} = 0 \qquad \Rightarrow \qquad x = \pm\sqrt{\frac{p}{3}}$

Maxima at $x = -\sqrt{\frac{p}{3}}$, minima at $x = \sqrt{\frac{p}{3}}$

57. Given $P(x) = x^4 + ax^3 + bx^2 + cx + d$ such that $x = 0$ is the only real root of $P'(x) = 0$. If $P(-1) < P(1)$, then in the interval $[-1,1]$: **[AIEEE-2009]**

(A) $P(-1)$ is the minimum and $P(1)$ is the maximum of P

(B) $P(-1)$ is not minimum but $P(1)$ is the maximum of P

(C) $P(-1)$ is the minimum but $P(1)$ is not the maximum of P

(D) Neither $P(-1)$ is the minimum nor $P(1)$ is the maximum of P

माना $P(x) = x^4 + ax^3 + bx^2 + cx + d$ के लिए $x = 0$ ही अकेला $P'(x) = 0$ का वास्तविक मूल है। यदि $P(-1) < P(1)$ है तब अंतराल $[-1,1]$ में : **[AIEEE-2009]**

(A) P का न्यूनतम $P(-1)$ और अधिकतम $P(1)$ है

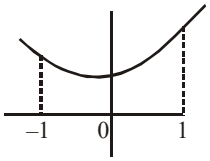
(B) P का न्यूनतम $P(-1)$ नहीं है परन्तु अधिकतम $P(1)$ है

(C) P का न्यूनतम $P(-1)$ है परन्तु अधिकतम $P(1)$ नहीं है

(D) न तो P का न्यूनतम $P(-1)$ है और न ही P का अधिकतम $P(1)$ है

Ans. B

Sol. Graph of $P(x)$ under given conditions. It is clear that $P(x)$ has max. at 1 but not minimum at -1.



58. Let $f : R \to R$ be defined by

$$f(x) = \begin{cases} k - 2x, & \text{if } x \le -1 \\ 2x + 3, & \text{if } x > -1 \end{cases}$$

If f has a local minimum at $x = -1$, then a possible value of k is : **[AIEEE-2010]**

माना $f : R \to R$ निम्न रूप में परिभाषित है

$$f(x) = \begin{cases} k - 2x, & \text{if } x \le -1 \\ 2x + 3, & \text{if } x > -1 \end{cases}$$ यदि फलन f का स्थानीय निम्नतम $x = -1$ पर है, तो k का संभावित मान है : **[AIEEE-2010]**

(A) 1 (B) 0 (C) $-\frac{1}{2}$ (D) -1

Ans. D

Sol. f has a local minimum at $x = -1$

$\therefore \lim\limits_{x \to -1} f(x) \geq f(-1)$

$k + 2 \leq 1$

$k \leq -1$

$\therefore k = -1$

59. For $x \in \left(0, \frac{5\pi}{2}\right)$, define $f(x) = \int\limits_0^x \sqrt{t} \sin t\,dt$. Then f has : **[AIEEE-2011]**

(A) local minimum at π and local maximum at 2π

(B) local maximum at π and local minimum at 2π

(C) local maximum at π and 2π

(D) local minimum at π and 2π

$x \in \left(0, \frac{5\pi}{2}\right)$, के लिए $f(x) = \int\limits_0^x \sqrt{t} \sin t\,dt$ को परिभाषित कीजिए। तो f का : **[AIEEE-2011]**

(A) स्थानीय निम्नतम मान π पर तथा स्थानीय उच्चतम मान 2π पर है

(B) स्थानीय उच्चतम मान π पर तथा स्थानीय निम्नतम मान 2π पर है

(C) स्थानीय उच्चतम मान π तथा 2π पर है

(D) स्थानीय निम्नतम मान π तथा 2π पर है

Ans. B

Sol. $f(x) = \sqrt{x} \sin x$

$f'(\pi)$ and $f'(2\pi)$ are 0.

$f'(x)$ $\overset{+}{\underset{\pi}{|}}$ $\overset{-}{\underset{2\pi}{|}}$ $+$

$\Rightarrow$ local maximum at $x = \pi$ and local minimum at $x = 2\pi$

60. If $x = -1$ and $x = 2$ are extreme points of $f(x) = \alpha \log|x| + \beta x^2 + x$ then : **[JEE(Main)2014]**

यदि $x = -1$ तथा $x = 2$, $f(x) = \alpha \log|x| + \beta x^2 + x$ के चरमबिन्दु है तो : **[JEE(Main)2014]**

(A) $\alpha = 2, \beta = -\frac{1}{2}$ (B) $\alpha = 2, \beta = \frac{1}{2}$ (C) $\alpha = -6, \beta = \frac{1}{2}$ (D) $\alpha = -6, \beta = -\frac{1}{2}$

Ans. A

Sol. $f(x) = \alpha \ell n|x| + \beta x^2 + x$

(1). $f'(x) = \frac{\alpha}{x} + 2\beta x + 1$.

$= \frac{2\beta x^2 + x + \ell}{x}$

Since $x = -1, 2$ are extreme points $\Rightarrow f'(x) = 0$ at these points.

Hence अतः $2\beta - 1 + \alpha = 0$

$8\beta + 2 + \alpha = 0$

$-6\beta - 3 = 0 \Rightarrow \beta = -\frac{1}{2}$ & $\alpha = 2$.

61. Let f(x) be a polynomial of degree four having extreme values at x = 1 and x = 2. If $\lim_{x\to 0}\left[1+\frac{f(x)}{x^2}\right] = 3$, then f(2) is equal to : **[JEE(Main)2015]**

माना f(x) घात 4 का एक बहुपद है जिसके x = 1 तथा x = 2 पर चरम मान है। यदि $\lim_{x\to 0}\left[1+\frac{f(x)}{x^2}\right] = 3$ है, तो f(2) बराबर है : **[JEE(Main)2015]**

(A) –8 (B) –4 (C) 0 (D) 4

Ans. C

Sol. $f(x) = ax^4 + bx^3 + 2x^2$

$f'(1) = 0$ $\quad 4a + 3b + 4 = 0$

$f'(2) = 0$ $\quad 32a + 12b + 8 = 0$

$a = \frac{1}{2}, b = -2$

$f(x) = \frac{x^4}{2} - x^3 + 2x^2$

$f(2) = 8 - 16 \times 8 = 0$

62. Let $f(x) = x^2 + \frac{1}{x^2}$ and $g(x) = x - \frac{1}{x}$, $x \in R - \{-1, 0, 1\}$. If $h(x) = \frac{f(x)}{g(x)}$, then the local minimum value of h(x) is : **[JEE(Main)2018]**

माना $f(x) = x^2 + \frac{1}{x^2}$ तथा $g(x) = x - \frac{1}{x}$, $x \in R - \{-1, 0, 1\}$ हैं। यदि $h(x) = \frac{f(x)}{g(x)}$ है, तो h(x) का स्थानीय न्यूनतम मान है : **[JEE(Main)2018]**

(A) $-2\sqrt{2}$ (B) $2\sqrt{2}$ (C) 3 (D) –3

Ans. B

Sol. $h(x)=\dfrac{x^2+\dfrac{1}{x^2}}{1-\dfrac{1}{x}}$

$h(x)=\dfrac{t^2+2}{t}=t+\dfrac{2}{t}$ where $t=x-\dfrac{1}{x}$

$h'(x)=\left(1-\dfrac{2}{t^2}\right)\left(\dfrac{dt}{dx}\right)$

$h'(x)=\left(\dfrac{t^2-2}{t}\right)\left(1+\dfrac{1}{x^2}\right)$

Minima at $t=\sqrt{2}$

$h(x)|_{Min}=\dfrac{2+2}{\sqrt{2}}=2\sqrt{2}$

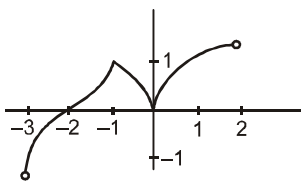
63. (a) The total number of local maxima and local minima of the function $f(x)=\begin{cases}(2+x)^3, & -3<x\le -1\\ x^{2/3}, & -1<x<2\end{cases}$ is :

(a) फलन $f(x)=\begin{cases}(2+x)^3, & -3<x\le -1\\ x^{2/3}, & -1<x<2\end{cases}$ के कुल स्थानीय उच्चिष्ठ तथा स्थानीय निम्निष्ठ की संख्या है :

(A) 0 (B) 1 (C) 2 (D) 3

Ans. C

Sol. $f(x)=\begin{cases}(2+x)^2, & -3<x\le -1\\ x^{2/3}, & -1<x<2\end{cases}$



one point of local maxima at $x=-1$

one point of local minima at $x=0$

$\therefore$ (C) is the correct option

GEOMETRY BASED QUESTIONS ON MAXIMA AND MINIMA

64. The sum of lengths of the hypotenuse and another side of a right angled triangle is given. The area of the triangle will be maximum if the angle between them is :

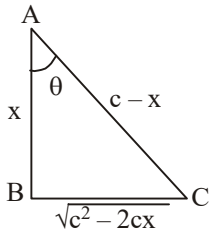
यदि किसी समकोण त्रिभुज में कर्ण तथा अन्य एक भुजा की लम्बाई का योग दिया गया है। तब त्रिभुज का क्षेत्रफल अधिकतम होगा यदि इनके मध्य का कोण है :

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{3}$ (D) $\frac{5\pi}{12}$

Ans. C

Sol. $A = \frac{x\sqrt{c^2 - 2cx}}{2}$

$f(x) = 4A^2 = x^2(c^2 - 2cx)$



65. The length of the shortest path that begins at the point (2, 5), touches the x-axis and then ends at a point on the circle $x^2 + y^2 + 12x - 20y + 120 = 0$, is :

न्यूनतम पथ की लम्बाई जिसका प्रारम्भिक बिन्दु (2, 5) है तथा x-अक्ष को स्पर्श करता है तथा वृत्त $x^2 + y^2 + 12x - 20y + 120 = 0$ के किसी बिन्दु पर अन्त होता है :

(A) 13 (B) $4\sqrt{10}$ (C) 15 (D) $6 + \sqrt{89}$

Ans. A

Sol. Circle with centre (–6, 10) and radius = $\sqrt{36 + 100 - 120} = 4$

Now let (a, 0) be a point on the x-axis.

If y is the distance from A to P and P to M

$$y = \sqrt{(a-2)^2 + 25} + \sqrt{(a+6)^2 + 100} - 4$$

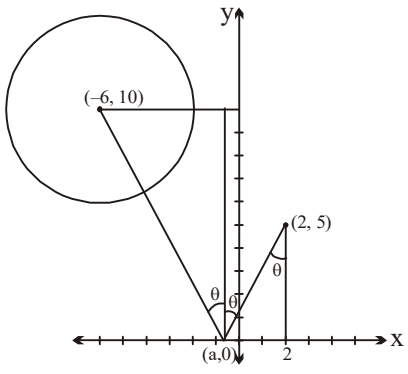
$$\frac{dy}{da} = \frac{2(a-2)}{2\sqrt{(a-2)^2 + 25}} + \frac{2(a+6)}{2\sqrt{(a+6)^2 + 100}}$$

$\frac{dy}{da}$ can be zero only if $a - 2 > 0$

and $a + 6 < 0$ not possible

or $a - 2 < 0$ and $a + 6 > 0$

hence $a \in (-6, 2)$



solving $\frac{dy}{da} = 0$, gives $a = 10$ (rejected) or $a = -\frac{2}{3}$

hence $y_{min} = \sqrt{\frac{64}{9} + 25} + \sqrt{\frac{256}{9} + 100} - 4$

$= \frac{17}{3} + \frac{\sqrt{1156}}{3} - 4 = \frac{17}{3} + \frac{34}{3} - 4 = 17 - 4 = 13$ **Ans.**

Aliter: $\frac{a+2}{5} = \frac{6-a}{10} \Rightarrow 10a + 20 = 30 - 5a \Rightarrow 15a = 10 \Rightarrow a = \frac{2}{3}$

Hence P $\left(\frac{-2}{3}, 0\right)$ for find $y_{min.}$

66. A triangular park is enclosed on two sides by a fence and on the third side by a straight river bank. The two sides having fence are of same length x. The maximum area enclosed by the park is :

एक त्रिभुजाकार पार्क दो तरफ से बाड़े (fence) द्वारा घिरा है तथा तीसरी ओर एक नदी के सीधे किनारे द्वारा घिरा है। यदि बाडे द्वारा घिरी भुजाओं की समान लम्बाई x है, तो पार्क का अधिकतम क्षेत्रफल है :

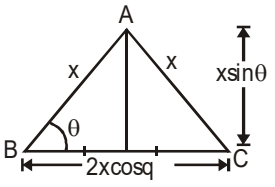
(A) $\sqrt{\frac{x^3}{8}}$ (B) $\frac{1}{2}x^2$ (C) πx^2 (D) $\frac{3}{2}x^2$

Ans. B

Sol. A triangular park

$\Delta = \frac{1}{2}(2x\cos\theta)(x\sin\theta)$

$= \frac{1}{2}x^2\sin2\theta$



$\boxed{\Delta_{max.} = \frac{x^2}{2}}$

67. A right triangle is drawn in a semicircle of radius $\frac{1}{2}$ with one of its legs along the diameter. The maximum area of the triangle is :

$\frac{1}{2}$ त्रिज्या वाले अर्द्धवृत्त में एक समकोण त्रिभुज बनाया जाता है इसप्रकार की उसका आधार व्यास पर है तब त्रिभुज का अधिकतम क्षेत्रफल है :

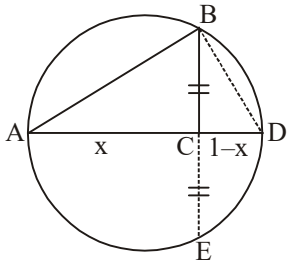
(A) $\frac{1}{4}$ (B) $\frac{3\sqrt{3}}{32}$ (C) $\frac{3\sqrt{3}}{16}$ (D) $\frac{1}{8}$

Ans. B

Sol. (BC)(CE) = x (1 – x) (property of circle) [AC = x ; CD = 1 – x] [12 & 13th test (09-10-2005)]

but BC = CE

$\therefore$ BC = $\sqrt{x(1-x)}$



$A = \frac{x\sqrt{x-x^2}}{2}$. Now maximise A.

A_{max} occurs at $x = \frac{3}{4}$

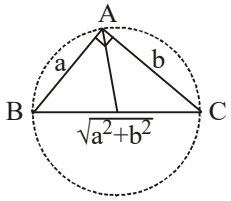
68. Two sides of a triangle are to have lengths 'a' cm & 'b' cm. If the triangle is to have the maximum area, then the length of the median from the vertex containing the sides 'a' and 'b' is :

किसी त्रिभुज की दो भुजाओं की लम्बाईयाँ 'a' cm तथा 'b' cm है। यदि त्रिभुज का क्षेत्रफल अधिकतम है तब भुजाओं 'a' तथा 'b' को मिलने वाले शीर्ष से डाली गई माध्यिका की लम्बाई है :

(A) $\frac{1}{2}\sqrt{a^2+b^2}$ (B) $\frac{2a+b}{3}$ (C) $\sqrt{\frac{a^2+b^2}{2}}$ (D) $\frac{a+2b}{3}$

Ans. A

Sol. A = (1/2)ab sinθ $\Rightarrow$ A is maximum if $\theta = 90^0$



69. A spherical balloon is filled with 4500π cubic meters of helium gas. If a leak in the balloon causes the gas to escape at the rate of 72π cubic meters per minute, then the rate (in meters per minute) at which the radius of the balloon decreases 49 minutes after the leakage began is : **[AIEEE-2012]**

एक गोलाकर गुब्बारे में 4500π घन मी. हीलियम गैस भरी गई है। यदि गुब्बारे में छेद के कारण 72π घन मी. प्रति मिनट की दर से गैस रिस रही है, तो गैस रिसने के प्रारम्भ होने के 49 मिनट बाद, जिस दर से गुब्बारे की त्रिज्या घट रही है (मीटर प्रति मिनट में), हैं : **[AIEEE-2012]**

(A) 9/2 (B) 9/7 (C) 7/9 (D) 2/9

Ans. D

Sol. $V = \frac{4}{3}\pi r^3$

Initially $r = 4500\,\pi$, $r = r_0$

$$4500\,\pi = \frac{4}{3}\pi\ r_0^3 \ \Rightarrow\ \boxed{r_0 = 15\text{m}}$$

Now $\frac{dV}{dt} = \frac{4}{3}\pi\,(3r^2)\frac{dr}{dt}$

$-72\,\pi = 4\pi\,r^2\frac{dr}{dt} \Rightarrow \frac{dr}{dt} = \frac{-18}{r^2}$ .....(i)

$\int r^2 dr = -\int 18\ dt \ \Rightarrow \frac{r^3}{3} = -18\,t + C$

At $t = 0$, $r = 15$ m

So, $\frac{(15)^3}{3} = -18(0) + C \Rightarrow C = 1125$

$\Rightarrow r^3 = -54t + 3375$ .....(ii)

At time $t = 49$ min $\quad r = 9$ m

from eq. (i)

$$\left(\frac{dr}{dt}\right)_{t=49} = \frac{-18}{(9)^2} = -2/9$$

(Negative sign shows decrement in radii)

70. A wire of length 2 units is cut into two parts which are bent respectively to form a square of side = x units and a circle of radius = r units. If the sum of the areas of the square and the circle so formed is minimum, then :

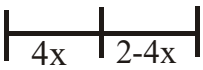
[JEE(Main)2016]

2 इकाई लम्बी एक तार को दो भागों में काट कर उन्हें क्रमशः x इकाई भुजा वाले वर्ग तथा r इकाई त्रिज्या वाले वृत्त के रूप में मोड़ा जाता है। यदि बनाये गये वर्ग तथा वृत्त के क्षेत्रफलों का योग न्यूनतम है, तो : **[JEE(Main)2016]**

(A) $x = 2r$ (B) $2x = r$ (C) $2x = (\pi + 4)r$ (D) $(4 - \pi)x = \pi r$

Ans. A

Sol.



Perimeter of square = 4x and circumference of circle = 2 – 4x

$2\pi r = (2 - 4x)$

$$A = x^2 + \pi\left(\frac{2-4x}{2\pi}\right)^2$$

$$\frac{dA}{dx} = 0$$

$$2x = \frac{2}{\pi}(2 - 4x)$$

$$2x = \frac{2(2\pi r)}{\pi}$$

$x = 2r$

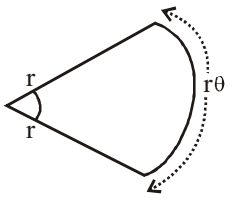
71. Twenty meters of wire is available for fencing off a flower-bed inthe form of a circular sector. Then the maximum area (in sq. m) of the flower-bed, is : **[JEE(Main)2017]**

एक फूलों की क्यारी, जो एक वृत के त्रिज्य खंड के रूप में है, की घेराबंदी करने के लिए बीस मीटर तार उपलब्ध है। तो फूलों की क्यारी का अधिकतम क्षेत्रफल (वर्ग मी. में) है : **[JEE(Main)2017]**

(A) 25 (B) 30 (C) 12.5 (D) 10

Ans. A

Sol.



$r + r + r\theta = 20$

$$\theta = \frac{20 - 2r}{r}$$

$$A = \frac{1}{2}r^2\theta$$

$$A = \frac{r(20 - r)}{r}$$

$$\frac{dA}{dr} = 0 \Rightarrow r = 5$$

$\Rightarrow \theta = 2$

$$\text{Area} = \frac{1}{2}r^2\theta = 25$$

POINT OF INFLECTION & CONCAVITY OF GRAPH

72. If the function $f(x) = x^4 + bx^2 + 8x + 1$ has a horizontal tangent and a point of inflection for the same value of x then the value of b is equal to :

यदि $f(x) = x^4 + bx^2 + 8x + 1$ की क्षैतिज स्पर्श रेखा और नति परिवर्तन (inflection point) बिन्दु दोनों एक ही बिन्दु x पर मिलते है तब b =

(A) – 1 (B) 1 (C) 6 (D) – 6

Ans. D

Sol. $f'(x) = 0$ and $f''(x) = 0$ for the same $x = x_1$ (say)

now $f'(x) = 4x^3 + 2bx + 8$

$f'(x_1) = 2[2x_1^3 + bx_1 + 4] = 0$ ....(1)

$f''(x_1) = 2[6x_1^2 + b] = 0$ ....(2)

from (2) $b = -6x_1^2$

substituting this value of b in (1)

$2x_1^3 + (-6x_1^3) + 4 = 0 \Rightarrow 4x_1^3 = 4 \Rightarrow x_1 = 1$. Hence $b = -6$

73. The function $f(x) = 2x^3 + \alpha x^2 + \beta x + \gamma$, where $\alpha, \beta, \gamma \in R$ has local minimum at $P\left(\log_3 t^2, f\left(\log_3 t^2\right)\right)$ and local maximum at $Q\left(\log_3 t, f\left(\log_3 t\right)\right)$. If $R\left(\frac{5}{2}, f\left(\frac{5}{2}\right)\right)$ is the point of inflection, then 't' is equal to :

यदि फलन $f(x) = 2x^3 + \alpha x^2 + \beta x + \gamma$, जहाँ $\alpha, \beta, \gamma \in R$ बिन्दु $P\left(\log_3 t^2, f\left(\log_3 t^2\right)\right)$ और $Q\left(\log_3 t, f\left(\log_3 t\right)\right)$ पर क्रमशः स्थानीय निम्नष्ठ और स्थानीय उच्चिष्ठ मान देता है। यदि $R\left(\frac{5}{2}, f\left(\frac{5}{2}\right)\right)$ फलन का नतिपरिवर्तन बिन्दु (Point of inflection) है तब 't' =

(A) $3^{2/5}$ (B) $3^{1/5}$ (C) $3^{5/3}$ (D) $3^{3/5}$

Ans. C

Sol. We have $f'(x) = 6x^2 + 2\alpha x + \beta \Rightarrow f'(\log_3 t^2) = 0$ and $f'(\log_3 t) = 0$

Now $\log_3 t^2 + \log_3 t = -\frac{2\alpha}{6}$ (sum of the roots)

$\Rightarrow 3\log_3 t = \frac{-\alpha}{3}$ ....(i)

Also, $f''(x) = 12x + 2\alpha$

$f''\left(\frac{5}{2}\right) = 0 \Rightarrow 12\left(\frac{5}{2}\right) + 2\alpha = 0 \Rightarrow \alpha = -15$

$\therefore 3\log_3 t = \frac{15}{3} \Rightarrow t = 3^{5/3}$

74. If the point (1, 3) serves as the point of inflection of the curve $y = ax^3 + bx^2$ then the value of 'a' and 'b' are :

(A) $a = 3/2$ and $b = -9/2$ (B) $a = 3/2$ and $b = 9/2$

(C) $a = -3/2$ and $b = -9/2$ (D) $a = -3/2$ and $b = 9/2$

यदि बिन्दु (1, 3) वक्र $y = ax^3 + bx^2$ पर नति परिवर्तन की तरह कार्य करता है, तो 'a' और 'b' का मान है :

(A) $a = 3/2$ तथा $b = -9/2$ (B) $a = 3/2$ तथा $b = 9/2$

(C) $a = -3/2$ तथा $b = -9/2$ (D) $a = -3/2$ तथा $b = 9/2$

Ans. D

Sol. $y = ax^3 + bx^2$

$y' = 3ax^2 + 2bx$

$y'' = 6ax + 2b$

for point of inflection $y'' = 0$

$x = \frac{-b}{3a}$

$3a + b = 0$ .....(i) (as $x = 1$)

point satisfy the curve also, so

$3 = a + b$ .....(ii)

from (i) & (ii)

$a = -\frac{3}{2}, \; b = \frac{9}{2}$

SUBJECTIVE / INTEGER TYPE QUESTIONS

TANGENT AND NORMAL

TANGENT AND NORMAL AT A POINT

1. Find the equation of the normal to the curve $y = (1+x)^y + \sin^{-1}(\sin^2 x)$ at $x = 0$.

वक्र $y = (1+x)^y + \sin^{-1}(\sin^2 x)$ के बिन्दु $x = 0$ पर अभिलम्ब का समीकरण ज्ञात कीजिये।

Ans. $x + y - 1 = 0$

Sol. Given curve is $y = (1+x)y + \sin^{-1}(\sin^2 x)$

$y = e^{y\log(1+x)} + \sin^{-1}(\sin^2 x)$

$$\Rightarrow \frac{dy}{dx} = e^{y\log(1+x)}\left(\frac{y}{1+x} + \log(1+x)\frac{dy}{dx}\right) + \frac{\sin 2x}{\sqrt{1-\sin^4 x}}$$

when $x = 0$, then $y = 1$.

So, the point is $(0, 1)$

$$\left(\frac{dy}{dx}\right)_{(0,1)} = 1$$

So, slope of the normal $= -1$

Hence, the equation of the normal is

$y - 1 = -1(x - 0) = -x$

$\Rightarrow x + y = 1$

2. Find the point of intersection of the tangents drawn to the curve $x^2y = 1 - y$ at the points where it is intersected by the curve $xy = 1 - y$.

वक्र $x^2y = 1 - y$ एवं वक्र $xy = 1 - y$ के प्रतिच्छेदन बिन्दुओं पर वक्र $x^2y = 1 - y$ पर स्पर्श रेखाओं का प्रतिच्छेदन बिन्दु ज्ञात कीजिये।

Ans. $(0, 1)$

Sol. Points of intersection of curve $x^2y = xy$ are

$(0, 1), (1, \frac{1}{2})$

The equation of tangent at $(0, 1)$

$2xy + x^2y' = -y'$

$y' = 0$

equation is $y - 1 = 0 \Rightarrow y = 1$

equation of tangent at $(1, \frac{1}{2})$ is $2xy + x^2y' = -y'$

$$1 + y' = -y' \Rightarrow y' = -\frac{1}{2}$$

$$y - \frac{1}{2} = -\frac{1}{2}(x - 1)$$

Put $y = 1$ $\quad \frac{1}{2} = -\frac{x}{2} + \frac{1}{2}$

$x = 0$

Point of intersection of tangent is (0, 1)

3. Find all the lines that pass through the point (1, 1) and are tangent to the curve represented parametrically as $x = 2t - t^2$ and $y = t + t^2$.

बिन्दु (1, 1) से गुजरने वाले सभी रेखाऐं ज्ञात कीजिये जो एक वक्र पर स्पर्श रेखाऐं है, जिसका प्राचलिक समीकरण $x = 2t - t^2$ तथा $y = t + t^2$ से दिया जाता है।

Ans. $x = 1$ when $t = 1$, $m \to \infty$; $5x - 4y = 1$ if $t \neq 1$, $t = 1/3$

Sol. Let the tangent at (α, β)

Now, $\frac{dy}{dx} = \frac{2t+1}{2-2t}$

$$\Rightarrow \frac{\beta - 1}{\alpha - 1} = \frac{2t+1}{2-2t}$$

$$\Rightarrow \frac{t + t^2 - 1}{2t - t^2 - 1} = \frac{2t+1}{2-2t}$$

$$\Rightarrow \frac{t^2 + t - 1}{t^2 - 2t + 1} = \frac{2y+1}{2(t-1)}$$

$$\Rightarrow \frac{t^2 + t - 1}{(t-1)^2} = \frac{2t+1}{2(t-1)}$$

$$\frac{t^2 + t - 1}{(t-1)} = \frac{2t+1}{2}$$

$\Rightarrow 2t^2 + 2t - 2 = 2y^2 - t - 1$

$\Rightarrow 2t - 2 = -t - 1$

$\Rightarrow 3t = 1$

$\Rightarrow t = \frac{1}{3}$

Now, $\frac{dy}{dx} = \frac{2t+1}{2-2t} = \frac{2+3}{6-2} = \frac{5}{4}$

Thus, the equation of the tanget is

$y - 1 = \frac{5}{4}(x - 1)$

$\Rightarrow 4y - 4 = 5(x - 1)$

$\Rightarrow 5x - 4y = 1$

4. A function is defined parametrically by the equations

$$f(t) = x = \begin{cases} 2t + t^2 \sin\frac{1}{t} & \text{if } t \neq 0 \\ 0 & \text{if } t = 0 \end{cases} \text{ and } g(t) = y = \begin{cases} \frac{1}{t}\sin t^2 & \text{यदि } t \neq 0 \\ o & \text{यदि } t = 0 \end{cases}$$

Find the equation of the tangent and normal at the point for $t = 0$ if exist.

एक फलन को प्राचलिक समीकरणों से परिभाषित किया गया

$$f(t) = x = \begin{cases} 2t + t^2 \sin\frac{1}{t} & \text{यदि } t \neq 0 \\ 0 & \text{यदि } t = 0 \end{cases} \text{ तथा } g(t) = y = \begin{cases} \frac{1}{t}\sin t^2 & \text{यदि } t \neq 0 \\ o & \text{यदि } t = 0 \end{cases}$$

समीकरण के $t = 0$ पर स्पर्श रेखा एवं अभिलम्ब का समीकरण लिखिए (यदि अस्तित्व में हो)

Ans. $T : x - 2y = 0$; $N : 2x + y = 0$

Sol. At $t = 0$ the point is origin

$\frac{dx}{dt} = \lim_{t\to 0} \frac{2t + t^2 \sin 1/t - 0}{t} = 2$

$\frac{dy}{dt} = \lim_{t\to 0} \frac{\frac{1}{t}\sin t^2}{t} = 1$

$$\frac{dy}{dx}=\frac{1}{2}$$

equation of tangent is $y-0=\frac{1}{2}(x-0)$

equation of normal is $y-0=-2(x-0)$

5. Prove that the segment of the normal to the curve $x=2a\sin t+a\sin t\cos^2 t$; $y=-a\cos^3 t$ contained between the co-ordinate axes is equal to 2a.

सिद्ध कीजिये कि वक्र $x=2a\sin t+a\sin t\cos^2 t$; $y=-a\cos^3 t$ पर अभिलम्ब द्वारा अक्षों के मध्य रेखा खण्ड 2a है।

Ans.

Sol. $$\frac{dy}{dx}=\frac{dy/dt}{dx/dt}=\frac{3a\cos^2 t\sin t}{2a\cos t+a\cos^3 t-2a\sin^2 t\cos t}$$

$$=\frac{3a\cos^2 t\sin t}{2a\cos t\left(1-\sin^2 t\right)+a\cos^3 t}$$

$$=\frac{3a\cos^2 t\sin t}{2a\cos^2 t+a\cos^3 t}$$

$$=\frac{3a\cos^2 t\sin t}{3a\cos^3 t}$$

$\frac{dy}{dx}=\tan t$, equation of normal is

$$y+a\cos^3 t=-\frac{\cot t}{\sin t}\left(x-2a\sin t-a\sin t\cos^2 t\right)$$

$$\therefore y\sin t+a\cos^3 t\sin t=-x\cos t+2a\sin t\cot t+a\sin t\cos^3 t$$

$$x\cot t+y\sin t=2a\sin t\cos t$$

It meets axis at A(2a sin t, 0) and B(0, 2a cost)

Then AB = 2a.

6. Determine a differentiable function $y=f(x)$ which satisfies $f'(x)=[f(x)]^2$ and $f(0)=-\frac{1}{2}$. Find also the equation of the tangent at the point where the curve crosses the y-axis.

अवकलनीय फलन $y = f(x)$ ज्ञात कीजिये यदि $f'(x) = [f(x)]^2$ तथा $f(0) = -\frac{1}{2}$ है। साथ ही उस बिन्दु पर स्पर्श रेखा बताइए जहाँ वक्र y-अक्ष को प्रतिच्छेद करता है।

Ans. $f(x) = -\frac{1}{x+2}$; $x - 4y = 2$

Sol.

7. The curve $y = ax^3 + bx^2 + cx + 5$, touches the x-axis at $P(-2, 0)$ & cuts the y-axis at a point Q where its gradient is 3. Find a, b, c.

वक्र $y = ax^3 + bx^2 + cx + 5$, x-अक्ष को $P(-2, 0)$ पर स्पर्श करता है तथा y-अक्ष को उस बिन्दु Q पर प्रतिच्छेद करता है जहाँ उसका ढाल 3 है। a, b, c का मान ज्ञात कीजिये।

Ans. $a = -1/2$; $b = -3/4$; $c = 3$

Sol. $y(x) = ax^3 + bx^2 + cx + 5$ is passing through (–2, 0) then $8a - 4b + 2c = 5$ _____(1)

$y'(x) = 3ax^2 + 2bx + c$ touches x-axis at (–2, 0)

$12a - 4b + c = 0$ _____(2)

again, for $x = 0$, $y(x) = 3$

$c = 3$ _____(3)

Solving eq. (1), (2) and (3) $a = -\frac{1}{2}$, $b = -\frac{3}{4}$

ULTIMATE CONCEPT

8. The tangent to $y = ax^2 + bx + \frac{7}{2}$ at (1, 2) is parallel to the normal at the point (–2, 2) on the curve $y = x^2 + 6x + 10$. Find the value of a and b.

$y = ax^2 + bx + \frac{7}{2}$ के (1, 2) पर स्पर्श रेखा वक्र $y = x^2 + 6x + 10$ के बिन्दु (–2, 2) पर अभिलम्ब के समानान्तर है। a तथा b के मान ज्ञात कीजिये।

Ans. $a = 1, b = \frac{-5}{2}$

Sol. $2 = a + b + \frac{7}{2}$ _____(1)

$\frac{dy}{dx} = 2x + 6$ at –2, 2

$\Rightarrow \frac{dy}{dx} = 2ax + b$

$\Rightarrow m = 2a + b$

$2a + b = -\frac{1}{2}$ _____(2)

from (1) and (2) $a = 1,\ b = -\frac{5}{2}$

9. Find all the tangents to the curve $y = \cos(x + y)$, $-2\pi \le x \le 2\pi$, that are parallel to the line $x + 2y = 0$.

वक्र $y = \cos(x + y)$, $-2\pi \le x \le 2\pi$ पर सभी स्पर्श रेखाओं का समीकरण लिखिए जो रेखा $x + 2y = 0$ के समान्तर है।

Ans. $x + 2y = \pi/2$ & $x + 2y = -3\pi/2$

Sol. Slope $= -\frac{1}{2}$

$\frac{dy}{dx} = -\sin(x + y)(1 + y')$

$\Rightarrow -\frac{1}{2} = -\sin(x + y)(1 - \frac{1}{2})$

$\Rightarrow \sin(x + y) = 1$

$\Rightarrow \cos(x + y) = 0 \quad \Rightarrow \quad y = 0$

$\cos x = 0$

$\therefore \quad x = \frac{\pi}{2}, \frac{3\pi}{2}, -\frac{\pi}{2}, -\frac{3\pi}{2}$

$\sin x = 1$ is possible for $x = \frac{\pi}{2}$ or $-\frac{3\pi}{2}$

Equation are : $y - 0 = -\frac{1}{2}\left(x - \frac{\pi}{2}\right)$

and $y-0=-\frac{1}{2}\left(x+\frac{3\pi}{2}\right)$

10. Tangent at a point P_1 [other than $(0,0)$] on the curve $y=x^3$ meets the curve again at P_2. The tangent at P_2 meets the curve at P_3 and so on. Show that the abscissae of $P_1, P_2, P_3, \ldots\ldots P_n$, form a GP. Also find the ratio $\frac{\text{area}\,(P_1P_2P_3)}{\text{area}\;\;(P_2P_3P_4)}$.

वक्र $y=x^3$ के बिन्दु P_1 [$(0,0)$ के अलावा] पर स्पर्श रेखा वक्र को P_2 पर मिलती है तथा P_2 पर स्पर्श रेखा वक्र को P_3 पर मिलती है, इस प्रकार आगे क्रम चलता है। सिद्ध कीजिये की $P_1, P_2, P_3, \ldots\ldots P_n$ के भुज गुणोत्तर श्रेणी में है। साथ ही अनुपात $\frac{\text{area}\,(P_1P_2P_3)}{\text{area}\;\;(P_2P_3P_4)}$ भी ज्ञात करें।

Ans. 1/16

Sol. Let any point P_1 on

$y=x^3$ be (h, h^3).

Then tangent at P_1 is

$y-h^3=3h^2(x-h)$ _____(1)

It meets $y=x^3$ at P_2.

On putting the value of y in Eq. (1)

$x^3-h^3=3h^2(x-h)$

$\Rightarrow (x-h)(x^2+xh+h^2)=3h^2(x-h)$

$\Rightarrow x^2+xh+h^2=3h^2$ or $x=h$

$\Rightarrow x^2+xh-2h^2=0$

$\Rightarrow (x-h)(x+2h)=0$

$\Rightarrow x=h$ or $x=-2h$

Therefore, $x=-2h$ is the point P_2,

Which implies $y=-8h^3$

Hence, point $P_2\equiv(-2h,-8h^3)$

Again, tangent at P_2 is

$$y + 8h^3 = 3(-2h)^2 (x + 2h)$$

It meets $y = x^3$ at P_3

$$\Rightarrow x^3 + 8h^3 = 12h^2 (x + 2h)$$

$$\Rightarrow x^2 - 2hx - 8h^2 = 0$$

$$\Rightarrow (x + 2h)(x - 4h) = 0$$

$$\Rightarrow x = 4h$$

$$\Rightarrow y = 64h^3$$

Therefore, $P_3 \equiv (4h, 64h^3)$

Similarly, we get $P_4 \equiv (-8h, -8^3 h^3)$

Hence, the abscissa are h, –2h, 4h, –8h, ... which form a GP.

Let $D' = \Delta P_1P_2P_3$ and $D'' = \Delta P_2P_3P_4$

$$\frac{D'}{D''} = \frac{\Delta P_1P_2P_3}{\Delta P_2P_3P_4} = \frac{\frac{1}{2}\begin{vmatrix} h & h^3 & 1 \\ -2h & -8h^3 & 1 \\ 4h & 64h^3 & 1 \end{vmatrix}}{\frac{1}{2}\begin{vmatrix} -2h & -8h^3 & 1 \\ 4h & 64h^3 & 1 \\ -8h & -512h^3 & 1 \end{vmatrix}}$$

$$= \frac{\frac{1}{2}\begin{vmatrix} h & h^3 & 1 \\ -2h & -8h^3 & 1 \\ 4h & 64h^3 & 1 \end{vmatrix}}{\frac{1}{2}\times(-2)\times(-8)\begin{vmatrix} h & h^3 & 1 \\ -2h & -8h^3 & 1 \\ 4h & 64h^3 & 1 \end{vmatrix}}$$

$\Rightarrow C_2 \frac{D'}{D''} = \frac{1}{16}$ which is the required ratio.

11. Find the gradient of the line passing through the point (2, 8) and touching the curve $y = x^3$.

बिन्दु (2, 8) से गुजरने वाली रेखा जो वक्र $y = x^3$ को स्पर्श करती है, की प्रवणता ज्ञात कीजिये।

Ans. 3, 12

Sol. $y' = 3x^2$

$$3x_1^2 = \frac{x_1^3 - 8}{x_1 - 2} \Rightarrow x_1 = -1 \text{ or } 2$$

$$y' = 3 \text{ or } y' = 12$$

12. The graph of a certain function f contains the point (0, 2) and has the property that for each number 'p' the line tangent to $y = f(x)$ at $(p, f(p))$ intersect the x-axis at $p + 2$. Find $f(x)$.

किसी फलन f के आरेख पर बिन्दु (0, 2) स्थित है तथा उसका एक गुणधर्म है कि प्रत्येक संख्या 'p' के लिये $y = f(x)$ पर बिन्दु $(p, f(p))$ पर स्पर्श रेखा x-अक्ष को $p + 2$ पर प्रतिच्छेद करती है। $f(x)$ ज्ञात कीजिए।

Ans. $2e^{-x/2}$

Sol. point (p,f(p)) slop = $f'(p)$

Eq. : $y - f(e) = f'(p)(x - p)$

$y = 0 \quad x = p + 2 \qquad -f(p) = f'(p)(p + 2 - p)$

$$-f(p) = f'(p).2$$

$$-\int \frac{f'(p)}{f(p)}.dp^2 \quad \int \frac{1}{2} dp \, s$$

$$-ln\, f(p) = \frac{p}{2} + c$$

$$-ln\, 2 = c$$

$$c = -ln2$$

$$-\ell n f(p) = \frac{p}{2} - \ell n 2$$

$$\ell n 2 - \ell n f(p) = \frac{p}{2}$$

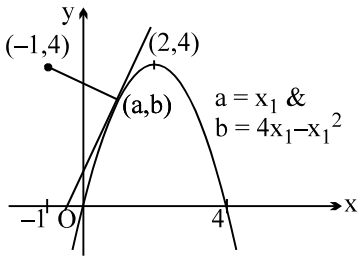
MINIMUM DISTANCE BETWEEN TWO CURVES*s*

13. Let P(a, b) be a point on the parabola $y = 4x - x^2$ and is the point nearest to the point A(–1, 4). Find (a + b).

यदि वक्र $y = 4x - x^2$ पर बिन्दु P(a, b) इस प्रकार से है कि यह A(–1, 4) से न्यूनतम दूरी पर है तो (a + b) ज्ञात कीजिये।

Ans. 4

Sol. Let the point P be $a = x_1$ and $b = 4x_1 - x_1^2$



$$\frac{4x_1 - x_1^2 - 4}{x_1 + 1} = -\frac{1}{\left(\frac{dy}{dx}\right)_{x=x_1}} = -\frac{1}{4 - 2x_1}$$

$(2x_1 - 4)(4x_1 - x_1^2 - 4) = x_1 + 1$

$-2x_1^3 + 12x_1^2 - 25x_1 + 15 = 0$

$2x_1^3 - 12x_1^2 + 25x_1 - 15 = 0$

$\therefore \quad x_1 = 1 \; ; \quad y_1 = 4 - 1 = 3$

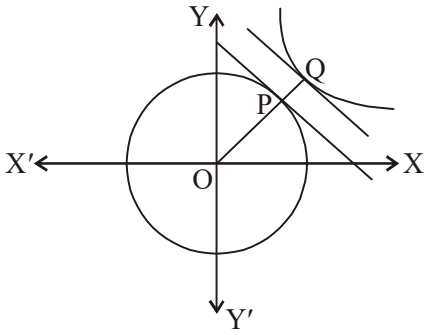
$\therefore \quad (a, b) = (1, 3) \Rightarrow \quad a + b = 4$

14. Find the shortest distance between curves $xy = 9$ and $x^2 + y^2 = 1$

वक्रों $xy = 9$ और $x^2 + y^2 = 1$ के बीच न्यूनतम दूरी ज्ञात कीजिए।

Ans. $3\sqrt{2} - 1$

Sol.



Given curves are $x^2 + y^2 = 2$ and $xy = 9$

Since both the tangents at P and Q are parallel, so their slopes are same

$$-\frac{x}{y} = -\frac{9}{x^2}$$

$\Rightarrow x^3 = 9y$

$\Rightarrow x^3 = 9.\frac{9}{x} = \frac{81}{x}$

$\Rightarrow x^4 = 81 \Rightarrow x = 3$

when x = 3, then y = 3

Thus, the point Q = (3, 3)

Therefore, the shortest distance

= PQ

= OQ – OP

$= \sqrt{(9+9)} - \sqrt{2}$

$= 3\sqrt{2} - \sqrt{2}$

$= 2\sqrt{2}$

15. Find the point on hyperbola $3x^2 - 4y^2 = 72$ which is nearest to the straight line $3x + 2y + 1 = 0$

अतिपरवलय $3x^2 - 4y^2 = 72$ पर वह बिन्दु ज्ञात कीजिए जो कि सरल रेखा $3x + 2y + 1 = 0$ से न्यूनतम दूरी पर है।

Ans. (– 6, 3)

Sol. $3x^2 - 4y^2 = 72$

$m = -\frac{3}{2}$

$6x - 8y\frac{dy}{dx} = 0$

$\frac{dy}{dx} = \frac{3x}{4y}$

$\frac{3x_0}{4y_0} = -\frac{3}{2}$

$x_0 = -2y_0$

$8y_0^2 = 72$

$y = \pm 3$

$3x + 2y + c = 0$

$c = -3x_0 - 2y_0$

$|c - 1|$ is min *i* $\mu m = 4y_0$

$|c - 1|$ is minimum atc = 12 and $y_0 = 3$

$x_0 + y_0 = 3 + 3 \,.\, (-2) = -3$

ANGLE OF INTERSECTION BETWEEN TWO CURVES

16. Find the angle of intersection of the curves $y = 2\sin^2 x$ and $y = \cos 2x$.

वक्र $y = 2\sin^2 x$ और वक्र $y = \cos 2x$ के प्रतिच्छेद कोण का मान ज्ञात कीजिये।

Ans. $\frac{\pi}{3}$

Sol. Angle between two curves is basically the angle between there tangents at the point of intersection.

Point of intersection of the curves will be $\cos 2x = 2\sin^2 x$

$\Rightarrow \cot^2 x - 1 = 2$

$\Rightarrow \cot x = \sqrt{3}$

$x = \frac{\pi}{6}$

Slope of both the curve at the point of intersection is given by

$m_1 = -3\sin 2x = -\sqrt{3}$

$m_2 = 2\sin x \cos x = \sqrt{3}$

Applying the formula, $\tan\theta = \left|\frac{m_1 - m_2}{1 + m_1 m_2}\right|$

This gives $\theta = \frac{\pi}{3}$

17. Find the angle of intersection of the curves $y^2 = 4ax$ and $x^2 = 32ay$.

वक्रो $y^2 = 4ax$ और $x^2 = 32ay$ के प्रतिच्छेद कोण का मान ज्ञात कीजिये।

Ans. $\frac{\pi}{2}, \tan^{-1}\left(\frac{3}{5}\right)$

Sol. $y^2 = 4x$ and $x^2 = 32y$ intersect at (16, 8) at an angle θ

Now slope of the tangent to $y^2 = 4x$ at (16, 8) is given by

$$m_1 = \left(\frac{dy}{dx}\right)(16,8) = \left(\frac{4}{2y}\right)(16,8) = \frac{1}{4}$$

Again slope of tangent to $x^2 = 32y$ at (16, 8)

is given by $m_2 = \left(\frac{dy}{dx}\right)(16,8) = \left(\frac{2x}{32}\right)(16,8) = 1$

We know $\tan\theta = \frac{m_2 - m_1}{m_2 + m_1}$

$$= \frac{1-\frac{1}{4}}{1+\frac{1}{4}} = \frac{3}{5}$$

$$\therefore \boxed{\theta = \tan^{-1}\left(\frac{3}{5}\right)}$$

18. Show that the curves $\frac{x^2}{a^2+K_1} + \frac{y^2}{b^2+K_1} = 1$ and $\frac{x^2}{a^2+K_2} + \frac{y^2}{b^2+K_2} = 1$ intersect orthogonally.

सिद्ध कीजिये कि वक्र $\frac{x^2}{a^2+K_1} + \frac{y^2}{b^2+K_1} = 1$ तथा $\frac{x^2}{a^2+K_2} + \frac{y^2}{b^2+K_2} = 1$ लम्बवत् प्रतिच्छेद करते हैं।

Ans.

Sol. $\frac{x^2}{a^2+k_1} + \frac{y^2}{b^2+k_1} = 1$ _____(1)

$\frac{x^2}{a^2+k_2} + \frac{y^2}{b^2+k_2} = 1$ _____(2)

$$m_1 \Rightarrow \frac{dy}{dx} = -\left(\frac{b^2+k_1}{a^2+k_1}\right)\frac{x}{y} \quad \text{_____(3) from}$$

(1)

$$m_1 \Rightarrow \frac{dy}{dx} = -\left(\frac{b^2+k_2}{a^2+k_2}\right)\frac{x}{y} \quad \text{_____(4) from}$$

(2)

$$m_1.m_2 \Rightarrow \frac{\left(b^2+k_1\right)\left(b^2+k_2\right)x^2}{\left(a^2+k_1\right)\left(a^2+k_2\right)y^2} \quad \text{_____(5)}$$

from eq(1) – eq(2)

$$x^2\left(\frac{1}{a^2+k_1}-\frac{1}{a^2+k_2}\right)+y^2\left(\frac{1}{b^2+k_1}-\frac{1}{b^2+k_1}\right)=0$$

$$\frac{\left(b^2+k_1\right)\left(b^2+k_2\right)}{\left(a^2+k_1\right)\left(a^2+k_2\right)} = \frac{-y^2}{x^2} \quad \text{_____(6)}$$

Putting value of eq (6) in eq (5)

$m_1.m_2 = -1$

Hence the angle between two curve is $\frac{\pi}{2}$.

19. If the two curves $C_1 : x = y^2$ and $C_2 : xy = k$ cut at right angles find the value of k.

यदि दो वक्र $C_1 : x = y^2$ तथा $C_2 : xy = k$ एक दूसरे को समकोण पर प्रतिच्छेद करते है तो k का मान ज्ञात कीजिये।

Ans. $\pm \frac{1}{2\sqrt{2}}$

Sol. By solving the equations $x = y^2$ and $xy = k$

$\Rightarrow y^3 = k,$

$\Rightarrow y = k^{\frac{1}{3}}$

$\Rightarrow x = k^{\frac{2}{3}}$

$\Rightarrow$ Point of intersection of the two curves is $\left(k^{\frac{2}{3}}, k^{\frac{1}{3}}\right)$

Slope of tangent to the curve $x = y^2$ is $\frac{dy}{dx} = \frac{1}{2y}$

$$\Rightarrow m_1 = \left(\frac{dy}{dx}\right)_{\left(k^{\frac{2}{3}}, k^{\frac{1}{3}}\right)} = \frac{1}{2k^{\frac{1}{3}}}$$

Slope of tangent to the curve $xy = k$ is $\Rightarrow \frac{dy}{dx} = \frac{-k}{x^2}$

$$\Rightarrow m_2 = \left(\frac{dy}{dx}\right)_{\left(k^{\frac{2}{3}}, k^{\frac{1}{3}}\right)} = \frac{-k}{k^{\frac{4}{3}}} = -\frac{1}{k^{\frac{1}{3}}}$$

$\because$ the curves are orthogonal $\Rightarrow m_1.m_2 = -1$

$$\Rightarrow \frac{1}{2k^{\frac{1}{3}}} \cdot -\frac{1}{k^{\frac{1}{3}}} = -1$$

$$\Rightarrow k^{\frac{2}{3}} = \frac{1}{2}$$

$$\Rightarrow k^2 = \frac{1}{8}$$

SOME IMPORTANT LENGTHS (TANGENT SUBTANGENT, NORMAL SUBNORMAL)

20. Find the value of n so that the subnormal at any point on the curve $xy^n = a^{n+1}$ may be constant.

वक्र $xy^n = a^{n+1}$ में n का मान ज्ञात कीजिये यदि उस पर अधोलम्ब की लम्बाई नियत है।

Ans. $n = -2$

Sol. $xy^n = a^{n+1}$, $a \neq 0$

$$y^n + nxy^{n-1}\frac{dy}{dx} = 0$$

i.e. $$\frac{dy}{dx} = -\frac{y}{nx}$$

$\therefore$ length of subnormal

$$= \left| y.\frac{dy}{dx} \right| = \left| \frac{y^2}{nx} \right| = \left| \frac{y^2.y^n}{an^{n+1}} \right|$$

$$= \left| \frac{y^{n+2}}{na^{n+1}} \right| \text{ is constant only if } n = -2.$$

21. Show that in the curve $y = a.\, ln\,(x^2 - a^2)$, sum of the length of tangent & subtangent varies as the product of the coordinates of the point of contact.

सिद्ध कीजिए की वक्र $y = a.\, ln\,(x^2 - a^2)$ में स्पर्श रेखा एवं अधःस्पर्शी की लम्बाई का योग बिन्दु के निर्देशांको के गुणन के सापेक्ष समानुपाती है।

Ans.

Sol. Given, $y = a \log\left(x^2 - a^2\right)$

$$\frac{dy}{dx} = a, \frac{2x}{x^2 - a^2} = \frac{2ax}{x^2 - a^2}$$

$$\therefore \text{Length of subtangent} = y\frac{dx}{dy}$$

$$= y.\frac{\left(x^2 - a^2\right)}{2ax}$$

$$\text{Length of tangent} = y\sqrt{1 + \left(\frac{dx}{dy}\right)^2}$$

$$= y\sqrt{1 + \left(\frac{x^2 - a^2}{2ax}\right)^2}$$

$$= y\sqrt{\frac{4a^2x^2 + \left(x^2 - a^2\right)}{4a^2x^2}}$$

$$= y.\frac{y}{2ax}\sqrt{\left(x^2 + a^2\right)^2} = y\frac{\left(x^2 + a^2\right)}{2ax}$$

Hence, sum of lengths of tangents and subtangents $= y\frac{\left(x^2-a^2\right)}{2ax} + y\frac{\left(x^2-a^2\right)}{2ax}$

$= \frac{y}{2ax}\left(x^2-a^2+x^2-a^2\right)$

$= \frac{xy}{a} = \frac{1}{a}(xy)$

$\therefore$ Sum of lengths of t and s a(xy).

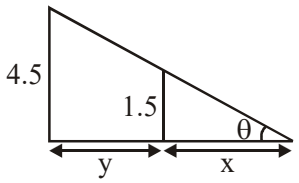
RATE MEASURE

22. A man 1.5 m tall walks away from a lamp post 4.5 m high at the rate of 4 km/hr.
(i) how fast is the farther end of the shadow moving on the pavement ?
(ii) how fast is his shadow lengthening?

एक आदमी, जिसकी लम्बाई 1.5 m है, एक प्रकाश स्तम्भ, जिसकी लम्बाई 4.5 m है, से दूर 4 km/hr की गति से जा रहा है।
(i) किस गति से उसकी परछाई का अंतिम छोर आगे बढ़ रहा है?
(ii) किसी दर से उसकी परछाई की लम्बाई बढ़ रही है?

Ans. **(i)** 6 km/h ; **(ii)** 2 km/hr

Sol.



$\frac{4.5}{x+y} = \frac{1.5}{x} \Rightarrow 3x = x + y \Rightarrow 2x = y$

$\frac{2dx}{dt} = \frac{dy}{dt}$

$\frac{dy}{dt} = 4$ is given then $\frac{dx}{dt} = 2$

(a) $\frac{d}{dt}(x+y) = \frac{dx}{dt} + \frac{dy}{dt} = 4+2 = 6$ km/hr.

(b) Shadow lengthening $= \frac{dx}{dt} = 2$ km/hr.

23. A particle moves along the curve $6y = x^3 + 2$. Find the points on the curve at which the y coordinate is changing 8 times as fast as the x coordinate.

एक कण वक्र $6y = x^3 + 2$ पर गमन करता है। यदि वक्र पर बिन्दु जिनके y निर्देशाक x निर्देशांक से 8 गुना तेजी से बदल रहा है तो वह बिन्दु ज्ञात कीजिये।

Ans. $(4, 11)$ & $(-4, -31/3)$

Sol. Let the coordinate on the curve be (x, y)

According to question

$$\frac{dy}{dx} = 8$$

$$\Rightarrow \frac{d}{dx}\left(\frac{x^3+2}{6}\right) = 8$$

$$\Rightarrow \frac{1}{6}.\left(3x^2\right) = 8$$

$$\Rightarrow x^2 = 16 \qquad \Rightarrow x = \pm 4$$

When $x = 4$

$$y = \left(\frac{x^3+2}{6}\right) = \left(\frac{64+2}{6}\right) = \frac{66}{6} = 11$$

When $x = -4$

$$y = \left(\frac{x^3+2}{6}\right) = \left(\frac{-64+2}{6}\right) = \frac{-62}{6} = -\frac{31}{3}$$

Therefore, required points are (4, 11) and $\left(-4, -\frac{31}{3}\right)$

24. An inverted cone has a depth of 10 cm & a base of radius 5 cm. Water is poured into it at the rate of 1.5 cm^3/min. Find the rate at which level of water in the cone is rising, when the depth of water is 4 cm.

एक उलटे शंकु की गहराई 10 cm है तथा आधार की त्रिज्या 5 cm है, में जल 1.5 cm^3/min की दर से भरा जा रहा है। जल की सतह की बढने की दर ज्ञात कीजिये जब जल की गहराई 4 cm हो।

Ans. $3/8\,\pi$ cm/min

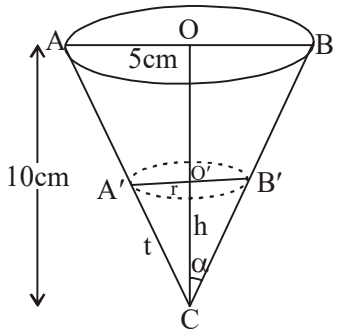
Sol. Let a be the semi-vertical angle of the cone CAB whose height CO is 10 cm and radius OB = 5 cm.

Then

$$\tan\alpha = \frac{5}{10} = \frac{1}{2}$$

$$\tan\alpha = \frac{O'B'}{CO'} = \frac{O'B'}{h}$$

$$\Rightarrow O'B' = h\tan\alpha$$



Let V be the volume of the water in the cone i.e. the volume of the cone CA 'B' after time t minutes and h be the height of water. Then,

$$V = \frac{1}{3}\pi(OB')^2(CO')$$

$$\Rightarrow V = \frac{1}{3}\pi h^3\tan^2\alpha$$

$$\Rightarrow V = \frac{\pi}{12}h^3 \qquad \left[\because \tan\alpha = \frac{1}{2}\right]$$

$$\therefore \frac{dv}{dt} = \frac{\pi}{12}3h^2\frac{dh}{dt} = \frac{\pi}{4}h^2\frac{dh}{dt}$$

$$\Rightarrow \frac{3}{2} = \frac{\pi h^2}{4}\frac{dh}{dt} \qquad \left[\because \frac{dv}{dt} = \frac{3}{2}\text{cm}^2/\text{minute(given)}\right]$$

$$\frac{dh}{dt} = \frac{6}{\pi r^2}$$

$$\Rightarrow \left(\frac{dh}{dt}\right)_{h=4} = \frac{6}{\pi(4)^2} = \frac{3}{8\pi}\text{cm/min}$$

$\therefore$ required are $= \frac{3}{8\pi}$ cm / sec .

25. A water tank has the shape of a right circular cone with its vertex down. Its altitude is 10 cm and the radius of the base is 15 cm. Water leaks out of the bottom at a constant rate of 1cu. cm/sec. Water is poured into the tank at a constant rate of C cu. cm/sec. Compute C so that the water level will be rising at the rate of 4 cm/sec at the instant when the water is 2 cm deep.

एक पानी की टंकी जो एक शंकु के आकार की हो जिसका शीर्ष नीचे की ओर है जिसकी ऊँचाई 10 cm है तथा आधार की त्रिज्या 15 cm है। उसमें से पानी का रिसाव एक निश्चित दर 1cu. cm/sec से हो रहा है। टंकी में एक नियत दर C cu. cm/sec से भरा जा रहा है। जब उसकी गहराई दो 2 cm है तो पानी की सतह 4 cm/sec की दर से बढ़ रही हो तो C का मान ज्ञात कीजिये।

Ans. $1 + 36\pi$ cu. cm/sec

Sol. r = 15cm

h = 10cm

Let r be radius of wooden-level at the instant when the water is 2cm deep

$\therefore \frac{r'}{h'} = \frac{r}{h}$

$\Rightarrow \frac{r'}{h'} = \frac{15}{10}$

$\Rightarrow r' = \frac{3}{2}h'$

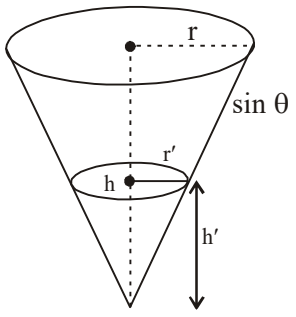
Volume of the right circular cone $= \frac{1}{3} \times R^2H$

$= \frac{1}{3}\pi \times \frac{9}{4}K^2h'$

$= \frac{3\pi}{4}K^3$

$\therefore \frac{dv}{dt} = \frac{3\pi}{4}.3K^2.\frac{dK}{dt}$

$= \frac{3\pi}{4} \times 3 \times (2)^2 \times 4$ Cu.cm/sec



$= 36\pi$ Cu.cm/sec

$\therefore$ value of c = $36\pi + 1$ Cu.cm/sec

26. Sand is pouring from a pipe at the rate of 12 cc/sec. The falling sand forms a cone on the ground in such a way that the height of the cone is always 1/6th of the radius of the base. How fast is the height of the sand cone increasing when the height is 4 cm.

एक नलिका से रेत 12 cc/sec की दर से भरी जा रही है नीचे गिरने वाली रेत जमीन पर एक शंकु का आकार इस प्रकार बना रही है कि शंकु की ऊँचाई सदैव त्रिज्या की 1/6th है। शंकु की ऊँचाई की वृद्धि दर ज्ञात कीजिये जब ऊँचाई 4 cm हो।

Ans. $1/48\,\pi$ cm/s

Sol. $h = \frac{1}{6}r$, $h = 4cm$

$$\frac{dv}{dt} = 12cm^3/\sec$$

$$v = \frac{1}{3}\pi r^2 h$$

$$v = \frac{1}{3}\pi(6h)^2 h$$

$$v = 12\pi h^3$$

$$\frac{dv}{dt} = 12\pi(3h)^2\frac{dh}{dt}$$

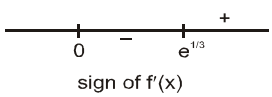
$$\frac{12}{12\pi(3\times16)} = \frac{dh}{dt} \qquad \Rightarrow \frac{dh}{dt} = \frac{1}{48\pi}\text{cm}/\sec$$

27. An open Can of oil is accidently dropped into a lake; assume the oil spreads over the surface as a circular disc of uniform thickness whose radius increases steadily at the rate of 10 cm/sec. At the moment when the radius is 1 meter, the thickness of the oil slick is decreasing at the rate of 4 mm/sec, how fast is it decreasing when the radius is 2 meters.

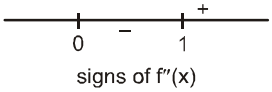
एक खुला हुआ तेल का केन दुर्घटनावश एक झील में गिर जाता है मान लिजिये की तेल झील की सतह पर एक वृत्ताकार चकती जिसकी एकसमान मोटाई है तथा त्रिज्या 10 cm/sec नियत दर से बढ़ रही है एक क्षण जब त्रिज्या 1 मीटर है तथा मोटाई 4 mm/sec की दर से घट रही है। मोटाई की घटने की दर क्या होगी जब त्रिज्या 2 मीटर है।

Ans. 0.05 cm/sec

Sol. $f'(x) = 16\,x^3(3\ell n\,x - 1)$



sign of f'(x)



signs of f''(x)

$f''(x) = 16 \cdot 3 \cdot x^2 \cdot 3 \cdot \ell nx$

$x = 1$ is point of inflection. $x = e^{1/3}$ is point of minima. In (0, 1) f(x) is concave downward. In $(1, \infty)$ f(x) is concave upward.

MONOTONOCITY

DETERMINE MONOTONOCITY

28. Find the intervals of monotonocity for the following functions & represent your solution set on the number line.

(a) $f(x) = 2.\ e^{x^2-4x}$ (b) $f(x) = e^x/x$ (c) $f(x) = x^2e^{-x}$ (d) $f(x) = 2x^2 - \ln|x|$

Also plot the graphs in each case and state their range.

निम्नलिखित फलनों के लिये एकदिष्टता के अन्तराल ज्ञात कीजिये तथा अपने हलों को संख्या रेखा पर निरूपित कीजिये।

(a) $f(x) = 2.\ e^{x^2-4x}$ (b) $f(x) = e^x/x$ (c) $f(x) = x^2e^{-x}$ (d) $f(x) = 2x^2 - \ln|x|$

साथ ही प्रत्येक स्थिति के लिये फलनों के आरेख भी बनाइयें, उनका परिसर भी लिखिये।

Ans. (a) I in $(2, \infty)$ & D in $(-\infty, 2)$

(b) I in $(1, \infty)$ & D in $(-\infty, 0) \cup (0, 1)$

(c) I in (0, 2) & D in $(-\infty, 0) \cup (2, \infty)$

(d) I for $x > \frac{1}{2}$ or $-\frac{1}{2} < x < 0$ & D for $x < -\frac{1}{2}$ or $0 < x < \frac{1}{2}$

Sol.

29. Find the intervals of monotonocity of the functions in $[0, 2\pi]$

(a) $f(x) = \sin x - \cos x$ in $x \in [0, 2\pi]$ (b) $g(x) = 2\sin x + \cos 2x$ in $(0 \le x \le 2\pi)$.

(c) $f(x) = \dfrac{4\sin x - 2x - x\cos x}{2 + \cos x}$

निम्नलिखित फलनों के लिये अन्तराल $[0, 2\pi]$ में एकदिष्टता के अन्तराल ज्ञात कीजिये।

(a) $f(x) = \sin x - \cos x$ in $x \in [0, 2\pi]$ (b) $g(x) = 2\sin x + \cos 2x$ in $(0 \le x \le 2\pi)$.

(c) $f(x) = \dfrac{4\sin x - 2x - x\cos x}{2 + \cos x}$

Ans. (a) I in $[0, 3\pi/4) \cup (7\pi/4, 2\pi]$ & D in $(3\pi/4, 7\pi/4)$

(b) I in $[0, \pi/6) \cup (\pi/2, 5\pi/6) \cup (3\pi/2, 2\pi]$ & D in $(\pi/6, \pi/2) \cup (5\pi/6, 3\pi/2)]$

(c) I in $[0, \pi/2) \cup (3\pi/2, 2\pi]$ and D in $(\pi/2, 3\pi/2)$

Sol.

30. Let $f(x) = x^3 - x^2 + x + 1$ and $g(x) = \begin{cases} \max\{f(t) : 0 \le t \le x\} & , 0 \le x \le 1 \\ 3 - x & , 1 < x \le 2 \end{cases}$. Discuss the conti. & differentiability of g(x) in the interval (0,2).

माना $f(x) = x^3 - x^2 + x + 1$ तथा $g(x) = \begin{cases} \max\{f(t) : 0 \le t \le x\} & , 0 \le x \le 1 \\ 3 - x & , 1 < x \le 2 \end{cases}$ g(x) की अन्तराल (0,2) में सान्तत्य एवं अवकलनीयता के बारे में बताइयें।

Ans. continuous but not diff. at $x = 1$

Sol. Here, $f(x) = x^3 - x^2 + x + 1$

$\Rightarrow f'(x) = 3x^2 - 2x + 1$, which is strictly

Increasing in (0, 2)

$\therefore g(x) = \begin{cases} f(x); & 0 \le x \le 1 \\ 3 - x; & 1 < x \le 2 \end{cases}$

[As f(x) is increasing so f(x) is maximum when $0 \le t \le x$]

So, $g(x) = \begin{cases} x^3 - x^2 + x + 1; & 0 \le x \le 1 \\ 3 - x; & 1 < x \le 2 \end{cases}$

Also, $g'(x) = \begin{cases} 3x^2 - 2x + 1; & 0 \le x \le 1 \\ -1; & 1 < x \le 2 \end{cases}$

Which clearly shows g(x) is continuous for all $x \in [0, 2]$,

but g(x) is not differentiable at $x = 1$

ALWAYS INCREASING /ALWAYS DECREASING BASED QUESTIONS

31. Let $f(x) = 1 - x - x^3$. Find all real values of x satisfying the inequality, $1 - f(x) - f^3(x) > f(1 - 5x)$ माना $f(x) = 1 - x - x^3$, x के सभी वास्तविक मान ज्ञात कीजिये जो असमिका $1 - f(x) - f^3(x) > f(1 - 5x)$ को संतुष्ट करते है।

Ans. $(-2, 0) \cup (2, \infty)$

Sol. $1 - f(x) - f^3(x) > f(1 - 5x)$

$\Rightarrow \quad f'(x) = -1 - 3x^2 < 0 \Rightarrow \quad f$ is decreasing

Now f(f(x)) > f(1 – 5x)

$\Rightarrow$ f(x) < 1 – 5x {$\because$ f is decreasing }

$1 - x - x^3 < 1 - 5x$.

32. Let $f(x)$ be a increasing function defined on $(0, \infty)$. If $f(2a^2 + a + 1) > f(3a^2 - 4a + 1)$. Find the range of a.

Ans. $(0, 1/3) \cup (1, 5)$

Sol. Increasing function $\Rightarrow 2a^2 + a + 1 > 3a^2 - 4a + 1$

$\Rightarrow a^2 - 5a < 0 \qquad \Rightarrow a \in (0, 5)$

$\Rightarrow a = 1, 2, 3, 4$ are the possible integral values.

From domain conditions $2a^2 + a + 1 > 0$;

$3a^2 - 4a + 1 > 0$ we can say that first satisfies but the second one makes it $a \in \left(0, \frac{1}{3}\right) \cup (1, \infty)$.

So possible integral values are a = 2, 3, 4

32. एक वर्द्धमान फलन $f(x)$ अन्तराल $(0, \infty)$ में परिभाषित है। यदि $f(2a^2 + a + 1) > f(3a^2 - 4a + 1)$ है तो a का परिसर ज्ञात कीजिये।

Ans. $(0, 1/3) \cup (1, 5)$

Sol.

33. Find the set of all values of the parameter 'a' for which the function, $f(x) = \sin 2x - 8(a + 1)\sin x + (4a^2 + 8a - 14)x$ increases for all $x \in R$ and has no critical points for all $x \in R$.

प्राचल 'a' के सभी मान ज्ञात कीजिये जिनके लिये फलन $f(x) = \sin 2x - 8(a + 1)\sin x + (4a^2 + 8a - 14)x$, सभी x के वास्तविक मानों के लिये वर्द्धमान हो एवं कोई भी क्रांतिक बिन्दु ना हो।

Ans. $a < -\left(2 + \sqrt{5}\right)$ or $a > \sqrt{5}$

Sol. $f(x) = \sin 2x - 8(a + 1) \sin x + (4a^2 + 8a - 14)x$

$f'(x) = 2 \cos 2x - 8(a + 1) \cos x + (4a^2 + 8a - 14)$

$f'(x) = 2(2\cos^2 x - 1) - 8(a + 1) \cos x + 4a^2 + 8a - 14$

$= 4\{\cos^2 x - 2(a + 1) \cos x\} + 4a^2 + 8a - 16$

$= 4\{\cos x - (a + 1)\}^2 - 20 > 0$

$= \{\cos x - (a + 1)\}^2 - (\sqrt{5})^2 > 0$

$f'(x) = \{\cos x - (a+1) - \sqrt{5}\}\ \{\cos x - (a+1) + \sqrt{5}\} > 0$

$\Rightarrow \cos x > a + 1 + \sqrt{5}$ or $\cos x < (a+1) - \sqrt{5}$

$\forall\ x \in R$

$a + 1 + \sqrt{5} < -1$ or $(a+1) - \sqrt{5} > 1$

$a < -2 - \sqrt{5}$ or $a > \sqrt{5}$

$a \in (-\infty, -2 - \sqrt{5}) \cup (\sqrt{5}, \infty)$

34. Find all the values of the parameter 'a' for which the function; $f(x) = 8ax - a\sin 6x - 7x - \sin 5x$ increases and has no critical points for all $x \in R$.

प्राचल 'a' के सभी मान ज्ञात कीजिये जिनके लिये फलन, $f(x) = 8ax - a\sin 6x - 7x - \sin 5x$ सभी x के वास्तविक मानों के लिये वर्द्धमान हो एवं कोई भी क्रांतिक बिन्दु ना हो।

Ans. $(6, \infty)$

Sol. $f(x) = 8ax - a\sin 6x - 7x - \sin 5x$

$f'(x) = 8a - 6a\cos 6x - 7 - 5\cos 5x > 0\ \forall\ x \in R$

$\Rightarrow a > \dfrac{7 + 5\cos 5x}{8 - 6\cos 6x}\ \forall\ x \in R$

$\Rightarrow$ RHS assumes maximum value for $x = 0$.

$\Rightarrow a > \dfrac{7+5}{8-6} \Rightarrow a > 6$

35. Find the set of values of 'a' for which x the function $f(x) = (a^2 + a - 6)\cos 2x + (a-2)x + \cos 1$ has no critical points.

प्राचल 'a' के सभी मान ज्ञात कीजिये जिनके लिये फलन, $f(x) = (a^2 + a - 6)\cos 2x + (a-2)x + \cos 1$ के कोई भी क्रांतिक बिन्दु ना हों।

Ans. $a \in \left(-\dfrac{7}{2}, -\dfrac{5}{2}\right)$

Sol. There is no critical point if $f'(x) \neq 0 : f'(x) = -2\left(a^2 + a - 6\right)\sin 2x + (a-2) \neq 0$

$\Rightarrow a \neq 2$

$$\sin 2x \neq \frac{1}{2(a+3)} \Rightarrow \begin{cases} \frac{1}{2(a+3)} < -1 \\ \frac{1}{2(a+3)} > 1 \end{cases} \Rightarrow ....$$

36. Find the values of 'a' for which the function $f(x) = \sin x - a\sin 2x - \frac{1}{3}\sin 3x + 2ax$ increases throughout the number line.

Ans. $[1, \infty)$

Sol. $f'(x) = \cos x - 2a\cos 2x - \cos x + 2a \geq 0$

$\Rightarrow \cos x - \cos 3x + 2a(1 - \cos 2x) \geq 0$

$\Rightarrow 2\sin 2x \sin x + 2a(2\sin^2 x) \geq 0$

$\Rightarrow 4\sin^2 x \cos x + 4a\sin^2 x \geq 0$

$\Rightarrow 4\sin^2 x(\cos x + a) \geq 0$

$a \geq -\cos x$

$a \geq 1$

36. 'a' के सभी मान ज्ञात कीजिये जिनके लिये फलन $f(x) = \sin x - a\sin 2x - \frac{1}{3}\sin 3x + 2ax$ सम्पूर्ण संख्या रेखा पर वर्द्धमान हो।

Ans. $[1, \infty)$

Sol. Let $f(x) = \ln(1 + x) - \frac{x}{1+x}$ then

$$f'(x) = \frac{1}{1+x} - \frac{1}{(1+x)^2} = \frac{x}{(1+x)^2} > 0$$

for $x > 0$, so $\ln(1 + x) > \frac{x}{1+x}$ for $x > 0$.

But the answer given in my book is $(-1, 0) \cup (0, \infty)$

37. Find the set of values of 'a' for which the function, $f(x) = \left(1 - \frac{\sqrt{21 - 4a - a^2}}{a + 1}\right)x^3 + 5x + \sqrt{7}$ is increasing at every point of its domain.

'a' के मानों का समुच्चय ज्ञात कीजिये जिनके लिये फलन $f(x) = \left(1 - \frac{\sqrt{21 - 4a - a^2}}{a + 1}\right)x^3 + 5x + \sqrt{7}$ अपने प्रान्त के सभी बिन्दु के लिये वर्द्धमान है।

Ans. $[-7, -1) \cup [2, 3]$

Sol. $f(x) = \left(1 - \frac{\sqrt{21 - 4b - b^2}}{b + 1}\right)x^3 + 5x + \sqrt{6}$

$f'(x) = 3\left(1 - \frac{\sqrt{21 - 4b - b^2}}{b + 1}\right)x^2 + 5$

$f(x)$ is increasing $\Rightarrow$ $f'(x) \geq 0$ $\forall\ x \in R$

$\Rightarrow$ $1 - \frac{\sqrt{21 - 4b - b^2}}{b + 1} \geq 0$ $\Rightarrow$ $\frac{\sqrt{(b + 7)(3 - b)}}{b + 1} \leq 1$

Case- I If $b + 1 > 0$, then $\frac{(b + 7)(3 - b)}{(b + 1)^2} \leq 1$ and $-7 \leq b \leq 3$

$\Rightarrow$ $b \leq -5$ and $b \geq 2$ $\Rightarrow$ $b \in [2, 3]$

Case-II If $b + 1 < 0$ $\Rightarrow$ $b \in [-7, -1)$

38. If $f(x) = 2e^x - ae^{-x} + (2a+1)x - 3$ monotonically increases for every $x \in R$ then find the range of values of 'a'.

यदि $f(x) = 2e^x - ae^{-x} + (2a+1)x - 3$ प्रत्येक $x \in R$ के लिये एकदिष्ट वर्द्धमान है तो 'a' के मानों का परिसर ज्ञात कीजिये।

Ans. $a \geq 0$

Sol. $f(x) = 2e^x - ae^{-x} + (2a + 1)\,x - 3$

$f'(x) = 2e^x + ae^{-x} + (2a + 1)$

Let (माना) $e^x = t$; $t > 0$

$\therefore$ $2t + \frac{a}{t} + (2a + 1)$ $\Rightarrow$ $\frac{2t^2 + (2a + 1)t + a}{t}$

$\Rightarrow$ Now (पुनः) $f'(x) \geq 0$ $\therefore$ $2t^2 + (2a + 1)\,t + a \geq 0$

Case-I (स्थिति-I) $D \leq 0$

$(2a + 1)^2 - 8a \le 0$

$\Rightarrow \quad (2a - 1)^2 \le 0 \quad \Rightarrow \quad a = \frac{1}{2}$

Case-II (स्थिति-II) 

$\Rightarrow$ (i) $D \ge 0 \Rightarrow a \in R$

(ii) $f(0) \ge 0 \Rightarrow a \ge 0$

(iii) $-\frac{b}{2a} < 0 \quad \Rightarrow \quad \frac{-(2a+1)}{4} < 0 \quad \Rightarrow \quad a > -\frac{1}{2} \qquad \therefore a \in [0, \infty)$

PROVING INEQUALITIES

39. Find the set of values of x for which the inequality $\ell n(1+x) > x/(1+x)$ is valid.

असमिका $\ell n(1+x) > x/(1+x)$ को सन्तुष्ट करने वाले x के मानों का समुच्चय ज्ञात कीजिये।

Ans. $(-1, 0) \cup (0, \infty)$

Sol.

40. Prove that, $x^2 - 1 > 2x \ell n\, x > 4(x - 1) - 2 \ell n\, x$ for $x > 1$.

$x > 1$ के लिये सिद्ध कीजिये $x^2 - 1 > 2x \ell n\, x > 4(x - 1) - 2 \ell n\, x$

Ans.

Sol. $x^2 - 1 > 2x \ell n\, x > 4(x - 1) - 2 \ell n\, x, \; x > 1$

(a) Consider $f(x) = x^2 - 1 - 2x \ell n\, x$

$f'(x) = 2\{x - 1 - \ell n\, x\}$

$f''(x) = 2\left\{1 - \frac{1}{x}\right\}$

$f''(x) > 0 \; \forall \; x > 1$

$\Rightarrow$ $f'(x)$ is increasing $\forall \; x > 1$

$\Rightarrow$ $f'(x) > f'(1) \quad \Rightarrow \quad f'(x) > 0 \; \forall \; x > 1$

$\Rightarrow$ $f(x)$ is increasing $\forall \; x > 1$

$\Rightarrow$ $f(x) > f(1) \Rightarrow \; f(x) > 0 \; \forall \; x > 1$

(b) Consider,

$g(x) = 2x \ell n\, x + 2 \ell n\, x - 4(x - 1)$

$g'(x) = 2(1 + \ell n\, x) + \frac{2}{x} - 4$

$$g''(x) = \frac{2}{x} - \frac{2}{x^2}$$

$$g''(x) = \frac{2(x-1)}{x^2} > 0 \qquad \forall\ x > 1$$

$\Rightarrow$ g'(x) is increasing $\forall\ x > 1$

$\Rightarrow$ $g'(x) > g'(1) \Rightarrow g'(x) > 0\ \forall\ x > 1$

$\Rightarrow$ g(x) is increasing $\forall\ x > 1$

$\Rightarrow$ $g(x) > g(1)\ \forall\ x > 1$

$\Rightarrow$ $g(x) > 0\ \forall\ x > 1$.

41. Let f, g be differentiable on R and suppose that $f(0) = g(0)$ and $f'(x) \le g'(x)$ for all $x \ge 0$. Show that $f(x) \le g(x)$ for all $x \ge 0$.

माना f, g दो R में परिभाषित अवकलनीय फलन है तथा $x \ge 0$ के लिये $f(0) = g(0)$ एवं $f'(x) \le g'(x)$ है सिद्ध कीजिये $x \ge 0$ के लिये $f(x) \le g(x)$ है।

Ans.

Sol. Let $h(x) : f(x) - g(x), x \in R$

The function h is differentiable in R as a difference of differentaible functions, with

$h'(x) = f'(x) - g'(x), x \in R$, So, according to hypothesis we obtain that :

$h'(x) = f'(x) - g'(x) < 0 \qquad \forall\ x \in R$

so h is (stricly) decreasing in R, and therefore for $x > 0$ we have that

$$h(x) < h(0)$$

Thus,

$f(x) - g(x) < f(0) - g(0) = 0 \qquad \Rightarrow f(x) < g(x)$

$\forall x > 0$

42. Show that exactly two real values of x satisfy the equation $x^2 = x \sin x + \cos x$.

सिद्ध कीजिये समीकरण $x^2 = x \sin x + \cos x$ को संतुष्ट करने वाले x के केवल दो वास्तविक मान है।

Ans.

Sol. Taking the difference $h(x) = x^2 - x \sin x - \cos x$ to show that $h'(x) > 0$ if $x > 0$ and $h'(x) < 0$ if $x < 0$ then since $h(0) < 0$ there exists exactly two sets of numbers $[a_1, b_1]$ and $[a_2, b_2]$ such that $h(a_1) > 0 > h(b_1)$ and

$h(a_2) < 0 < h(b_2)$. Therefore by the intermediate value theorem and since h is continuous there exists exactly two numbers c_1, c_2 such that $h(c_1) = h(c_2) = 0$ implying these numbers hold true in the proposed statement.

GREATEST AND LEAST VALUE OF A FUNCTION IN AN INTERVAL

43. Find the greatest & the least values of the following functions in the given interval if they exist :

(a) $f(x) = \sin^{-1}\frac{x}{\sqrt{x^2+1}} - \ln x \text{ in} \left[\frac{1}{\sqrt{3}}, \sqrt{3}\right]$ (b) $f(x) = 12x^{4/3} - 6x^{1/3}, x \in [-1, 1]$

(c) $y = x^5 - 5x^4 + 5x^3 + 1 \text{ in } [-1, 2]$

निम्नलिखित फलनों के दिये गये अन्तराल में महत्तम एवं न्यूनतम मान ज्ञात कीजिये। (यदि अस्तित्व में हो तो) :

(a) $f(x) = \sin^{-1}\frac{x}{\sqrt{x^2+1}} - \ln x \text{ in} \left[\frac{1}{\sqrt{3}}, \sqrt{3}\right]$ (b) $f(x) = 12x^{4/3} - 6x^{1/3}, x \in [-1, 1]$

(c) $y = x^5 - 5x^4 + 5x^3 + 1 \text{ in } [-1, 2]$

Ans. **(a)** $(\pi/6)+(1/2)\ln 3, (\pi/3) - (1/2)\ln 3$

(b) Maximum at $x = 1$ and $f(-1) = 18$; Minimum at $x = 1/8$ and $f(1/8) = -9/4$

(c) 2 & -10

Sol. (a) $f'(x)\frac{1}{\sqrt{1-\frac{x^2}{(x^2+1)}}}\left(\frac{\left(\sqrt{x^2+1}\right).1 - \frac{x}{2\sqrt{x^2+1}}.2x}{x^2+1}\right) - \frac{1}{x}$

$$= \frac{1}{1+x^2} - \frac{1}{x} = \frac{x-1-x^2}{x(1+x^2)}$$

Let $g(x) = x - 1 - x^2$

$g'(x) = 1 - 2x \text{ for } x \in \left[\frac{1}{\sqrt{3}}, \sqrt{3}\right]$

$g'(x) < 0 \Rightarrow f(x)$ decreasing function

maximum f(x) is $f\left(\frac{1}{\sqrt{3}}\right) = \sin\left(\frac{1}{2}\right) - \ell n\left(\frac{1}{\sqrt{3}}\right)$

$$= \frac{\pi}{6} + \frac{1}{2}\ell n 3$$

minimum f(x) is $= \sin\left(\frac{\sqrt{3}}{2}\right) - \ell n\left(\sqrt{3}\right) = \frac{\pi}{3} - \frac{1}{2}\ell n3$

(b) Consider, $f(x) = 12x^{4/3} - 6x^{1/3}$ $\quad x \in [-1, 1]$

$f'(x) = 16^{1/3} - 2x^{-2/3}$

$f'(x) > 0 \Rightarrow 16x^{1/3} > 2x^{-2/3} \quad \Rightarrow 16x > 2 \Rightarrow x > \frac{1}{8}$

$f'(x) < 0 \Rightarrow 16x^{1/3} < 2x^{-2/3} \Rightarrow x < \frac{1}{8}$

$\therefore$ f' increases on $\left(\frac{1}{8}, 1\right)$ and decreases on $\left(-1, \frac{1}{8}\right)$

Also $f(1) = 18$

$f\left(\frac{1}{8}\right) = \frac{9}{4}$

$\therefore$ Min value $= -\frac{9}{4}$ and max value u= 18

(c) $\therefore y'(x) = 5x^4 - 20x^3 + 15x^2$

$= 5x^2[x^2 - 4x + 3]$

$= 0$

$x = 0, 1, 3$

Now $y(0) = 1$ and $y(1) = 2$

3 does not belong in the given range.

Now $y(-1) = 10$

Hence, in the interval $[-1, 2]$ it attains the maximum value, $y = 2$ at $x = 1$

44. Find the minimum value of the function $f(x) = x^{\frac{3}{2}} + x^{\frac{-3}{2}} - 4\left(x + \frac{1}{x}\right)$ for all permissible real x.

x के सभी सम्भावित मानों के लिये फलन $f(x) = x^{\frac{3}{2}} + x^{\frac{-3}{2}} - 4\left(x + \frac{1}{x}\right)$ का न्यूनतम मान ज्ञात कीजिये।

Ans. – 10

Sol. $f(x)=x^{\frac{3}{2}}+x^{-\frac{3}{2}}-4\left(x+\frac{1}{x}\right)$

$$f(x)=\left(\sqrt{x}+\frac{1}{\sqrt{x}}\right)^3-3\left(\sqrt{x}+\frac{1}{\sqrt{x}}\right)$$

$$-4\left[\left(\sqrt{x}+\frac{1}{\sqrt{x}}\right)^2-2\right]$$

Let $\sqrt{x}+\frac{1}{\sqrt{x}}=t\left(x>0\right)$

Let $g(t)=t^3-3t-4t^2+8$

$g(t)=t^3-4t^2-3t+8$

$g'(t)=3t^2-8t-3=(t-3)(3t+1)$

$g'(t)=0\Rightarrow t=3\left(t\neq-\frac{1}{3}\right)$

$g''(t)=6t-8$

$g''(3)=10>0\Rightarrow g(3)$ is minimum

$g(3) = 27 – 9 – 36 + 8 = –10$

45. If $b > a$, find the minimum value of $|(x-a)^3|+|(x-b)^3|$, $x \in R$.

यदि $b > a$, तो $|(x-a)^3|+|(x-b)^3|$, $x \in R$ का न्यूनतम मान बताइयें।

Ans. $(b – a)^3/4$

Sol. Let $a < x < b$.

Hence, since $f(x) = x^3$ is a convex function, by Jensen we obtain :

$$\left|(x-a)^3\right|+\left|(x-b)^3\right|=(x-a)^3+(b-x)^3\geq 2\left(\frac{x-a+b-x}{2}\right)^3=\frac{(6-a)^3}{4}$$

Let $x \geq b$. Hence,

$$\left|(x-a)^3\right| + (x-b)^3 = (x-a)^3 + (x-b)^3 \geq (b-a)^3 > \frac{(b-a)^3}{4}.$$

ROLLE'S THERORM

46. Let $f(x) = 4x^3 - 3x^2 - 2x + 1$, use Rolle's theorem to prove that there exist c, $0 < c < 1$ such that $f(c) = 0$.

माना $f(x) = 4x^3 - 3x^2 - 2x + 1$, रोले प्रमेय का प्रयोग करते हुये सिद्ध कीजिये कि c, $0 < c < 1$, अस्तित्व में है जिसके लिये $f(c) = 0$ है।

Ans.

Sol. Let $g(x) = x^4 - x^3 - x^2 + x$. Then $g'(x) = f(x)$.

$g(0) = g(1) = 0$.

By Rolle's Theorem, there exist c such that $0 < c < 1$ and $g'(c) = 0$.

47. Let f and g be functions, continuous in [a, b] and differentiable on [a, b]. If $f(a) = f(b) = 0$ then show that there is a point $c \in (a, b)$ such that $g'(c) f(c) + f'(c) = 0$.

माना f तथा g अन्तराल [a, b] में सतत् तथा अन्तराल [a, b] में अवकलनीय है। यदि $f(a) = f(b) = 0$ तो दर्शाइये कि एक बिन्दु $c \in (a, b)$ इस प्रकार है कि $g'(c) f(c) + f'(c) = 0$.

Ans.

Sol.

48. Assume that f is continuous on [a, b], $a > 0$ and differentiable on an open interval (a, b). Show that if $\frac{f(a)}{a} = \frac{f(b)}{b}$, then there exist $x_0 \in (a, b)$ such that $x_0 f'(x_0) = f(x_0)$.

माना f अन्तराल [a, b], $a > 0$ में सतत् फलन है तथा (a, b) में अवकलनीय है तो सिद्ध कीजिये यदि $\frac{f(a)}{a} = \frac{f(b)}{b}$, है तो एक $x_0 \in (a, b)$ अस्तित्व में होगा जिसके लिये $x_0 f'(x_0) = f(x_0)$ है।

Ans.

Sol. Let $g(x) = \frac{f(x)}{x}$, $x \in [a, b]$

By Rolle's theorem, $g'(x_0) = 0$

$$\Rightarrow \quad \frac{f'(x_0)x_0 - 1.f(x_0)}{x_0^2} = 0$$

$$\Rightarrow \quad f'(x_0) = \frac{f(x_0)}{x_0}$$

49. f(x) and g (x) are differentiable functions for $0 \le x \le 2$ such that f(0) = 5, g (0) = 0, f(2) = 8, g (2) = 1. Show that there exists a number c satisfying $0 < c < 2$ and f' (c) = 3 g' (c).

f(x) तथा g (x) अन्तराल $0 \le x \le 2$ में अवकलनीय फलन है जिनके लिये f(0) = 5, g (0) = 0, f(2) = 8, g (2) = 1 है। सिद्ध कीजिये एक संख्या c इस प्रकार अस्तित्व में है जो $0 < c < 2$ तथा f' (c) = 3 g' (c) को संतुष्ट करती है।

Ans.

Sol. Let $n(x) = f(x) - kg(x)$

$n'(x) = f'(x) - k\,g'(x)$

$h'(c) = 0$

$h(0) = h(g)$

$f(0) - kg(0) = f(g) - kg(g)$

$S - kxo = 8 - k(1)$

$k = 3$ H.P.

50. If p (x) = $51x^{101} - 2323x^{100} - 45x + 1035$, using Rolle's theorem, prove that at least one root of p(x) lies between $(45^{1/100}, 46)$.

यदि p (x) = $51x^{101} - 2323x^{100} - 45x + 1035$, रोले प्रमेय का प्रयोग करते हुये सिद्ध कीजिये कि p(x) का कम से कम एक शून्य अन्तराल $(45^{1/100}, 46)$ में स्थित है।

Ans.

Sol. Let $g(x) = \int p(x)\,dx + K$

$$g(x) = \frac{x^{102}}{2} - 23\,x^{101} - \frac{45x^2}{2} + 1035x + K$$

$$= \frac{x^{102} - 46x^{101} - 45x^2 + 2070x}{2} + K$$

$$= \frac{x(x^{100} - 45)(x - 46)}{2} + K$$

$g(45^{1/100}) = g(46)$

$\Rightarrow$ $g'(x) = 0$ has exactly one root in $(45^{1/100}, 46)$

51. Let $f: R \to R$ be a twice differentiable function such that $f(a) = 0, f(b) = 2, f(c) = -1, f(d) = 2, f(e) = 0,$ where $a < b < c < d < e$. Determine the minimum number of zeroes of. $g(x) = (f'(x))^2 + f(x).f''(x)$ in the interval [a, e].

दिया है $f: R \to R$ एक द्विक अवकलनीय फलन इस प्रकार से है कि $f(a) = 0, f(b) = 2, f(c) = -1, f(d) = 2, f(e) = 0,$ जहाँ $a < b < c < d < e$ तो अन्तराल [a, e] में $g(x) = (f'(x))^2 + f(x).f''(x)$ के शून्यों की न्यूनतम संख्या ज्ञात करो।

Ans. 6

Sol. $g(x) = \frac{d}{dx}(f(x).\ f'(x))$

to get the zero of g(x) we take function

$h(x) = f(x).\ f'(x)$

between any two roots of h(x) there lies at least one root of $h'(x) = 0$ i.e. $g(x) = 0$

Now $h(x) = 0 \quad \Rightarrow \quad f(x) = 0$ or $f'(x) = 0$

As $f(x) = 0$ has 4 minimum solutions.

and $f'(x) = 0$ minimum 3 solutions.

$\therefore$ $h(x) = 0$ minimum 7 solutions.

$\Rightarrow$ $h'(x) = g(x) = 0$ has minimum 6 solutions.

LMVT

52. Let f be continuous on [a, b] and differentiable on (a, b). If $f(a) = a$ and $f(b) = b$, show that there exist distinct c_1, c_2 in (a, b) such that $f'(c_1) + f'(c_2) = 2$.

माना f अन्तराल [a, b] में सतत् तथा अन्तराल (a, b) में अवकलनीय है यदि $f(a) = a$ एवं $f(b) = b$ है तो सिद्ध कीजिये ऐसे दो भिन्न c_1, c_2, अन्तराल (a, b) में, अस्तित्व में है जिनके लिये $f'(c_1) + f'(c_2) = 2$ है।

Ans.

Sol. By LMVT in $\left[a, \frac{a+b}{2}\right]$

$$\frac{f\left(\frac{a+b}{2}\right) - f(a)}{\frac{a+b}{2} - a} = f'(c_1),\ c_1 \in \left(a, \frac{a+b}{2}\right) \ldots\ldots (i)$$

By LMVT in $\left[\frac{a+b}{2},\ b\right]$

$$\frac{f(b)-f\left(\frac{a+b}{2}\right)}{b-\frac{a+b}{2}} = f'(c_2),\ c_2 \in \left(\frac{a+b}{2}, b\right) \text{.....(ii)}$$

(i) + (ii) $\Rightarrow$ $f'(c_1) + f'(c_2) = 2$

53. For what value of a, m and b does the function $f(x) = \begin{cases} 3 & x = 0 \\ -x^2 + 3x + a & 0 < x < 1 \\ mx + b & 1 \le x \le 2 \end{cases}$ satisfy the hypothesis of the mean value theorem for the interval [0, 2].

a, m तथा b के किन मानों के लिये फलन $f(x) = \begin{cases} 3 & x = 0 \\ -x^2 + 3x + a & 0 < x < 1 \\ mx + b & 1 \le x \le 2 \end{cases}$ माध्य मान प्रमेय को अन्तराल [0, 2] में संतुष्ट करता है।

Ans. a = 3, b = 4 and m = 1

Sol. Using Largrange mean value theorem

Let f : [a, b] $\to$ R be a continuous function and differentiable on (a, b). Then there exists c $\in$ (a, b) such that

$$f(c) = \frac{f(b)-f(a)}{b-a}$$

Hence f(x) is continuous

f(0) = 3 = a

f(1) = 2 = m + b

Hence b = 4

f(x) is differentiable at 1

So $f'(1) = -2 = m$

54. Let a > 0 and f be continuous in [–a, a]. Suppose that $f'(x)$ exists and $f'(x) \le 1$ for all x $\in$ (–a, a). If $f(a) = a$ and $f(-a) = -a$, show that f(0) = 0.

माना a > 0 है तथा अन्तराल [–a, a] में फलन f सतत् है। माना $f'(x)$ अस्तित्व में है एवं $f'(x) \le 1 \ \forall \ x \in (-a, a)$. यदि $f(a) = a$ तथा $f(-a) = -a$ तो सिद्ध कीजिये $f(0) = 0$ है।

Ans.

Sol. As f is continuous in [–a, a] and differential in (–a, a)

So using Largrange's mean value theorem we get some $c \in (-a, a)$ such that $f'(c) = \frac{f(-a) - f(a)}{-a - a} = 1$

And f(c) = c + d from f(a) = a we getd = 0

Hence f(c) = c

f(0) = 0

55. Let f be a differentiable function for all x and that $|f'(x)| \le 2$ for all x. If $f(1) = 2$ and $f(4) = 8$ then compute the value of $f^2(2) + f^2(3)$.

माना f सभी x के लिये अवकलनीय फलन है तथा सभी x के लिये $|f'(x)| \le 2$ है। यदि $f(1) = 2$ तथा $f(4) = 8$ है तो $f^2(2) + f^2(3)$ का मान ज्ञात करें।

Ans. 52

Sol. Since, f is differentiable for all x.

$\Rightarrow$ f is differentiable in (1, 4)

Also, f is continuous on [1, 4]

So, by Lagrange mean value theorem in [1, 4], there exists $c \in (0, 4)$ such that $f'(c) = \frac{f(4) - f(1)}{3}$

$f'(c) = 2$

$\Rightarrow f'(x) = 2$

$\Rightarrow f(x) = 2x + k$

f(1) = 2 + k

k = 0

$\Rightarrow f(x) = 2x$

MAXIMA AND MINIMA AT A POINT

56. Find the greatest & least value for the function :

(a) $y = x + \sin 2x,\ 0 \le x \le 2\pi$ (b) $y = 2\cos 2x - \cos 4x,\ 0 \le x \le \pi$

फलन का महत्तम एवं न्यूनतम मान बताइये :

(a) $y = x + \sin 2x,\ 0 \le x \le 2\pi$ (b) $y = 2\cos 2x - \cos 4x,\ 0 \le x \le \pi$

Ans. (a) Max at $x = 2\pi$, Max value $= 2\pi$, Min. at $x = 0$, Min value $= 0$

(b) Max at $x = \pi/6$ & also at $x = 5\pi/6$ and Max value $= 3/2$, Min at $x = \pi/2$, Min value $= -3$

Sol.

57. Let P(x) be a polynomial of degree 5 having extremum at $x = -1, 1$ and $\underset{x\to 0}{\text{Lim}}\left(\frac{P(x)}{x^3} - 2\right) = 4$. If M and m are the maximum and minimum value of the function $y = P'(x)$ on the set $A = \left\{x \middle| x^2 + 6 \le 5x\right\}$ then find $\frac{m}{M}$.

माना P(x) एक 5 घातीय बहुपद है जिसके चरम बिन्दु $x = -1, 1$ है तथा $\underset{x\to 0}{\text{Lim}}\left(\frac{P(x)}{x^3} - 2\right) = 4$ है। यदि M तथा m फलन $y = P'(x)$ के समुच्चय $A = \left\{x \middle| x^2 + 6 \le 5x\right\}$ में अधिकतम तथा न्यूनतम मान है तो $\frac{m}{M}$ का मान ज्ञात कीजिये।

Ans. 6

Sol. $P'(x) = -18x^4 + 18x^2 = 18\left(x^2 - x^4\right)$

and $P''(x) = 18\left(2x - 4x^3\right) = 36 \times \left(1 - 2x^2\right)$

Also, $A = \left\{x : x^2 + 6 \le 5x\right\}$

$\Rightarrow x \in [2,3]$

Clearly, $P''(x) < 0\ \forall\ x \in [2,3]$

So, $y = P'(x)$ is decreasing function in [2, 3]

$\therefore M = P'\max(x = 2) = 18(4 - 16) = -18 \times 12$

and $m = P'_{min}(x = 3) = 18(9 - 81) = -18 \times 72$

Hence, $\frac{m}{M} = \frac{-18 \times 72}{-18 \times 12} = 6$

58. Of all the lines tangent to the graph of the curve $y = \frac{6}{x^2 + 3}$, find the equations of the tangent lines of minimum and maximum slope.

वक्र $y = \frac{6}{x^2 + 3}$ कि सभी स्पर्श रेखाओं के लिये अधिकतम एवं न्यूनतम प्रवणता की स्पर्श रेखाओं का समीकरण बताइये।

Ans. $3x + 4y - 9 = 0$; $3x - 4y + 9 = 0$

Sol. The given function is $f(x) = y = \frac{6}{x^2 + 3}$.

$$\Rightarrow f'(x) = -\frac{12x}{(x^2 + 3)^2}$$

Let $g(x) = f'(x) = -\left(\frac{12x}{(x^2 + 3)^2}\right)$

When the tangent line has the minimum or maximum slope the first derivative of the function representing the tangent lines (i.e. g(x)) should be equal to zero.

$$\Rightarrow g'(x) = -\left(\frac{12(x^2 + 3)^2 - 48x^2(x^2 + 3)}{(x^2 + 3)^4}\right) = 0.$$

$$\Rightarrow 12(x^2 + 3)^2 - 48x^2(x^2 + 3) = 0$$

$$\Rightarrow 36(x^2 + 3)(1 - x^2) = 0$$

Assuming that the domain of the function is the set of real numbers, we get, $x = \pm 1$.

It can be seent that $g(-1) = -\left(\frac{-12}{(1+3)^2}\right) = \frac{3}{4}$

and

$\Rightarrow$ The value of the derivative of the function f(x) is maximum at $x = -1$ and minimum at $x = 1$

when $x = -1$, $\quad y = \frac{6}{1+3} = \frac{3}{2}$

when $x = 1$, $\quad y = \frac{6}{1+3} = \frac{3}{2}$

The tangent the having the maximum slope has a slope of $\frac{3}{4}$ and passes through point $\left(-1, \frac{3}{2}\right)$

$\Rightarrow$ The equation of the tangent line having the maximum slope is $\frac{y - \frac{3}{2}}{x+1} = \frac{3}{4}$.

$\Rightarrow 3x - 4y + 9 = 0$

The tangent line having the minimum slope has a slope of $-\frac{3}{4}$ and passes through point $\left(1, \frac{3}{2}\right)$.

$\Rightarrow 3x + 4y - 9 = 0$

59. If $y = \frac{ax + b}{(x-1)(x-4)}$ has a turning value at $(2, -1)$ find a & b and show that the turning value is a maximum.

यदि $y = \frac{ax + b}{(x-1)(x-4)}$ का चरम बिन्दु $(2, -1)$ है तो a तथा b का मान ज्ञात कीजिये सिद्ध कीजिये कि चरम बिन्दु उच्चिष्ठ है।

Ans. $a = 1, b = 0$

Sol. $(2, -1) \quad \Rightarrow \quad -1 = \frac{2a + b}{(+1)(-2)}$

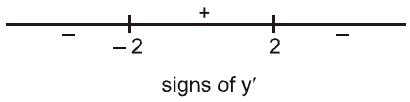
$2a + b = 2$

$y' = \frac{a(x-1)(x-4) - (ax+b)(2x-5)}{(x-1)^2(x-4)^2}$

$y' = 0$ at $x = 2 \quad \Rightarrow \quad b = 0 \quad \Rightarrow \quad a = 1$

$y = \frac{x}{(x-1)(x-4)}$

$$y' = \frac{(2+x)(2-x)}{(x-1)^2(x-4)^2}$$



At $x = 2$, y' changes sign from positive to negative

$\Rightarrow$ $x = 2$ is point of maxima.

60. Let $f(x)$ be a cubic polynomial which has local maximum at $x = -1$ and $f'(x)$ has a local minimum at $x = 1$. If $f(-1) = 10$ and $f(3) = -22$, then find the distance between its two horizontal tangents.

माना $f(x)$ तीन घात का बहुपद है जिसका स्थानीय उच्चिष्ठ $x = -1$ पर है एवं $f'(x)$ का स्थानिय निम्ननिष्ठ $x = 1$ पर है। यदि $f(-1) = 10$ एवं $f(3) = -22$ तो फलन की दो क्षैतिज स्पर्श रेखाओं के मध्य दूरी ज्ञात कीजिये।

Ans. 32

Sol. $x = 1 \quad \Rightarrow \quad 3 = a + b$

$$\left.\frac{d^2y}{dx^2}\right|_{x=1} = 0 \quad \Rightarrow \quad 3a + b = 0$$

$$a = -\frac{3}{2}, \qquad b = \frac{9}{2}$$

61. Consider the function $f(x) = \begin{cases} \sqrt{x}\ln x & \text{when } x > 0 \\ 0 & \text{for } x = 0 \end{cases}$:

(a) Find whether f is continuous at $x = 0$ or not.

(b) Find the minima and maxima if they exist.

(c) Does $f'(0)$ exist ? Find $\lim\limits_{x\to 0} f'(x)$.

(d) Find the inflection points of the graph of $y = f(x)$.

माना फलन $f(x) = \begin{cases} \sqrt{x}\ln x & \text{when } x > 0 \\ 0 & \text{for } x = 0 \end{cases}$:

(a) ज्ञात कीजिये कि f बिन्दु $x = 0$ पर सतत् है अथवा नहीं

(b) उच्चिष्ठ एवं निम्निष्ठ मान ज्ञात कीजिये यदि अस्तित्व में हो तो

(c) क्या $f'(0)$ अस्तित्व में है ? $\lim_{x\to 0} f'(x)$ ज्ञात कीजिये।

(d) आरेख $y = f(x)$ के नति परिवर्तन बिन्दु ज्ञात कीजिये।

Ans. **(a)** f is continuous at $x = 0$; **(b)** $x = e^{-2}, 0$; **(c)** does not exist, does not exist ; **(d)** pt. of inflection $x = 1$

Sol. $f(x) = \sqrt{x}\ln x$

$$f'(x) = \frac{1}{\sqrt{x}} + \frac{\ln x}{2\sqrt{x}}$$

$$f''(x) = -\frac{1}{2x\sqrt{x}} + \frac{\frac{1}{\sqrt{x}} + \frac{\ln x}{2x\sqrt{x}}}{x}$$

For point of inflection $f''(x) = 0 \Rightarrow x = 1$

Since $x = k \to k = 1s$

62. Let p(x) be a polynomial of degree 4 having extremum at $x = 1, 2$ and $\lim_{x\to 0}\left(1+\frac{p(x)}{x^2}\right) = 2$. Then the value of p(2) is. **[JEE-2009]**

माना p(x) एक चार घात का बहुपद है जिसका चरम बिन्दु $x = 1, 2$ पर है तथा $\lim_{x\to 0}\left(1+\frac{p(x)}{x^2}\right) = 2$ है। तो p(2) का मान है। **[JEE-2009]**

Ans. 0

Sol. $\lim_{x\to 0}\left(1+\frac{p(x)}{x^2}\right) = 2$

$$\Rightarrow \lim_{x\to 0}\frac{p(x)}{x^2} = 1$$

Let $p(x) = ax^4 + bx^3 + cx^2$

$$\because \lim_{x\to 0}\frac{p(x)}{x^2} = 1 \Rightarrow c = 1$$

$p(x) = ax^4 + bx^3 + x^2$

Now, $p'(x) = 4ax^3 + 3bx^2 + 2x$

$\because$ p'(1) = 0, p'(2) = 0

$\Rightarrow 4a + 3b + 2 = 0$

$32a + 12b + 4 = 0 \Rightarrow a = \frac{1}{4}$, $b = -1$

$\Rightarrow p(x) = \frac{1}{4}x^4 - x^3 + x^2 \Rightarrow p(2) = 4 - 8 + 4 = 0$

63. The maximum value of the function $f(x) = 2x^3 - 15x^2 + 36x - 48$ on the set $A = \{x \mid x^2 + 20 \le 9x\}$ is.

[JEE-2009]

फलन $f(x) = 2x^3 - 15x^2 + 36x - 48$ का समुच्चय $A = \{x \mid x^2 + 20 \le 9x\}$ में अधिकतम मान है। **[JEE-2009]**

Ans. 7

Sol. $f(x) = 2x^3 - 15x^2 + 36x - 48$

Set $A = \{x \mid x^2 + 20 \le 9x\}$

$\because x^2 - 9x + 20 \le 0$

$(x-5)(x-4) \le 0$

$\Rightarrow x \in [4, 5]$

Now, $f'(x) = 6x^2 - 30x + 36 = 0$

$\Rightarrow x^2 - 5x + 6 = 0$

$x = 2, 3$ and $f(x) \uparrow$ in $x \in (-\infty, 2) \cup (3, \infty)$

$\Rightarrow$ In the set A, f(x) is increasing

$\Rightarrow f(x)_{max} = f(5)$

$= 2.125 - 15.25 + 36.5 - 48$

$= 7$

64. Let f be a function defined on R (the set of all real numbers) such that
$f'(x) = 2010(x - 2009)(x - 2010)^2(x - 2011)^3(x - 2012)^4$, for all $x \in R$.
If g is a function defined on R with values in the interval $(0, \infty)$ such that $f(x) = \ln(g(x))$, for all $x \in R$, then the number of points in R at which g has a local maximum is. **[JEE-2010]**

माना R (सभी वास्तविक संख्याओं का समुच्चय) में परिभाषित फलन इस प्रकार है कि
$f'(x) = 2010(x - 2009)(x - 2010)^2(x - 2011)^3(x - 2012)^4$, सभी $x \in R$ के लिये यदि g फलन R परिभाषित है जिसके मान अन्तराल $(0, \infty)$ में स्थित है इस प्रकार है कि $f(x) = \ln(g(x))$, सभी $x \in R$ के लिये, तो R में स्थित उन बिन्दुओं की संख्या जिसके लिये g का एक स्थानीय उच्चिष्ठ हो। **[JEE-2010]**

Ans. (b) 1

Sol. $f'(x)=2010(x-2009)(x-2010)^2(x-2011)^3(x-2012)^4 \forall x \in R$

$f(x) = \ell n(g(x)) \forall x \in R$

$g(x) = e^{f(x)}$

$g'(x) = 0 \Rightarrow e^{f(x)}.f'(x) = 0 \Rightarrow f'(x) = 0$

increasing (+) 2009 decreasing (−) 2010 decreasing (−) 2011 increasing (+) 2012 increasing (+)

local maximum at x = 2009, hence only 1 point.

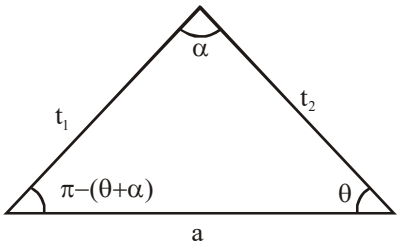
GEOMETRY BASED QUESTIONS ON MAXIMA AND MINIMA

65. Find the maximum perimeter of a triangle on a given base 'a' and having the given vertical angle α.

दिये गए आधार 'a' तथा दिये गये शीर्ष कोण α के लिये त्रिभुज का अधिकतम परिमाप ज्ञात कीजिये।

Ans. $P_{max} = a\left(1+\cos ec\frac{\alpha}{2}\right)$

Sol.



$$\frac{a}{\sin\alpha} = \frac{t_1}{\sin\theta} = \frac{t_2}{\sin(\theta+\alpha)}$$

$t_1 = \frac{a}{\sin\alpha}\sin\theta$, $t_2 = \frac{a}{\sin\alpha}\sin(\theta+\alpha)$

to maximize perimeter we maximize $t_1 + t_2$

$t_1 + t_2 = f(\theta) = 2\frac{a}{\sin\alpha}\{\sin(\theta)+\sin(\theta+\alpha)\}$

$= \frac{a}{\sin\alpha}\left\{\sin\left(\theta+\frac{\alpha}{2}\right)\cos\frac{\alpha}{2}\right\}$

for maximum $\sin\left(\theta+\frac{\alpha}{2}\right) = 1$

66. The length of three sides of a trapezium are equal, each being 10 cms. Find the maximum area of such a trapezium.

एक समलम्ब चतुर्भुज की तीन भुजाये समान है एवं 10 से.मी. के बराबर है। चतुर्भुज का अधिकतम क्षेत्रफल ज्ञात कीजिये।

Ans. $75\sqrt{3}$ sq. units

Sol. Consider the trapezim ABCD, where AD // BC, AB = CD = AD = 10 am

Let BC be x

Draw AM $\perp$ BC and DN $\perp$ BC

$\therefore$ MN = 10 cm

Let BC = x cm

Now, $CN = \left(\frac{x-10}{2}\right)cm = \left(\frac{x}{2}-5\right)cm$

Using Pythagoras theorem in ΔDNC,

$$DN = \sqrt{CD^2 - CN^2} = \sqrt{(10)^2 - \left(\frac{x}{2}-5\right)^2}\ cm$$

Thus, Area of trapezim $ABCD = \frac{1}{2}(BC + AD) \times DN = \frac{1}{2}(x+10)\sqrt{(10)^2 - \left(\frac{x}{2}-5\right)^2}\ cm^2$

Let $A = \frac{1}{2}(x+10)\sqrt{(10)^2 - \left(\frac{x}{2}-5\right)^2}\ cm^2$

$$\Rightarrow A^2 = \frac{1}{4}(x+10)^2\left[(10)^2 - \left(\frac{x}{2}-5\right)^2\right]cm^2$$

$$\Rightarrow A^2 = \frac{1}{4}(x+10)^2\left[10+\left(\frac{x}{2}-5\right)\right]\left[10-\left(\frac{x}{2}-5\right)\right]$$

$$\Rightarrow A = \frac{1}{4}(x+10)^2\left(\frac{x}{2}+5\right)\left(15-\frac{x}{2}\right)$$

$$\Rightarrow A^2 = \frac{1}{16}(x+10)^3(30-x)$$

$\Rightarrow$ Let $P = 16A^2 = (x + 10)^3(30 - x)$

More, area is more $\frac{dp}{dx} = (x+10)^3(-1) + 3(x+10)^2(30-x)$

$$= (x+10)^2\left[-(x+10) + 3(30-x)\right]$$

$$= (x+10)^2 (80-4x)$$

$$= 4(x+10)^2 (20-x)$$

$$\frac{d^2P}{dx^2} = 4\left[2(x+10)(20-x)+(x+10)^2(-1)\right]$$

$$= 4(x+10)\,[2(20-x)-(x-10)]$$

$$= 4(x+10)(30-3x)$$

$$= 12(x+10)(10-x)$$

$\frac{dP}{dx} = 0$ Implies that $x + 10 = 0$ or $20 - x = 0$

$\Rightarrow x = -10$ or $x = 20$

x cannot be negative

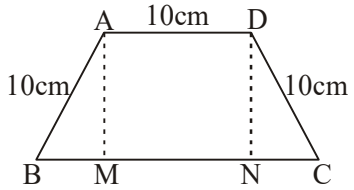
$\therefore x = 210$

Now, for $x = 20$, $\frac{d^2P}{dx^2} = 12 \times 30 \times (-10) = -3600 < 0$

Thus, P is maximum area of the trapezium $= \frac{1}{4}\sqrt{(20+10)^3 (30-20)}$

$$= \frac{1}{4} \times \sqrt{(30)^3 \times 10}\ \text{cm}^2$$

$$= \frac{1}{4} \times 30 \times 10\sqrt{3}\ \text{cm}^2 = 75\sqrt{3}\ \text{cm}^2$$

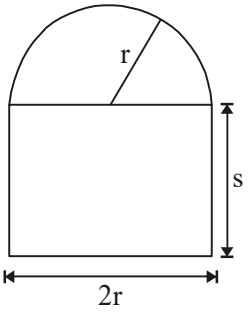


67. The plan view of a swimming pool consists of a semicircle of radius r attached to a rectangle of length '2r' and width 's'. If the surface area A of the pool is fixed, for what value of 'r' and 's' the perimeter 'P' of the pool is minimum.

एक तरणताल का तलीय दर्शन में एक अर्धवृत (r त्रिज्या) के साथ एक '2r' लम्बाई एवं 's' चौड़ाई का आयत जुड़ा हुआ है। यदि ताल का पृष्ठीय क्षेत्रफल A नियत है, तो 'r' तथा 's' के किन मानों के लिये परिमाप 'P' अधिकतम होगा।

Ans. $r = \sqrt{\frac{2A}{\pi+4}}$, $s = \sqrt{\frac{2A}{\pi+4}}$

Sol. $A = \frac{\pi r^2}{2} + rs \quad P = 2s + 2r + \pi r$



$$P = \left(\frac{\pi r}{2} + 2s\right) + 2r + \frac{\pi r}{2} \qquad P = \frac{A}{r} + 2r + \frac{\pi r}{2}$$

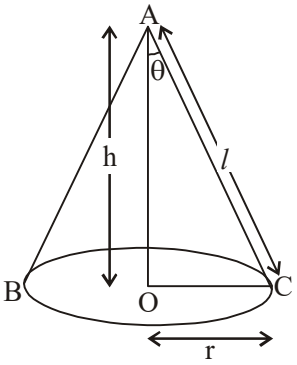
$$\text{Now } \frac{dP}{dr} = 0 \Rightarrow r = \sqrt{\frac{2A}{\pi+4}} \quad \Rightarrow s = \sqrt{\frac{2A}{\pi+4}}$$

68. For a given curved surface of a right circular cone when the volume is maximum, prove that the semi vertical angle is $\sin^{-1}\frac{1}{\sqrt{3}}$.

दिये गये पृष्ठीय क्षेत्रफल के लिये एक सम वृतीय शंकु का आयतन जब अधिकतम हो तो सिद्ध कीजिये कि अर्द्ध शीर्ष कोण $\sin^{-1}\frac{1}{\sqrt{3}}$ होगा।

Ans.

Sol.



With usual notation, given that total surface area $S = \pi r l + \pi r^2$

$$\Rightarrow S = \pi r\sqrt{r^2+h^2} + \pi r^2 \quad \left(\because l = \sqrt{r^2+h^2}\right)$$

$$\Rightarrow \frac{S}{\pi r} - r = \sqrt{r^2+h^2} \Rightarrow \frac{S^2}{\pi^2 r^2} - \frac{2S}{\pi} = h^2$$

$$\Rightarrow h = \sqrt{\frac{S^2}{\pi^2 r^2} - \frac{2S}{\pi}} \quad \left(\because \frac{S^2}{\pi^2 r^2} > \frac{2S}{\pi}\right) \qquad \text{_____(1)}$$

and volume $V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi r^2 \sqrt{\frac{S^2}{\pi^2 r^2} - \frac{2S}{\pi}}$

$$\Rightarrow V = \frac{r}{3}\sqrt{S^2 - 2S\pi r^2}, r^2 < \frac{S}{2\pi} \text{ i.e., } O < r < \sqrt{\frac{S}{2\pi}}$$

Since, V is maximum, then V^2 is maximum

Now, $V^2 = \frac{S^2 r^2}{9} - \frac{2S\pi r^4}{9}, O < r < \sqrt{\frac{S}{2\pi}}$

$$\therefore \frac{d}{dr}\left(V^2\right) = \frac{2S^2}{9} = \frac{8S\pi r^3}{9}$$

and $\frac{d}{dr^2}\left(V^2\right) = \frac{2S^2}{9} - \frac{24S\pi r^2}{9}$

For maxima put $\frac{dV}{dr} = O$

$$\Rightarrow \frac{2rS^2}{9} - \frac{8S\pi r^3}{9} = O \Rightarrow r^2 = \frac{S}{4\pi}$$

$$\Rightarrow \frac{d^2\left(V^2\right)}{dr^2} < O \text{ for } r = \sqrt{\frac{S}{4\pi}}$$

From equ. (i) $h = \sqrt{\frac{S^2}{\pi^2 r^2} - \frac{2S}{\pi}} = \sqrt{\frac{S^2(4\pi)}{\pi^2 S} - \frac{2S}{\pi}} = \sqrt{\frac{2S}{\pi}}$

If θ is semi-vertical angle of the cone when the volume is maximum, then in right triangle AOC,

$$\sin\theta = \frac{r}{r^2 + h^2} = -\frac{\sqrt{\frac{S}{4\pi}}}{\sqrt{\frac{S}{4\pi} + \frac{S\pi}{2}}} = \frac{1}{\sqrt{1+8}} \text{ i.e. } \theta = \sin^{-1}\left(\frac{1}{3}\right)$$

69. By the post office regulations, the combined length & girth of a parcel must not exceed 3 metre. Find the volume of the biggest cylindrical (right circular) packet that can be sent by the parcel post.

पोस्ट ऑफिस के नियमो के अनुसार एक पार्सल की लम्बाई एवं घेरा का योग दोनों 3 मीटर से अधिक नही होनी चाहिये समवृताकार बेलन का अधिकतम आयतन ज्ञात कीजिये जिसे पार्सल पोस्ट द्वारा भेजा जा सकता हो।

Ans. $1/\pi$ cu m

Sol. Let the dimension of the box with square base be x, x, h

Volume of the rectangular box $V = x^2h$

$x^2h = 1000$

Area of top $= x^2$

Area of base $= x^2$

Area of faces $= 4xh$

$$E = 15x^2 + 25x^2 + 20(4xh) + 300$$

$$E = 40x^2 + 80x\left(\frac{1000}{x^2}\right) + 300$$

$$= 40x^2 + \frac{80\times1000}{x} + 300$$

For maxima or minima, $\frac{dE}{dx} = 0$

$$\therefore 80x - \frac{80\times1000}{x^2} = 0$$

$$\therefore x^3 = 1000$$

$x = 10$

$$\frac{d^2E}{dx^2} = 80 + \frac{2\times80\times1000}{x^3} = +\text{ive}$$

$\therefore$ Min. when $x = 10$

$$h = \frac{1000}{x^2} = 10$$

Hence the box should be a cube of edge 10 feet.

70. A running track of 440 ft. is to be laid out enclosing a football field, the shape of which is a rectangle with semi circle at each end. If the area of the rectangular portion is to be maximum, find the length of its sides. (Use : $\pi \approx 22/7$.)

एक फुटबाल के मैदान के चारो ओर एक 440 ft. लम्बाई की धावक पट्टिका बनाई गई है। फुटबाल का मैदान आयताकार है जिसके छोर अर्धवृताकार है। यदि आयताकार भाग का क्षेत्रफल अधिकतम कर दिया जाये तो इसके लम्बाईयाँ क्या होगी। (Use : $\pi \approx 22/7$.)

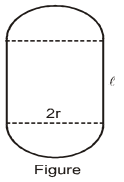
Ans. 110', 70'

Sol. $2\ell + 2\pi r = 440$

$A = \ell\, 2r = -2\pi r^2 + 440r$

$\frac{dA}{dr} = -4\pi r + 440 = 0$

at $r = \frac{110}{\pi}$



Figure

71. A closed rectangular box with a square base is to be made to contain 1000 cubic feet. The cost of the material per square foot for the bottom is 15 paise, for the top 25 paise and for the sides 20 paise. The labour charges for making the box are Rs. 3/-. Find the dimensions of the box when the cost is minimum.

एक बंद आयताकार डिब्बे का आधार वर्ग है एवं आयतन 1000 cubic feet है। आधार को बनाने में काम में लिये गये माल की प्रति स्वायर फीट की दर 15 पैसे है, छत के माल की 25 पैसे है तथा दिवारों के लिये 20 पैसे है। मजूदरी का चार्ज 3 रू. है। न्यूनतम लागत के बक्से की विमाये बताइयें।

Ans. side 10', height 10'

Sol.

72. Find the area of the largest rectangle with lower base on the x-axis & upper vertices on the curve $y = 12 - x^2$.

ऐसे आयत का अधिकतम क्षेत्रफल ज्ञात कीजिये जिसका आधार x-अक्ष पर एवं दोनों शीर्ष वक्र $y = 12 - x^2$ पर स्थित है।

Ans. 32 sq. units

Sol. Let A be area of rectangle.

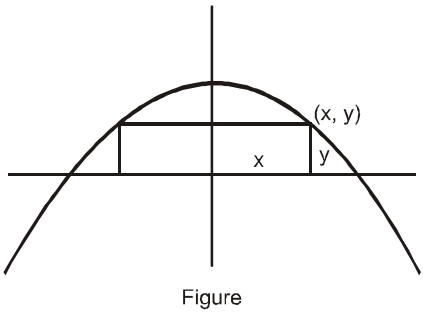
$A = (2x)(y)$

$= 2x(12 - x^2)$

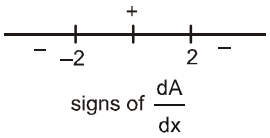
$\frac{dA}{dx} = 6(2 + x)(2 - x)$

At $x = 2$, A has largest value.

Largest $A = 32$



Figure

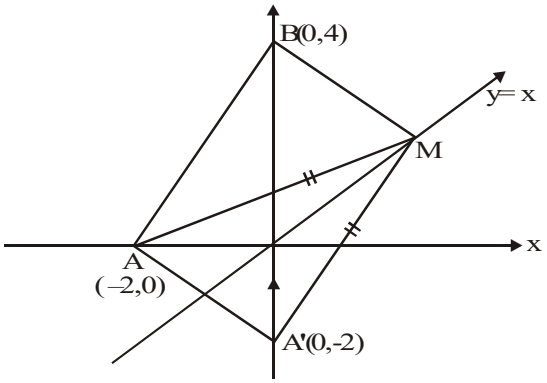


73. Given two points A(–2, 0) & B (0, 4) and a line y = x. Find the co-ordinates of a point M on this line so that the perimeter of the Δ AMB is least.

दिया गया है दो बिन्दु A(–2, 0) और B (0, 4) तथा एक रेखा y = x है। रेखा पर स्थित बिन्दु M के निर्देशांक बताइये जब Δ AMB का परिमाप न्यूनतम हो।

Ans. (0 , 0)

Sol.



To minimize the perimeter.
AM + MB is to be minimized.
i.e. A'M + MB is to be minimized.
[where A' is image of A in y = x.]
Obviously A'M + MB is minimized.
when A',M and B are collinear.
i.e. M coincide with origin.
$\therefore$ M ≡ (0, 0)

POINT OF INFLECTION & CONCAVITY OF GRAPH

74. Consider the function $y = f(x) = \ln(1 + \sin x)$ with $-2\pi \le x \le 2\pi$. Find :
(a) the zeroes of $f(x)$
(b) inflection points if any on the graph
(c) local maxima and minima of $f(x)$
(d) asymptotes of the graph
(e) sketch the graph of $f(x)$ and compute the value of the definite integral $\int_{-\pi/2}^{\pi/2} f(x)\,dx$.

माना कि फलन $y = f(x) = l\text{n}\,(1 + \sin x)$ में $-2\pi \le x \le 2\pi$ है। ज्ञात कीजिये :

(a) $f(x)$ के शून्य

(b) आरेख पर स्थित नति परिवर्तन बिन्दु यदि होतो

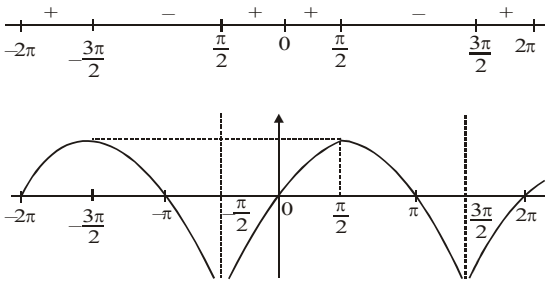
(c) $f(x)$ का स्थानीय उच्चिष्ठ एवं निम्निष्ठ बिन्दु

(d) वक्र की अनन्त स्पर्शी रेखायें

(e) $f(x)$ का आरेख बनाइये साथ ही निश्चित समाकलन $\int_{-\pi/2}^{\pi/2} f(x)\,dx$ की गणना कीजिये।

Ans. (a) $x = -2\pi, -\pi, 0, \pi, 2\pi$, (b) no inflection point, (c) maxima at $x = \pi/2$ and $-3\pi/2$ and no minima, (d) $x = 3\pi/2$ and $x = -\pi/2$, (e) $-\pi\, l\text{n}\, 2$

Sol. $f'(x) = \dfrac{\cos x}{1+\sin x}$



75. Prove that $\sin x + 2x \ge \dfrac{3x\cdot(x+1)}{\pi}\ \forall\ x \in \left[0, \dfrac{\pi}{2}\right]$. (Justify the inequality, if any used). **[JEE 2004]**

सिद्ध कीजिये $\sin x + 2x \ge \dfrac{3x\cdot(x+1)}{\pi}\ \forall\ x \in \left[0, \dfrac{\pi}{2}\right]$. (असमिका के लिये न्यायसंगत तर्क दीजिये, यदि कही उपयोग हो तो).

Ans.

Sol. Let $f(x) = \sin x + 2x$ & $g(x) = \dfrac{3x^2+3x}{\pi}$

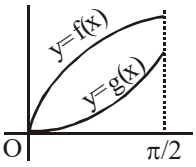
$f'(x) = \cos x + 2 \qquad g'(x) = \dfrac{6x+3}{\pi}$

$f''(x) = -\sin x \qquad g''(x) = \dfrac{6}{\pi}$

$\Rightarrow$ 'f' is increasing & concave down in $\left[0, \dfrac{\pi}{2}\right]$

and 'g' is increasing & concave up in $\left[0, \frac{\pi}{2}\right]$

& $f\left(\frac{\pi}{2}\right) > g\left(\frac{\pi}{2}\right)$.



from the graph $f(x) \geq g(x) \; \forall \; x \in \left[0, \frac{\pi}{2}\right]$

76. The number of distinct real roots of $x^4 - 4x^3 + 12x^2 + x - 1 = 0$ is. **[JEE-2011]**

समीकरण $x^4 - 4x^3 + 12x^2 + x - 1 = 0$ के भिन्न वास्तविक मूलों की संख्या होगी। **[JEE-2011]**

Ans. 2

Sol. Let $f(x) = x^4 - 4x^3 + 12x^2 + x - 1$

$f'(x) = 4x^3 - 12x^2 + 24x$

$f''(x) = 12x^2 - 24x + 24$

$= 12(x^2 - 2x + 2) > 0$

$\Rightarrow$ $f'(x)$ is strictly increasing function

$\because$ $f'(x)$ is cubic polynomial

hence number of roots of $f'(x) = 0$ is 1

$\Rightarrow$ Number of maximum roots of $f(x) = 0$ are 2

Now $f(0) = -1$, $f(1) = 9$, $f(-1) = 15$

$\Rightarrow$ $f(x)$ has exactly 2 distinct real roots.

NATURE OF F(X) FROM GRAPH OF F'(X)

77. The graph of the derivative f' of a continuous function f is shown with $f(0) = 0$. If:

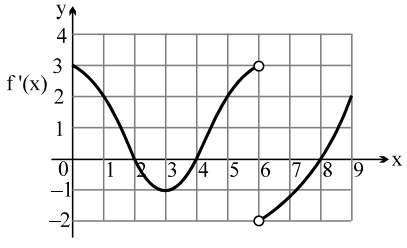
(i) f is monotomic increasing in the interval [a, b)∪(c, d)∪(e, f] and decreasing in (p, q)∪(r, s).

(ii) f has a local minima at $x = x_1$ and $x = x_2$.

(iii) f is concave up in $(l, m) \cup (n, t]$

(iv) f has inflection point at $x = k$

(v) number of critical points of $y = f(x)$ is 'w'.



Find the value of $(a + b + c + d + e) + (p + q + r + s) + (l + m + n) + (x_1 + x_2) + (k + w)$.

सतत् फलन f के अवकलज f' का आरेख बनाया गया है जहाँ $f(0) = 0$ है। यदि :

(i) f अन्तराल $[a, b)\cup(c, d)\cup(e, f]$ में एकदिष्ठ वर्द्धमान है एवं अन्तराल $(p, q)\cup(r, s)$ में एकदिष्ठ ह्रासमान है।

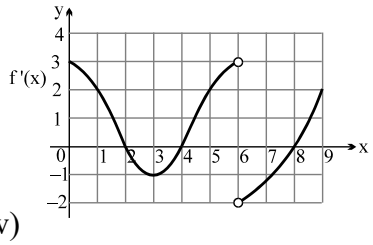
(ii) $x = x_1$ एवं $x = x_2$ पर f का स्थानीय निम्निष्ठ है।

(iii) अन्तराल $(l, m) \cup (n, t]$ में f की नति उपर है (Concave up)

(iv) f का नति परिवर्तन बिन्दु $x = k$ पर है

(v) $y = f(x)$ के चरम बिन्दुओं की संख्या 'w' है।

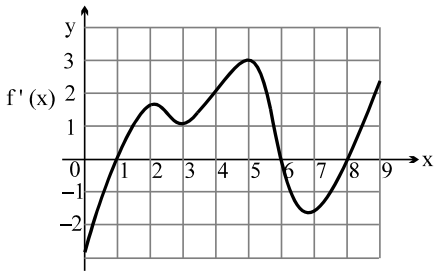
$(a + b + c + d + e) + (p + q + r + s) + (l + m + n) + (x_1 + x_2) + (k + w)$ का मान ज्ञात कीजिये।



Ans. 74

Sol.

78. The graph of the derivative f' of a continuous function f is shown with $f(0) = 0$:



(i) On what intervals is f increasing or decreasing?

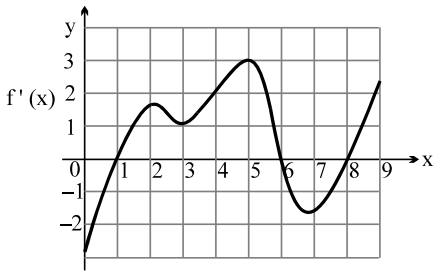
(ii) At what values of x does f have a local maximum or minimum?

(iii) On what intervals is f concave upward or downward?

(iv) State the x-coordinate(s) of the point(s) of inflection.

(v) Assuming that $f(0) = 0$, sketch a graph of f.

सतत् फलन f के अवकलज f' का आरेख जहाँ $f(0) = 0$ है, दिया गया है :



(i) किन अन्तरालों में f वर्द्धमान या ह्रासमान है ?

(ii) x के किन मानों के लिये f का स्थानीय उच्चिष्ठ या स्थानीय निम्निष्ठ प्राप्त होता है ?

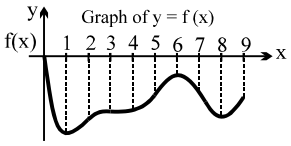
(iii) किन अन्तरालों में f की नति उपर (concave upward) या नति नीचे (downward) है?

(iv) नति परिवर्तन बिन्दुओं के x-निर्देशांक बताइयें।

(v) $f(0) = 0$ मानते हुये f का आरेख बनाइये।

Ans. **(i)** I in $(1, 6) \cup (8, 9)$ and D in $[0, 1) \cup (6, 8)$; **(ii)**.Min. at $x = 1$ and $x = 8$; L.Max. $x = 6$

(iii) CU in $(0, 2) \cup (3, 5) \cup (7, 9)$ and CD in $(2, 3) \cup (5, 7)$; **(iv)** $x = 2, 3, 5, 7$

(v) Graph is 

Sol.

ANALYSIS OF CUBIC EQATION

79. The value of'a' for which $f(x) = x^3 + 3(a-7)x^2 + 3(a^2-9)x - 1$ have a positive point of maximum lies in the interval $(a_1, a_2) \cup (a_3, a_4)$. Find the value of $a_2 + 11a_3 + 70a_4$.

'a' वह मान जिनके लिये $f(x) = x^3 + 3(a-7)x^2 + 3(a^2-9)x - 1$ के उच्चिष्ठ मान के धनात्मक बिन्दु विद्यमान है, अन्तराल $(a_1, a_2) \cup (a_3, a_4)$ में स्थित है। $a_2 + 11a_3 + 70a_4$ का मान ज्ञात कीजिये।

Ans. 320

Sol. $f(x) = x^3 + 3(a-7)x^2 + 3(a^2-9)x - 1$

$f'(x) = 3x^2 + 6(a-7)x + 3(a^2-9)$

$D > 0, f'(0) > 0$

$D > 0 \Rightarrow 36(a-7)^2 - 36(a^2-9) > 0$

$\Rightarrow a < \frac{29}{7}$

$f'(0) = 3(a^2-9) > 0$

$\Rightarrow (a-3)(a+3) > 0$

$\Rightarrow a < -3$ or $a > 3$

Hence $a \in (-\infty, -3) \cup \left(3, \frac{29}{7}\right)$

COMPREHENSION TYPE QUESTIONS

Paragraph for Question nos. 80 to 82

Let $f(x) = \frac{1}{1+x^2}$. Let m be the slope, a be the x-intercept and b be the y-intercept of a tangent to $y = f(x)$, then

माना $f(x) = \frac{1}{1+x^2}$. माना $y = f(x)$ पर स्पर्श रेखा का ढाल m, x-अन्तःखण्ड a और y-अन्तःखण्ड b है, तो

80. Abscissa of the point of contact of the tangent for which m is greatest, is :

स्पर्श रेखा के स्पर्शी बिन्दु का भुज जिसके लिए m महत्तम है :

(A) $\frac{1}{\sqrt{3}}$ (B) 1 (C) – 1 (D) $-\frac{1}{\sqrt{3}}$

Ans. D

Sol. $f'(x) = -\frac{2x}{\left(1+x^2\right)^2}$

$f''(x) = \frac{-2+6x^2}{\left(1+x^2\right)^3}$

$f''(x) = 0$ if $x = \pm\frac{1}{\sqrt{3}}$

Hence, $f'(x)$ is greatest at $x = -\frac{1}{\sqrt{3}}$.

81. Greatest value of b is :

b का अधिकतम मान है :

(A) $\frac{9}{8}$ (B) $\frac{3}{8}$ (C) $\frac{1}{8}$ (D) $\frac{5}{8}$

Ans. A

Sol.

82. Abscissa of the point of contact of tangent for which $\frac{1}{a}$ is greatest, is :

स्पर्श रेखा के स्पर्शी बिन्दु का भुज जिसके लिए $\frac{1}{a}$ महत्तम है :

(A) $\frac{1}{\sqrt{3}}$ (B) 1 (C) – 1 (D) $-\frac{1}{\sqrt{3}}$

Ans. A

Sol.

Paragraph for Question nos. 83 to 85

Consider the cubic $f(x) = 8x^3 + 4ax^2 + 2bx + a$ where $a, b \in R$.

माना घन फलन $f(x) = 8x^3 + 4ax^2 + 2bx + a$ जहाँ $a, b \in R$.

83. For $a = 1$ if $y = f(x)$ is strictly increasing $\forall\ x \in R$ then maximum range of values of b is :

यदि $a = 1$ के लिये $y = f(x)$ एकदिष्ट वर्धमान है $\forall\ x \in R$ तो b के मानों का अधिकतम परास है :

(A) $\left(-\infty, \frac{1}{3}\right]$ (B) $\left(\frac{1}{3}, \infty\right)$ (C) $\left[\frac{1}{3}, \infty\right)$ (D) $(-\infty, \infty)$

Ans. C

Sol. $a = 1$

$f(x) = 8x^3 + 4x^2 + 2bx + 1$

$f'(x) = 24x^2 + 8x + 2b = 2(12x^2 + 4x + b)$

for increasing function, $f'(x) \geq 0 \quad \forall\ x \in R$

$\therefore \quad D \leq 0 \ \Rightarrow \ 16 - 48b \leq 0 \ \Rightarrow \ b \geq \frac{1}{3} \ \Rightarrow$ (C)

84. For $b = 1$, if $y = f(x)$ is non monotonic then the sum of all the integral values of $a \in [1, 100]$, is :

यदि $b = 1$ के लिये, $y = f(x)$ एकदिष्ट नहीं है तब $a \in [1, 100]$ में सभी पूर्णांक मानों का योग होगा :

(A) 4950 (B) 5049 (C) 5050 (D) 5047

Ans. B

Sol. if $b = 1$

$f(x) = 8x^3 + 4ax^2 + 2x + a$

$f'(x) = 24x^2 + 8ax + 2$ or $2(12x^2 + 4ax + 1)$

for non monotonic $f'(x) = 0$ must have distinct roots

hence $D > 0$ i.e. $16a^2 - 48 > 0 \ \Rightarrow \ a^2 > 3; \quad \therefore \quad a > \sqrt{3}$ or $a < -\sqrt{3}$

$\therefore \quad a \in 2, 3, 4, ...$

sum = 5050 – 1 = 5049

85. If the sum of the base 2 logarithms of the roots of the cubic f (x) = 0 is 5 then the value of 'a' is :

f(x) = 0 के मूलों के आधार 2 पर लघुगणक का योग 5 है तो 'a' बराबर :

(A) – 64 (B) – 8 (C) – 128 (D) – 256

Ans. D

Sol. If x_1, x_2 and x_3 are the roots then $\log_2 x_1 + \log_2 x_2 + \log_2 x_3 = 5$

$\log_2(x_1x_2x_3) = 5$

$x_1x_2x_3 = 32$

$-\frac{a}{8} = 32 \quad \Rightarrow \quad a = -256$

Paragraph for Question nos. 86 to 88

If a continuous function f defined on the real line R, assumes positive and negative values in R then the equation $f(x) = 0$ has a root in R. For example, if it is known that a continuous function f on R is positive at some point and its minimum value is negative then the equation $f(x) = 0$ has a root in R.
Consider $f(x) = ke^x - x$ for all real x where k is a real constant. **[JEE 2007]**

यदि संख्या रेखा R पर परिभाषित f एक सतत् फलन है, जो R में ऋणात्मक एवं धनात्मक मान लेता है, तो समीकरण $f(x) = 0$ के R में एक मूल होता है। उदाहरण के लिये यदि f, R में किसी बिन्दु पर धनात्मक है तथा उसका न्यूनतम मान शून्य है तो समीकरण $f(x) = 0$ का R में एक हल होगा।

e ku fy ft ; s $f(x) = ke^x - x$ सभी वास्तविक x के लिये है, जहाँ k एक वास्तविक नियतांक है।

86. The line y = x meets $y = ke^x$ for $k \le 0$ at :

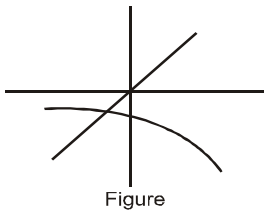
(A) no point (B) one point (C) two points (D) more than two points

रेखा y = x वक्र $y = ke^x$ को $k \le 0$ के लिये मिलती है :

(A) किसी बिन्दु पर नही (B) एक बिन्दु पर (C) दो बिन्दुओं पर (D) दो से अधिक बिन्दुओं पर

Ans. B

Sol.



Figure

87. The positive value of k for which $ke^x - x = 0$ has only one root is :

k का धनात्मक मान जिसके लिये $ke^x - x = 0$ का केवल एक मूल है :

(A) 1/e (B) 1 (C) e (D) $\log_e 2$

Ans. A

Sol. Consider $y = ke^x$ and $y = x$

Let (α, ke^α) be a point on $y = ke^x$

if it lies on $y = x$ also then $\alpha = ke^\alpha$

$$\frac{dy}{dx} = ke^x$$

$$\therefore \quad \left.\frac{dy}{dx}\right|_{x=\alpha} = ke^\alpha = \alpha = 1 \quad \{\because y = x \text{ is tangent to } y = ke^x \text{ at one point}\}$$

$$\therefore \quad 1 = ke \qquad \text{i.e. } k = 1/e$$

88. For $k > 0$, the set of all values of k for which $ke^x - x = 0$ has two distinct roots is :

$k > 0$ के लिये k मानों का समुच्चय जिसके लिये $ke^x - x = 0$ के दो भिन्न मूल है :

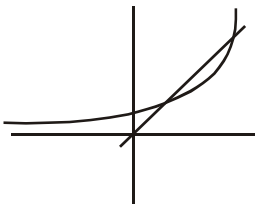
(A) $(0, 1/e)$ (B) $(1/e, 1)$ (C) $(1/e, \infty)$ (D) $(0, 1)$

Ans. A

Sol. Consider $y = ke^x$ and $y = x$

from above question $e^x = \frac{x}{k}$ if we decrease the value of k from $\frac{1}{e}$, then

slope of $y = \frac{x}{k}$ increases



$\therefore$ $y = e^x$ and $y = \frac{x}{k}$ intersect at two distinct points

$$\therefore \quad k \in \left(0, \frac{1}{e}\right)$$

Paragraph for Question nos. 89 to 91

Consider the function $f(x) = \frac{x^2}{x^2 - 1}$

माना फलन $f(x) = \frac{x^2}{x^2 - 1}$

89. The interval in which f is increasing is :

वह अन्तराल जहाँ f वर्धमान है, है :

(A) $(-1, 1)$ (B) $(-\infty, -1) \cup (-1, 0)$

(C) $(-\infty, \infty) - \{-1, 1\}$ (C) $(0, 1) \cup (1, \infty)$

Ans. B

Sol. $y = \frac{x^2}{x^2-1}$; not defined at $x = \pm 1$ [13th, 04-11-2007] [12th, quiz]

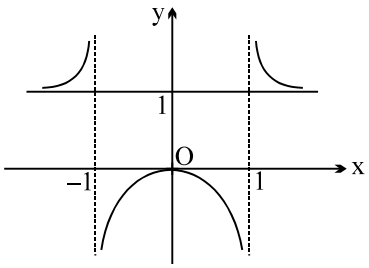
$= 1 + \frac{1}{x^2-1}$; $y' = -\frac{2x}{(x^2-1)^2}$

$\frac{dy}{dx} = 0 \quad \Rightarrow \quad x = 0$ (point of maxima)

as $x \to 1^+,\ y \to \infty$; $x \to 1^-,\ y \to -\infty$

|||ly $x \to -1^+,\ y \to -\infty$; $x \to -1^-,\ y \to \infty$

The graph of $y = \frac{x^2}{x^2-1}$ is as shown



verify all alternatives from the graph.

90. If f is defined from $R - \{-1, 1\} \to R$ then f is :

(A) injective but not surjective
(B) surjective but not injective
(C) injective as well as surjective
(D) neither injective nor surjective.

यदि f, $R - \{-1, 1\} \to R$ पर परिभाषित है, तब

(A) f एकैकी है लेकिन आच्छादक नहीं
(B) आच्छादक है लेकिन एकैकी नहीं
(C) एकैकी और आच्छादक दोनों
(D) न तो एकैकी ना ही आच्छादक

Ans. D

Sol.

91. f has :

(A) local maxima but no local minima
(B) local minima but no local maxima
(C) both local maxima and local minima
(D) neither local maxima nor local minima

f में :

(A) स्थानिय उच्चिष्ठ है लेकिन स्थानिय निम्निष्ठ नहीं
(B) स्थानिय निम्निष्ठ है लेकिन स्थानिय उच्चिष्ठ नहीं
(C) दोनों स्थानीय निम्निष्ठ और उच्चिष्ठ रखता है
(D) न तो स्थानीय निम्निष्ठ है ना ही स्थानिय उच्चिष्ठ

Ans. A

Sol.

Paragraph for Question nos. 92 to 94

Consider a rational function $f(x) = \dfrac{x^2 - 6x + 4}{x^2 + 2x + 4}$ and a quadratic function $g(x) = (1+m)x^2 - 2(1+3m)x - 2(1+m)$ where m is a parameter.

माना परिमेय फलन $f(x) = \dfrac{x^2 - 6x + 4}{x^2 + 2x + 4}$ और द्विघात फलन $g(x) = (1+m)x^2 - 2(1+3m)x - 2(1+m)$ जहाँ m प्राचाल है।

92. Which one of the following statement does not hold good for the rational function f(x)?

(A) It is a continuous function
(B) It has only one asymptote
(C) It has exactly one maxima and one minima
(D) f(x) is monotonic in $(0, \infty)$

परिमेय फलन f(x) के लिये निम्न में से असत्य है?

(A) यह सतत् फलन है
(B) f(x) की केवल एक अनन्त स्पर्शी है
(C) f(x) का केवल एक उच्चिष्ठ और एक ही निम्निष्ठ है
(D) f(x), $(0, \infty)$ में एकदिष्ठ है

Ans. D

Sol. $f(x) = \dfrac{x^2 - 6x + 4}{x^2 + 2x + 4} = 1 - \dfrac{8x}{x^2 + 2x + 4}$

$$f'(x) = -8\left[\frac{(x^2+2x+4) - x(2x+2)}{(x^2+2x+4)^2}\right] = -8\left[\frac{-x^2+4}{(x^2+2x+4)^2}\right] = \frac{8(x^2-4)}{(x^2+2x+4)^2}$$

$f'(x) = 0 \quad \Rightarrow \quad x = 2 \text{ or } -2$

$$f(2) = \frac{4-12+4}{4+4+4} = \frac{-4}{12} = \frac{-1}{3}$$

$$f(-2) = \frac{4+12+4}{4-4+4} = 5$$

Hence range of f(x) is $\left[-\dfrac{1}{3}, 5\right]$

the graph of $y = f(x)$ is as shown

Let $y = \dfrac{8x}{x^2 + 2x + 4}$

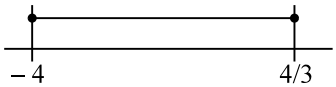
$x^2y + 2(y-4)x + 4y = 0$

$x \in R \quad \Rightarrow \quad D \geq 0$

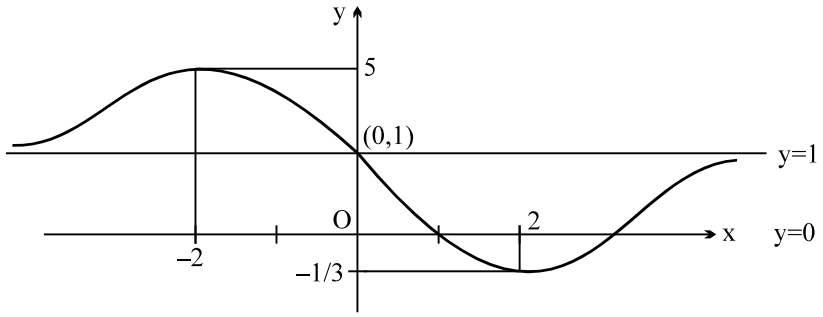
$4(y-4)^2 - 16y^2 \geq 0$

$(y-4)^2 - (2y)^2 \geq 0$

$(3y-4)(y+4) \leq 0$



hence $-\dfrac{1}{3} \leq f(x) \leq 5$



(i) obviously f is continuous and has $y = 1$ as its asymptote with one maxima and one minima and f is non monotonic in $(0, \infty)$ $\Rightarrow$ (D) is incorrect

(ii) $\cot\left(\cot^{-1} f(x)\right) = k$

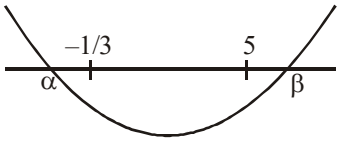
$f(x) = k$

$y = k$ and $y = f(x)$

hence only $y = 0$ cuts the graph at two distinct points $\Rightarrow$ (A)

(iii) now $g(x) = 0$

$$x^2 - \frac{2(1+3m)}{1+m}x - \frac{2(1+m)}{1+m} = 0$$



or $g(x) = x^2 - \frac{2(1+3m)}{1+m}x - 2 = 0$

as range of $f(x)$ is $\left[-\frac{1}{3}, 5\right]$

so one root of $g(x)$ is less than $\left(-\frac{1}{3}\right)$ and other is greater than 5

$$g\left(-\frac{1}{3}\right) < 0 \quad \text{and} \quad g(5) < 0$$

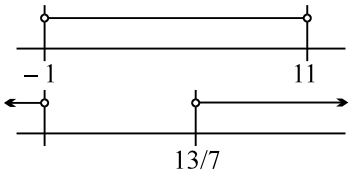
now $g\left(-\frac{1}{3}\right) = \frac{1}{9} + \frac{2}{3}\frac{(1+3m)}{1+m} - 2 < 0 = 1 + \frac{6(1+3m)}{1+m} - 18 < 0 = \frac{6(1+3m)}{1+m} - 17 < 0$

$$= \frac{6+18m-17-17m}{1+m} < 0 = \frac{m-11}{1+m} < 0$$

$g(5) < 0$

$25 - \frac{10(1+3m)}{1+m} - 2 < 0$; $\quad 23 - \frac{10(1+3m)}{1+m} < 0$; $\quad \frac{23+23m-10-30m}{1+m} < 0$

$\frac{13-7m}{1+m} < 0$; $\quad \frac{7m-13}{m+1} > 0$



$m \in \{2, 3, 4, 5, 6, 7, 8, 9, 10\}$

93. If $\cot\left(\cot^{-1} f(x)\right) = k$, has exactly two distinct real solutions then the integral value of 'k' can be :

यदि $\cot\left(\cot^{-1} f(x)\right) = k$, के दो वास्तविक और भिन्न–भिन्न हल है तब 'k' का पूर्णांक मान होगा :

(A) 0 (B) – 1 (C) 1 (D) 5

Ans. A

Sol.

94. If the range of the function f(x) lies between the roots of g (x) = 0 then the number of integral values of m equals :

यदि f(x) का परिसर g (x) = 0 के मूलो के मध्य विद्यमान है तब m के पूर्णांक मानों की संख्या है :

(A) 7 (B) 8 (C) 9 (D) 10

Ans. C

Sol.

Paragraph for Question nos.95 to 97

Consider the function $f: (-\infty, \infty) \to (-\infty, \infty)$ defined by $f(x) = \dfrac{x^2 - ax + 1}{x^2 + ax + 1}$, $0 < a < 2$

माना फलन $f: (-\infty, \infty) \to (-\infty, \infty)$ परिभाषित है। $f(x) = \dfrac{x^2 - ax + 1}{x^2 + ax + 1}$, $0 < a < 2$

95. Which of the following is true?

निम्न में से कौन सत्य है?

(A) $(2 + a)^2 f''(1) + (2 - a)^2 f''(-1) = 0$ (B) $(2 - a)^2 f''(1) - (2 + a)^2 f''(-1) = 0$

(C) $f'(1) f'(-1) = (2 - a)^2$ (D) $f'(1) f'(-1) = -(2 + a)^2$

Ans. A

Sol. $f(x) = \dfrac{x^2 - ax + 1}{x^2 + ax + 1}$

$$f'(x) = \frac{2a(x^2-1)}{(x^2+ax+1)^2} \text{ and } f''(x) = \frac{4a(-x^3+3x+a)}{(x^2+ax+1)^3}$$

$$f''(1) = \frac{4a}{(a+2)^2} \text{ and } f''(-1) = \frac{-4a}{(a-2)^2}$$

$\therefore (a+2)^2 f''(1) + (2-a)^2 f''(-1) = 0$

96. Which of the following is true?

(A) f(x) is decreasing on (–1, 1) and has a local minimum at x = 1

(B) f(x) is increasing on (–1, 1) and has a local maximum at x = 1

(C) f(x) is increasing on (–1, 1) but has neither a local maximum and nor a local minimum at x=1.

(D) f(x) is decreasing on (–1, 1) but has neither a local maximum and nor a local minimum at x = 1.

निम्न में से कौन सत्य है ?

(A) f(x) अन्तराल (–1, 1) में ह्रासमान है तथा स्थानीय निम्निष्ठ x = 1 पर है

(B) f(x) अन्तराल (–1, 1) में वर्द्धमान है तथा स्थानीय उच्चिष्ठ x = 1 पर है

(C) f(x) अन्तराल (–1, 1) में वर्द्धमान है लेकिन x = 1 कोई स्थानीय उच्चिष्ठ अथवा निम्निष्ठ नही है

(D) f(x) अन्तराल (–1, 1) में ह्रासमान है लेकिन x = 1 पर कोई स्थानीय उच्चिष्ठ अथवा निम्निष्ठ नही है

Ans. A

Sol. As when $x \in (-1, 1)$, $f'(x) < 0$

so f(x) is decreasing on (–1, 1) at x = 1

$$f''(1) = \frac{4a}{(a+2)^2} > 0 \qquad \text{so local minima at } x = 1.$$

97. Let $g(x) = \int_0^{e^x} \frac{f'(t)}{1+t^2} dt$

Which of the following is true? **[JEE-2008]**

(A) g'(x) is positive on $(-\infty, 0)$ and negative on $(0, \infty)$

(B) g'(x) is negative on $(-\infty, 0)$ and positive on $(0, \infty)$

(C) g'(x) changes sign on both $(-\infty, 0)$ and $(0, \infty)$

(D) g'(x) does not change sign on $(-\infty, \infty)$

माना $g(x) = \int_0^{e^x} \frac{f'(t)}{1+t^2} dt$ निम्न में से कौन सत्य है? **[JEE-2008]**

(A) g'(x), $(-\infty, 0)$ में धनात्मक है तथा $(0, \infty)$ में ऋणात्मक है

(B) g'(x), $(-\infty, 0)$ में ऋणात्मक है तथा $(0, \infty)$ में धनात्मक है

(C) g'(x) दोनो अन्तरालो $(-\infty, 0)$ तथा $(0, \infty)$ में अपने चिन्ह बदलता है

(D) g'(x) कही भी $(-\infty, \infty)$ में अपना चिन्ह नहीं बदलता है

Ans. B

Sol. $g(x) = \int_0^{e^x} \frac{f'(t)}{1+t^2} dt$

$$g'(x) = \frac{f'(e^x)}{1+e^{2x}} e^x = \frac{2a(e^{2x}-1)e^x}{(e^{2x}+ae^x+1)^2(1+e^{2x})}$$

$g'(x) > 0$ when $x > 0$

$g'(x) < 0$ when $x < 0$

ONE OR MORE THAN ONE CORRECT TYPE QUESTIONS

98. The local maximum value of $x(1-x)^2$, $0 \le x \le 2$ is :

$x(1-x)^2$, $0 \le x \le 2$ का स्थानीय उच्चिष्ठ मान है :

(A) 2 (B) 4/27 (C) 5 (D) 0

Ans. AB

Sol. Let $f(x) = x(1-x)^2$

$$f'(x) - 1(1-x)^2 + x \times 2(1-x) \times -1$$

$$= (1-x)^2 - 2x(1-x)$$

$$f''(x) = -2(1-x) - 2[1-x+x\times -1]$$

$$= -2 + 2x - 2[1-x-x]$$

$$= -2 + 2x - 2 + 4x$$

$$= 6x - 4$$

For critical points, $F'(x) = 0$

$$\Rightarrow (1-x)^2 - 2x(1-x) = 0$$

$$\Rightarrow (1-x)(1-x-2x) = 0$$

$$\Rightarrow (1-x)(1-3x) = 0$$

$$\Rightarrow (x-1)(3x-1) = 0$$

$x = 1, \quad \frac{1}{3}$

$f'(1) = 6(1) - 4 = 6 - 4 = 2 > 0$ minimum

$f''\left(\frac{1}{3}\right) = 6 \times \frac{1}{3} - 4 = 2 - 4 = -2 < 0$ minimum

$f\left(\frac{1}{3}\right) = \frac{1}{3}\left(1 - \frac{1}{3}\right)^2 = \frac{1}{3} \times \frac{4}{9} = \frac{4}{27}$

99. The function $f(x) = \int_0^x \sqrt{1 - t^4}\, dt$ is such that :

(A) it is defined on the interval $[-1, 1]$
(B) it is an increasing function
(C) it is an odd function
(D) the point $(0, 0)$ is the point of inflection

फलन $f(x) = \int_0^x \sqrt{1 - t^4}\, dt$ के लिये :

(A) $[-1, 1]$ में परिभाषित है
(B) वर्धमान फलन है
(C) एक विषम फलन है
(D) $(0, 0)$ फलन का नति परिवर्तन बिन्दु है

Ans. ABCD

Sol. $f'(x) = \sqrt{1 - x^4} > 0$ in $(-1, 1) \Rightarrow f$ is $\uparrow$

Now $f(x) + f(-x) = \int_0^x \sqrt{1 - t^4}\, dt + \int_0^{-x} \sqrt{1 - t^4}\, dt \Rightarrow \int_0^x \sqrt{1 - t^4}\, dt + \left(- \int_0^y \sqrt{1 - y^4}\, dy \right)$ $(t = -y)$

$= 0 \Rightarrow f(x)$ is odd

again $f''(x) = \frac{-4x^3}{2\sqrt{1 - x^4}}$ which vanishes at $x = 0$ and changes sign $\Rightarrow (0, 0)$ is inflection since f is well defined in $[-1, 1] \Rightarrow$ A, B, C, D

100. Let $f(x) = \frac{x - 1}{x^2}$ then which of the following is/are correct :

(A) f(x) has maxima but no minima
(B) f(x) increases in the interval $(0, 2)$ and decreases in the interval $(-\infty, 0) \cup (2, \infty)$
(C) f(x) is concave down in $(-\infty, 0) \cup (0, 3)$

(D) $x = 3$ is the point of inflection

माना $f(x) = \frac{x-1}{x^2}$ तो निम्न में से सत्य है :

(A) f(x) का उच्चिष्ठ है लेकिन निम्निष्ठ नहीं

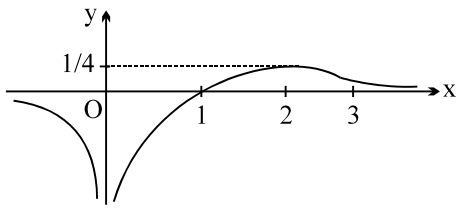
(B) f(x) ; (0, 2) में वर्धमान है और $(-\infty, 0) \cup (2, \infty)$ में

(C) f(x) ; $(-\infty, 0) \cup (0, 3)$ में नीचे की ओर अवत्तल है

(D) $x = 3$ नति परिवर्तन बिन्दु है

Ans. ABCD

Sol. $f'(x) = \frac{2-x}{x^3}$ and $f''(x) = \frac{x-3}{x^4}$. Now interpret



101. Which of the following functions fail to satisfy the condition of Rolle's theorem on the interval $[-1, 1]$, where [x] denotes the greatest integer less or equal to x and {x} denotes the fractional part of x respectively :

निम्न में से कौनसे फलन अन्तराल $[-1, 1]$ में, रोले प्रमेय की शर्तों को संतुष्ट नहीं करते है, जहाँ [x] महत्तम पूर्णांक फलन है एवं {x} भिन्नात्मक भाग फलन है :

(A) $f(x) = |x|[x]$

(B) $f(x) = \begin{cases} \frac{\tan x}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$

(C) $f(x) = \{x\} + \{-x\}$

(D) $f(x) = |x| - |\sin x|$

Ans. ABC

Sol. (A) $f(x) = \begin{cases} 1, & x = 1 \\ 0, & 0 \le x < 1 \\ x & -1 \le x < 0 \end{cases}$ $\Rightarrow$ not differentiable at $x = 0$ in $(-1, 1)$

(B) $f(0) = 0$ and $\lim_{x \to 0} \frac{\tan x}{x} = 1$ $\Rightarrow$ not continuous at $x = 0$

(C) $f(x) = \begin{cases} 1, & x \notin I \\ 0, & x \in I \end{cases}$ $\Rightarrow$ not continuous at $x = 0$ [12th, 29-08-2010, P-2]

(D) $f(x) = \begin{cases} x - \sin x & 0 \le x \le 1 \\ -x + \sin x & -1 \le x < 0 \end{cases}$ $\Rightarrow$ continuous & differentiable at $x = 0$]

102. Which of the following is/are correct?

(A) Between any two roots of $e^x \cos x = 1$, there exists atleast one root of $\tan x = 1$

(B) Between any two roots of $e^x \sin x = 1$, there exists atleast one root of $\tan x = -1$

(C) Between any two roots of $e^x \cos x = 1$, there exists atleast one root of $e^x \sin x = 1$

(D) Between any two roots of $e^x \sin x = 1$, there exists atleast one root of $e^x \cos x = 1$

निम्न में से कौन–कौन से कथन सत्य है?

(A) $e^x \cos x = 1$ के किन्ही दो मूलो के मध्य, $\tan x = 1$ का कम से कम एक मूल विद्यमान होगा

(B) $e^x \sin x = 1$ के किन्ही दो मूलो के मध्य, $\tan x = -1$ का कम से कम एक मूल विद्यमान होगा

(C) $e^x \cos x = 1$ के किन्ही दो मूलो के मध्य, $e^x \sin x = 1$ का कम से कम एक मूल विद्यमान होगा

(D) $e^x \sin x = 1$ के किन्ही दो मूलों के मध्य, $e^x \cos x = 1$ का कम से कम एक मूल विद्यमान होगा

Ans. ABC

Sol. (A) Let $f(x) = e^x \cos x - 1$

$\Rightarrow$ $f'(x) = e^x(\cos x - \sin x) = 0$

$\Rightarrow$ $\tan x = 1$, between two roots of $f(x)$. (Rolle's theorem)

(B) Let $g(x) = e^x \sin x - 1$,

$g'(x) = e^x(\sin x + \cos x) = 0$

$\Rightarrow$ $\tan x = -1$, between two roots of $g(x)$.

(C) Let $h(x) = e^{-x} - \cos x$,

$h'(x) = -e^{-x} + \sin x = 0$

$\Rightarrow$ $e^{-x} = \sin x$, between two roots of $h(x)$

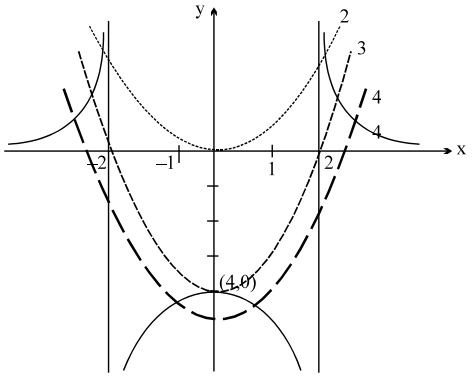
103. A parabola of the form $y = ax^2 + bx + c$ with $a > 0$ intersects the graph of $f(x) = \dfrac{1}{x^2 - 4}$. Number of possible distinct intersection(s) of these graph is :

परवलय $y = ax^2 + bx + c$; $a > 0$ और $f(x) = \dfrac{1}{x^2 - 4}$ के सम्भव प्रतिच्छेदन बिन्दुओं की संख्या :

(A) 0 (B) 2 (C) 3 (D) 4

Ans. BCD

Sol. Figure is self explanatory



104. If $f(x) = \begin{cases} 3x^2 + 12x - 1 & , \ -1 \le x \le 2 \\ 37 - x & , \ 2 < x \le 3 \end{cases}$ then :

(A) f(x) is increasing on $[-1, 2]$ (B) f(x) is continuous on $[-1, 3]$

(C) $f'(2)$ does not exist (D) f(x) has the maximum value at $x = 2$

यदि $f(x) = \begin{cases} 3x^2 + 12x - 1 & , \ -1 \le x \le 2 \\ 37 - x & , \ 2 < x \le 3 \end{cases}$ तो :

(A) f(x); $[-1, 2]$ में वर्धमान है (B) f(x) ; $[-1, 3]$ में सतत् है

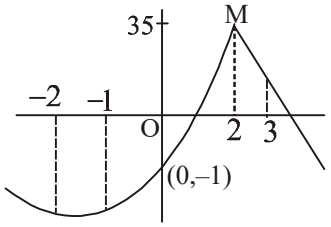
(C) $f'(2)$ विद्यमान नही है (D) f(x) का $x = 2$ पर अधिकतम मान प्राप्त होगा

Ans. ABCD

Sol. Note that f(x) is continuous at x= 2

and f is decreasing for (2, 3) and increasing for $[-1, 2]$.

At $x = 2$ f has a maxima



105. Assume that inverse of the function f is denoted by g. Then which of the following statement hold good?

(A) If f is increasing then g is also increasing (B) If f is decreasing then g is increasing

(C) The function f is injective (D) The function g is onto

माना f का प्रतिलोम फलन g है तो निम्न में से सत्य कथन होगें ?

(A) यदि f वर्धमान है तब g भी वर्धमान होगा (B) यदि f ह्यसमान है तब g वर्घमान होगा

(C) फलन f एकैकी होगा (D) फलन g आच्छादक होगा

Ans. ACD

Sol. If f and g are inverse then $(fog)(x) = x$

$f'[g(x)]\, g'(x) = 1$

if f is increasing $\Rightarrow$ $f' > 0$ $\Rightarrow$ sign of g ' is also + ve $\Rightarrow$ (A) is correct

If f is decreasing $\Rightarrow$ $f' < 0$ $\Rightarrow$ sign of g ' is – ve $\Rightarrow$ (B) is false

since f has an inverse $\Rightarrow$ f is bijective $\Rightarrow$ f is injective $\Rightarrow$ (C) is correct

inverse of a bijective mapping is bijective

$\Rightarrow$ g is also bijective $\Rightarrow$ g is onto $\Rightarrow$ (D) is correct

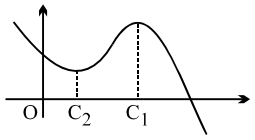
106. If f(x) is continuous and derivable $\forall\ x \in R$ and $f'(c) = 0$ for exactly 2 real values of 'c' then the number of real and distinct values of 'd' for which $f(d) = 0$ can be :

यदि f(x), $\forall\ x \in R$ में सतत् व अवकलनीय है और $f'(c) = 0$, 'c' के दो मानो के लिये तो 'd' के वास्तविक अलग–अलग मानों की संख्या जिसके लिये $f(d) = 0$:

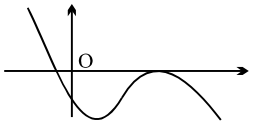
(A) 1 (B) 2 (C) 3 (D) 4

Ans. ABC

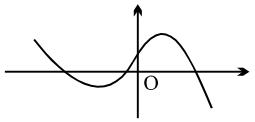
Sol. If $f'(x) = 0$ has n real roots $\Rightarrow$ $f(x) = 0$ has atmost $(n+1)$ roots

(A) 

$f(x) = 0$ has 1 real root

(B) 

$f(x) = 0$ has 2 real roots

(C) 

$f(x) = 0$ has 3 real roots $\Rightarrow$ A, B, C

107. Consider the function $f: R \to R$ defined as $f(x) = x + \sin x$. Which of the following is/are the correct statement(s)?

(A) The function is strictly increasing at every point on R except at 'x' equal to an odd integral multiple of π where the derivative of $f(x)$ is zero and where the function f is not strictly increasing.

(B) The function is bounded in every bounded interval but unbounded on whole real line.

(C) The graph of the function $y = f(x)$ lies in the first and third quadrants only.

(D) The graph of the function $y = f(x)$ cuts the line $y = x$ at infinitely many points.

$f: R \to R$, $f(x) = x + \sin x$ तो निम्न में से सत्य है :

(A) $x = (2n+1)\pi$; $n \in I$ को छोड़कर, जहाँ $f'(x) = 0$, तथा एक दिष्ट वर्धमान नही, $f(x)$ सभी $x \in R$ के लिये एकदिष्ठ वर्धमान है

(B) $f(x)$ सभी परिबद्ध अन्तरालों में परिबद्ध है लेकिन $x \in R$ पर अपरिबद्ध

(C) $y = f(x)$ का ग्राफ केवल प्रथम और तृतीय चतुर्थांश में है

(D) $y = f(x)$ का ग्राफ $y = x$ को अनन्त बिन्दुओं पर प्रतिच्छेद करता है

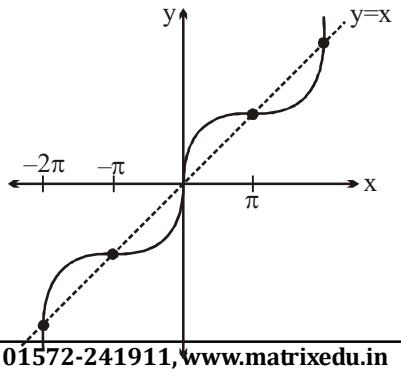
Ans. BCD

Sol. We have $f'(x) = 1 + \cos x \Rightarrow$ f is strictly increasing and has inflection point at $x = (2n+1)\pi$

Also there is no x for which $f(x)$ is not increasing.

$\Rightarrow$ (A) is not correct

As f is continuous $\forall\, x \in R$ hence bounded in every closed interval.



As f is odd hence symmetric w.r.t origin and its graphs lies in 1st and 3rd quadrant and y = x cut the graph at infinitely many points.

$\Rightarrow$ B, C, D are correct

108. Equation of a line which is tangent to both the curves $y = x^2 + 1$ and $y = -x^2$ is :

उस सरल रेखा का समीकरण जो वक्रो $y = x^2 + 1$ और $y = -x^2$ दोनों की स्पर्श रेखा है :

(A) $y = \sqrt{2}x + \frac{1}{2}$ (B) $y = \sqrt{2}x - \frac{1}{2}$ (C) $y = -\sqrt{2}x + \frac{1}{2}$ (D) $y = -\sqrt{2}x - \frac{1}{2}$

Ans. AC

Sol. Let the tangent line be $y = ax + b$

The equation for its intersection with the upper parabola is

$x^2 + 1 = ax + b$

$x^2 - ax + (1 - b) = 0$

This has a double root when $a^2 - 4(1 - b) = 0$ or $a^2 + 4b = 4$

For the lower parabola

$ax + b = -x^2$

$x^2 + ax + b = 0$

This has a double root when $a^2 - 4b = 0$

subtract these two equations to get $8b = 4$ or $b = 1/4$

add them to get $2a^2 = 4$ or $a = \pm\sqrt{2}$

The tangent lines are $y = \sqrt{2}x + \frac{1}{2}$ and $y = -\sqrt{2}x + \frac{1}{2}$

Alternatively: Let the common tangent is

$y = mx + c$

Solving with first curve we get

$x^2 + 1 = mx + c$

$x^2 - mx + 1 - c = 0$

$D = 0$

$\boxed{m^2 = 4(1-c)}$ .....(A)

Solving with second curve

$mx + c = -x^2 \Rightarrow x^2 + mx + c = 0$

$D = 0$

$\boxed{m^2 = 4c}$ .....(B)

From A and B

$8c = 4$, $c = ½$

$m = \pm\sqrt{2}$

The tangent lines are $y = \sqrt{2}x + \frac{1}{2}$ and $y = -\sqrt{2}x + \frac{1}{2}$

109. Which of the following pair(s) of curves touch each other?

(A) $y^2 = 4x$ and $x^2 + y^2 = 6x - 1$ (B) $xy = 4$ and $x^2 + y^2 = 8$

(C) $y = 2x - 3$ and $y = x^3 - x + 1$ (D) $y = 6 + x - x^2$ and $y = \frac{x+2}{x-2}$

निम्न में से कौन–कौन से वक्र एक दूसरे को स्पर्श करते है?

(A) $y^2 = 4x$ तथा $x^2 + y^2 = 6x - 1$ (B) $xy = 4$ तथा $x^2 + y^2 = 8$

(C) $y = 2x - 3$ तथा $y = x^3 - x + 1$ (D) $y = 6 + x - x^2$ तथा $y = \frac{x+2}{x-2}$

Ans. AB

Sol. (A) $y^2 = 4x$; $x^2 + y^2 = 6x - 1$

solving, $x^2 + 4x = 6x - 1$

$x^2 - 2x + 1 = 0 \Rightarrow (x-1)^2 = 0 \Rightarrow x = 1$

touch at (1, 2) and (1, – 2) $\Rightarrow$ (A) is correct

(B) $xy = 4$; $x^2 + y^2 = 8$

$x^2 + y^2 = 16 \Rightarrow x^2(8 - x^2) = 16$; put $x^2 = t > 0$

$t^2 - 8t + 16 = 0 \Rightarrow (t-4)^2 = 0$

$t = 4 \Rightarrow x^2 = 4 \Rightarrow x = 2$ or $-2 \Rightarrow$ (2, 2) or (–2, – 2)

(C) solving the points of intersection are (1, 1); (– 2, – 5); also $\frac{dy}{dx}$ is same at (1, 1).

Hence they touch at (1, 1)

110. The co-ordinates of a point on the parabola $2y = x^2$ which is nearest to the point (0, 3) is :

परवलय $2y = x^2$ पर स्थित बिन्दु के निर्देशांक जो कि बिन्दु (0, 3) से न्यूनतम दूरी पर है :

(A) (2, 2) (B) $\left(-\sqrt{2}, 1\right)$ (C) $\left(\sqrt{2}, 1\right)$ (D) (– 2, 2)

Ans. AD

Sol. Let (a, b) is the point on the parabola from which its distance from (0, 3) is minimum

$\Rightarrow a^2 = 2b$

Assume d is the distance

$d^2 = a^2 + (b-3)^2 = 2b + (b-3)^2$

For minimum value of $d\frac{d\left(d^2\right)}{db^2} = 0 = 2 + 2\left(b - 3\right)$

$\Rightarrow b = 2$

and $a^2 = 2b = 4$ $\quad a = \pm 2$

Hence require points on the parabola are (2, –2) and (2, 2)

111. In the curve $x = t^2 + 3t - 8$, $y = 2t^2 - 2t - 5$, at point (2, – 1) :

(A) length of subtangent is 7/6 (B) slope of tangent = 6/7

(C) length of tangent = $\sqrt{(85)} / 6$ (D) None of these

वक्र $x = t^2 + 3t - 8$, $y = 2t^2 - 2t - 5$ के बिन्दु (2, – 1) पर :

(A) अधः स्पर्शी की लम्बाई 7/6 हैं (B) स्पर्श रेखा की प्रवणता = 6/7

(C) स्पर्श रेखा की लम्बाई = $\sqrt{(85)} / 6$ (D) इनमें से कोई नहीं

Ans. ABC

Sol. $x = t^2 + 3t - 8$, $\quad y = 2t^2 - 2t - 5$

$\frac{dx}{dt} = 2t + 3$, $\quad \frac{dy}{dt} = 4t - 2$

$\Rightarrow \frac{dy}{dx} = \left(\frac{dy}{dt}\right)\cdot\left(\frac{dt}{dx}\right) = \left(\frac{4t-2}{2t+3}\right)$

The slope of tangent, $\frac{dy}{dx}$ at $(2, -1) = \left(\frac{dy}{dt}\right)_{(-2,1)}$

when $x = 2$, $t^2 + 3t - 8 = 2$ $\quad \Rightarrow t^2 + 3t - 10 = 0$

$\Rightarrow t^2 + 5t - 2t - 10 = 0$ $\quad \Rightarrow t(t + 5) - 2(t + 5) = 0$

$\Rightarrow (t - 2)(t + 5) = 0$ $\quad \Rightarrow t = 2$, or -5

$\therefore \frac{dy}{dx} = \left(\frac{4(2)-2}{2(2)+3}\right) = \frac{6}{7}$

When $x = 2$, $y = -1$

$\therefore -1 = 2t^2 - 2t - 5$ $\quad \Rightarrow 2t^2 - 2t - 4 = 0$ $\quad \Rightarrow 2t^2 - 4t + 2t - 4 = 0$

$\Rightarrow 2t\left(t - 2\right) + 2\left(t - 2\right) = 0$ $\quad \Rightarrow t = 2, -1$

So the common value of t = 2

$\therefore$ Slope of tangent at $(2, -1) = \boxed{\frac{6}{7}}$

Length of tangent $\sqrt{\left(\frac{y_1}{m}\right)^2 + (y_1)^2}$

Here, $y_1 = -1$ and $m = \frac{6}{7}$

$\therefore$ Length $= \sqrt{\left(\frac{7}{6}\right)^2 + (-1)^2} = \sqrt{\frac{49+36}{6}}$

112. The function $f(x) = 2\,\ell n\,(x-2) - x^2 + 4x + 1$ increases in the intervals :

फलन $f(x) = 2 \log (x-2) - x^2 + 4x + 1$ किस अन्तराल में वर्द्धमान है :

(A) $(1, 2)$ (B) $(2, 3)$ (C) $\left[\frac{5}{2}, 3\right]$ (D) $(2, 4)$

Ans. BC

Sol. $f'(x) > 0$ for monotonically increasing function.

Hence $\frac{2}{x-2} - 2x + 4 > 0$

or

$\frac{2}{x-2} > 2x - 4$

or

$\frac{1}{x-2} > x - 2$

or

$(x-2)^2 < 1$

$-1 < x - 2 < 1$

or

$1 < x < 3$

Hence $X \in (1, 3)$

However $x - 2 > 0$ since input of logarithmic function cannot be negative

$(x - 2) > 0$ or $x > 2$

Hence $x \in (2, 3)$

113. If $f(x) = 2x + \cot^{-1} x + \log\left(\sqrt{1+x^2} - x\right)$, then f(x) :

(A) increases in $[0, \infty)$ (B) decreases in $[0, \infty)$

(C) neither increases nor decreases in $[0, \infty)$ (D) increases in $(-\infty, \infty)$

यदि $f(x) = 2x + \cot^{-1} x + \log\left(\sqrt{1+x^2} - x\right)$ हो, तो फलन f(x) :

(A) अन्तराल $[0, \infty)$ में वर्द्धमान है (B) अन्तराल $[0, \infty)$ में ह्रासमान है

(C) अन्तराल $[0, \infty)$ में न वर्द्धमान न ही ह्रासमान है (D) अन्तराल $(-\infty, \infty)$ में वर्द्धमान है

Ans. AD

Sol. $f'(x) = 2 - \frac{1}{1+x^2} - \frac{1}{\sqrt{x^2+1}} = 1 - \frac{1}{1+x^2} + 1 - \frac{1}{\sqrt{x^2+1}}$

$$= \frac{x^2}{1+x^2} + \left(1 - \frac{1}{\sqrt{x^2+1}}\right) \geq 0$$

114. Let $g(x) = 2f(x/2) + f(1 - x)$ and $f''(x) < 0$ in $0 \leq x \leq 1$ then g(x) :

(A) decreases in $\left[0, \frac{2}{3}\right)$ (B) decreases $\left(\frac{2}{3}, 1\right]$

(C) increases in $\left[0, \frac{2}{3}\right)$ (D) increases in $\left(\frac{2}{3}, 1\right]$

यदि अन्तराल $0 \leq x \leq 1$ में $g(x) = 2f(x/2) + f(1 - x)$ एवं $f''(x) < 0$ हो, तो g(x) है :

(A) $\left[0, \frac{2}{3}\right)$ में ह्रासमान (B) $\left(\frac{2}{3}, 1\right]$ में ह्रासमान

(C) $\left[0, \frac{2}{3}\right)$ में वर्द्धमान (D) $\left(\frac{2}{3}, 1\right]$ में वर्द्धमान

Ans. BC

Sol. $g(x) = 2f\left(\frac{x}{2}\right) + f(1-x)$ and

$g'(x) = f'(x/2) - f'(1-x)$

Now $g(x)$ is increasing if $g'(x) \geq 0$

$f'\left(\frac{x}{2}\right) \geq f'(1-x)$

[$\because$ $f''(x) < 0$ i.e. $f'(x)$ is decreasing]

$\Rightarrow \quad \frac{x}{2} \leq 1 - x \quad \Rightarrow \quad x \leq 2 - 2x$

$\Rightarrow \quad 3x \leq 2 \quad \Rightarrow \quad x \leq 2/3 \quad \Rightarrow \quad 0 \leq x \leq \frac{2}{3}$

$\Rightarrow$ $g(x)$ increases in $0 \leq x \leq 2/3$

and $g'(x) \leq 0$ for decreasing

$\Rightarrow \quad f'\left(\frac{x}{2}\right) \leq f'(1-x) \quad \Rightarrow \quad \frac{x}{2} \geq 1 - x$

$\Rightarrow \quad x \geq 2/3$

$\Rightarrow \quad 2/3 \leq x \leq 1$

115. If $f(x) = \tan^{-1}x - (1/2)\,\ell n\, x$. Then :

(A) the greatest value of $f(x)$ on $\left[\frac{1}{\sqrt{3}}, \sqrt{3}\right]$ is $\pi/6 + (1/4)\,\ell n\, 3$

(B) the least value of $f(x)$ on $\left[\frac{1}{\sqrt{3}}, \sqrt{3}\right]$ is $\pi/3 - (1/4)\,\ell n\, 3$

(C) $f(x)$ decreases on $(0, \infty)$

(D) $f(x)$ increases on $(-\infty, 0)$

यदि $f(x) = \tan^{-1}x - (1/2)\,\ell n\, x$. तो :

(A) $f(x)$ का $\left[\frac{1}{\sqrt{3}}, \sqrt{3}\right]$ में महत्तम मान $\pi/6 + (1/4)\,\ell n\, 3$ होगा

(B) $f(x)$ का $\left[\frac{1}{\sqrt{3}}, \sqrt{3}\right]$ में न्यूनतम मान $\pi/3 - (1/4)\,\ell n\, 3$ होगा

(C) $f(x)$ अन्तराल $(0, \infty)$ में ह्यसमान है

(D) $f(x)$ अन्तराल $(-\infty, 0)$ में वर्धमान है

Ans. ABC

Sol. $f'(x) = \frac{1}{1+x^2} - \frac{1}{2}\frac{1}{x}, x > 0$

$= \frac{-(x-1)^2}{2x(1+x^2)} \le 0 \ \ \forall \ x > 0.$

f(x) is decreasing $\forall\, x > 0.$

On $\left[\frac{1}{\sqrt{3}}, \sqrt{3}\right]$, greatest value is

$f\left(\frac{1}{\sqrt{3}}\right) = \frac{\pi}{6} - \frac{1}{2}\ell n\left(\frac{1}{\sqrt{3}}\right)$ and least value is

$f\left(\sqrt{3}\right) = \frac{\pi}{3} - \frac{1}{2}\ell n\sqrt{3}\,.$

116. If $f(x) = \log(x-2) - \frac{1}{x}$, then :

(A) f(x) is M.I. for $x \in (2, \infty)$
(B) f(x) is M.I. for $x \in [-1, 2]$
(C) f(x) is always concave downwards
(D) $f^{-1}(x)$ is M.I. wherever defined

यदि $f(x) = \log(x-2) - \frac{1}{x}$, तो :

(A) $x \in (2, \infty)$ के लिए f(x) एकदिष्ट वर्द्धमान है
(B) $x \in [-1, 2]$ के लिए f(x) एकदिष्ट वर्द्धमान है
(C) f(x) हमेशा नीचे की ओर अवत्तल है
(D) जहाँ भी $f^{-1}(x)$ परिभाषित है, एकदिष्ट वर्द्धमान है

Ans. ACD

Sol. $f(x) = \ell n(x-2) - \frac{1}{x}$

$f'(x) = \frac{1}{x-2} + \frac{1}{x^2} = \frac{x^2+x-2}{x^2(x-2)} = \frac{x^2+2x-x-2}{x^2(x-2)}$

$= \frac{(x+2)(x-1)}{x^2(x-2)}.$

As $\ell n(x-2)$ is defined when $x > 2$

$\Rightarrow$ f(x) is M.I. for $x \in (2, \infty)$

$\Rightarrow$ $f^{-1}(x)$ is M.I. wherever defined

Also $f''(x) = \frac{-1}{(x-2)^2} - \frac{2}{x^3} < 0$

$\Rightarrow$ f(x) is always concave downward

117. The function $f(x) = x^{1/3}(x-1)$:

(A) has 2 inflection points

(B) is strictly increasing for $x > 1/4$ and strictly decreasing for $x < 1/4$

(C) is concave down in $(-1/2,\ 0)$

(D) Area enclosed by the curve lying in the fourth quadrant is 9/28

फलन $f(x) = x^{1/3}(x-1)$ के लिये :

(A) 2 नति परिवर्तन बिन्दु है

(B) $x > 1/4$ के लिये एकदिष्ट वर्धमान और $x < 1/4$ में एकदिष्ट ह्यसमान है

(C) $(-1/2,\ 0)$ में उत्तलता नीचे की ओर होगी

(D) चतुर्थ चतुर्थांशः में परिबद्ध क्षेत्र का क्षेत्रफल 9/28 है

Ans. ABCD

Sol. $y = x^{1/3}(x-1)$

$$\frac{dy}{dx} = \frac{4}{3}x^{1/3} - \frac{1}{3}\cdot\frac{1}{x^{2/3}} = \frac{1}{3x^{2/3}}[4x-1]$$

hence f is $\uparrow$ for $x > \frac{1}{4}$

and f $\downarrow$ for $x < \frac{1}{4}$

] $x^{2/3}$ is always positive and $x = 1/4$ the curves has a local minima

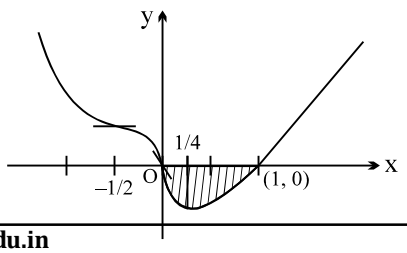
now $f'(x) = \frac{4}{3}x^{1/3} - \frac{1}{3}\cdot x^{-2/3}$ (non existent at $x = 0$, vertical tangent)

$$f''(x) = \frac{4}{9}\cdot\frac{1}{x^{2/3}} + \frac{1}{3}\cdot\frac{2}{3}\cdot\frac{1}{x^{5/3}}$$

$$= \frac{2}{9x^{2/3}}\left[2 + \frac{1}{x}\right] = \frac{2}{9x^{2/3}}\left[\frac{2x+1}{x}\right]$$

$\therefore$ $f''(x) = 0$ at $x = -\frac{1}{2}$ (inflection point)

graph of f(x) is as



$$A = \int_0^1 \left(x^{4/3} - x^{1/3}\right)dx = \frac{3}{7}x^{3/7} - \frac{3}{4}x^{4/3}\Big]_0^1$$

$$= \left|\frac{3}{7}-\frac{3}{4}\right| = 3\left|\frac{4-7}{28}\right| = \frac{9}{28} \Rightarrow \text{(D)}$$

118. $f''(x) > 0$ for all $x \in [-3, 4]$, then which of the following are always true?

(A) f(x) has a relative minimum on (–3, 4)

(B) f(x) has a minimum on [–3, 4]

(C) f(x) is concave upwards on [–3, 4]

(D) if f(3)= f (4) then f(x) has a critical point on [–3, 4]

यदि $f''(x) > 0; \forall\ x \in [-3, 4]$, तो निम्न में से कौनसा/से सदैव सत्य होगें?

(A) f(x) ; (–3, 4) में सापेक्षतः निम्निष्ठ है

(B) f(x); [–3, 4] में निम्निष्ठ देगा

(C) f(x); [–3, 4] में ऊपर की ओर अवतल है

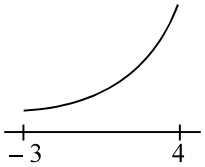
(D) यदि f(3) = f(4) तो [–3, 4] में f(x) का क्रान्तिक बिन्दु होगा

Ans. BCD

Sol. (A) f(x) has no relative minimum on (–3, 4)

(B) f(x) is continuous function on [–3, 4]

$\Rightarrow$ f(x) has min. and max. on [–3, 4] by IVT



(C) $f''(x) > 0 \Rightarrow$ f(x) is concave upwards on [–3, 4]

(D) $f(3) = f(4)$

By Rolle's theorem

$\exists\ c \in (3, 4)$, where $f'(c) = 0$

$\Rightarrow$ $\exists$ critical point on [– 3, 4]

119. The function $f(x) = \begin{cases} x+2 & \text{if } x < -1 \\ x^2 & \text{if } -1 \le x < 1 \\ (x-2)^2 & \text{if } x \ge 1 \end{cases}$:

(A) is continuous for all $x \in R$

(B) is continuous but not differentiable $\forall\ x \in R$

(C) is such that f'(x) change its sign exactly twice

(D) has two local maxima and two local minima

यदि फलन $f(x) = \begin{cases} x+2 & \text{if } x < -1 \\ x^2 & \text{if } -1 \le x < 1 \\ (x-2)^2 & \text{if } x \ge 1 \end{cases}$:

(A) $\forall\ x \in R$ सतत् है

(B) $\forall\ x \in R$ सतत् है लेकिन अवकलनीय नहीं

(C) f'(x) का चिन्ह दो बार बदलता है

(D) f(x) के दो स्थानिय निम्निष्ठ और दो स्थानिय उच्चिष्ठ बिन्दु है

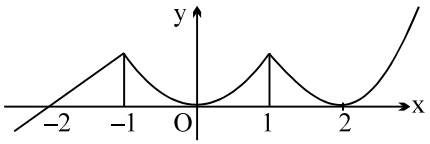
Ans. ABD

Sol. f is obvious continuous $\forall\ x \in R$ and not derivable at – 1 and 1

f' (x) changes sign 4 times at – 1, 0, 1, 2

local maxima at 1 and – 1

local minima at x = 0 and 2



120. Consider the function $f(x) = \left[\cos\left(\tan^{-1}\left(\sin\left(\cot^{-1} x\right)\right)\right)\right]^2$. Which of the following is correct?

(A) range of f is (0, 1)

(B) f is even

(C) f'(0) = 0

(D) the line y = 1 is asymptotes to the graph y = f(x)

माना $f(x) = \left[\cos\left(\tan^{-1}\left(\sin\left(\cot^{-1} x\right)\right)\right)\right]^2$. निम्न में से सत्य कथन है?

(A) f का परिसर (0, 1)

(B) f सम है

(C) f'(0) = 0

(D) y = 1 रेखा, y = f(x) की अनन्तस्पर्शी है

Ans. BCD

Sol. Domain is $x \in R$

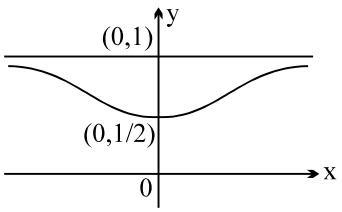
Also $f(x) = \left[\cos\left(\tan^{-1}(\sin\theta)\right)\right]^2$ where $\cot\theta = x$

$$= \left[\cos\left(\tan^{-1}\left(\frac{1}{\sqrt{1+x^2}}\right)\right)\right]^2 = (\cos\phi)^2 \text{ where } \tan\phi = \frac{1}{\sqrt{1+x^2}}$$

$$= \left(\frac{\sqrt{1+x^2}}{\sqrt{2+x^2}}\right)^2$$

$$g(x) = \frac{1+x^2}{2+x^2} = 1 - \frac{1}{2+x^2}$$

range is $\left[\frac{1}{2}, 1\right)$; $f'(x) = \frac{2x}{(2+x^2)^2}$



hence f' (0) = 0

also $\lim_{x\to\infty} f(x) = 1$

hence (B), (C), (D)

121. The function $f(x) = \int_{-1}^{x} t\left(e^t - 1\right)(t-1)(t-2)^3(t-3)^5 dt$ has a local minimum at x =

फलन $f(x) = \int_{-1}^{x} t\left(e^t - 1\right)(t-1)(t-2)^3(t-3)^5 dt$, x के किस भाग पर स्थानिय निम्निष्ठ है :

(A) 0 (B) 1 (C) 2 (D) 3

Ans. BD

Sol. (A) may be false since if $0 < f(x) < 1$, $f^2(x) < f(x)$ and $\int_a^b f(x)\,dx > \int_a^b f^2(x)\,dx$

(B) may be false since if $f(x) < 0$, $\frac{d}{dx} f^2(x) = 2 f(x) f'(x) < 0$ when $f'(x) > 0$ and so $f^2(x)$ is decreasing while $f(x)$ is increasing

(C) may be false since a function can be negative and increasing.

(D) may be false since a function may not be differentiable at x = c for which it attains its minimum.

122. Consider a real valued continuous function $f(x)$ defined on the interval [a, b]. Which of the following statements does not hold(s) good?

(A) If $f(x) \geq 0$ on [a, b] then $\int_a^b f(x)\,dx \leq \int_a^b f^2(x)\,dx$

(B) If $f(x)$ is increasing on [a, b], then $f^2(x)$ is increasing on [a, b].

(C) If $f(x)$ is increasing on [a, b], then $f(x) \geq 0$ on (a, b).

(D) If $f(x)$ attains a minimum at $x = c$ where $a < c < b$, then $f'(c) = 0$

माना $f(x)$, [a, b] में परिभाषित वास्तविक मान वाला सम फलन है तो निम्न में से असत्य कथन है ?

(A) यदि $f(x) \geq 0$, [a, b] तो $\int_a^b f(x)\,dx \leq \int_a^b f^2(x)\,dx$

(B) यदि $f(x)$, [a, b] में वर्धमान है तो $f^2(x)$ भी [a, b] में वर्धमान होगा

(C) यदि $f(x)$, [a, b] में वर्धमान है तो $f(x) \geq 0$; $x \in (a, b)$

(D) यदि $f(x)$, $x = c$ पर निम्निष्ठ है जहाँ $a < c < b$ तो $f'(c) = 0$

Ans. ABCD

Sol.

123. If $\lim\limits_{x\to a} f(x) = \lim\limits_{x\to a}[f(x)]$ (a is a finite quantity), where $[\cdot]$ denotes greatest integer function and f (x) is a non constant continuous function, then :

(A) $\lim\limits_{x\to a}$ f(x) is an integer

(B) $\lim\limits_{x\to a}$ f(x) need not be an integer

(C) f(x) has a local minimum at x = a

(D) f(x) has a local maximum at x = a

यदि $\lim\limits_{x\to a} f(x) = \lim\limits_{x\to a}[f(x)]$ (a परिमित संख्या है), जहाँ $[\cdot]$ महत्तम पूर्णांक फलन है यदि f (x) एक नियतांक सतत फलन नही है, तो :

(A) $\lim\limits_{x\to a}$ f(x) एक पूर्णांक होगा

(B) $\lim\limits_{x\to a}$ f(x) एक पूर्णांक नही होगा

(C) f(x); x = a पर स्थानीय निम्निष्ठ है

(D) x = a पर f(x) का स्थानीय उच्चिष्ठ मान प्राप्त होगा

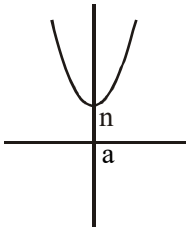
Ans. AC

Sol. Clearly $\lim\limits_{x\to a}[f(x)]$ is an integer and LHL and RHL should be same for existence of $\lim\limits_{x\to a} f(x)$

Let $\lim\limits_{x\to a}[f(x)] = n\ (n \in I)$

Clearly LHL and RHL both should be just greater than n as f(x) is cont. at x = a

$\therefore$ f(x) has local minimum at x = a



124. Let the function $g : (-\infty, \infty) \to \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ be given by $g(u) = 2\tan^{-1}(e^u) - \frac{\pi}{2}$. Then, g is :

(A) even and is strictly increasing in (0, ∞)

(B) odd and is strictly decreasing in (– ∞, ∞)

(C) odd and is strictly increasing in (– ∞, ∞)

(D) neither even nor odd, but is strictly increasing in (– ∞, ∞)

माना फलन $g : (-\infty, \infty) \to \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ इस प्रकार है कि $g(u) = 2\tan^{-1}(e^u) - \frac{\pi}{2}$ तो g एक :

(A) सम तथा (0, ∞) में निरन्तर वर्द्धमान फलन है

(B) विषम तथा (– ∞, ∞) में निरन्तर ह्रासमान फलन है

(C) विषम तथा (– ∞, ∞) में निरन्तर वर्द्धमान फलन है

(D) ना ही सम ना ही विषम है, लेकिन (– ∞, ∞) में निरन्तर वर्द्धमान फलन है

Ans. C

Sol. $g(u) = 2\tan^{-1}(e^u) - \frac{\pi}{2}$

$g(u) = \tan^{-1}(e^u) - \cot^{-1}(e^u)$

$g(u) = \tan^{-1}(e^u) - \tan^{-1}(e^{-u})$ odd function.

$g'(u) = \dfrac{e^u}{1+e^{2u}} + \dfrac{e^{-u}}{1+e^{-2u}} > 0$ Strictly increasing function

125. For the function $f(x) = x\cos\dfrac{1}{x},\ x \geq 1$: **[JEE-2009]**

(A) for at least one x in the interval $[1, \infty)$, $f(x+2) - f(x) < 2$

(B) $\lim\limits_{x\to\infty} f'(x) = 1$

(C) for all x in the interval $[1, \infty)$, $f(x+2) - f(x) > 2$

(D) $f'(x)$ is strictly decreasing in the interval $[1, \infty)$

फलन $f(x) = x\cos\dfrac{1}{x},\ x \geq 1$, के लिये : **[JEE-2009]**

(A) अन्तराल $[1, \infty)$ में x का कम से कम एक मान होगा कि $f(x+2) - f(x) < 2$

(B) $\lim\limits_{x\to\infty} f'(x) = 1$

(C) अन्तराल $[1, \infty)$ में x के सभी मानों के लिये, $f(x+2) - f(x) > 2$ होगा

(D) $f'(x)$ अन्तराल $[1, \infty)$ में ह्रासमान है

Ans. BCD

Sol. $f(x) = x\cos\dfrac{1}{x},\ x \geq 1$

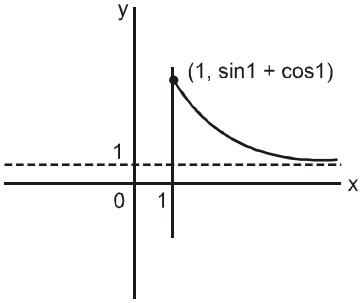
$\Rightarrow \quad f'(x) = \dfrac{1}{x}\sin\dfrac{1}{x} + \cos\dfrac{1}{x}$

$\Rightarrow \quad f''(x) = -\dfrac{1}{x^3}\cos\left(\dfrac{1}{x}\right)$

Now $\lim\limits_{x\to\infty} f'(x) = 0 + 1 = 1 \quad \Rightarrow$ option 'B' is correct

$x \in [1, \infty) \quad \Rightarrow \quad \dfrac{1}{x} \in (0, 1]$

$\Rightarrow \quad f''(x) < 0 \quad \Rightarrow$ option 'D' is correct



As $f'(1) = \sin 1 + \cos 1 > 1$

$f'(x)$ is strictly decreasing and $\lim\limits_{x\to\infty} f'(x) = 1$

so graph of $f'(x)$ is as below

Now in $[x, x+2]$, $x \in [1, \infty)$, $f(x)$ is continuous and differentiable

so by LMVT, $f'(x) = \dfrac{f(x+2) - f(x)}{2}$

as $f'(x) > 1$ for all $x \in [1, \infty)$

$\Rightarrow \quad \frac{f(x+2)-f(x)}{2} > 1 \quad \Rightarrow \quad f(x+2) - f(x) > 2$

for all $x \in [1, \infty)$

MATCH THE COLUMN TYPE QUESTION(S)

126. Match the following column :

Column – I	**Column – II**
(A) $f(x) = \frac{\sin x}{e^x}$, $x \in [0,\pi]$	(P) Condition of Rolle's theorem are satisfied
(B) $f(x) = \text{sgn}((e^x - 1)\,\ell nx)$, $x \in \left[\frac{1}{2}, \frac{3}{2}\right]$	(Q) Condition of LMVT are satisfied
(C) $f(x) = (x-1)^{2/5}$, $x \in [0,3]$	(R) At least one condition of Rolle's theorem is not satisfied
(D) $f(x) = \begin{cases} x\left(\frac{e^{\frac{1}{x}}-1}{e^{\frac{1}{x}}+1}\right), & x \in [-1,1]-\{0\} \\ 0, & x = 0 \end{cases}$	(S) At least one condition of LMVT is not satisfied.

निम्न कॉलम को सुमेलित कीजिए :

Column – I	**Column – II**
(A) $f(x) = \frac{\sin x}{e^x}$, $x \in [0,\pi]$	(P) रोले प्रमेय के प्रतिबन्ध सन्तुष्ट होते है
(B) $f(x) = \text{sgn}((e^x - 1)\,\ell nx)$, $x \in \left[\frac{1}{2}, \frac{3}{2}\right]$	(Q) LMVT के प्रतिबन्ध सन्तुष्ट होते है
(C) $f(x) = (x-1)^{2/5}$, $x \in [0,3]$	(R) रोले प्रमेय की कम से कम एक प्रतिबन्ध सन्तुष्ट नहीं होता है
(D) $f(x) = \begin{cases} x\left(\frac{e^{\frac{1}{x}}-1}{e^{\frac{1}{x}}+1}\right), & x \in [-1,1]-\{0\} \\ 0, & x = 0 \end{cases}$	(S) LMVT का कम से कम एक प्रतिबन्ध सन्तुष्ट नहीं होता है

Ans. (A) P,Q (B) R,S (C) R,S (D) R, S

Sol. (A) $f(x)$ is continuous and differentiable $f(0) = f(\pi)$
Hence condition in Rolle's theorem and LMVT are satisfied.

(B) $f(1^-) = -1$, $f(1) = 0$, $f(1^+) = 1$

f(x) is not continuous at x = 1, belonging to $\left[\frac{1}{2}, \frac{3}{2}\right]$

Hence, atleast one condition in LMVT and Rolle's theorem is not satisfied

(C) $f'(x) = \frac{2}{5}(x-1)^{-3/5}, x \neq 1$

At x = 1, f(x) is not differentiable.

Hence at least one condition in LMVT and Rolle's theorem is not satisfied.

(D) At x = 0

$$\text{L.H.D.} = \lim_{x \to 0^-} t \frac{x\left(\frac{e^{\frac{1}{x}}-1}{e^{\frac{1}{x}}+1}\right)-0}{x-0} = \frac{0-1}{0+1} = -1$$

R.H.D. = 1

At x = 0, f(x) is not differentiable

Hence at least one condition in LMVT and Rolle's theorem is not satisfied.

127. In the following [x] denotes the greatest integer less than or equal to x. Match the functions in Column I with the properties in Column II :

Column – I	**Column – II**
(A) $x\|x\|$	(P) continuous in (–1, 1)
(B) $\sqrt{\|x\|}$	(Q) differentiable in (–1, 1)
(C) $x + [x]$	(R) strictly increasing in (–1, 1)
(D) $\|x-1\| + \|x+1\|$	(S) non differentiable at least at one point in (–1, 1)

स्तम्भ का मिलान कीजिये : निम्न में [x], x के महत्तम पूर्णांक फलन को दर्शाता है। स्तम्भ– I में फलन है तथा स्तम्भ–II उनके गुणधर्म :

Column – I	**Column – II**
(A) $x\|x\|$	(P) (–1, 1) में सतत्
(B) $\sqrt{\|x\|}$	(Q) (–1, 1) में अवकलनीय
(C) $x + [x]$	(R) (–1, 1) में वर्द्धमान
(D) $\|x-1\| + \|x+1\|$	(S) (–1, 1) में कम से कम एक बिन्दु पर अवकलनीय नहीं

Ans. (A) P,Q,R (B) P,S (C) R,S (D) P, Q

Sol.

Exercise J-M

1. The number of points on the curve $y = 54x^5 - 135x^4 - 70x^3 + 180x^2 + 210x$ at which the normal lines are parallel to $x + 90y + 2 = 0$ is : **[JEE(Main)Jan.2023]**

वक्र $y = 54x^5 - 135x^4 - 70x^3 + 180x^2 + 210x$ पर उन बिंदुओं, जिन पर अभिलंब, रेखा $x + 90y + 2 = 0$ के समांतर है, की संख्या है : **[JEE(Main)Jan.2023]**

(A) 3 (B) 0 (C) 4 (D) 2

Ans. C

Sol. Normal of the line is parallel to line $x + 90y + 2 = 0$

$$m_N = -\frac{1}{90}$$

$$-\left(\frac{dx}{dy}\right)_{(x_1y_1)} = -\frac{1}{90} \Rightarrow \left(\frac{dy}{dx}\right)_{(x_1y_1)} = 90$$

Now,

$$\frac{dy}{dx} = 270x^4 - 540x^3 - 210x^2 + 360x + 210 = 90$$

$$\Rightarrow x = 1, 2, \frac{-2}{3}, \frac{-1}{3}$$

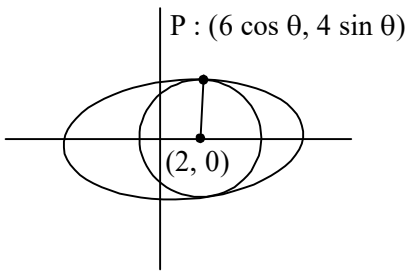
(D) normals

2. Let C be the largest circle centred at (2, 0) and inscribed in the ellipse $\frac{x^2}{36} + \frac{y^2}{16} = 1$. If $(1, \alpha)$ lies on C, then $10\alpha^2$ is equal to _____. **[JEE(Main)Jan.2023]**

माना C सबसे बड़ा वृत्त हैख्जिसका केन्द्र (2, 0) पर है तथा जो दीर्घवृत $\frac{x^2}{36} + \frac{y^2}{16} = 1$ के अन्तर्गत है। यदि बिंदु $(1, \alpha)$ वृत्त C पर है, r ks $10\alpha^2$ बराबर है _____ **[JEE(Main)Jan.2023]**

Ans. 118

Sol.



equation of ellipse $\frac{x^2}{36} + \frac{y^2}{16} = 1$

$$\frac{x}{18}+\frac{yy'}{8}=0$$

$$y'=\frac{-x}{18}.\frac{8}{y}$$

$$Y_N=\frac{9y}{4x}=9\cdot\frac{4\sin\theta}{4\cdot 6\cos\theta}=\frac{3}{2}\frac{\sin\theta}{\cos\theta}$$

$$Y_N(\text{ Slope })=\frac{4\sin\theta}{6\cos\theta-2}=\frac{2\sin\theta}{3\cos\theta-1}$$

$$\frac{3}{2}\frac{\sin\theta}{\cos\theta}=\frac{2\sin\theta}{3\cos\theta-1}$$

$$\cos\theta=\frac{3}{5}$$

$$P:\left(\frac{18}{5},\frac{16}{5}\right)$$

$$S=\sqrt{\left(\frac{8}{5}\right)^2+\left(\frac{16}{5}\right)^2}$$

Now $(x-2)^2+y^2=r^2$

$$(1,\alpha)\to 1+\alpha^2=\frac{8^2}{5^2}+\frac{16^2}{5^2}=12.8$$

$\alpha^2 = 11.8$
$10\alpha^2 = 118$

3. If the equation of the normal to the curve $y=\frac{x-a}{(x+b)(x-2)}$ at the point (1,–3) is $x-4y=13$, then the value of $a+b$ is equal to _____. **[JEE(Main)Jan.2023]**

यदि वक्र $y=\frac{x-a}{(x+b)(x-2)}$ के बिंदु (1,-3) पर अभिलंब का समीकरण x - $4y$ = 13 है, तो $x=\frac{5}{2}$ पर $a+b$ का मान बराबर है।

[JEE(Main)Jan.2023]

Ans. 4

Sol. $y=\frac{x-a}{(x+b)(x-2)}$

At point (1, –3),

$$-3=\frac{1-9}{(1+b)(1-2)}$$

$\Rightarrow 1 - a = 3(1 + b)$ ...(1)

Now, $y=\frac{x-a}{(x+b)(x-2)}$

$$\Rightarrow \frac{dy}{dx} = \frac{(x+b)(x-2)\times(1)-(x-a)(2x+b-2)}{(x+b)^2(x-2)^2}$$

At (1, –3) slope of normal is $\frac{1}{4}$ hence $\frac{dy}{dx} = -4$,

So, $-4 = \frac{(1+b)(-1)-(1-a)b}{(1+b)^2(-1)^2}$

Using equation (1)

$$\Rightarrow -4 = \frac{(1+b)(-1)-3(b+1)b}{(1+b)^2}$$

$$\Rightarrow -4 = \left|\frac{(-1)-3b}{(1+b)}\right| (b \neq -1)$$

$\Rightarrow b = -3$

So, $a = 7$

Hence, $a + b = 7 - 3 = 4$

4. Let $f: (0, 1) \to R$ be a function defined by $f(x) = \frac{1}{1-e^{-x}}$, and $g(x) = (f(-x) - f(x))$. Consider two statements

(I) g is an increasing function in (0, 1)
(II) g is one-one in (0, 1)

Then, : **[JEE(Main)Jan.2023]**

(A) Both (I) and (II) are true (B) Only (I) is true

(C) Neither (I) nor (II) is true (D) Only (II) is true

माना $f: (0, 1) \to R$, $f(x) = \frac{1}{1-e^{-x}}$ द्वारा परिभाषित है तथा $g(x) = (f(-x) - f(x))$ है। दो कथनों का विचार कीजिए।

(I) (0, 1) में g एक वर्धमान फलन है
(II) (0, 1) में g एकैकी है

तो, : **[JEE(Main)Jan.2023]**

(A) (I) तथा (II) दोनों सत्य है (B) केवल (I) सत्य है

(C) न तो (I) न ही (II) सत्य है (D) केवल (II) सत्य है

Ans. A

Sol. $g(x) = f(-x) - f(x) = \frac{1}{1-e^{x}} - \frac{1}{1-e^{-x}}$

$$\Rightarrow g(x) = \frac{1+e^{x}}{1-e^{x}}$$

$$g'(x) = \frac{\left(1-e^{x}\right)\left(e^{x}\right)-\left(1+e^{x}\right)\left(-e^{x}\right)}{\left(1-e^{x}\right)^2}$$

$$g'(x)=\frac{2e^x}{\left(1-e^x\right)^2}>0$$

g (x) is increasing as well as one-one.

5. Let $f:\ R\to R$ be a function such that $f(x)=\frac{x^2+2x+1}{x^2+1}$. Then **[JEE(Main)Jan.2023]**

(A) $f(x)$ is one-one in $[1,\infty)$ but not in $(-\infty,\infty)$

(B) $f(x)$ is one-one in $(-\infty,\infty)$

(C) $f(x)$ is many-one in $(1,\infty)$

(D) $f(x)$ is many-one in $(-\infty,-1)$

माना $f:\ R\to R$ एक फलन $f(x)=\frac{x^2+2x+1}{x^2+1}$ है। तब **[JEE(Main)Jan.2023]**

(A) $[1,\infty)$ में $f(x)$ एकैकी है परन्तु $(-\infty,\infty)$ में एकैकी नहीं है।

(B) $(-\infty,\infty)$ में $f(x)$ एकैकी है।

(C) $(1,\infty)$ में $f(x)$ बहु–एक है।

(D) $(-\infty,-1)$ में $f(x)$ बहु–एक है।

Ans. A

Sol. $\because f(x)=\frac{(x+1)^2}{(x+1)}$

$$\because f'(x)=\frac{\left(x^2+1\right).2(x+1)-(x+1)^2(2x)}{\left(x^2+1\right)^2}=-\frac{2(x+1)(x-1)}{\left(x^2+1\right)^2}$$

clearly f(x) is one – one is $(-\infty, -1)$ and also in $(1, \infty)$ but f(x) is not one – one is $(-\infty, \infty)$

6. Let f and g be twice differentiable functions on R such that

$$f''(x)=g''(x)+6x,\quad f'(1)=4g'(1)-3=9,\quad f(2)=3g(2)=12$$

Then which of the following is NOT true? **[JEE(Main)Jan.2023]**

(A) $g(-2)-f(-2)=20$

(B) If $-1<x<2$, then $|f(x)-g(x)|<8$

(C) There exists $x_0\in(1,3/2)$ such that $f(x_0)=g(x_0)$

(D) $|f'(x)-g'(x)|<6\Rightarrow -1<x<1$

माना फलन f तथा g, R पर दो बार अवकलनीय हैं तथा

$$f''(x)=g''(x)+6x,\quad f'(1)=4g'(1)-3=9,\quad f(2)=3g(2)=12$$

हैं। तो निम्न में से कौन सा सत्य नहीं है ? **[JEE(Main)Jan.2023]**

(A) $g(-2) - f(-2) = 20$ है

(B) यदि $-1 < x < 2$ है, तो $|f(x) - g(x)| < 8$ है

(C) $x_0 \in (1, 3/2)$ का अस्तित्व है, जिसके लिए $f(x_0) = g(x_0)$ है

(D) $|f'(x) - g'(x)| < 6 \Rightarrow -1 < x < 1$

Ans. B

Sol. $f''(x) = g''(x) + 6x$

$\Rightarrow \int f''(x)dx = \int g''(x)dx + \int 6xdx + C$

$\Rightarrow f'(x) = g'(x) + 3x^2 + C$ .....(1)

$x = 1 \Rightarrow f'(1) = g'(1) + 3 + C$

$\Rightarrow 9 = 3 + 3 + C \Rightarrow C = 3$

from (1) $\int f'(x)dx = \int g'(x)dx + \int 3x^2dx + \int 3dx + k$

$\Rightarrow f(x) = g(x) + x^3 + 3x + K$

$x = 2 \Rightarrow f(2) = g(2) + 8 + 6 + K$

$\Rightarrow 12 = 4 + 8 + 6 + K$

$\Rightarrow K = -6$

$\Rightarrow f(x) = g(x) + x^3 + 3x - 6$

Let $h(x) = f(x) - g(x) = x^3 + 3x - 6$

$\Rightarrow h'(x) = f'(x) - g'(x) = 3x^2 + 3 > 0 \ \forall \ x$

$\Rightarrow h'(x)$ is S.I. $\Rightarrow h(x) = 0$ has no solution

$\Rightarrow h(x) \in (h(-1), h(2)) \Rightarrow h(x) \in (-8, 8)$

$|f'(x) - g'(x)| < 6 \Rightarrow |3x^2 + 3| < 6 \Rightarrow 3x^2 + 3 < 6$

$\Rightarrow x^2 - 1 < 0 \Rightarrow x \in (-1, 1)$

$h(-2) = f(-2) - g(-2) = -20$

7. Let $x = 2$ be local minima of the function $f(x) = 2x^4 - 18x^2 + 8x + 12$, $x \in (-4, 4)$. If M is local maximum value of the function f in $(-4, 4)$, then M = **[JEE(Main)Jan.2023]**

माना फलन $f(x) = 2x^4 - 18x^2 + 8x + 12$, $x \in (-4, 4)$ का एक स्थानीय निम्नतम $x = 2$ पर है। यदि $(-4, 4)$ में फलन f का स्थानीय उच्चतम M है, तो M = **[JEE(Main)Jan.2023]**

(A) $12\sqrt{6} - \frac{33}{2}$ (B) $18\sqrt{6} - \frac{31}{2}$ (C) $18\sqrt{6} - \frac{33}{2}$ (D) $12\sqrt{6} - \frac{31}{2}$

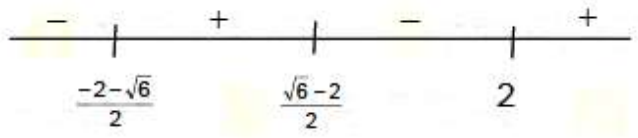
Ans. A

Sol. $f(x) = 2x^4 - 18x^2 + 8x + 12$

$f'(x) = 8x^3 - 36x + 8$

$= 4(x - 2)(2x^2 + 4x - 1)$

Number line of $f'(x)$ is given by



so maxima point is $x = \frac{\sqrt{6}-2}{2}$

and maximum value of $f(x) = 12\sqrt{6} - \frac{33}{2}$

8. Let the function $f(x) = 2x^3 + (2p - 7)x^2 + 3(2p - 9)x - 6$ have a maxima for some value of $x < 0$ and a minima for some value of $x > 0$. Then, the set of all values of p is : **[JEE(Main)Jan.2023]**

माना फलन $f(x) = 2x^3 + (2p - 7)x^2 + 3(2p - 9)x - 6$ का एक उच्चिष्ठ किसी $x < 0$ पर है तथा एक निम्निष्ठ किसी $x > 0$ पर है। तो p के सभी मानों का समुच्चय है **[JEE(Main)Jan.2023]**

(A) $\left(\frac{9}{2}, \infty\right)$ (B) $\left(-\infty, \frac{9}{2}\right)$ (C) $\left(-\frac{9}{2}, \frac{9}{2}\right)$ (D) $\left(0, \frac{9}{2}\right)$

Ans. B

Sol. $f'(x) = 6x^2 + 2(2p-7)x + 3(2p-9)$

$f'(0) < 0$

$3(2p-9) < 0$

$2p - 9 < 0$

$p < \frac{9}{2} \Rightarrow p \in \left(-\infty, \frac{9}{2}\right)$

9. If the functions $f(x) = \frac{x^3}{3} + 2bx + \frac{ax^2}{2}$ and $g(x) = \frac{x^3}{3} + ax + bx^2$, $a \neq 2b$ have a common extreme point, then $a + 2b + 7$ is equal to : **[JEE(Main)Jan.2023]**

यदि फलनों $f(x) = \frac{x^3}{3} + 2bx + \frac{ax^2}{2}$ तथा $g(x) = \frac{x^3}{3} + ax + bx^2$, $a \neq 2b$ का एक उभयनिष्ठ चरम बिंदु है, तब $a + 2b + 7$ बराबर है : **[JEE(Main)Jan.2023]**

(A) 4 (B) $\frac{3}{2}$ (C) 6 (D) 3

Ans. C

Sol. $f'(x) = x^2 + 2b + ax$

$g'(x) = x^2 + a + 2bx$

$(2b - a) - x(2b - a) = 0$

$\therefore x = 1$ is the common root

Put $x = 1$ in $f'(x) = 0$ or $g'(x) = 0$

$1 + 2b + a = 0$

$7 + 2b + a = 6$

10. A wire of length 20 m is to be cut into two pieces. A piece of length l_1 is bent to make a square of area A_1 and the other piece of length l_2 is made into a circle of area A_2. If $2A_1 + 3A_2$ is minimum then $(\pi l_1) : l_2$ is equal to : **[JEE(Main)Jan.2023]**

20 मीटर लंबी एक तार को दो भागों में काटा गया है। l_1 लंबाई के एक भाग को मोड़ कर A_1 क्षेत्रफल का एक वर्ग बनाया गया है तथा l_2 लंबाई के दूसरे भाग से A_2 क्षेत्रफल का एक वृत्त बनाया गया है। यदि $2A_1 + 3A_2$ न्यूनतम है, तो $(\pi l_1) : l_2$ बराबर है : **[JEE(Main)Jan.2023]**

(A) 4 : 1 (B) 1 : 6 (C) 6 : 1 (D) 3 : 1

Ans. C

Sol. $\ell_1 + \ell_2 = 20 \Rightarrow \dfrac{d\ell_2}{d\ell_1} = -1$

$A_1 = \left(\dfrac{\ell_1}{4}\right)^2$ and $A_2 = \pi\left(\dfrac{\ell_2}{2\pi}\right)^2$

Let $S = 2A_1 + 3A_2 = \dfrac{\ell_1^2}{8} + \dfrac{\ell_2^2}{4\pi}$

$\dfrac{ds}{d\ell} = 0 \Rightarrow \dfrac{2\ell_1}{8} + \dfrac{6\ell_2}{4\pi} \cdot \dfrac{d\ell_2}{d\ell_1} = 0$

$\Rightarrow \dfrac{\ell_1}{4} = \dfrac{6\ell_2}{4\pi} \Rightarrow \dfrac{\pi\ell_1}{\ell_2} = 6$

11. Let $f(x) = 2x + \tan^{-1} x$ and $g(x) = \log_e\left(\sqrt{1+x^2} + x\right)$, $x \in [0, 3]$. Then : **[JEE(Main)Jan.2023]**

(A) $\min f'(x) = 1 + \max g'(x)$

(B) there exist $0 < x_1 < x_2 < 3$ such that $f(x) < g(x)$, $\forall\ x \in (x_1, x_2)$

(C) there exists $\hat{x} \in [0,3]$ such that $f'(\hat{x}) < g'(\hat{x})$

(D) $\max f(x) > \max g(x)$

माना $f(x) = 2x + \tan^{-1} x$ एवं $g(x) = \log_e\left(\sqrt{1+x^2} + x\right)$, $x \in [0, 3]$ है। तब **[JEE(Main)Jan.2023]**

(A) $\min f'(x) = 1 + \max g'(x)$

(B) $0 < x_1 < x_2 < 3$ का अस्तित्व है, जिनके लिए $f(x) < g(x)$, $\forall\ x \in (x_1, x_2)$ है।

(C) $\hat{x} \in [0,3]$ का अस्तित्व है, जिसके लिए $f'(\hat{x}) < g'(\hat{x})$ है।

(D) $\max f(x) > \max g(x)$

Ans. D

Sol. $f(x)=2x+\tan^{-1}x$ and $g(x)=\ln\left(\sqrt{1+x^2}+x\right)$

and $x \in [0, 3]$

$$g'(x)=\frac{1}{\sqrt{1+x^2}}$$

Now, $0 \le x \le 3$

$0 \le x^2 \le 9$

$1 \le 1 + x^2 \le 10$

So, $2+\frac{1}{10} \le f'(x) \le 3$

$\frac{21}{10} \le f'(x) \le 3$ and $\frac{1}{\sqrt{10}} \le g'(x) \le 1$

option (4) is icorrect

From above, $g'(x) < f'(x)\ \forall\ x \in [0, 3]$

Option (1) is incorrect

f'(x) & g'(x) both positive so f(x) & g(x) both are increasing

So, max (f(x) at x = 3 is $6 + \tan^{-1} 3$)

Max (g(x) at x = 3 is $\ln\left(3+\sqrt{10}\right)$

And $6 + \tan^{-1} 3 > \ln\left(3+\sqrt{10}\right)$

Option (4) is correct.

12. The sum of the absolute maximum and minimum values of the function $f(x) = |x^2 - 5x + 6| - 3x + 2$ in the interval [-1,3] is equal to : **[JEE(Main)Jan.2023]**

अंतराल [-1,3] में फलन $f(x) = |x^2 - 5x + 6| - 3x + 2$ के निरपेक्ष उच्चतम और निरपेक्ष निम्नतम मानों का योग बराबर है:

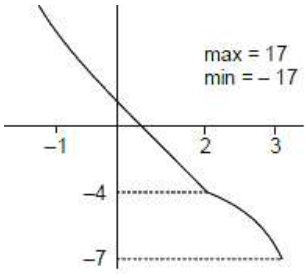
[JEE(Main)Jan.2023]

(A) 12 (B) 24 (C) 10 (D) 13

Ans. C

Sol. $f(x) = |x^2 - 5x + 6| - 3x + 2$

$$f(x)=\begin{cases} x^2-8x+8 & ; \quad x[-1,2] \\ -x^2+2x-4 & ; \quad x \in [2,3] \end{cases}$$



$$f'(x)=\begin{cases} 2x-8; & x\in[-1,2] \\ -2x+2; & x\in[2,3] \end{cases}$$

$f(2) = 4 - 16 + 8 = -4 = 4$

$f(-1) = 17$ (max)

$f(3) = -9 + 6 - 4 = -7$ (min)

Abs. Sum = 10

13. Let a tangent to the curve $9x^2 + 16y^2 = 144$ intersects the coordinate axes at the points A and B. Then, the minimum length of the line segment AB is _____. **[JEE(Main)Jan.2023]**

माना वक्र $9x^2 + 16y^2 = 144$ की एक स्पर्श रेखा निर्देशांक अक्षों को बिंदुओं A तथा B पर मिलती है। तो रेखाखंड AB की न्यूनतम लंबाई है _____ **[JEE(Main)Jan.2023]**

Ans. 7

Sol. $\frac{x^2}{16}+\frac{y^2}{9}=1$

Equation of tangent $\frac{x}{4}\cos\theta+\frac{y}{3}\sin\theta=1$

A (4sec θ, 0) B(0, 3(cosec θ)

$$AB=\sqrt{16\sec^2\theta+9\operatorname{cosec}^2\theta}=\sqrt{25+\left(4\tan\theta-3\cot\theta\right)^2+24}\geq\sqrt{49}\geq 7$$

$AB_{min} = 7$

14. Let the quadratic curve passing through the point (–1, 0) and touching the line y = x at (1, 1) be y = f(x). Then the x-intercept of the normal to the curve at the point (α, α + 1) in the first quadrant is _____. **[JEE(Main)April,2023]**

माना बिंदु (–1, 0) से होकर जाने वाला तथा रेखा y = x को (1, 1) पर स्पर्श करने वाला द्विघातीय वक्र y = f(x) है। तो प्रथम चतुर्थांश में बिंदु (α, α + 1) पर वक्र के अभिलंब का x-अंतःखंड है : **[JEE(Main)April,2023]**

Ans. 11

Sol. $f(x)=(x+1)(ax+b)$

$1 = 2a + 2b$ ....(1)

$f'(x)=(ax+b)+a(x+1)$

$1 = (3a + b)$ ....(2)

$\Rightarrow b = 1/4, a = 1/4$

$$f(x)=\frac{(x+1)^2}{4}$$

$$f'(x)=\frac{x}{2}+\frac{1}{2}\quad \alpha+1=\frac{(\alpha+1)^2}{4},\alpha>-1$$

$\alpha + 1 = 4$

$\alpha = 3$

normal at (3, 4)

$y-4=-\frac{1}{2}(x-3)$

$y=0 \qquad x=8+3$

15. Let $g(x)=f(x)+f(1-x)$ and $f''(x)>0, x\in(0,1)$. If g is decreasing in the interval $(0, \alpha)$ and increasing in the interval $(\alpha, 1)$, then $\tan^{-1}(2\alpha)+\tan^{-1}\left(\frac{1}{\alpha}\right)+\tan^{-1}\left(\frac{\alpha+1}{\alpha}\right)$ is equal to : **[JEE(Main)April,2023]**

माना $g(x)=f(x)+f(1-x)$ तथा $f''(x)>0, x\in(0,1)$ है। यदि g अंतराल $(0, \alpha)$ में ह्रासमान है तथा अंतराल $(\alpha, 1)$ में वर्धमान है, तो $\tan^{-1}(2\alpha)+\tan^{-1}\left(\frac{1}{\alpha}\right)+\tan^{-1}\left(\frac{\alpha+1}{\alpha}\right)$ बराबर है : **[JEE(Main)April,2023]**

(A) π (B) $\frac{3\pi}{4}$ (C) $\frac{5\pi}{4}$ (D) $\frac{3\pi}{2}$

Ans. A

Sol. $g'(x)=f'(x)-f'(1-x)$

If $x>1-x$

$\Rightarrow f'(x)>f'(1-x) \because f'(x)$ is strictly increasing

$\Rightarrow g'(x)>0 \Rightarrow$ so g(x) is increasing when $x>1-x \Rightarrow x>\frac{1}{2}$

$\Rightarrow \alpha=\frac{1}{2}$

similarly when $x<1-x \Rightarrow g(x)$ is decreeing

Now

$\tan^{-1}(2\alpha)+\tan^{-1}\left(\frac{1}{\alpha}\right)+\tan^{-1}\left(\frac{\alpha+1}{\alpha}\right)$

$=\tan^{-1}(1)+\tan^{-1}(2)+\tan^{-1}(3)=\pi(\because 1+2+3=1.2.3)$

16. Let $f:[2,4]\to\mathbb{R}$ be a differentiable function such that

$(x\log_e x)f'(x)+(\log_e x)f(x)+f(x)\geq 1, x\in[2,4]$ with $f(2)=\frac{1}{2}$ and $f(4)=\frac{1}{4}$.

Consider the following two statements :

(A) : $f(x)\leq 1$, for all $x\in[2, 4]$

(B) : $f(x)\geq\frac{1}{8}$, for all $x\in[2, 4]$

Then : **[JEE(Main)April,2023]**

(A) Only statement (A) is true (B) Both the statements (A) and (B) are true

(C) Neither statement (A) nor statement (B) is true (D) Only statement (B) is true

माना $f:[2,4]\to\mathbb{R}$ एक अवकलनीय फलन है, जिसके लिए

$\left(x\log_e x\right)f'(x)+\left(\log_e x\right)f(x)+f(x)\ge 1, x\in[2,4]$, $f(2)=\frac{1}{2}$ तथा $f(4)=\frac{1}{4}$ है।

निम्न में दो कथनों का विचार कीजिए :

(A) : सभी $x\in[2,4]$ के लिए $f(x)\le 1$ है।

(B) : सभी $x\in[2,4]$ के लिए $f(x)\ge\frac{1}{8}$ है।

तो : **[JEE(Main)April,2023]**

(A) केवल कथन (A) सत्य है (B) (A) तथा (B) दोनों कथन सत्य हैं

(C) न तो कथन (A) न ही कथन (B) सत्य है (D) केवल कथन (B) सत्य है

Ans. B

Sol. $\frac{d}{dx}\left(x\,\ell n(x).f(x)-x\right)\ge 0,\forall x\in[2,4]$

let $g(x)=x\,\ell n\left(x\right)f(x)-x,\quad \forall x\in[2,4]$

$\Rightarrow g'(x)\ge 0,\forall x\in[2,4]$

$\Rightarrow g'(x)$ is increasing funtion in [2,4]

$\Rightarrow g(2)\le g(x)\le g(4)$

$\Rightarrow 2\cdot\ell n\cdot\frac{1}{2}-2\le x\,\ell n\left(x\right)f(x)-x\le 4\cdot\ell n(4)\cdot\frac{1}{4}-4$

$\Rightarrow \ln 2-2+x\le x.\ell nx.f(x)\le 2(\ell n2-2)+x$

$\Rightarrow \frac{\ell n2-2}{x\ell nx}+\frac{1}{\ell nx}\le f(x)\le\frac{2(\ln 2-2)}{x\ell nx}+\frac{1}{\ell nx}$

Since numerator is constant

$\Rightarrow \frac{\ell n2-2}{4\ell n4}+\frac{1}{\ell\ln 4}\le f(x)\le\frac{2(\ell n2-2)}{2\ell n2}+\frac{1}{\ell n2}$

$\Rightarrow \frac{1}{8}+\frac{1}{4\ln 2}\le f(x)\le 1-\frac{1}{\ell n2}$

$2\Rightarrow \frac{1}{8}<f(x)<1,\forall x\in[2,4]$

17. The number of points, where the curve $y=x^5-20x^3+50x+2$ crosses the x-axis, is _____.

[JEE(Main)April,2023]

उन बिंदुओं, जहाँ वक्र $y=x^5-20x^3+50x+2$, x-अक्ष को काटता है, की संख्या है _____

[JEE(Main)April,2023]

Ans. 5

Sol. $y = x^5 - 20x^3 + 50x + 2$

$$\frac{dy}{dx} = 5x^4 - 60x^2 + 50 = 5\left(x^4 - 12x^2 + 10\right)$$

$$\frac{dy}{dx} = 0 \Rightarrow x^4 - 12x^2 + 10 = 0$$

$$\Rightarrow x^2 = \frac{12 \pm \sqrt{144 - 40}}{2}$$

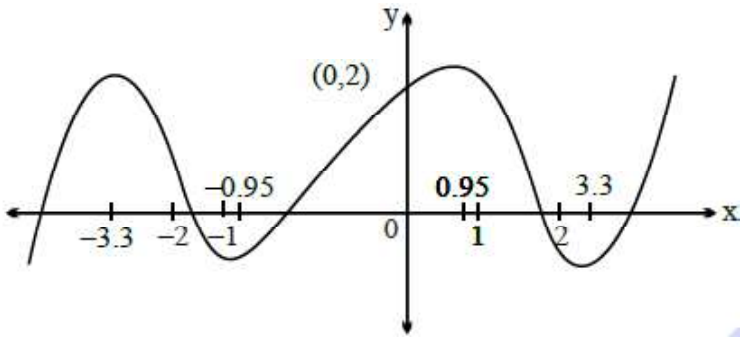
$$\Rightarrow x^2 = 6 \pm \sqrt{26} \Rightarrow x^2 \approx 6 \pm 5.1$$

$$\Rightarrow x^2 \approx 11.1, 0.9$$

$$\Rightarrow x \approx \pm 3.3, \pm 0.95$$

$f(0) = 2, f(1) = +\text{ ve}, f(2) = -\text{ ve}$

$f(-1) = -\text{ve}, f(-2) = +\text{ve}$



18. If a_α is the greatest term in the sequence $a_n = \frac{n^3}{n^4 + 147}, n = 1, 2, 3, \ldots$ then α is equal to _____.

[JEE(Main)April,2023]

यदि अनुक्रम $a_n = \frac{n^3}{n^4 + 147}, n = 1, 2, 3, \ldots$ का अधिकतम पद a_α है, तो α बराबर है _____

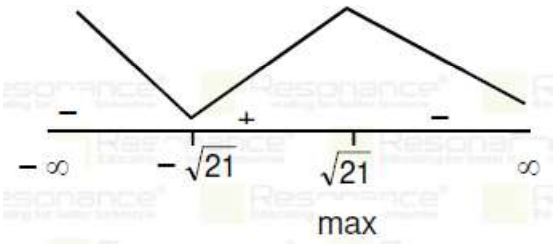
[JEE(Main)April,2023]

Ans. 5

Sol. $$f'(n) = \frac{3n^2\left(n^4 + 147\right) - 4n^3\left(n^3\right)}{\left(n^4 + 147\right)^2}$$

$$f'(n) = \frac{n^2\left[3n^4 + 441 - 4n^4\right]}{\left(n^4 + 147\right)^2}$$

$$f'(n) = \frac{-n^2\left(n^4 - 441\right)}{\left(n^4 + 147\right)^2} = \frac{-n^2\left(n^2 + 21\right)\left(n^2 - 21\right)}{\left(n^4 + 147\right)^2} = \frac{-n^2\left(n^2 + 21\right)}{\left(n^4 + 147\right)^2}(n + \sqrt{21})(n - \sqrt{21})$$



When $n = 4 \quad a_4 = \frac{4^3}{4^4 + 147} = \frac{64}{256 + 147} = \frac{64}{403}$

When $n = 5 \quad a_5 = \frac{125}{625 + 147} = \frac{125}{772}$

a_5 is $\max = \frac{125}{772}$ hence $n = 5$

19. The number of points, where the curve $f(x) = e^{8x} - e^{6x} - 3e^{4x} - e^{2x} + 1, x \in \mathbb{R}$ cuts x-axis, is equal to _____.

उन बिंदुओं, जहाँ वक्र $f(x) = e^{8x} - e^{6x} - 3e^{4x} - e^{2x} + 1, x \in \mathbb{R}$, x-अक्ष को काटता है, की संख्या है _____

Ans. B

Sol. $f(x) = e^{8x} - e^{6x} - 3e^{4x} - e^{2x} + 1, x \in \mathbb{R}$

$\lim\limits_{x \to -\infty} f(x) = 1, \quad f(0) = -3, \quad \lim\limits_{x \to \infty} f(x) \to \infty$

One root of f(x) = 0 is negative and other root of f(x) = 0 is positive

Curve y = f(x) cuts x-axis at two points

20. A square piece of tin of side 30 cm is to be made into a box without top by cutting a square from each corner and folding up the flaps to form a box. If the volume of the box is maximum, then its surface area (in cm^2) is equal to :

[JEE(Main)April,2023]

30 cm भुजा के टिन के एक एक वर्गाकार टुकड़े के प्रत्येक कोने पर एक वर्ग काटकर तथा इस प्रकार बनें टिन के फलकों को मोड़ कर ढक्कन रहित एक संदूक बनाना है। यदि संदूक का आयतन उच्चतम हैं, तो इसका पृष्ठीय क्षेत्रफल (वर्ग cm में) बराबर है :

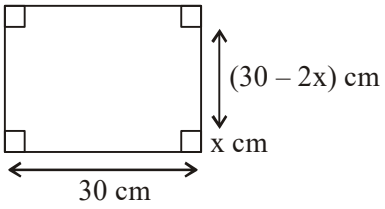
[JEE(Main)April,2023]

(A) 675 (B) 1025 (C) 900 (D) 800

Ans. D

Sol. Let the side of the square to be cut off be x cm.

Then, the length and breadth of the box will be (30 – 2x) cm each and the height of the box is x cm therefore,



The volume V(x) of the box is given by

$V(x) = x(30 - 2x)^2$

$\frac{dv}{dx} = (30 - 2x)^2 + 2x(30 - 2x)(-2)$

$0 = (30 - 2x)^2 - 4x(30 - 2x)$

$0 = (30 - 2x)[(30 - 2x) - 4x]$

$0 = (30 - 2x)(30 - 6x)$

$x = 15, 5$

$x \neq 15$ (Not possible)

$\{V = 0\}$

Surface area without top of the box $= \ell b + 2(bh + h\ell)$

$= (30 - 2x)(30 - 2x) + 2[(30 - 2x)x + (30 - 2x)x]$

$= [(30 - 2x)^2 + 4\{(30 - 2x)x\}$

$= [(30 - 10)^2 + 4(5)(30 - 10)]$

$= 400 + 400$

$= 800\ cm^2$

21. If the local maximum value of the function $f(x) = \left(\frac{\sqrt{3}e}{2\sin x}\right)^{\sin^2 x}$, $x \in \left(0, \frac{\pi}{2}\right)$, is $\frac{k}{e}$, then $\left(\frac{k}{e}\right)^8 + \frac{k^8}{e^5} + k^8$ is equal to : **[JEE(Main)April,2023]**

यदि फलन $f(x) = \left(\frac{\sqrt{3}e}{2\sin x}\right)^{\sin^2 x}$, $x \in \left(0, \frac{\pi}{2}\right)$, का स्थानीय उच्चतम मान $\frac{k}{e}$ है, तो $\left(\frac{k}{e}\right)^8 + \frac{k^8}{e^5} + k^8$ बराबर है : **[JEE(Main)April,2023]**

(A) $e^5 + e^6 + e^{11}$ (B) $e^3 + e^6 + e^{10}$ (C) $e^3 + e^5 + e^{11}$ (D) $e^3 + e^6 + e^{11}$

Ans. A

Sol. Let $y = \left(\frac{\sqrt{3}e}{2\sin x}\right)^{\sin^2 x}$

$$\ln y = \sin^2 x \cdot \ln\left(\frac{\sqrt{3}e}{2\sin x}\right)$$

$$\frac{1}{y}y' = \ln\left(\frac{\sqrt{3}e}{2\sin x}\right)2\sin x\cos x + \sin^2 x \frac{2\sin x}{\sqrt{3}e}\frac{\sqrt{3}e}{2}(-\operatorname{cosec} x \cot x)$$

$$\Rightarrow \sin x \cos\left[2\ln\left(\frac{\sqrt{3}e}{2\sin x}\right) - 1\right] = 0$$

$$\Rightarrow \ln\left(\frac{3e}{4\sin^2 x}\right) = 1 \Rightarrow \frac{3e}{4\sin^2 x} = e \Rightarrow \sin^2 x = \frac{3}{4}$$

$$\Rightarrow \sin x = \frac{\sqrt{3}}{2}\left(\text{as } x \in \left(0, \frac{\pi}{2}\right)\right)$$

$$\Rightarrow \text{local max value} = \left(\frac{\sqrt{3e}}{\sqrt{3}}\right)^{3/4} = e^{3/8} = \frac{k}{e}$$

$$\Rightarrow k^8 = e^{11}$$

$$\Rightarrow \left(\frac{k}{e}\right)^8 + \frac{k^8}{e^5} + k^8 = e^3 + e^6 + e^{11}$$

22. $\max_{0\le x\le \pi}\left\{x - 2\sin x\cos x + \frac{1}{3}\sin 3x\right\} =$ **[JEE(Main)April,2023]**

(A) π (B) $\frac{5\pi + 2 + 3\sqrt{3}}{6}$ (C) 0 (D) $\frac{\pi + 2 - 3\sqrt{3}}{6}$

Ans. B

Sol. $f(x) = x - \sin 2x + \frac{1}{3}\sin 3x$

$f'(x) = 1 - 2\cos 2x + \cos 3x = 0$

$x = \frac{5\pi}{6}, \frac{\pi}{6}$

$\therefore f''(x) = 4\sin 2x - 3\sin 3x$

$f''\left(\frac{5\pi}{6}\right) < 0$

$\Rightarrow \left(\frac{5\pi}{6}\right)$ is point of maxima

$\Rightarrow f\left(\frac{5\pi}{6}\right) = \frac{5\pi}{6} + \frac{\sqrt{3}}{2} + \frac{1}{3}$

23. Let $g(x) = 3f\left(\frac{x}{3}\right) + f(3-x)$ and $f''(x) > 0$ for all $x \in (0, 3)$. If g is decreasing in $(0, \alpha)$ and increasing in $(\alpha, 3)$, then 8α is : **[JEE(Main)Jan.2024]**

माना सभी $x \in (0, 3)$ के लिए $g(x) = 3f\left(\frac{x}{3}\right) + f(3-x)$ तथा $f''(x) > 0$ हैं। यदि $(0, \alpha)$ में g ह्रासमान है तथा $(\alpha, 3)$ में वध
र्मिान है, तो 8α बराबर है : **[JEE(Main)Jan.2024]**

(A) 20 (B) 0 (C) 24 (D) 18

Ans. D

Sol. Since $f''(x) > 0 \Rightarrow f'(x)$
is increasing in $(0, 3)$

Now $g'(x) = f'\left(\frac{x}{3}\right) - f'(3-x)$

For g(x)

to be increasing $g'(x) = f'\left(\frac{x}{3}\right) - f'(3-x) > 0$

$\Rightarrow f'\left(\frac{x}{3}\right) > f'(3-x)$

$\Rightarrow \frac{x}{3} > 3-x$

$\Rightarrow x > 9-3x \Rightarrow x > \frac{9}{4}$

For g(x) be be decreasing

$g'(x) < 0 \quad \Rightarrow x < \frac{9}{4} \qquad \Rightarrow \alpha = \frac{9}{4}$

24. Consider the function $f : \left[\frac{1}{2}, 1\right] \to \mathbb{R}$ defined by $f(x) = 4\sqrt{2}x^3 - 3\sqrt{2}x - 1$. Consider the statements :

(I) The curve y = f(x) intersects the x-axis exactly at one point

(II) The curve y = f(x) intersects the x-axis at $x = \cos\frac{\pi}{12}$.

Then : **[JEE(Main)Jan.2024]**

(A) Only (I) is correct (B) Both (I) and (II) are the correct

(C) Both (I) and (II) are incorrect (D) Only (II) is correct

$f(x) = 4\sqrt{2}x^3 - 3\sqrt{2}x - 1$ द्वारा परिभाषित फलन $f : \left[\frac{1}{2}, 1\right] \to \mathbb{R}$ के लिए कथनों :

(I) वक्र y = f(x), x-अक्ष को मात्र एक बिन्दु पर काटता है।

(II) वक्र y = f(x), x-अक्ष को $x = \cos\frac{\pi}{12}$ पर काटता है।

में से : **[JEE(Main)Jan.2024]**

(A) केवल (I) सही है (B) (I) तथा (II) दोनों सही हैं

(C) (I) तथा (II) दोनों गलत हैं (D) केवल (II) सही हैं

Ans. B

Sol. Let $x = \frac{\pi}{12}$

$\therefore 3x = \frac{\pi}{4}$

$\cos(3x) = \frac{1}{\sqrt{2}}$

$\Rightarrow 4\cos^3 x - 3\cos x = \frac{1}{\sqrt{2}}$

$\Rightarrow 4\sqrt{2}\cos^3 x - 3\sqrt{2}\cos x - 1 = 0$

$\therefore x = \left(\frac{\pi}{12}\right)$ is the solution of above equation statement II is correct

$f'(x) = 12\sqrt{2}x^2 - 3\sqrt{2}$

Which is positive for $x \in \left[\frac{1}{2}, 1\right]$

Hence, only one solution in $\left[\frac{1}{2}, 1\right]$

$\therefore$ Statement I is also correct

25. The function $f(x) = \frac{x}{x^2 - 6x - 16}$, $x \in R - \{-2, 8\}$: **[JEE(Main)Jan.2024]**

(A) decreases in $(-\infty, -2) \cup (-2, 8) \cup (8, \infty)$

(B) increases in $(-\infty, -2) \cup (-2, 8) \cup (8, \infty)$

(C) decreases in $(-2,8)$ and increases in $(-\infty, -2) \cup (8, \infty)$

(D) decreases in $(-\infty,-2)$ and increases in $(8, \infty)$

फलन $f(x) = \frac{x}{x^2 - 6x - 16}$, $x \in R - \{-2, 8\}$: **[JEE(Main)Jan.2024]**

(A) $(-\infty, -2) \cup (-2, 8) \cup (8, \infty)$ में ह्रासमान है

(B) $(-\infty, -2) \cup (-2, 8) \cup (8, \infty)$ में वर्धमान है

(C) $(-2,8)$ में ह्रासमान है तथा $(-\infty, -2) \cup (8, \infty)$ वर्धमान है

(D) $(-\infty,-2)$ में ह्रासमान है तथा $(8, \infty)$ में वर्धमान है

Ans. A

Sol. $f(x) = \frac{x}{x^2 - 6x - 16}$

Now

$$f'(x)=\frac{-\left(x^2+16\right)}{\left(x^2-6x-16\right)^2}$$

$f'(x)<0$

Thus f(x) is decreasing in $(-\infty,-2)\cup(-2,8)\cup(8,\infty)$

26. Let $g:R\to R$ be a non constant twice differentiable function such that $g'\left(\frac{1}{2}\right)=g'\left(\frac{3}{2}\right)$. If a real valued function f is defined as $f(x)=\frac{1}{2}[g(x)+g(2-x)]$, then : **[JEE(Main)Jan.2024]**

(A) f" (x) = 0 for no x in (0, 1) (B) f" (x) = 0 for exactly one x in (0, 1)

(C) $f'\left(\frac{3}{2}\right)+f'\left(\frac{1}{2}\right)=1$ (D) f" (x) = 0 for atleast two x in (0, 2)

माना $g:R\to R$ एक परिवर्तनीय तथा दो बार अवकलनीय फलन है और $g'\left(\frac{1}{2}\right)=g'\left(\frac{3}{2}\right)$ है। यदि एक वास्तविक मान फलन f, $f(x)=\frac{1}{2}[g(x)+g(2-x)]$ द्वारा परिभाषित है, तो : **[JEE(Main)Jan.2024]**

(A) (0,1) में कोई भी x नहीं है, जिसके लिए f" (x) = 0 है

(B) (0,1) में ठीक एक x के लिए f" (x) = 0 है

(C) $f'\left(\frac{3}{2}\right)+f'\left(\frac{1}{2}\right)=1$

(D) (0,2) में कम से कम दो x के लिए f" (x) = 0 है

Ans. D

Sol. $f'(x)=\frac{1}{2}\left(g'(x)-g'(2-x)\right)$

Since $f'\left(\frac{1}{2}\right)=f'(1)=f'\left(\frac{3}{2}\right)=0$ (Rolle Theorem)

$\Rightarrow$ f"(x) = 0 for atleast 2 values of $x\in(0,2)$

27. If the function $f:(-\infty,-1]\to(a,b]$ defined by $f(x)=e^{x^3-3x+1}$ is one – one and onto, then the distance of the point P (2b + 4, a + 2) from the line $x+e^{-3}y=4$ is : **[JEE(Main)Jan.2024]**

यदि $f(x)=e^{x^3-3x+1}$ द्वारा परिभाषित फलन $f:(-\infty,-1]\to(a,b]$ एकैकी तथा आच्छादी है, तो रेखा $x+e^{-3}y=4$ से बिन्दु P(2b + 4, a + 2) की दूरी है : **[JEE(Main)Jan.2024]**

(A) $\sqrt{1+e^6}$ (B) $2\sqrt{1+e^6}$ (C) $4\sqrt{1+e^6}$ (D) $3\sqrt{1+e^6}$

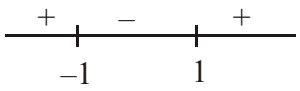
Ans. B

Sol. $f:(-\infty,-1)\to(a,b]$

$f(x)=e^{x^3-3x+1}$

$f'(x)=e^{x^3-3x+1}\left(3x^2-3\right)$

$f'(x)=e^{x^3-3x+1}3(x-1)(x+1)$



$(-\infty,-1]\to\uparrow F^n$

So $x\to-\infty$

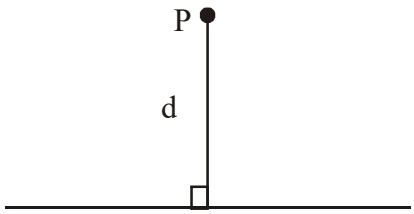
$\Rightarrow f(x)=e^{-\infty}\to 0=a$

$x\to-1$

$\Rightarrow F(-1)=e^{-1+3+1}=e^3=b$

$(a,b]\Rightarrow\left(0,e^3\right]$

$P\left(2e^3+4,2\right)$



$$d=\left|\frac{2e^3+4+e^{-3}(2)-4}{\sqrt{1+\left(e^{-3}\right)^2}}\right|$$

$$d=\frac{2\left(e^3+e^{-3}\right)}{\sqrt{\frac{e^6+1}{e^6}}}$$

$$v=\frac{\frac{2}{e^3}\left(1+e^6\right)}{\frac{\sqrt{1+e^6}}{e^3}}$$

$d=2\sqrt{1+e^6}$

28. If $5f(x)+4f\left(\frac{1}{x}\right)=x^2-2$, $\forall\, x\neq 0$ and $y=9x^2\,f(x)$, then y is strictly increasing in :

[JEE(Main)Jan.2024]

यदि $5f(x)+4f\left(\frac{1}{x}\right)=x^2-2$, $\forall\ x\neq 0$ है तथा $y=9x^2\ f(x)$ है, तो किस अन्तराल में y निरंतर वर्धमान है :

[JEE(Main)Jan.2024]

(A) $\left(0,\frac{1}{\sqrt{5}}\right)\cup\left(\frac{1}{\sqrt{5}},\infty\right)$ (B) $\left(-\frac{1}{\sqrt{5}},0\right)\cup\left(0,\frac{1}{\sqrt{5}}\right)$

(C) $\left(-\infty,\frac{1}{\sqrt{5}}\right)\cup\left(0,\frac{1}{\sqrt{5}}\right)$ (D) $\left(-\frac{1}{\sqrt{5}},0\right)\cup\left(\frac{1}{\sqrt{5}},\infty\right)$

Ans. D

Sol. $5f(x)+4f\left(\frac{1}{x}\right)=x^2-2$ ...(1)

Substitute $x\to\frac{1}{x}$

$5f\left(\frac{1}{x}\right)+4f(x)=\frac{1}{x^2}-2$ ...(2)

$(1)\times 5-(2)\times 4$

$\Rightarrow 9f(x)=5x^2-10-\frac{4}{x^2}+8$

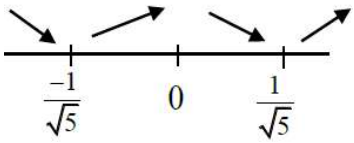
$\Rightarrow 9f(x)=\frac{5x^2-2x^2-4}{x^2}$

$\Rightarrow 9x^2f(x)=5x^4-2x^2-4$

$y=5x^4-2x^2-4$

$\frac{dy}{dx}=20x^3-4x>0$ (given)

$\Rightarrow 4x\left(5x^2-1\right)>0$



$\Rightarrow$ y is increasing in $\left(\frac{-1}{\sqrt{5}},0\right)\cup\left(\frac{1}{\sqrt{5}},\infty\right)$

29. Let $f(x)=2^x-x^2$, $x\in R$. If m and n are respectively the number of points at which the curves $y=f(x)$ and $y=f'(x)$ intersects the x-axis, then the value of $m+n$ is _____. **[JEE(Main)Jan.2024]**

माना $f(x)=2^x-x^2$, $x\in R$ है। यदि वक्र $y=f(x)$, $y=f'(x)$, x-अक्ष को क्रमशः m तथा n बिन्दुओं पर काटते हैं, तो $m+n$ का मान

है _____. **[JEE(Main)Jan.2024]**

Ans. 5

Sol. $f(x) = 2^x - x^2$

$f(x) = 0 \quad \Rightarrow 2^x = x^2$

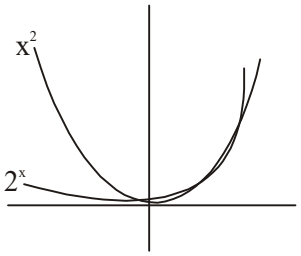
$m = 3$

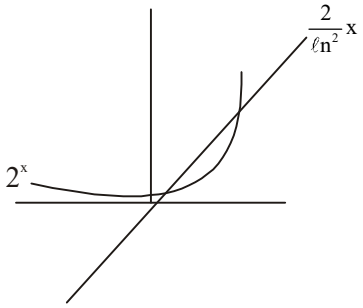
$f'(x) = 2^x \ell n2 - 2x = 0$

$2^x = \frac{2}{\ell n2} x$

$n = 2$

$m + n = 5$





30. In an A.P., the sixth term $a_6 = 2$. If the product $a_1a_4a_5$ is the greatest, then the common difference of the A.P. is equal to : **[JEE(Main)Jan.2024]**

एक A.P. में छठा पद $a_6 = 2$ है। यदि गुणनफल $a_1a_4a_5$ अधिकतम है, तो A.P. का सार्व अंतर बराबर है : **[JEE(Main)Jan.2024]**

(A) 5/8 (B) 8/5 (C) 2/3 (D) 3/2

Ans. B

Sol. Let first term = a

Common difference = d

Given, $a + 5d = 2$ _____(1)

Product (P) = $a_1a_5a_6 = a(a + 4d)(a + 3d)$

From (1)

$P = (2 - 5d)(2 - d)(2 - 2d)$

$\frac{dP}{dd} = (2-5d)(2-d)(-2) + (2-5d)(2-2d)(-1) + (-5)(2-d)(2-2d)$

$= -2[(d - 2)(5d - 2) + (d - 1)(5d - 2) + 5(d - 1)(d - 2)]$

$= -2[5d^2 + 4 - 12d + 5d^2 + 2 - 7d + 5d^2 + 10 - 15d]$

$= -2[15d^2 - 34d + 16]$

$\Rightarrow d = \frac{8}{5} \text{ or } \frac{2}{3}$

at $\left(\frac{8}{5}\right)$, at Product attains maxima

Answer is $\frac{8}{5}$

31. The function $f(x) = 2x + 3(x)^{\frac{2}{3}}, x \in R$, has : **[JEE(Main)Jan.2024]**

(A) exactly one point of local minima and no point of local maxima

(B) exactly two points of local maxima and exactky one point of local minima

(C) exactly one point of local maxima and exactly one point of local minima

(D) exactly one point of local maxima and no point of local minima

मफलन $f(x) = 2x + 3(x)^{\frac{2}{3}}, x \in R$: **[JEE(Main)Jan.2024]**

(A) का स्थानीय निम्निष्ठ का मात्र एक बिन्दु है तथा स्थानीय उच्चिष्ठ का कोई बिन्दु नहीं है

(B) के स्थानीय उच्चिष्ठ के मात्र दो बिन्दु है तथा स्थानीय निम्निष्ठ का मात्र एक बिन्दु है

(C) के स्थानीय उच्चिष्ठ का मात्र एक बिन्दु है तथा स्थानीय निम्निष्ठ का मात्र एक बिन्दु है

(D) के स्थानीय उच्चिष्ठ का मात्र एक बिन्दु है तथा स्थानीय निम्निष्ठ का मात्र एक बिन्दु है

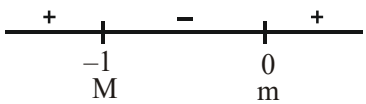
Ans. C

Sol. $f(x) = 2x + 3(x)^{\frac{2}{3}}$

$f'(x) = 2 + 2x^{\frac{-1}{3}}$

$= 2\left(1 + \frac{1}{x^{\frac{1}{3}}}\right)$

$= 2\left(\frac{x^{\frac{1}{3}} + 1}{x^{\frac{1}{3}}}\right)$



So, maxima (M) at $x = -1$ and minima(m) at $x = 0$

32. The maximum area of a triangle whose one vertex is at (0, 0) and the other two vertices lie on the curve $y = -2x^2 + 54$ at points (x, y) and (– x, y), where $y > 0$, is : **[JEE(Main)Jan.2024]**

त्रिभुज, जिसका एक शीर्ष (0, 0) है तथा अन्य दो शीर्ष वक्र $y = -2x^2 + 54$ के बिंदु (x, y) तथा (– x, y) हैं, जहाँ $y > 0$ है, का अधिकतम क्षेत्रफल है : **[JEE(Main)Jan.2024]**

(A) 122 (B) 88 (C) 92 (D) 108

Ans. D

Sol. (–x, y) (x, y)

(0,0) or (0,0)

(–x, y) (x, y)

$\text{Area} = \frac{1}{2}(2x)\left(54-2x^2\right)$

$A = 54x - 2x^3 \Rightarrow \frac{dA}{dx} = 54 - 6x^2$

$x = \pm 3$

Maximum area $A = (54 - 18)3 = 108$

33. Let $f(x) = (x+3)^2(x-2)^3, x \in [-4,4]$. If M and m are the maximum and minimum values of f, respectively in [–4, 4], then the value of M – m is : **[JEE(Main)Jan.2024]**

माना $f(x) = (x+3)^2(x-2)^3, x \in [-4,4]$ हैं। यदि f के [–4, 4] में अधिकतम तथा न्यूनतम मान क्रमशः M तथा m, हैं, तो

M – m का मान है : **[JEE(Main)Jan.2024]**

(A) 392 (B) 600 (C) 108 (D) 608

Ans. D

Sol. $f'(x) = (x+3)^2 \cdot 3(x-2)^2 + (x-2)^3 2(x+3)$

$= 5(x+3)(x-2)^2(x+1)$

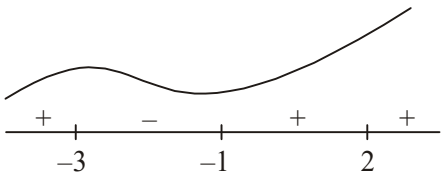
$f'(x) = 0, x = -3, -1, 2$

$f(-4) = -216$

$f(-3) = 0, f(4) = 49 \times 8 = 392$

$M = 392, m = -216$

$M - m = 392 + 216 = 608$



34. Let S = (–1, ∞) and f : S → R be defined as $f(x) = \int_{-1}^{x} \left(e^t - 1\right)^{11} (2t-1)^5 (t-2)^7 (t-3)^{12} (2t-10)^{61} dt$.

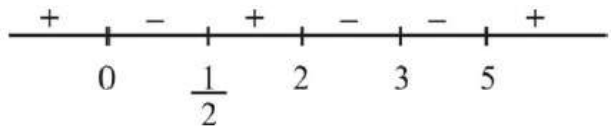
Let p = Sum of squares of the values of x, where f(x) attains local maxima on S, and q = Sum of the values of x, where f(x) attains local minima on S. Then, the value of $p^2 + 2q$ is _____. **[JEE(Main)Jan.2024]**

माना S = (–1, ∞) है तथा $f(x) = \int_{-1}^{x} \left(e^t - 1\right)^{11} (2t-1)^5 (t-2)^7 (t-3)^{12} (2t-10)^{61} dt$ द्वारा परिभाषित फलन

$f : S \to R$ है। माना p = x के उन मानों, जहाँ f, S पर स्थानीय उच्चतम प्राप्त करता है, के वर्गों का योग है तथा q = x के उन मानों, जहाँ f, S पर स्थानीय न्यूनतम प्राप्त करता है, का योग है। तो $p^2 + 2q$ का मान है_____. **[JEE(Main)Jan.2024]**

Ans. 27

Sol. $f'(x) = (e^x - 1)^{11} (2x - 1)^5 (x - 2)^7 (x - 3)^{12} (2x - 10)^{61}$



Local minima at $x = \frac{1}{2}, x = 5$

Local maxima at $x = 0, x = 2$

$\therefore p = 0 + 4 = 4,\ q = \frac{1}{2} + 5 = \frac{11}{2}$

Then $p^2 + 2q = 16 + 11 = 27$

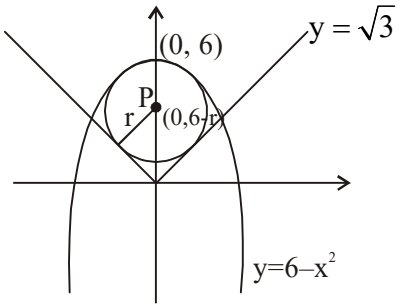
35. Let C be the circle of minimum area touching the parabola $y = 6 - x^2$ and the lines $y = \sqrt{3}|x|$. Then, which one of the following points lies on the circle C ? **[JEE(Main)April,2024]**

माना C न्यूनतम क्षेत्रफल का कोई वृत्त है जो परवलय $y = 6 - x^2$ और रेखा $y = \sqrt{3}|x|$ को स्पर्श कर रहा है। तब निम्नलिखित में से कौनसा बिन्दु वृत्त C पर स्थित होगा? **[JEE(Main)April,2024]**

(A) (1, 2) (B) (2, 2) (C) (1, 1) (D) (2, 4)

Ans. D

Sol.



Equation of circle $x^2 + (y - 6 + r)^2 = r^2$

distance between P and line = r

$\left|\frac{6 - r}{2}\right| = r$

$r = 2$

Equation is $x^2 + (y - 4)^2 = 4$ Point (2, 4) is lie on circle

36. Let maximum and minimum values of $\left(\sqrt{8x - x^2 - 12} - 4\right)^2 + (x - 7)^2$, $x \in R$ be M and m, respectively. Then $M^2 - m^2$ is equal to _____. **[JEE(Main)April,2024]**

माना $\left(\sqrt{8x - x^2 - 12} - 4\right)^2 + (x - 7)^2$, $x \in R$ के उच्चतम तथा निम्नतम मान क्रमशः M तथा m हैं। तो $M^2 - m^2$ बराबर है

Ans. 1600

Sol. $\left(\sqrt{8x-x^2-12}-4\right)^2+(x-7)^2=(y-4)^2+(x-7)^2$

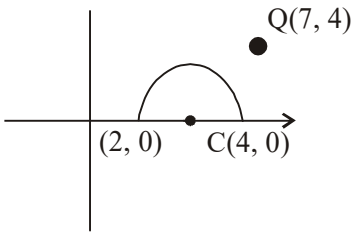
Where $y=\sqrt{8x-x^2-12}>0$

$\Rightarrow x^2+y^2-8x+12=0$

$\Rightarrow (x-4)^2+y^2=4$

Hence, $\left(\sqrt{8x-x^2-12}-4\right)^2+(x-7)^2=(PQ)^2$

Where P (x, y) lies on circle $y=\sqrt{8x-x^2-12}$ and Q (7, 4)



Now, minimum value of $(PQ)^2$ is :

$m=(CQ-2)^2$ where C is centre

$=(5-2)^2=9$

and maximum value of $(PQ)^2$ is :

$M=(7-2)^2+(4-0)^2=25+16=41$

$\Rightarrow \quad M^2-m^2=(41)^2-(9)^2=50\times 32=1600$

37. For the function $f(x)=\sin x+3x-\frac{2}{\pi}\left(x^2+x\right)$, where $x\in\left[0,\frac{\pi}{2}\right]$ consider the following two statements :

(I) f is increasing in $\left(0,\frac{\pi}{2}\right)$

(II) f' is decreasing in $\left(0,\frac{\pi}{2}\right)$

Between the above two statements : [JEE(Main)April,2024]

(A) only (I) is true (B) only (II) is true

(C) both (I) and (II) are true (D) neither (I) nor (II) is true

फलन

$f(x)=\sin x+3x-\frac{2}{\pi}\left(x^2+x\right)$, जहाँ $x\in\left[0,\frac{\pi}{2}\right]$, के लिए, निम्न दो कथनों पर विचार करें।

(I) f अंतराल $\left(0,\frac{\pi}{2}\right)$ में वर्धमान है

(II) f' अंतराल $\left(0,\frac{\pi}{2}\right)$ में ह्रासमान है

उपरोक्त दो कथनों में से : **[JEE(Main)April,2024]**

(A) केवल (I) सत्य है (B) केवल (II) सत्य है

(C) (I) तथा (II) दोनों सत्य है (D) न तो (I) और न ही (II) सत्य है

Ans. C

Sol. $f(x)=\sin x+3x-\frac{2}{\pi}\left(x^2+x\right),$ Where $x\in\left[0,\frac{\pi}{2}\right]$

$f'(x)=\cos x+3-\frac{2}{\pi}(2x+1)$

$=\cos x-\frac{4x}{\pi}-\frac{2}{\pi}+3$

as $x\in\left[0,\frac{\pi}{2}\right]$

$\frac{4x}{\pi}\in[0,2]$

$\Rightarrow 3-\frac{2}{\pi}-\frac{4x}{\pi}>0$

and also $\cos x>0$ when $x\in\left[0,\frac{\pi}{2}\right]$

$\Rightarrow f'(x)>0$

$\Rightarrow$ F(x) is increasing

Now, $f''(x)=-\sin x-\frac{4}{\pi}<0\ \forall\ x\in\left[0,\frac{\pi}{2}\right]$

Hence, f'(x) is decreasing

$\therefore$ Both statements (I) and (II) are true

38. The interval in which the function $f(x)=x^x, x>0$, is strictly increasing is : **[JEE(Main)April,2024]**

वह अंतराल, जिसमें फलन $f(x)=x^x, x>0$, निरंतर वर्धमान है, है : **[JEE(Main)April,2024]**

(A) $\left[\frac{1}{e^2},1\right)$ (B) $\left(0,\frac{1}{e}\right]$ (C) $\left[\frac{1}{e},\infty\right)$ (D) $(0,\infty)$

Ans. C

Sol. $f(x) = x^x$

$f'(x) = x^x (1 + \ln x) \geq 0$

$x > 0 \Rightarrow 1 + \ln x \geq 0$

$x \geq \frac{1}{e}$

$f'(x) \geq 0$ for $x \in \left[\frac{1}{e}, \infty\right)$

39. The number of critical points of the function $f(x) = (x-2)^{\frac{2}{3}}(2x+1)$ is : **[JEE(Main)April,2024]**

फलन $f(x) = (x-2)^{\frac{2}{3}}(2x+1)$ के क्रांतिक बिन्दुओं की संख्या होगी : **[JEE(Main)April,2024]**

(A) 0 (B) 1 (C) 2 (D) 3

Ans. C

Sol. $f(x) = (x-2)^{\frac{2}{3}}(2x+1)$

$f'(x) = 2(x-2)^{\frac{2}{3}} + (2x+1) \times \frac{2}{3}(x-2)^{-\frac{1}{3}}$

$f'(x) = 2(x-2)^{\frac{2}{3}} + \frac{2(2x+1)}{3(x-2)^{\frac{1}{3}}}$

$f'(x) = \frac{6(x-2) + 2(2x+1)}{3(x-2)^{\frac{1}{3}}}$

$f'(x) = \frac{10x-10}{3(x-2)^{\frac{1}{3}}} = \frac{10(x-1)}{3(x-2)^{\frac{1}{3}}}$

Critical points 1,2

$\Rightarrow$ 2 critical points

40. For the function $f(x) = (\cos x) - x + 1$, $x \in R$, between the following two statements :

(S1) $f(x) = 0$ for only one value of x in $[0, \pi]$.

(S2) $f(x)$ is decreasing in $\left[0, \frac{\pi}{2}\right]$ and increasing in $\left[\frac{\pi}{2}, \pi\right]$. **[JEE(Main)April,2024]**

(A) Only (S2) is correct. (B) Only (S1) is correct.

(C) Both (S1) and (S2) are correct. (D) Both (S1) and (S2) are incorrect.

फलन $f(x) = (\cos x) - x + 1, x \in R$ के लिए निचे दिये गये दो कथनों में से :

(S1) $[0, \pi]$ में x के केवल एक मान के लिए $f(x) = 0$ है

(S2) $\left[0, \frac{\pi}{2}\right]$ में f(x) ह्रासमान और $\left[\frac{\pi}{2}, \pi\right]$ में वृद्धमान है **[JEE(Main)April,2024]**

(A) केवल (S2) सत्य होगा (B) केवल (S1) सत्य होगा

(C) (S1) तथा (S2) दोनों सत्य होंगे (D) (S1) तथा (S2) दोनों असत्य होंगे

Ans. B

Sol. $f(x) = \cos x - x + 1$

$f'(x) = -\sin x - 1$

f is decreasing $\forall x \in R$

$f(x) = 0$

$f(0) = 2, f(\pi) = -\pi$

f is strictly decreasing in $[0, \pi]$ and $f(0).f(\pi) < 0$

$\Rightarrow$ Only one position of $f(x) = 0$

S1 is correct and S2 is incorrect.

41. Let $f : R \to R$ be a thrice diffrentiable function such that $f(0) = 0, f(1) = 1, f(2) = -1, f(3) = 2$ and $f(4) = -2$. Then, the minimum number of zeroes of $(3f'f'' + ff''')(x)$ is _____. **[JEE(Main)April,2024]**

माना त्रिअवकलनीय एक फलन $f : R \to R$ इस प्रकार है कि $f(0) = 0, f(1) = 1, f(2) = -1, f(3) = 2$ है, तथा $f(4) = -2$, तो $(3f'f'' + ff''')(x)$ के न्यूनतम शून्यों की संख्या है _____ **[JEE(Main)April,2024]**

Ans. 5

Sol. $\because f : R \to R$ and $f(0) = 0, f(1) = 1, f(2) = -1, f(3) = 2$ and $f(4) = -2$ then

f(x) has atleast 4 real roots.

Then f' (x) has atleast 3 real roots and f"(x) has atleast 2 real roots.

$$\frac{d}{dx}\left(f^3 \cdot f''\right) = 3f^2 \cdot f' \cdot f'' + f^3 \cdot f'''$$

$$= f^2\left(3f' \cdot f'' + f \cdot f'''\right)$$

Here $f^3 \cdot f''$ has atleast 6 roots.

Then its differentiation has atleast 5 distinct roots.

42. Let the set of all values of p, for which $f(x)=\left(p^2-6p+8\right)\left(\sin^2 2x-\cos^2 2x\right)+2(2-p)x+7$ does not have any critical point, be the interval (a, b). Then 16ab is equal to _____. **[JEE(Main)April,2024]**

माना p के सभी मानों, जिनके लिए $f(x)=\left(p^2-6p+8\right)\left(\sin^2 2x-\cos^2 2x\right)+2(2-p)x+7$ का एक भी क्रांतिक बिन्दु नहीं हैं, का समुच्चय, अंतराल (a, b) है। तो 16ab बराबर है _____. **[JEE(Main)April,2024]**

Ans. 252

Sol. $f(x) = -(p^2-6p+8)\cos 4n + 2(2-p)n + 7$

$f^1(x) = +4(p^2-6p+8)\sin 4x + (4-2p) \neq 0$

$$\sin 4x \neq \frac{2p-4}{4(p-4)(p-2)}$$

$$\sin 4x \neq \frac{2(p-2)}{4(p-4)(p-2)}$$

$p \neq 2$

$$\sin 4x \neq \frac{1}{2(p-4)}$$

$$\Rightarrow \left|\frac{1}{2(p-4)}\right| > 1$$

on solving we get

$$\therefore p \in \left(\frac{7}{2}, \frac{9}{2}\right)$$

Hence $a=\frac{7}{2}, b=\frac{9}{2}$

$\therefore$ 16ab = 252

43. Let $f(x)=3\sqrt{x-2}+\sqrt{4-x}$ be a real valued function. If α and β are respectively the minimum and the maximum values of f, then $\alpha^2+2\beta^2$ is equal to : **[JEE(Main)April,2024]**

माना $f(x)=3\sqrt{x-2}+\sqrt{4-x}$ वास्तविक मूल्य फलन है। यदि α तथा β क्रमशः f के न्यूनतम तथा उच्चतम मान हैं, तो $\alpha^2+2\beta^2$ बराबर है : **[JEE(Main)April,2024]**

(A) 24 (B) 38 (C) 44 (D) 42

Ans. D

Sol. $f'(x)=\frac{3}{2\sqrt{x-2}}-\frac{1}{2\sqrt{4-x}}: f'(x)=0 \qquad \Rightarrow x=\frac{19}{5}\in[2,4]$

$f(2)=\sqrt{2}, f\left(\frac{19}{5}\right)=2\sqrt{5}, f(4)=3\sqrt{2}$

$\therefore \alpha=\sqrt{2}, \beta=2\sqrt{5}$

$\alpha^2 + 2\beta^2 = 2 + 40 = 42$

44. Let a rectangle ABCD of sides 2 and 4 be inscribed in another PQRS such that the vertices of the rectangle ABCD lie on the sides of the rectangle PQRS. Let a and b be the sides of the rectangle PQRS when its area is maximum. Then $(a + b)^2$ is equal to : **[JEE(Main)April,2024]**

माना भुजाओं 2 तथा 4 का एक आयत ABCD एक अन्य आयत PQRS के अंतर्गत इस प्रकार है कि आयत ABCD के शीर्ष आयत PQRS की भुजाओं पर हैं। माना आयत PQRS की भुजाएँ a तथा b है, जब इसका क्षेत्रफल है। तो $(a + b)^2$ बराबर है : **[JEE(Main)April,2024]**

(A) 80 (B) 72 (C) 60 (D) 64

Ans. B

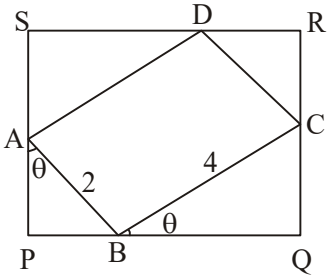
Sol. $\text{Area} = (4\cos\theta + 2\sin\theta)(2\cos\theta + 4\sin\theta)$

$= 8\cos^2\theta + 16\sin\theta\cos\theta + 4\sin\theta\cos\theta + 8\sin^2\theta$

$= 8 + 20\sin\theta\cos\theta$

$= 8 + 10\sin 2\theta$

Max Area $= 8 + 10 = 18$ $(\sin 2\theta = 1)$ $\theta = 45°$



$(a + b)^2 = (4\cos\theta + 2\sin\theta + 2\cos\theta + 4\sin\theta)^2$

$= (6\cos\theta + 6\sin\theta)^2$

$= 36(\sin\theta + \cos\theta)^2$

$= 36\left(\sqrt{2}\right)^2$

$= 72$

45. If the function $f(x)=\left(\frac{1}{x}\right)^{2x}; x>0$ attains the maximum value at $x=\frac{1}{e}$ then : **[JEE(Main)April,2024]**

यदि फलन $f(x)=\left(\frac{1}{x}\right)^{2x}; x>0,\ x=\frac{1}{e}$ पर उच्चतम मान प्राप्त करता है, तो : **[JEE(Main)April,2024]**

(A) $e^{2\pi} < (2\pi)^e$ (B) $e^{\pi} > \pi^e$ (C) $e^{\pi} < \pi^e$ (D) $(2e)^{\pi} > \pi^{(2e)}$

Ans. B

Sol. $f(x)=\left(\frac{1}{x}\right)^{2x}$

$f\left(\frac{1}{e}\right)>f\left(\frac{1}{\pi}\right)$

$e^{2/e}>\pi^{2/\pi}$

$e^{\pi}>\pi^{e}$

46. Let $f(x)=4\cos^3 x+3\sqrt{3}\cos^2 x-10$. The number of points of local maxima of f in interval $(0, 2\pi)$ is :
[JEE(Main)April,2024]

माना $f(x)=4\cos^3 x+3\sqrt{3}\cos^2 x-10$, अन्तराल $(0, 2\pi)$ में f के स्थानीय उच्चिष्ठ बिन्दुओं की संख्या होगी :
[JEE(Main)April,2024]

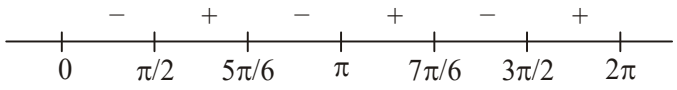
(A) 4 (B) 1 (C) 2 (D) 3

Ans. C

Sol. $f'(x)=-12\cos^2 x\sin x+6\sqrt{3}\cos x\times(-\sin x)$

$=-6\sin x\left[2\cos^2 x+\sqrt{3}\cos x\right]$

$-6\sin x\cos x\left(2\cos x+\sqrt{3}\right)$



local maxima at $\frac{5\pi}{6}$ and $\frac{7\pi}{6}$ so number of local maxima points are 2

47. If the function $f(x)=2x^3-9ax^2+12a^2x+1$, $a>0$ has a local maximum at $x=\alpha$ and a local minimum at $x=\alpha^2$, then α and α^2 are the roots of the equation :
[JEE(Main)April,2024]

यदि फलन $f(x)=2x^3-9ax^2+12a^2x+1$, $a>0$ का $x=\alpha$ पर एक स्थानीय उच्चतम है, α तथा α^2 निम्न में से किस समीकरण के मूल है :
[JEE(Main)April,2024]

(A) $8x^2+6x-1=0$ (B) $x^2+6x+8=0$ (C) $x^2-6x+8=0$ (D) $8x^2-6x+1=0$

Ans. C

Sol. $f'(x)=6x^2-18ax+12a^2$

$=6(x^2-3a+2a^2)$

$=6(x-a)(x-2a)=0$

$x = a, 2a$

$a = \alpha,$ $\quad 2a = \alpha^2$ $\quad \Rightarrow \alpha = 0,2$

$a > 0$ $\quad \therefore \alpha = 2$

$\alpha^2 = 4$

$\therefore x^2 - 6x + 8 = 0$ is the required quadratic equation.

48. A variable line L passes through the point (3, 5) and intersects the positive coordinate axes at the points A and B. The minimum area of the triangle OAB, where O is origin, is : **[JEE(Main)April,2024]**

एक चर रेखा L, बिन्दु (3, 5) से होकर जाती है तथा धनात्मक निर्देशांक अक्षों को बिन्दुओं A और B पर काटती है। यदि O मूलबिन्दु है, तो त्रिभुज OAB का न्यूनतम क्षेत्रफल है : **[JEE(Main)April,2024]**

(A) 35 (B) 25 (C) 40 (D) 30

Ans. D

Sol. As a line passes through P(a, b) cuts the coordinate axis at A and B then area of ΔOAB will be minimum iff P is mid point of AB

$\Rightarrow$ Equation line AB is $\frac{x}{2a} + \frac{y}{2b} = 1$

Hence equation of required line is $\frac{x}{6} + \frac{y}{10} = 1$

$\Rightarrow 5x + 3y = 30$

and area of Δ is $= \frac{1}{2}(6)(10) = 30$

49. Let A be the region enclosed by the parabola $y^2 = 2x$ and the line $x = 24$. Then the maximm area of the rectangle inscribed in the region A is _____. **[JEE(Main)April,2024]**

माना परवलय $y^2 = 2x$ तथा रेखा $x = 24$ से घिरा क्षेत्र A है। तो क्षेत्र A के अंतगर्त आयत का अधिकतम है _____.

[JEE(Main)April,2024]

Ans. 128

Sol. $y^2 = 2x$

$x = 24$

Area $= (24 - x)2y$

$\left(24 - \frac{y^2}{2}\right)2y$

$A = 48y - y^3$

$\frac{dA}{dy} = 48 - 3y^2$, $y = 4$, $x = 8$

Area = $(24 - 8)(2 \times 4)$

$= 128$

50. Let the set of all positive values of λ, for which the point of local minimum of the function $(1 + x(\lambda^2 - x^2))$ satisfies $\frac{x^2 + x + 2}{x^2 + 5x + 6} < 0$, be (α, β). Then $\alpha^2 + \beta^2$ is equal to _____. **[JEE(Main)April,2024]**

माना λ के सभी धनात्मक मानों, जिनके लिए फलन $\frac{x^2 + x + 2}{x^2 + 5x + 6} < 0$ को संतुष्ट करता है, का समुच्चय (α, β) है। तो $\alpha^2 + \beta^2$ बराबर है _____ **[JEE(Main)April,2024]**

Ans. 39

Sol. $f(x) = 1 + x(\lambda^2 - x^2)$

$f(x) = -x^3 + \lambda^2 x + 1$

$f'(x) = -3x^2 + \lambda^2$

$x = \pm\frac{\lambda}{\sqrt{3}}$

$-\frac{\lambda}{\sqrt{3}}$ should satisfy the given condition

$\frac{x^2 + x + 2}{x^2 + 5x + 6} < 0$

$\frac{1}{(x+2)(x+3)} < 0$



$x \in (-3, -2)$

$-3 < -\frac{\lambda}{\sqrt{3}} < -2$

$-3\sqrt{3} < -\lambda < -2\sqrt{3}$

$2\sqrt{3} < \lambda < 3\sqrt{3}$

$\left.\begin{aligned}\alpha &= 2\sqrt{3}\\ \beta &= 3\sqrt{3}\end{aligned}\right\}$

$\left(2\sqrt{3}\right)^2 + \left(3\sqrt{3}\right)^2$

$12 + 27$

$= 39$

TANGENT & NORMAL

1.01 The slope of the tangent to the curve $(y - x^5)^2 = x(1+x^2)^2$ at the point (1,3) is. **[JEE(Adv.)2014]**

वक्र $(y - x^5)^2 = x(1+x^2)^2$ पर स्थित बिन्दु (1,3) पर स्पर्श रेखा का ढाल होगा। **[JEE(Adv.)2014]**

Ans. 8

Sol. $(y - x^5)^2 = x(1 + x^2)^2$

$$2(y - x^5)\left(\frac{dy}{dx} - 5x^4\right) = (1 + x^2)^2 + 2x(1 + x^2)\,2x$$

at point (1, 3)

$$\therefore \quad 2(3 - 1)\left(\frac{dy}{dx} - 5\right) = 4 + 8$$

$$\frac{dy}{dx} - 5 = \frac{12}{4} = 3$$

$$\frac{dy}{dx} = 8$$

MONOTONOCITY

Paragraph for Question nos. 2.01 to 2.02

Let $f(x) = (1 - x)^2 \sin^2 x + x^2$ for all $x \in R$ and let $g(x) = \int_1^x \left(\frac{2(t-1)}{t+1} - \ln t\right) f(t)\,dt$ for all $x \in (1, \infty)$.

माना $f(x) = (1 - x)^2 \sin^2 x + x^2$, सभी $x \in R$ के लिये तथा $g(x) = \int_1^x \left(\frac{2(t-1)}{t+1} - \ln t\right) f(t)\,dt$ सभी $x \in (1, \infty)$

2.01 Which of the following is true? **[JEE-2012]**

(A) g is increasing on $(1, \infty)$

(B) g is decreasing on $(1, \infty)$

(C) g is increasing on (1, 2) and decreasing on $(2, \infty)$

(D) g is decreasing n (1, 2) and increasing on $(2, \infty)$

निम्न में से कौनसा कथन सत्य है? **[JEE-2012]**

(A) g, $(1, \infty)$ में वर्धमान है

(B) g, $(1, \infty)$ में ह्रासमान है

(C) g, (1, 2) में वर्धमान है तथा $(2, \infty)$ में ह्रासमान है

(D) g, (1, 2) में ह्रासमान है तथा $(2, \infty)$ में वर्धमान है

Ans. B

Sol. $f(x) = (1 - x)^2 \sin^2 x + x^2$

P : $f(x) + 2x = 2(1 + x^2)$

$\Rightarrow (1 - x)^2 \sin^2 x + x^2 + 2x = 2 + 2x^2$

$\Rightarrow (1 - x)^2 \sin^2 x - x^2 + 2x - 2 = 0$

$(1 - x)^2 \cos^2 x + 1 = 0$

which is not possible.

$\therefore$ P is false.

Q : $2f(x) + 1 = 2x(1 + x)$

$2x^2 + 2(1 - x)^2 \sin^2 x + 1 = 2x^2 + 2x$

$2(1 - x)^2 \sin^2 x - 2x + 1 = 0.$

Let $h(x) = 2(1 - x)^2 \sin^2 x - 2x + 1,$

clearly $h(1) = -1$

and $h(x) = 2(x^2 - 2x + 1)\sin^2 x - 2x + 1$

$$= x^2\left[2\left(1 - \frac{2}{x} + \frac{1}{x^2}\right).\sin^2 x - \frac{2}{x} + \frac{1}{x^2}\right]$$

$\therefore h(x) \to \infty$ as $x \to \infty$.

$\therefore$ By intermediate value theorem

$h(x) = 0$ has a root which is greater than 1.

Hence Q is true.

2.02 Consider the statements : **[JEE-2012]**

P : There exists some $x \in R$ such that $f(x) + 2x = 2(1 + x^2)$

Q : There exists some $x \in R$ such that $2f(x) + 1 = 2x(1 + x)$

Then :

(A) both P and Q are true (B) P is true and Q are false

(C) P is false and Q is true (D) both P and Q are false

निम्न कथनों पर विचार कीजिये : **[JEE-2012]**

P : किसी $x \in R$ के लिये $f(x) + 2x = 2(1 + x^2)$ होगा

Q : किसी $x \in R$ के लिये $2f(x) + 1 = 2x(1 + x)$ होगा

तो :

(A) P तथा Q दोनों सत्य है (B) P सत्य है तथा Q असत्य है

(C) P असत्य है तथा Q सत्य है (D) P तथा Q दोनों असत्य है

Ans. C

Sol. $g(x)=\int_1^x\left(\frac{2(t-1)}{(t+1)}-\ell nt\right)f(t)dt$

$$g'(x)=\left(\frac{2(x-1)}{x+1}-\ell nx\right)f(x)$$

$f(x)>0 \quad \forall\, x\in R$

Suppose.

$$h(x)=\frac{2(x-1)}{x+1}-\ell nx$$

$$h(x)=2-\left(\frac{4}{x+1}+\ell nx\right)$$

$$h'(x)=\frac{4}{(x+1)^2}-\frac{1}{x}$$

$$h'(x)=-\frac{(x-1)^2}{x(x+1)^2}$$

$h'(x)<0$

So h(x) is decreasing

so $h(x)<h(1)$. $\quad \forall\, x>1$

$h(x)<0 \qquad \forall\, x>1$

So $g'(x)=h(x)\,f(x)$

$g'(x)<0 \quad \forall\, x>1$

g(x) is decreasing in $(1,\infty)$.

2.03 The number of points in $(-\infty,\infty)$, for which $x^2-x\sin x-\cos x=0$, is : **[JEE(Adv.)2013]**

$(-\infty,\infty)$ में बिन्दुओं की संख्या, जिनके लिए $x^2-x\sin x-\cos x=0$, है : **[JEE(Adv.)2013]**

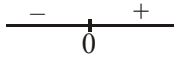
(A) 6 (B) 4 (C) 2 (D) 0

Ans. C

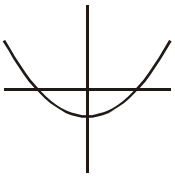
Sol. $f(x)=x^2-x\sin x-\cos x$

$f'(x)=2x-x\cos x-\sin x+\sin x$

$=x(2-\cos x)$



$\therefore$ graph of $f(x)$ will be



$\therefore$ $f(x)$ is zero for 2 values of x

$\therefore$ (c)

2.04 For every pair of continuous functions f, g $[0,1] \to R$ such that max $\{f(x): x \in [0,1]\}$ = max $\{g(x): x \in [0,1]\}$, the correct statement(s) is(are) : **[JEE(Adv.)2014]**

(A) $(f(c))^2 + 3f(c) = (g(c))^2 + 3g(c)$ for some $c \in [0,1]$

(B) $(f(c))^2 + f(c) = (g(c))^2 + 3g(c)$ for some $c \in [0,1]$

(C) $(f(c))^2 + 3f(c) = (g(c))^2 + g(c)$ for some $c \in [0,1]$

(D) $(f(c))^2 = (g(c))^2$ for some $c \in [0,1]$

प्रत्येक सत्त फलन f, g $[0,1] \to R$ के लिये max $\{f(x): x \in [0,1]\}$ = max $\{g(x): x \in [0,1]\}$, सही कथन होंगे : **[JEE(Adv.)2014]**

(A) $(f(c))^2 + 3f(c) = (g(c))^2 + 3g(c)$, किसी $c \in [0,1]$ के लिए

(B) $(f(c))^2 + f(c) = (g(c))^2 + 3g(c)$, किसी $c \in [0,1]$ के लिए

(C) $(f(c))^2 + 3f(c) = (g(c))^2 + g(c)$, किसी $c \in [0,1]$ के लिए

(D) $(f(c))^2 = (g(c))^2$, किसी $c \in [0,1]$ के लिए

Ans. AD

Sol. Consider

$h(x) = f(x) - g(x)$ Assume $a < b$

$h(a) = \lambda - g(a) > 0$

$h(b) = f(b) - \lambda < 0$

else if $a > b$ $h(a) < 0$ and $h(b) > 0$.

By intermediate value theorem $\Rightarrow h(c) = 0$ .....(1)

(A) $(f(c))^2 + 3f(c) = (g(c))^2 + 3g(c)$

$(f(c) - g(c))\,(f(c) + g(c) + 3) = 0$

So there exist a 'c' : $f(c) - g(c)$

from (1).

Hence A is correct.

(D) Similarly $(f(c))^2 = (g(c))^2$

$(f(c) - g(c))\,(f(c) + g(c)) = 0$

$\Rightarrow$ (D) is correct.

B & C are wrong as by counter eg

If $f(x) = g(x) = \lambda \neq 0$, then

B $\rightarrow \lambda^2 + \lambda = \lambda^2 + 3\lambda$ is not possible.

C $\rightarrow \lambda^2 + 3\lambda = \lambda^2 + \lambda$ is not possible.

2.05 Let f, g : $[-1, 2] \rightarrow R$ be continuous functions which are twice differentiable on the interval $(-1, 2)$. Let the values of f and g at the points –1, 0 and 2 be as given in the following table : **[JEE(Adv.)2015]**

	x = – 1	**x = 0**	**x = 2**
f (x)	3	6	0
g (x)	0	1	–1

In each of the intervals $(-1, 0)$ and $(0,2)$ the function $(f - 3g)''$ never vanishes. Then the correct statement(s) is (are)

(A) $f'(x) - 3g'(x) = 0$ has exactly three solutions in $(-1, 0) \cup (0,2)$

(B) $f'(x) - 3g'(x) = 0$ has exactly one solutions in $(-1, 0)$

(C) $f'(x) - 3g'(x) = 0$ has exactly one solutions in $(0,2)$

(D) $f'(x) - 3g'(x) = 0$ has exactly two solutions in $(-1, 0)$ and exactly two solutions in $(0,2)$

माना कि f, g : $[-1, 2] \rightarrow R$ सतत् फलन हैं जो की अन्तराल $(-1, 2)$ में दो बार अवकलनीय (twice differentiable) है। माना कि f और g के मान, बिन्दुओं –1, 0 और 2 पर निम्न सारणी में दर्शाए गए है : **[JEE(Adv.)2015]**

	x = – 1	**x = 0**	**x = 2**
f (x)	3	6	0
g (x)	0	1	–1

यदि प्रत्येक अन्तराल $(-1, 0)$ और $(0,2)$ में फलन $(f - 3g)''$ कभी भी शून्य का मान नहीं लेता है, तब सही कथन है (हैं)

(A) $(-1, 0) \cup (0,2)$ में, $f'(x) - 3g'(x) = 0$ के तीन ही हल (exactly three solutions) हैं

(B) $(-1, 0)$ के अन्तराल में $f'(x) - 3g'(x) = 0$ का एक ही हल है।

(C) $(0,2)$ में, $f'(x) - 3g'(x) = 0$ का एक ही हल (exactly one solutions) है

(D) $f'(x) - 3g'(x) = 0$ के अन्तराल $(-1, 0)$ में दो ही हल (exactly two solutions) है और $(0,2)$ में दो ही हल हैं

Ans. BC

Sol. Let $h(x) = f(x) - 3g(x)$

$\left.\begin{matrix} h(-1) = 3 \\ h(0) = 3 \end{matrix}\right\} \Rightarrow h'(x) = 0$ has atleast one root in $(-1, 0)$ and atleast one root in $(0, 2)$

$h(2) = 3$

But since $h''(x) = 0$ has no root in $(-1, 0)$ & $(0, 2)$ therefore $h'(x) = 0$ has exactly 1 root in $(-1, 0)$ & exactly 1 root in $(0, 2)$

Paragraph for question nos. 2.06 to 2.08

By appropriately matching the information given in the three columns of the following table.

Let $f(x) = x + \log_e x - x \log_e x,\ x \in (0, \infty)$

- Column 1 contains information about zeros of f(x), f'(x) and f''(x).
- Column 2 contains information about the limiting behavior of f(x), f'(x) and f''(x) at infinity.
- Column 3 contains information about the increasing/decreasing nature of f(x), f'(x)

Column-1	**Column-2**	**Column-3**
(I) $f(x) = 0$ for some $x \in (1, e^2)$	(i) $\lim_{x\to\infty} f(x) = 0$	(P) f is increasing in (0, 1)
(II) $f'(x) = 0$ for some $x \in (1, e)$	(ii) $\lim_{x\to\infty} f(x) = -\infty$	(Q) f is decreasing in (e, e^2)
(III) $f'(x) = 0$ for some $x \in (0, 1)$	(iii) $\lim_{x\to\infty} f'(x) = -\infty$	(R) f' is increasing in (0, 1)
(IV) $f''(x) = 0$ for some $x \in (1, e)$	(iv) $\lim_{x\to\infty} f''(x) = 0$	(S) f' is decreasing in (e, e^2)

नीचे दी गयी टेबल के तीन कॉलमों में उपलब्ध सूचना का उपयुक्त ढंग से सुमेल कर प्रश्नों 9 से 11 के उत्तर दीजिये।

εκοκδ $f(x) = x + \log_e x - x \log_e x,\ x \in (0, \infty)$ है।

- कॉलम 1 में f(x), f'(x) तथा f''(x) के शून्यों की सूचना दी गयी है।
- कॉलम 2 में f(x), f'(x) तथा f''(x) के अनन्त की तरफ सीमा पर व्यवहार की सूचना दी गयी हैं।
- कॉलम 3 में f(x) तथा f'(x) के वर्धमान/ह्रासमान होने की प्रकृति की सूचना दी गयी हैं।

(I) $f(x) = 0$ किसी $x \in (1, e^2)$ के लिये	(i) $\lim_{x\to\infty} f(x) = 0$	(P) f (0, 1) में वर्धमान है
(II) $f'(x) = 0$ किसी $x \in (1, e)$ के लिये	(ii) $\lim_{x\to\infty} f(x) = -\infty$	(Q) f (e, e^2) में ह्रासमान है
(III) $f'(x) = 0$ किसी $x \in (0, 1)$ के लिये	(iii) $\lim_{x\to\infty} f'(x) = -\infty$	(R) f' (0, 1) में वर्धमान है
(IV) $f''(x) = 0$ किसी $x \in (1, e)$ के लिये	(iv) $\lim_{x\to\infty} f''(x) = 0$	(S) f' (e, e^2) में ह्रासमान है

2.06 Which of the following options is the only CORRECT combination ? **[JEE(Adv.)2017]**

निम्न में से कौनसा विकल्प केवल सही संयोजन है? **[JEE(Adv.)2017]**

(A) (II) (ii) (Q) (B) (III) (iii) (R) (C) (IV) (iv) (S) (D) (I) (i) (P)

Ans. A

Sol. $f(x) = x + \ln x - x\ln x$

$$f'(x) = \frac{1 - x\ln x}{x}$$

$$f''(x) = -\frac{(1+x)}{x^2} < 0$$

$\Rightarrow f'(x)$ is $\downarrow\ \forall\ x \in (0, \infty)$

$$\lim_{x\to 0} f'(x) = \infty$$

$$f'(x) = 1$$

$$f'(e) = \frac{1-e}{e} < 0$$

$f'(x) < 0\ \forall\ x > e$

$f(1) = 1 \quad f(e^2) = e^2 + 2 - 2e^2 = 2 - e^2 < 0$

For column I (I) & (II) are true

For column II (ii), (iii) and (iv) are true

For column III P, Q, S are true

2.07 Which of the following options is the only CORRECT combintaion? **[JEE(Adv.)2017]**

निम्न में से कौनसा विकल्प केवल सही संयोजन है? **[JEE(Adv.)2017]**

(A) (III) (iv) (P) (B) (II) (iii) (S) (C) (I) (ii) (R) (D) (IV) (i) (S)

Ans. B

Sol. $f(x) = x + \ln x - x\ln x$

$$f'(x) = \frac{1 - x\ln x}{x}$$

$$f''(x) = -\frac{(1+x)}{x^2} < 0$$

$\Rightarrow f'(x)$ is $\downarrow \ \forall \ x \in (0, \infty)$

$\lim\limits_{x\to 0} f'(x) = \infty$

$f'(x) = 1$

$$f'(e) = \frac{1-e}{e} < 0$$

$f'(x) < 0 \ \forall \ x > e$

$f(1) = 1 \quad f(e^2) = e^2 + 2 - 2e^2 = 2 - e^2 < 0$

For column I (I) & (II) are true

For column II (ii), (iii) and (iv) are true

For column III P, Q, S are true

2.08 Which of the following option is the only INCORRECT combintaion? **[JEE(Adv.)2017]**

निम्न में से कौनसा विकल्प केवल गलत संयोजन है? **[JEE(Adv.)2017]**

(A) (I) (iii) (P) (B) (II) (iii) (P) (C) (III) (i) (R) (D) (II) (iv) (Q)

Ans. C

Sol. $f(x) = x + \ln x - x\ln x$

$$f'(x) = \frac{1 - x\ln x}{x}$$

$$f''(x) = -\frac{(1+x)}{x^2} < 0$$

$\Rightarrow f'(x)$ is $\downarrow \ \forall \ x \in (0, \infty)$

$\lim\limits_{x\to 0} f'(x) = \infty$

$f'(x) = 1$

$f'(e) = \dfrac{1-e}{e} < 0$

$f'(x) < 0 \ \forall \ x > e$

$f(1) = 1 \quad f(e^2) = e^2 + 2 - 2e^2 = 2 - e^2 < 0$

For column I (I) & (II) are true

For column II (ii), (iii) and (iv) are true

For column III P, Q, S are true

2.09 If $f: \mathbb{R} \to \mathbb{R}$ is a twice differentiable function such that $f''(x) > 0$ for all $x \in \mathbb{R}$, and $f\left(\frac{1}{2}\right) = \frac{1}{2}, f(1) = 1$, then :

[JEE(Adv.)2017]

यदि $f: \mathbb{R} \to \mathbb{R}$ एक इस प्रकार का द्विअवकलनीय फलन है, कि सभी $x \in \mathbb{R}$ के लिये $f''(x) > 0$ एवं $f\left(\frac{1}{2}\right) = \frac{1}{2}, f(1) = 1$ है, तब :

[JEE(Adv.)2017]

(A) $\frac{1}{2} < f'(1) \le 1$ (B) $0 < f'(1) \le \frac{1}{2}$ (C) $f'(1) \le 0$ (D) $f'(1) > 1$

Ans. D

Sol. $f'(x) > 0$ for all $x \in R$, $f(1/2) = 1/2$, $f(1) = 1$
$\Rightarrow f'(x)$ increases
Let $g(x) = f(x) - x$, $x \in [1/2, 1)$
$f'(x) = 1$ has atleast one real root in $(1/2, 1)$
Hence $f'(x)$ increases $\Rightarrow f'(1) > 1$

2.10 For every twice differentiable function $f: R \to [-2, 2]$ with $(f(0))^2 + (f'(0))^2 = 85$, which of the following statement(s) is (are) TRUE? **[JEE(Adv.)2018]**

(A) There exist $r, s \in R$, where $r < s$, such that f is one-one on the open interval (r, s)

(B) There exists $x_0 \in (-4, 0)$ such that $|f'(x_0)| \le 1$

(C) $\lim\limits_{x\to\infty} f(x) = 1$

(D) There exists $\alpha \in (-4, 4)$ such that $f(\alpha) + f''(\alpha) = 0$ and $f'(\alpha) \ne 0$

प्रत्येक द्विअवकलनीय फलन $f: R \to [-2, 2]$ with $(f(0))^2 + (f'(0))^2 = 85$, के लिये निम्नलिखित में से कौन सा (से) कथन सत्य है (हैं)? **[JEE(Adv.)2018]**

(A) ऐसे r, s ∈ R, जहाँ r < s, का अस्तित्व है जिनके लिये f खुले अन्तराल (r, s) पर एकैक है

(B) ऐसे $x_0 \in (-4, 0)$ का अस्तित्व है जिसके लिये $|f'(x_0)| \le 1$

(C) $\lim_{x\to\infty} f(x) = 1$

(D) ऐसे $\alpha \in (-4, 4)$ का अस्तित्व है जिसके लिये $f(\alpha) + f''(\alpha) = 0$ और $f'(\alpha) \ne 0$

Ans. ABD

Sol. $f^2(0) + (f'(0))^2 = 85$ $\qquad f: R \to [-2, 2]$

(A) This is true for every continuous function

(b) $f'(c) = \dfrac{f(-4) - f(0)}{-4 - 0}$

$$|f'(c)| = \left|\frac{f(-4) - f(0)}{4}\right|$$

$-2 \le f(-4) \le 2$

$-2 \le f(0) \le 2$

$4 \le f(-4) - f(0) \le 4$

This $|f'(c)| \le 1$

(C) $\lim_{x\to\infty} f(x) = 1$

Note f(x) should have a bound ∞ which can be concluded by considering

$$f(x) = 2\sin\left(\frac{\sqrt{85}x}{2}\right)$$

$$f(x) = \sqrt{85}\cos\left(\frac{\sqrt{85}x}{2}\right)$$

$$f(x) = \sqrt{85}\cos\left(\frac{\sqrt{85}x}{2}\right)$$

$f^2(0) + (f'(0)^2) = 85$

and $\lim_{x\to\infty} f(x)$ does not exist.

(D) Consider $H(x) = f^2(X) + (f'(x)^2)$

$H(0) = 85$

By (B) choice there exists some x_0 such that $(f'(x_0))^2 \le 1$ for some x_0 in $(-4, 0)$

hence $H(x0) = f^2(x_0) + (f'(x_0))^2 \le 4 + 1$

$H(x^0) \le 5$

Hence let $p \in (-4, 0)$ for which $H(p) = 5$

(note that we have considered p as largest such negative number)

similarly let q be smallest positive number $\in (0, 4)$ such that $H(q) = 5$

Hence by Rolle's theorem is (p, q)

$H'(c) = 0$ for some $c \in (-4, 4)$ and since H(x) is greater than 5 as we move from x = p to x = q and $f^2(x) \le 4$

$\Rightarrow (f'(x))2 \ge 1$ in (p, q)

Thus H' (c) = 0 $\Rightarrow$ f'f + f'f' = 0
so f + f' = 0 and f' $\neq$ 0

2.11 Let $f: R \to R$ be defined by $f(x) = \frac{x^2 - 3x - 6}{x^2 + 2x + 4}$ Then which of the following statements is (are)TRUE?

[JEE(Adv.)2021]

(A) f is decreasing in the interval (–2, –1)

(B) f is increasing in the interval (1, 2)

(C) f is onto

(D) Range of f is $\left[-\frac{3}{2}, 2\right]$

माना $f: R \to R$ निम्न प्रकार परिभाषित है :

$$f(x) = \frac{x^2 - 3x - 6}{x^2 + 2x + 4}$$

तब निम्न में से कौनसा/कौनसे कथन सही है/ है ? **[JEE(Adv.)2021]**

(A) f, अन्तराल (–2, –1) में ह्रासमान है

(B) f, अन्तराल (1, 2) में वर्धमान है

(C) f आच्छादक है

(D) f का परिसर $\left[-\frac{3}{2}, 2\right]$ है

Ans. AB

Sol. f: R $\to$ R

$$f(x) = \frac{x^2 - 3x - 6}{x^2 + 2x + 4}$$

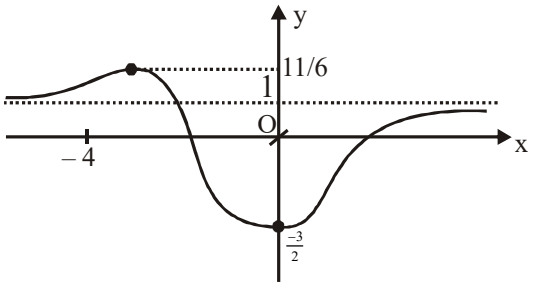
$$f'(x) = \frac{(2x-3)(x^2+2x+4) - (2x+2)(x^2-3x-6)}{(x^2+2x+4)^2}$$

$$f'(x) = \frac{5x(x+4)}{(x^2+2x+4)^2} \quad \Rightarrow \quad \underset{-4}{+} \quad \underset{0}{-} \quad +$$

f(x) is decreasing in (–2, –1) $\Rightarrow$ (A) is correct
f(x) is increasing in (1, 2) $\Rightarrow$ (B) is correct

Now, $f(-4) = \frac{16 + 12 - 6}{16 - 8 + 4} = \frac{11}{6}$; $f(0) = \frac{-3}{2}$

$\lim_{x \to \infty} f(x) = 1$ & $\lim_{x \to \infty} f(x) = 1$



Range is $\left[\frac{-3}{2}, \frac{11}{6}\right] \Rightarrow$ (D) is incorrect

Range $\neq$ Co-domain $\Rightarrow$ (C) is incorrect

2.12 Let S be the set of all twice differentiable functions f from R to R such that $\frac{d^2f}{dx^2}(x) > 0$ for all $x \in (-1,1)$. For f $\in$ S, let X_f be the number of points $x \in (-1, 1)$ for which f(x) = x. Then which of the following statements is (are) true ? **[JEE(Adv.)2023]**

माना कि S उन दो बार अवकलनीय (twice differentiable) फलनों $f : \mathbb{R} \to \mathbb{R}$ का समुच्चय है जो इस प्रकार हैं कि सभी $x \in (-1,1)$ के लिए $\frac{d^2f}{dx^2}(x) > 0$ है। फलन $f \in S$ के लिए, माना कि X_f उन बिन्दुओं $x \in (-1, 1)$ की संख्या है जिनके लिए f(x) = x है। तब निम्न में से कौनसा (से) कथन सत्य है (हैं)? **[JEE(Adv.)2023]**

(A) There exists a function $f \in S$ such that $X_f = 0$
एक फलन $f \in S$ का अस्तित्व है जिसके लिए $x_f = 0$
(B) For every function $f \in S$, we have $X_f \leq 2$
सभी फलनों $f \in S$ के लिए $X_f \leq 2$ है
(C) There exists a function $f \in S$ such that $X_f = 2$
एक फलन $f \in S$ का अस्तित्व है जिसके लिए $X_f = 2$
(D) There does **NOT** exist any function f in such that $X_f = 1$
S में ऐसे किसी फलन f का अस्तित्व नहीं है जिसके लिए $X_f = 1$

Ans. ABC

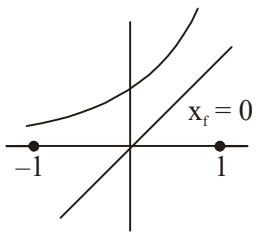
Sol. S = set of all twice differentiable functions $f: R \to R$

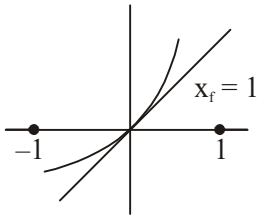
$\frac{d^2f}{dx^2} > 0$ in (–1, 1)

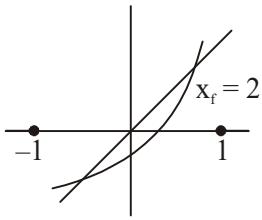
Graph 'f' is Concave upward

Number of solutions of $f(x) = x \to x_f$

(1)



(2) 

(3) 

$\Rightarrow$ Graph of $y = f(x)$ can intersect graph of $y = x$ at atmost two points $\Rightarrow 0 \le x_f \le 2$

Aliter

$$\frac{d^2 f(x)}{dx^2} > 0$$

Let $\phi(x) = f(x) - x$

$$\phi''(x) > 0$$

$\therefore$ $\phi'(x) > 0$ has atmost 1 root in $x \in (-1, 1)$

$\therefore$ $\phi'(x) > 0$ has atmost 2 root in $x \in (-1, 1)$

$\therefore$ $x_f \le 2$

2.13. Let the function $f : R \to R$ be defined by $f(x) = \frac{\sin x}{e^{\pi x}} \frac{\left(x^{2023} + 2024x + 2025\right)}{\left(x^2 - x + 3\right)} + \frac{2}{e^{\pi x}} \frac{\left(x^{2023} + 2024x + 2025\right)}{\left(x^2 - x + 3\right)}$

Then the number of solutions of $f(x) = 0$ in R is_____. **[JEE(Adv.)2024]**

माना कि फलन (function) $f : R \to R$, $f(x) = \frac{\sin x}{e^{\pi x}} \frac{\left(x^{2023} + 2024x + 2025\right)}{\left(x^2 - x + 3\right)} + \frac{2}{e^{\pi x}} \frac{\left(x^{2023} + 2024x + 2025\right)}{\left(x^2 - x + 3\right)}$ द्वारा परिभाषित है। तब R में, $f(x) = 0$ के हलों (solutions) की संख्या_____ है। **[JEE(Adv.)2024]**

Ans. 1

Sol. $f(x) = \frac{\sin x}{e^{\pi x}} \frac{\left(x^{2023} + 2024x + 2025\right)}{\left(x^2 - x + 3\right)} + \frac{2}{e^{\pi x}} \frac{\left(x^{2023} + 2024x + 2025\right)}{\left(x^2 - x + 3\right)}$

$$\Rightarrow \frac{\sin x}{e^{\pi x}} \frac{\left(x^{2023}+2024x+2025\right)}{\left(x^2-x+3\right)} [\sin x + 2] = 0$$

Now $\sin x + 2 = 0$

$\sin x = -2$ (No sol.)

So we need to solve

$x^{2023} + 2024x + 2025 = 0$

Let $g(x) = x^{2023} + 2024x + 2025$

Differentiate

$2023x^{2022} + 2024$ (Always positive)

So It cuts x -axis only once.

So exactly 1 solution.

MAXIMA & MINIMA

3.01 If $f(x) = \int_0^x e^{t^2}(t-2)(t-3)\,dt$ for all $x \in (0, \infty)$, then : **[JEE-2012]**

(A) f has a local maximum at x = 2 (B) f is decreasing on (2, 3)

(C) there exists some $c \in (0, \infty)$ such that $f''(c) = 0$ (D) f has a local minimum at x = 3

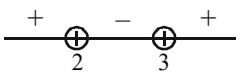
यदि $f(x) = \int_0^x e^{t^2}(t-2)(t-3)\,dt$, सभी $x \in (0, \infty)$ के लिए, तो : **[JEE-2012]**

(A) f का x = 2 पर स्थानीय उच्चिष्ठ है (B) f, (2, 3) में ह्यसमान है

(C) किसी $c \in (0, \infty)$ के लिए $f''(c) = 0$ होगा (D) f का x = 3 स्थानीय निम्निष्ठ है

Ans. ABCD

Sol. $f(x) = \int_0^x e^{t^2}(t-2)(t-3)dt$



$\Rightarrow f'(x) = e^{x^2}(x-2)(x-3)$

$\therefore f'(2) = f'(3) = 0$

$\Rightarrow f''(c) = 0$ for same $c \in (2,3)$ (by Rolle's theorem)

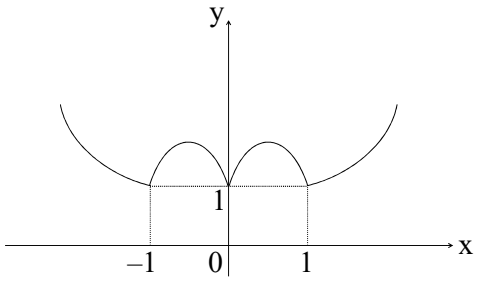
3.02 Let $f: R \to R$ be defined as $f(x) = |x| + |x^2 - 1|$. The total number of points at which f attains either a local maximum or a local minimum is. **[JEE-2012]**

माना फलन $f: R \to R$ इस प्रकार है कि $f(x) = |x| + |x^2 - 1|$ तो f के स्थानीय निम्निष्ठ या स्थानीय उच्चिष्ठ वाले बिन्दुओं की संख्या होगी। **[JEE-2012]**

Ans. 5

Sol. $f(x) = |x| + |(x + 1)(x - 1)|$

$$\Rightarrow \quad f(x) = \begin{cases} x^2 - x - 1 & x < -1 \\ -x^2 - x + 1 & -1 \le x < 0 \\ -x^2 + x + 1 & 0 \le x < 1 \\ x^2 + x - 1 & x \ge 1 \end{cases}$$



$\therefore$ f has 5 points where it attains either a local maximum or local minimum.

3.03 Let p(x) be a real polynomial of least degree which has a local maximum at x = 1 and a local minimum at x = 3. If p(1) = 6 and p(3) = 2, then p'(0) is. **[JEE-2012]**

माना कि p(x) न्यूनतम घात का एक वास्तविक बहुपद है जिसका स्थानीय उच्चिष्ठ x = 1 पर तथा स्थानीय निम्निष्ठ x = 3 पर है। यदि p(1) = 6 तथा p(3) = 2, तो p'(0) का मान होगा। **[JEE-2012]**

Ans. 9

3.04 The function $f(x) = 2|x| + |x + 2| - ||x + 2| - 2|x||$ has a local minimum or a local maximum at x = **[JEE(Adv.)2013]**

फलन $f(x) = 2|x| + |x + 2| - ||x + 2| - 2|x||$ का एक स्थानीय न्यूनतम या एक स्थानीय अधिकतम जिन x के मान पर है, वह है : **[JEE(Adv.)2013]**

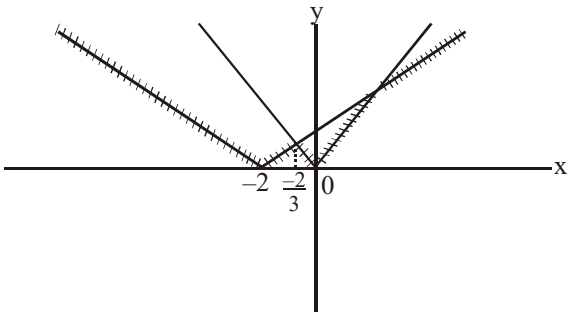
(A) – 2 (B) $\frac{-2}{3}$ (C) 2 (D) 2/3

Ans. AB

Sol. $f(x) = (a + b) - |b - a|$

$= \begin{cases} 2a\ ,\ a \le b \\ 2b\ ,\ a > b \end{cases} = 2 \min(a, b)$

where $a = 2|x|$, $b = |x + 2|$



$\therefore$ Local maxima and minima at $x = -2, -\frac{2}{3}$ & 0

Paragraph for question nos. 3.05 to 3.07

Let $f : (0, 1) \to R$ (the set of all real numbers) be a function. Suppose the function f is twice differentiable, $f(0) = f(1) = 0$ and satisfies $f''(x) - 2f'(x) + f(x) \ge e^x$, $x \in [0, 1]$.

माना कि $f : (0, 1) \to R$ (सभी वास्तविक संख्याओं का समुच्चय) एक फलन है। मान लीजिये फलन f दो बार अवकलनीय है, $f(0) = f(1) = 0$ तथा $f''(x) - 2f'(x) + f(x) \ge e^x$, $x \in [0, 1]$ को सन्तुष्ट करता है।

3.05 Which of the following is true for $0 < x < 1$? **[JEE(Adv.)2013]**

निम्न में से कौन $0 < x < 1$ के लिए सत्य है? **[JEE(Adv.)2013]**

(A) $0 < f(x) < \infty$ (B) $-\frac{1}{2} < f(x) < \frac{1}{2}$ (C) $-\frac{1}{4} < f(x) < 1$ (D) $-\infty < f(x) < 0$

Ans. D

Sol. $e^{-x}(f''(x) - 2f'(x) + f(x)) \ge 1$

$D((f'(x) - f(x))e^{-x}) \ge 1$

$\Rightarrow$ $D((f'(x) - f(x)e^{-x}) \ge 0$

$\Rightarrow$ $(f'(x) - f(x))e^{-x}$ is an increasing function.

As we know that $e^{-x}f(x)$ has local minima at $x = \frac{1}{4}$

$e^{-x}(f'(x) - f(x)) = 0$ at $x = \frac{1}{4}$

Let $F(x) = e^{-x}(f'(x) - f(x))$

$F(x) < 0$ in $\left(0, \frac{1}{4}\right)$

$e^{-x}(f'(x) - f(x) < 0$ in $\left(0, \frac{1}{4}\right)$

$f'(x) < f(x)$ in $\left(0, \frac{1}{4}\right)$

option C

3.06 If the function e^{-x} f(x) assumes its minimum in the interval [0, 1] at $x = \frac{1}{4}$, which of the following is true?

[JEE(Adv.)2013]

यदि फलन e^{-x} f(x), अन्तराल [0, 1] में अपना न्यूनतम मान $x = \frac{1}{4}$ पर लेता है, तब निम्न में से कौन सत्य है?

[JEE(Adv.)2013]

(A) $f'(x) < f(x), \frac{1}{4} < x < \frac{3}{4}$

(B) $f'(x) > f(x), 0 < x < \frac{1}{4}$

(C) $f'(x) < f(x), 0 < x < \frac{1}{4}$

(D) $f'(x) < f(x), \frac{3}{4} < x < 1$

Ans. C

Sol. $D(e^{-x}(f'(x) - f(x)) \geq 0 \ \forall \ x \in (0, 1)$

$D(D(e^{-x}f(x)) \geq 0 \ \forall \ x \in (0, 1)$

$D^2(e^{-x}f(x)) \geq 0$

Let $F(x) = e^{-x}f(x)$

F''(x) > 0 means it is concave upward.

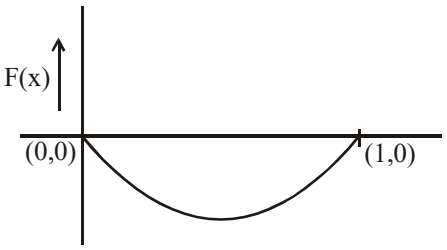
F(0) = F(1) = 0

$F(x) < 0 \ \forall \ x \in (0, 1)$

$e^{-x}f(x) < 0 \ \forall \ x \in (0, 1)$

$f(x) < 0$

Option D is possible



3.07 A rectangular sheet of fixed perimeter with sides having their lengths in the ratio 8 : 15 is converted into an open rectangular box by folding after removing squares of equal area from all four corners. If the total area of removed squares is 100, the resulting box has maximum volume. Then the lengths of the sides of the rectangular sheet are :

[JEE(Adv.)2013]

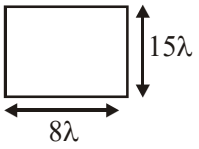
एक निश्चित परिमाप की आयताकार चादर को, जिसकी भुजाओं की लम्बाईयाँ 8 : 15 के अनुपात में हैं, सभी चारों किनारों से समान

क्षेत्रफल के वर्ग निकाल कर एक खुली आयताकार पेटी में परिवर्तित किया जाता है। यदि निकाले गये वर्गों का कुल क्षेत्रफल 100 है, तब परिणामी पेटी का आयतन महत्तम है। तब आयताकार चादर की भुजाओं की लम्बाईयाँ निम्न हैं : **[JEE(Adv.)2013]**

(A) 24 (B) 32 (C) 45 (D) 60

Ans. AC

Sol. Where $P = 8\lambda + 15\lambda + 8\lambda + 15\lambda$ and λ is constant



Let removed length from each sides is x

Removed area is $4x^2 = 100 \Rightarrow x = 5$

$V = (8\lambda - 2x)(15\lambda - 2x)x$

$V = 120l^2x - 46\lambda x^2 + 4x^3$

$$\frac{dv}{dx} = 120\lambda^2 - 92\lambda x + 12x^2 = 0$$

Put $x = 5 \Rightarrow 120\lambda^2 - 460\lambda + 300 = 0$

$12\lambda^2 - 40\lambda + 30 = 0$

$6\lambda^2 - 23\lambda + 15 = 0$

$(\lambda - 3)(6\lambda - 5) = 0$

$\lambda = 3$ and $\lambda = \frac{5}{6}$

$$\frac{d^2v}{dx^2} = -92\lambda + 24x = 120 - 92\lambda$$

at $\lambda = 3 \Rightarrow \frac{d^2v}{dx^2} < 0$

at $\lambda = \frac{5}{6} \Rightarrow \frac{d^2v}{dx^2} > 0$ (rejected)

3.08 Let $a \in R$ and let $f : R \to R$ be given by $f(x) = x^5 - 5x + a$. Then : **[JEE(Adv.)2014]**

(A) f(x) has three real roots if $a > 4$ (B) f(x) has only one real root if $a > 4$

(C) f(x) has three real roots if $a < -4$ (D) f(x) has three real roots if $-4 < a < 4$

माना कि $a \in R$ तथा $f : R \to R$ निम्न के द्वारा $f(x) = x^5 - 5x + a$. परिभाषित है। तब : **[JEE(Adv.)2014]**

(A) $a > 4$ के लिये f(x) के तीन वास्तविक मूल (real roots) है

(B) $a > 4$ के लिये f(x) का केवल एक वास्तविक मूल है

(C) $a < -4$ के लिये f(x) के तीन वास्तविक मूल है

(D) $-4 < a < 4$ के लिये f(x) के तीन वास्तविक मूल है

Ans. BD

Sol. $f(x) = x^5 - 5x + a = 0$

$x^5 - 5x = -a$

$x(x^4 - 5) = -a$

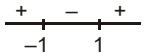
$x(x^2 - \sqrt{5})(x^2 + \sqrt{5}) = -a$

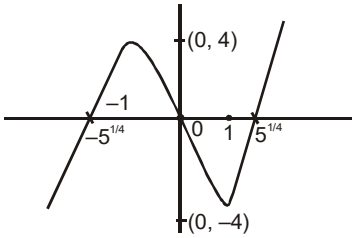
$x(x - 5^{1/4})(x + 5^{1/4})(x^2 + \sqrt{5}) = -a$ ..(1)

$f'(x) = 5x^4 - 5 = 0$

$(x^2 - 1)(x^2 + 1) = 0$

$(x - 1)(x + 1)(x^2 + 1) = 0$





3.09 A cylindrical container is to be made from certain solid material with the following constraints. It has a fixed inner volume of V mm^3, has a 2 mm thick wall and is open at the top. The bottom of the container is a solid circular disc of thickness 2 mm and is of radius equal to the outer radius of the container. If the volume of the material used to make the container is minimum when the inner radius of the container is 10 mm, then the value of $\frac{V}{250\pi}$ is.

[JEE(Adv.)2015]

निम्न व्यवरोधों को संतुष्ट करते हुए एक बेलनाकार पात्र एक ठोस पदार्थ से बनाया है। पात्र का आन्तरिक आयतन V घन मिमी नियत है तथा इसकी दीवार 2 मिमी मोटाई की हैं एवं पात्र ऊपर से खुला है। पात्र का निचला तल 2 मिमी मोटाई वाला ठोस वृतीय डिस्क है तथा जिसकी त्रिज्या, पात्र की बाहरी त्रिज्या के बराबर है। यदि पात्र की आंतरिक त्रिज्या 10 मिमी होने पर पदार्थ के न्यूनतम आयतन की आवश्यकता होती हो, तब $\frac{V}{250\pi}$ का मान है। **[JEE(Adv.)2015]**

Ans. 4

Sol. Volume of material $V = \pi r^2 h$

$\Rightarrow \quad V_1 = \pi(r + 2)^2 2 + \pi(r + 2)^2 h - \pi r^2 h \quad \Rightarrow \quad V_1 = 2\pi(r + 2)^2 + \pi h(4 + 4r)$

$\Rightarrow \quad V_1 = 2\pi(r + 2)^2 + 4\pi h(r + 1) \quad \Rightarrow \quad V_1 = 2\pi\left((r + 2)^2 + \frac{2(r + 1)V}{\pi r^2}\right)$

$$\Rightarrow \quad \frac{dV_1}{dr} = 2\pi\left(2(r+2)+\frac{2v}{\pi}\left(\frac{-1}{r^2}-\frac{2}{r^3}\right)\right)=0 \quad \Rightarrow \quad 24+\frac{2V}{\pi}\left(\frac{-2-10}{10^3}\right)=0$$

$$\Rightarrow \quad \frac{24v}{10^3\pi}=24 \quad \Rightarrow \quad v=10^3\pi \quad \Rightarrow \quad \frac{v}{250\pi}=4$$

3.10 Let $f:\mathbb{R}\to(0,\infty)$ and $g:\mathbb{R}\to\mathbb{R}$ be twice differentiable functions such that f'' and g'' are continuous functions of $\mathbb{R}$. Suppose $f'(2)=g(2)=0$, $f''(2)\neq 0$ and $g'(2)\neq 0$. If $\lim\limits_{x\to 2}\frac{f(x)g(x)}{f'(x)g'(x)}=1$, then. **[JEE(Adv.)2016]**

(A) f has a local minimum at x = 2
(B) f has a local maximum at x = 2
(C) f'' (2) > f(2)
(D) f(x) – f'' (x) = 0 for at least one $x\in\mathbb{R}$

माना कि $f:\mathbb{R}\to(0,\infty)$ और $g:\mathbb{R}\to\mathbb{R}$ ऐसे दो बार अवकलनीय फलन है कि $\mathbb{R}$ पर f'' और g '' संतत फलन है। मान लीजिये कि $f'(2)=g(2)=0$, $f''(2)\neq 0$ और $g'(2)\neq 0$ है। यदि $\lim\limits_{x\to 2}\frac{f(x)g(x)}{f'(x)g'(x)}=1$ है, तब : **[JEE(Adv.)2016]**

(A) x = 2 पर f का स्थानीय निम्नतम है
(B) x = 2 पर f स्थानीय उच्चतम है
(C) f'' (2) > f(2)
(D) कम से कम एक $x\in\mathbb{R}$ के लिए f(x) – f'' (x) = 0

Ans. AD

Sol. $\lim\limits_{x\to 2}\frac{f(x)g(x)}{f'(x)g'(x)}=1$

$$\lim_{x\to\infty}\frac{f(x)g'(x)+f'(x)g(x)}{f'(x)g''(x)+f''(x)g'(x)}=1 \qquad \Rightarrow \lim_{x\to 2}\frac{f(x)}{f''(x)}=1$$

$\Rightarrow f(2)=f''(2)=+ve \quad (\because$ co-domain $(0,\infty))$

Hence f has local maximum at x = 2.

3.11 If $f(x)=\begin{vmatrix}\cos(2x) & \cos(2x) & \sin(2x)\\ -\cos x & \cos x & -\sin x\\ \sin x & \sin x & \cos x\end{vmatrix}$, then : **[JEE(Adv.)2017]**

(A) f(x) attains its maximum at x = 0
(B) f(x) attains its minimum at x = 0
(C) f'(x) = 0 at more than three points in $(-\pi,\pi)$
(D) f'(x) = 0 at exactly three points in $(-\pi,\pi)$

यदि $f(x)=\begin{vmatrix}\cos(2x) & \cos(2x) & \sin(2x)\\ -\cos x & \cos x & -\sin x\\ \sin x & \sin x & \cos x\end{vmatrix}$, तब : **[JEE(Adv.)2017]**

(A) $x = 0$ पर $f(x)$ का उच्चिष्ठ है (B) $x = 0$ पर $f(x)$ का निम्निष्ठ है

(C) $(-\pi, \pi)$ में तीन से अधिक बिन्दुओं पर $f'(x) = 0$ है (D) $(-\pi, \pi)$ में ठीक तीन बिन्दुओं पर $f'(x) = 0$ है

Ans. AC

Sol. $\begin{vmatrix}\cos 2x & \cos 2x & \sin 2x\\ -\cos x & \cos x & -\sin x\\ \sin x & \sin x & \cos x\end{vmatrix}$

$= \cos2x - \cos2x(-\cos^2x + \sin^2x) + \sin2x(-2\sin x\cos x)$

$f(x) = \cos4x + \cos2x$

$\therefore\ f(x) = 2\cos^2 2x + \cos2x - 1$

Let $\cos2x = t$

$\Rightarrow\ f(x) = 2t^2 + t - 1$ and $t \in [-1, 1]$

$f(x)$ attains its minima at $t = -\frac{1}{4} \in [-1,1]$

$f(x), t = -\frac{1}{4} \in [-1,1]$

$\therefore\ f(x)|_{min} = \frac{2}{16} - \frac{1}{4} - 1 = \frac{-9}{8}$

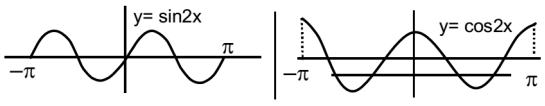
$\therefore\ f(x)|_{max} = 2 + 1 - 1 = 2 \ldots\ldots(\text{when } \cos 2x = 1)$

$f'(x) = -4\sin 4x - 2\sin 2x$

$f'(x) = 0 \Rightarrow 4\sin 4x + 2\sin 2x = 0$

$\Rightarrow 8\sin 2x\cos 2x + 2\sin 2x = 0$

$\Rightarrow 2\sin 2x(4\cos 2x + 1) = 0 \Rightarrow \sin 2x = 0$ or $\cos 2x = -\frac{1}{4}$



3.12 Let f: R $\rightarrow$R be given by

$$f(x)=\begin{cases} x^5+5x^4+10x^3+10x^2+3x+1, & x<0; \\ x^2-x+1, & 0\le x<1; \\ \frac{2}{3}x^3-4x^2+7x-\frac{8}{3}, & 1\le x<3; \\ (x-2)\log_e(x-2)-x+\frac{10}{3}, & x\ge 3 \end{cases}$$

Then which of the following options is/are correct : **[JEE(Adv.)2019]**

(A) f is increasing on $(-\infty,0)$ (B) f' has a local maximum at $x=1$

(C) f' is NOT differentiable at $x=1$ (D) f is onto

माना कि $f: R \to R$ निम्न प्रकार से दिया है।

$$f(x)=\begin{cases} x^5+5x^4+10x^3+10x^2+3x+1, & x<0; \\ x^2-x+1, & 0\le x<1; \\ \frac{2}{3}x^3-4x^2+7x-\frac{8}{3}, & 1\le x<3; \\ (x-2)\log_e(x-2)-x+\frac{10}{3}, & x\ge 3 \end{cases}$$

तब निम्न में से कौन सा (से) विकल्प सही है : **[JEE(Adv.)2019]**

(A) f अंतराल $(-\infty,0)$ में वर्धमान है (B) f' का एक स्थानीय उच्चतम $x=1$ पर है

(C) $x=1$ पर f' अवकलनीय नहीं है (D) f आच्छादक है

Ans. BCD

Sol.
$$f(x)=\begin{cases} (x+1)^5-2x & x<0 \\ x^2-x+1 & 0\le x<1 \\ \frac{2}{3}x^3-4x^2+7x-\frac{8}{3} & 1\le x<3 \\ (x-2)\ln(x-2)-x+\frac{10}{3} & x\ge 3 \end{cases}$$

$f(x)$ is continuous at $x=0,\ x=1, x=3$

$\Rightarrow$ continuous for all $x\in R$

$$f'(x)=\begin{cases}5(x+1)^4-2 & x<0\\ 2x-1 & 0\le x<1\\ 2x^3-8x+7 & 1<x<3\\ \ln(x-2) & x>3\end{cases}$$

$f'(x)$ is continuous at $x=1$,

Discountinuous at $x=0,3$

Option (1) for $x<0$ $f'(x)$ is changing its sign

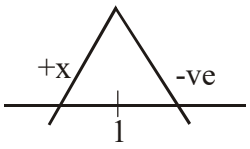
$\Rightarrow$ not increasing in $(0,\infty)$

Option (2) at $x=1$

$$f''(x)=\begin{cases}2 & 0<x<1\\ 4x-8 & 1<x<3\end{cases}$$

$f''(1^+)=-ve, f''(1^-)=+ve$

f' has a local max^m at $x=1$



Option (3) $f''(1^+)=-4\neq f''(1^-)=2$

f' is not differentiable at $x=1$

Option (4) for $x\in R$

for $x<0$, $(x+1)^5-2x\in(-\infty,1)$

for $x\ge 3$, $(x-2)\ln(x-2)-x+\frac{10}{3}\in\left[\frac{1}{3},\infty\right)$

Range of $f(x)=(-\infty,\infty)$

3.13 Let $f(x)=\frac{\sin\pi x}{x^2}, x>0$

Let $x_1<x_2<x_3<.....<x_n<....$ be all the points of local maxmimum of f and $y_1<y_2<y_3<....<y_n<.....$ be all the points of local minimum of f, then which of the following options is are correct : **[JEE(Adv.)2019]**

(A) $x_{n+1}-x_n>2$ for every n

(B) $x_1<y_1$

(C) $x_n\in\left(2n,2n+\frac{1}{2}\right)$ for every n

(D) $|x_n-y_n|>1$ for every n

माना कि $f(x) = \frac{\sin \pi x}{x^2}, x > 0$

माना कि f के सभी स्थानीय उच्चतम (local maximum) बिन्दु $x_1 < x_2 < x_3 < .....< x_n < ....$ है और f के सभी स्थानीय न्यूनतम (local minimum) बिन्दु $y_1 < y_2 < y_3 < .... < y_n < .....$ हैं। तब निम्न में से कौन सा(से) विकल्प सही है (हैं)?

[JEE(Adv.)2019]

(A) प्रत्येक n के लिए $x_{n+1} - x_n > 2$ है

(B) $x_1 < y_1$

(C) प्रत्येक n के लिए $x_n \in \left(2n, 2n + \frac{1}{2}\right)$ है

(D) प्रत्येक n के लिए $|x_n - y_n| > 1$ है

Ans. ACD

Sol. $f(x) = \frac{\sin \pi x}{x^2}, x > 0$

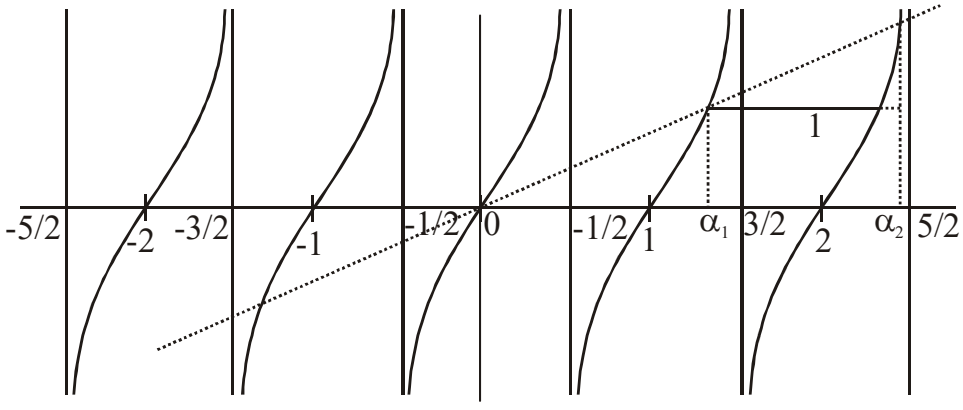
$$f'(x) = \frac{x^2 \cos(\pi x)\pi - \sin(\pi x).2x}{x^4}$$

$$= \frac{\left[\pi x \cos(\pi x) - 2\sin(\pi x)\right]}{x^3}$$

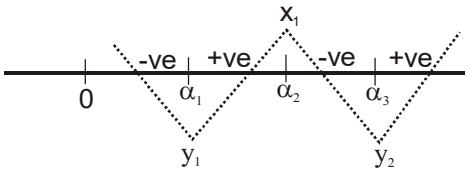
$f'(x) = 0$

$\Rightarrow \pi x \cos \pi x = 2 \sin \pi x$

$$\tan \pi x = \frac{\pi x}{2}$$



At 0^+ $\quad f'(x) = -ve$



$\alpha_1 = x_1$

$\alpha_2 = y_1$ (and so on)

$\alpha_3 = x_2$

$x_1 > y_1$

$x_1 \in (2,5/2)$
using periodicity
$x_n \in (2n,2n+1/2)$
$x_{n+1} - x_n > 2 \ \forall$ n correct
$|x_n - y_n| > 1 \ \forall$ n correct

Question Stem for Question Nos. 3.14 and 3.15

Let $f_1 : (0,\infty) \to R$ and $f_2 : (0,\infty) \to R$ be defined by $f_1(x) = \int_0^x \prod_{j=1}^{21} (t-j)^j \, dt, \quad x > 0$ and

$f_2(x) = 98(x-1)^{50} - 600(x-1)^{49} + 2450, \ x > 0,$ where, for any positive integer n and real numbers $a_1, a_2, ..., a_n$, $\prod_{i=1}^{n} a_i$ denotes the product of $a_1, a_2, ..., a_n$. Let m_i and n_i, respectively, denote the number of points of local minima and the number of points of local maxima of function f_i, i = 1, 2, in the interval $(0,\infty)$.

माना $f_1 : (0,\infty) \to R$ तथा $f_2 : (0,\infty) \to R$, $f_1(x) = \int_0^x \prod_{j=1}^{21} (t-j)^j \, dt, \quad x > 0$ तथा

$f_2(x) = 98(x-1)^{50} - 600(x-1)^{49} + 2450, \ x > 0,$ द्वारा पभिाषित है, जहाँ किसी धनात्मक पूर्णांक n तथा वास्तविक संख्याओं $a_1, a_2, ..., a_n$ के लिए $\prod_{i=1}^{n} a_i$ है, जो $a_1, a_2, ..., a_n$ के गुणनफल को निरूपित करता है। माना m_i व n_i अंतराल $(0,\infty)$ में फलन f_i, i = 1, 2 के क्रमशः स्थानीय निम्निष्ठ बिन्दुओं तथा स्थानीय उच्चिष्ठ बिन्दुओं की संख्या को निरूपित करते हैं। तब

3.14 The value of $2m_1 + 3n_1 + m_1n_1$ is_____. **[JEE(Adv.)2021]**

$2m_1 + 3n_1 + m_1n_1$ का मान _____है। **[JEE(Adv.)2021]**

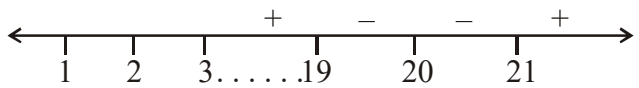
Ans. 57

3.15 The value of $6m_2 + 4n_2 + 8m_2n_2$ is_____. **[JEE(Adv.)2021]**

$6m_2 + 4n_2 + 8m_2n_2$ का मान _____है। **[JEE(Adv.)2021]**

Ans. 6

Sol. $f_1'(x) = \prod_{j=1}^{21} (x-j)^j = (x-1)^1 (x-2)^2 (x-3)^3 .....(x-21)^{21}$



Points of minima $\Rightarrow$ x = 21, 17, 13, 9, 5, 1 $\Rightarrow m_1 = 6$

Points of maxima $\Rightarrow x = 19, 15, 11, 7, 3 \Rightarrow n_1 = 5$

$2m_1 + 3n_1 + m_1n_1 = 57.$

$f_2'(x) = 98.50(x-1)^{49} - 600.49(x-1)^{48}$

$f_2'(x) = 4900(x-1)^{48}(x-7)$

$f_2'(x)$ $\xleftarrow{\quad - \quad - \quad +\quad}$ (1, 7)

Points of minima $\Rightarrow x = 7$ $\qquad m_2 = 1$

$n_2 = 0$

$6m_2 + 4n_2 + 8m_2n_2 = 6.$

3.16 Let $\alpha = \sum_{k=1}^{\infty} \sin^{2k}\left(\frac{\pi}{6}\right)$. Let $g:[0,1] \to R$ be the function defined by $g(x) = 2^{\alpha x} + 2^{\alpha(1-x)}$.

Then, which of the following statements is/are TRUE ? **[JEE(Adv.)2022]**

(A) The minimum value of g(x) is $2^{7/6}$

(B) The maximum value of g(x) is $1 + 2^{1/3}$

(C) The function g(x) attains its maximum at more than one point

(D) The function g(x) attains its minimum at more than one point

माना कि $\alpha = \sum_{k=1}^{\infty} \sin^{2k}\left(\frac{\pi}{6}\right)$ है। माना कि फलन $g:[0,1] \to R$, $g(x) = 2^{\alpha x} + 2^{\alpha(1-x)}$ द्वारा परिभाषित है। तब निम्न कथनों में से कौनसा सत्य है ? **[JEE(Adv.)2022]**

(A) g(x) का निम्नतम मान $2^{7/6}$ है

(B) g(x) का उच्चतम मान $1 + 2^{1/3}$ है

(C) फलन g(x) अपना उच्चतम मान एक से ज्यादा बिंदुओं पर प्राप्त करता है

(D) फलन g(x) अपना निम्नतम मान एक से ज्यादा बिंदुओं पर प्राप्त करता है

Ans. (A, B, C)

Sol. $\alpha = \sum_{k=1}^{\infty}\left(\frac{1}{2}\right)^{2k} = \sum_{k=1}^{\infty}\left(\frac{1}{4}\right)^{k} = \frac{1/4}{1-\frac{1}{4}} = \frac{1}{3}$

Hence, $g(x) = 2^{x/3} + 2^{(1-x)/3}$

Now, $g'(x) = \frac{\ln 2}{3}\frac{\left(2^{2x/3} - 2^{1/3}\right)}{2^{x/3}}$

$g'(x) = 0$ at $x = \frac{1}{2}$

And, derivative changes sign from negative to positive at $x = \frac{1}{2}$, hence $x = \frac{1}{2}$ is point of local minimum as well as absolute minimum of g(x) for $x \in [0, 1]$

Hence, minimum value of $g(x) = g\left(\frac{1}{2}\right)$

$= 2^{1/6} + 2^{1/6} = 2^{7/6}$

$\Rightarrow$ Option (A) is correct

Maximum value of g(x) is either equal to g(0) or g(1).

$g(0) = 1 + 2^{1/3}$

$g(1) = 2^{1/3} + 1$

Hence (B) and (C) are also correct

3.17. Let **R** denote the set of all real numbers. Let $f : \mathbf{R} \to \mathbf{R}$ be defined by

$$f(x) = \begin{cases} \frac{6x + \sin x}{2x + \sin x} & \text{if} \quad x \neq 0, \\ \frac{7}{3} & \text{if} \quad x = 0. \end{cases}$$

Then which of the following statements is (are) TRUE? **[JEE(Adv.)2025]**

(A) The point x = 0 is a point of local maxima of *f*

(B) The point x = 0 is a point of local minima of *f*

(C) Number of points of local maxima of *f* in the interval $[\pi, 6\pi]$ is 3

(D) Number of points of local minima of *f* in the interval $[2\pi, 4\pi]$ is 1

मान लीजिए **R** सभी वास्तविक संख्याओं (real numbers) के समुच्य (set) को दर्शाता है। मान लीजिए फलन (function) $f : \mathbf{R} \to \mathbf{R}$,

$$f(x) = \begin{cases} \frac{6x + \sin x}{2x + \sin x} & \text{if} \quad x \neq 0, \\ \frac{7}{3} & \text{if} \quad x = 0. \end{cases}$$

द्वारा परिभाषित है। तब निम्नलिखित कथनों में से कौनसा (से) सत्य है (हैं)? **[JEE(Adv.)2025]**

(A) बिन्दु x = 0, *f* का एक स्थानीय उच्चतम का बिन्दु (point of local maxima) है

(B) बिन्दु x = 0, *f* का एक स्थानीय निम्नतम का बिन्दु (point of local minima) है।

(C) अंतराल (interval) $[\pi, 6\pi]$ में *f* के स्थानीय उच्चतम के बिन्दुओं (points of local maxima) की 3 संख्या है

(D) अंतराल (interval) $[2\pi, 4\pi]$ में *f* के स्थानीय निम्नतम के बिन्दुओं (points of local minima) की 1 संख्या है

Ans. BCD

3.18. Let R denote the set of all real numbers. For a real number x, let [x] denote the greatest integer less than or equal

to x. Let n denote a natural number.

Match each entry in **List-I** to the correct entry in **List-II** and choose the correct option.

[JEE(Adv.)2025]

List-I	**List-II**
(P) The minimum value of n for which the function $f(x)=\left[\frac{10x^3-45x^2+60x+35}{n}\right]$ is continuous on the interval $[1,2]$, is	(1) 8
(Q) The minimum value of n for which $g(x)=(2n^2-13n-15)(x^3+3x)$, $x \in R$, is an increasing function on R, is	(2) 9
(R) The smallest natural number n which is greater than 5, such that $x=3$ is a point of local minima of $h(x)=(x^2-9)^n(x^2+2x+3)$, is	(3) 5
(S) Number of $x_0 \in R$ such that $l(x)=\sum_{k=0}^{4}\left(\sin\lvert x-k\rvert+\cos\left\lvert x-k+\frac{1}{2}\right\rvert\right)$ $x \in R$, is NOT differentiable at x_0, is	(4) *6*
	(5) 10

The correct option is :

मान लीजिए कि R सभी वास्तविक संख्याओं के समुच्चय को दर्शाता है। किसी भी वास्तविक संख्या x के लिए, मान लीजिए कि x से छोटा या उसके बराबर महत्तम पूर्णांक को [x] से दर्शाते हैं। मान लीजिए कि n एक प्राकृत संख्या को दर्शाता है।

सूची-I की प्रत्येक प्रविष्टि का **सूची-II** की सही प्रविष्टि से मिलान कीजिये और सही विकल्प का चयन करें।

[JEE(Adv.)2025]

सूची-I	**सूची-II**
(P) n का न्यूनतम मान, जिसके लिए अंतराल [1, 2] में फलन $f(x)=\left[\frac{10x^3-45x^2+60x+35}{n}\right]$ संतत है, है	(1) 8
(Q) n का न्यूनतम मान जिसके लिए, R पर, $g(x)=(2n^2-13n-15)(x^3+3x)$, $x \in R$, कएक वर्धमान फलन है, है	(2) 9
(R) सबसे छोटी प्राकृत संख्या n, जो 5 से अधिक है और जिसके लिए, $x=3$ $h(x)=(x^2-9)^n(x^2+2x+3)$ का एक स्थानीय निम्नतम का बिन्दु है, है	(3) 5
(S) ऐसे सभी $x_0 \in R$ की संख्या, कि x_0 पर $l(x)=\sum_{k=0}^{4}\left(\sin\lvert x-k\rvert+\cos\left\lvert x-k+\frac{1}{2}\right\rvert\right)$ $x \in R$, अवकलनीय नहीं है, है	(4) *6*
	(5) 10

सही विकल्प है :

(A) (P) $\rightarrow$ (1); (Q) $\rightarrow$ (3); (R) $\rightarrow$ (2); (S) $\rightarrow$ (5)

(B) (P) $\rightarrow$ (2); (Q) $\rightarrow$ (1); (R) $\rightarrow$ (4); (S) $\rightarrow$ (3)

(C) (P) $\rightarrow$ (5); (Q) $\rightarrow$ (1); (R) $\rightarrow$ (4); (S) $\rightarrow$ (3)

(D) (P) $\rightarrow$ (2); (Q) $\rightarrow$ (3); (R) $\rightarrow$ (1); (S) $\rightarrow$ (5)

Ans. B

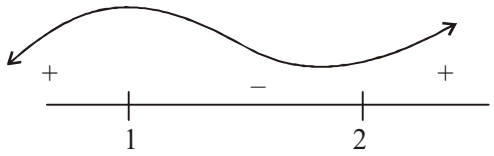
Sol. (P) $f(x)=\left[\frac{10x^3-45x^2+60x+35}{n}\right]$

Let $g(x) = 10x^3 - 45x^2 + 60x + 35$

$g'(x) = 30x^2 - 90x + 60$

$= 30 (x^2 - 3x + 2)$

$= 30 (x - 1) (x - 2)$



$g(x) \downarrow$ing is $[1, 2]$

$$f(1)=\left[\frac{g(1)}{n}\right]=\left[\frac{10-45+60+35}{n}\right]=\left[\frac{60}{n}\right]$$

$$f(2)=\left[\frac{g(2)}{n}\right]=\left[\frac{55}{n}\right]$$

for $n = 9$ both $f(1) = f(2) = 6$

$P \rightarrow 2$

(Q) $g(x) = (2n^2 - 13n - 15) (x^3 + 3x)$

$g'(x) = (2n^2 - 13n - 15) (3x^2 + 3)$

$= (2n^2 - 15n + 2n - 15) (3x^2 + 3)$

$= \{n(2n - 15) + (2n - 15)\} (3x^2 + 3)$

$= (n + 1) (2n - 15) (3x^2 + 3)$



for $g(x)$ to be $\uparrow$ing minimum $\quad n \rightarrow 1$

(R) $h(x) = (x^2 - 9)^n (x^2 + 2x + 3)$

$h(3) = 0$, for $n > 5$ at $n = 6$

$h(3 + \delta) > h(3)$ & $h(3 - \delta) > h(3)$

$\rightarrow h(x)$ has local minima at $x = 3$ for $n = 6$

$R \rightarrow 4$

(S) $\ell(x) = \sum_{k=0}^{4} \sin|x-k| + \cos\left|x-k+\frac{i}{2}\right|$

$$= \sin|x| + \cos\left|x+\frac{1}{2}\right| + \sin|x-1| + \cos\left|x-\frac{1}{2}\right| + \ldots\ldots + \sin|x-4| + \cos\left|x-\frac{7}{2}\right|$$

As $\sin|x-\alpha|$ is Non Diff at $x = \alpha$

But $\cos|x-\alpha|$ is diff at $x = \alpha$

$\Rightarrow$ l(n) Non Diff at $x = 0, 1, 2, 3, 4$ (5 Points)

$S \rightarrow 3$

3.19. Consider the function $f: (0, \infty) \rightarrow (-\infty, \infty)$ given by

$$f(x) = \sqrt{x}\log_e(x) - x + 1.$$

Then which one of the following statements is **TRUE**? **[JEE(Adv.)2026]**

(A) The derivative of the function f is decreasing in the interval (0, 1)

(B) The function f has a local maximum at some point $a \in (0, \infty)$

(C) The function f has a local minimum at some point $b \in (0, \infty)$

(D) The function f has NEITHER a point of local maximum NOR a point of local minimum in the interval $(0, \infty)$

फलन $f: (0, \infty) \rightarrow (-\infty, \infty)$, जो

$$f(x) = \sqrt{x}\log_e(x) - x + 1$$

द्वारा दिया गया है, पर विचार कीजिए।

तब निम्नलिखित कथनों में से कौन–सा एक कथन **सत्य** है? **[JEE(Adv.)2026]**

(A) अंतराल (0, 1) में, फलन f का अवकलज ह्रासमान है

(B) किसी बिन्दु $a \in (0, \infty)$ पर, फलन f का एक स्थानीय उच्चतम है

(C) किसी बिन्दु $b \in (0, \infty)$ पर, फलन f का एक स्थानीय निम्नतम है

(D) अंतराल $(0, \infty)$ में, फलन f का न तो कोई स्थानीय उच्चतम का बिन्दु है और न ही कोई स्थानीय निम्नतम का बिन्दु है

Ans. D

Sol. $f(x) = \sqrt{x}\ln(x) - x + 1$ for $x > 0$.

$$f'(x) = \frac{\ln(x)+2}{2\sqrt{x}} - 1$$

$$f'(x) = \frac{\ln(x) - 2\sqrt{x} + 2}{2\sqrt{x}}$$

Option (A)

To check if the derivative f'(x) is decreasing or increasing in the interval (0, 1), we find the second derivative

f"(x) : $f'(x) = \frac{1}{2}x^{-1/2}\ln(x) + x^{-1/2} - 1$

$$f''(x) = \frac{1}{2}\left(-\frac{1}{2}x^{-3/2}\ln(x) + x^{-1/2}\cdot\frac{1}{x}\right) - \frac{1}{2}x^{-3/2}$$

$$f''(x) = -\frac{\ln(x)}{4x^{3/2}}$$

Thus, f"(x) > 0 for all x ∈ (0, 1), which means f'(x) is increasing in the interval (0, 1).

Therefore, statement (A) is FALSE.

Finding Local Extrema (Options B, C, and D)

Local extrema occur at critical points where f'(x) = 0. Let's set the numerator of f'(x) to 0:

$\ln(x) - 2\sqrt{x} + 2 = 0$

Let us define a function $g(x) = \ln(x) - 2\sqrt{x} + 2$ and find its behavior

$$g'(x) = \frac{1}{x} - \frac{1}{\sqrt{x}} = \frac{1-\sqrt{x}}{x}$$

Let's analyze the sign of g'(x) for x > 0:

For x ∈ (0, 1), $\sqrt{x} < 1$ $\Rightarrow$ g '(x) > 0 (the function g(x) is increasing).

For x = 1, g'(1) = 0.

For x ∈ (1, ∞), $\sqrt{x} > 1$ $\Rightarrow$ g'(x) < 0 (the function g(x) is decreasing).

This shows that g(x) attains its absolute maximum at x = 1. and $g(1) = \ln(1) - 2\sqrt{1} + 2 = 0 - 2 + 2 = 0$

Since the maximum value of g(x) is 0 at x = 1, for all other values of $x \neq 1$, $g(x) < 0$.

For x = 1, $f'(1) = \frac{g(1)}{2\sqrt{1}} = 0$

For all $x \in (0, \infty)$ where $x \neq 1$, $f'(x) < 0$

Since $f'(x) \leq 0$ for the entire domain, the function f(x) is strictly decreasing everywhere except at the stationary point x = 1. Because f'(x) does not change its sign (it remains negative before and after x = 1), the point x = 1 is a point of inflection, not a local maximum or minimum.

Therefore, f(x) has neither a point of local maximum nor a point of local minimum in $(0, \infty)$.

SINGLE CORRECT TYPE QUESTIONS

1. The graphs $y = 2x^3 - 4x + 2$ and $y = x^3 + 2x - 1$ intersect in exactly 3 distinct points. The slope of the line passing through two of these points :

(A) is equal to 4 (B) is equal to 6 (C) is equal to 8 (D) is not unique

यदि वक्र $y = 2x^3 - 4x + 2$ और $y = x^3 + 2x - 1$, 3 बिन्दुओं पर प्रतिच्छेद करते है। तब इन बिन्दुओं में से दो से होकर जाने वाली रेखा की प्रवणता होगी :

(A) 4 (B) 6 (C) 8 (D) अद्वितीय नहीं

Ans. C

Sol. Let (x_1, y_1) and (x_2, y_2) are two of these points given $y = x^3 + 2x - 1$ and $y = 2x^3 - 4x + 2$

$\therefore \quad y_1 = 2x_1^3 - 4x_1 + 2 \quad ....(1)$

and $\quad 2y_1 = 2x_1^3 + 4x_1 - 2 \quad ....(2)$

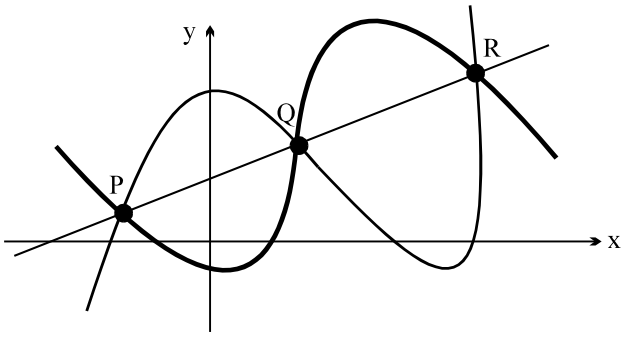
(2) – (1)

$y_1 = 8x_1 - 4 \quad ....(3)$

|||ly $\quad y_2 = 8x_2 - 4 \quad ....(4)$

$y_2 - y_1 = 8(x_2 - x_1)$

$\frac{y_2 - y_1}{x_2 - x_1} = 8$



2. A function $y = f(x)$ has a second order derivative $f''(x) = 6(x - 1)$. If its graph passes through the point (2,1) and at that point the tanget to the graph is $y = 3x - 5$, then the function, is :

एक फलन $y = f(x)$ का दूसरे क्रम का अवकलज $f''(x) = 6(x - 1)$ है। यदि इसका आरेख बिन्दु (2,1) से होकर जाता है तथा वहाँ आरेख पर स्पर्शरेखा का समीकरण $y = 3x - 5$ है, तो फलन है :

(A) $(x-1)^2$ (B) $(x-1)^3$ (C) $(x+1)^3$ (D) $(x+1)^2$

Ans. B

Sol. $f''(x) = 6(x-1)$

$\Rightarrow f'(x) = 3x^2 - 6x + k \quad$ _____(1)

Slope of tangent at (2, 1) is 3.

$\Rightarrow 3 = 3 \times 2^2 - 6 \times 2 + k$

$\Rightarrow k = 3 \quad$ _____(2)

From (1) and (2)

$f(x) = x^3 - 3x^2 + 3x + c$

Using f(x) = y = 1 when x = 2, we get

$1 = 2 + c \Rightarrow c = -1$

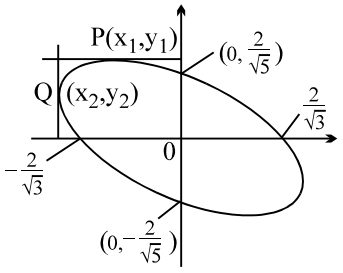
Hence, the function is

$f(x) = x^3 - 3x^2 + 3x - 1$

$\Rightarrow f(x) = (x-1)^3$

3. The lines $y = -\frac{3}{2}x$ and $y = -\frac{2}{5}x$ intersect the curve $3x^2 + 4xy + 5y^2 - 4 = 0$ at the points P and Q respectively.

The tangents drawn to the curve at P and Q :



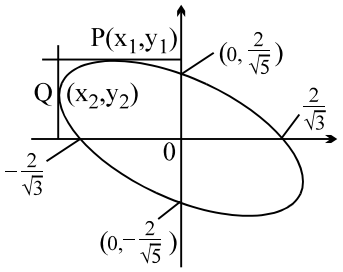
(A) intersect each other at angle of 45°

(B) are parallel to each other

(C) are perpendicular to each other

(D) none of these

रेखाऐं $y = -\frac{3}{2}x$ तथा $y = -\frac{2}{5}x$ वक्र $3x^2 + 4xy + 5y^2 - 4 = 0$ को बिन्दुओं P और Q पर क्रमशः प्रतिच्छेद करती है तब P और Q पर वक्र की स्पर्श रेखाऐं :



(A) एक दूसरे को 45° के कोण पर प्रतिच्छेद करती है

(B) समान्तर है

(C) लम्बवत् है

(D) इनमें से कोई नहीं

Ans. C

Sol. $3x^2 + 4xy + 5y^2 - 4 = 0$

Differentiate

$6x + 4xy' + 4y + 10yy' = 0$

$y'(4x + 10y) = -(6x + 4y)$

$y' = \frac{-(6x + 4y)}{4x + 10y}$

Slope of the tangent at intersection by

$$y = -\frac{3}{2}x$$

$$y' = \frac{-\left(6x + 4\left(-\frac{3x}{2}\right)\right)}{4x + 10\left(-\frac{3x}{2}\right)}$$

$$y' = 0$$

Parallel to x-axis.

Slope of the tangent at intersection by

$$y = \frac{-2x}{5}$$

$$y' = \frac{-\left(6x + 4\left(-\frac{2x}{5}\right)\right)}{4x + 10\left(\frac{-2x}{5}\right)}$$

$$y' = \frac{-\left(6x + 4\left(-\frac{3x}{2}\right)\right)}{0}$$

Parallel to y-axis.

So both tangent are perpendicular.

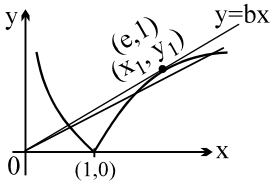
4. The set of values of p for which the equation $|\ln x| - px = 0$ possess three distinct roots is :

π का हल समुच्चय जबकि समीकरण $|\ln x| - px = 0$ के तीन भिन्न मूल हे, हैं :

(A) $\left(0, \frac{1}{e}\right)$ (B) (0, 1) (C) (1,e) (D) (0,e)

Ans. A

Sol. Hint: $1/x_1 = p = y_1/x_1 \Rightarrow y_1 = 1$ & $x_1 = e$.
Hence slope of $y = px$ i.e. $p \in (0, 1/e)$ for three roots



5. A curve is represented parametrically by the equations $x = t + e^{at}$ and $y = -t + e^{at}$ when $t \in R$ and $a > 0$. If the curve touches the axis of x at the point A, then the coordinates of the point A are :

यदि किसी वक्र का प्राचल समीकरण $x = t + e^{at}$ तथा $y = -t + e^{at}$ जहाँ $t \in R$ और $a > 0$ है। यदि वक्र x-अक्ष को बिन्दु A बिन्दु पर स्पर्श करता है तब A के निर्देशांक होगें :

(A) (1, 0) (B) (1/e, 0) (C) (e, 0) (D) (2e, 0)

Ans. D

Sol. $x = t + e^{at}$; $y = -t + e^{at}$

$\frac{dx}{dt} = 1 + ae^{at}$; $\frac{dy}{dt} = -1 + ae^{at}$; $\frac{dy}{dx} = \frac{-1+ae^{at}}{1+ae^{at}}$

at the point A, $y = 0$ and $\frac{dy}{dx} = 0$ for some $t = t_1$

$\therefore$ $ae^{at_1} = 1$ ....(1) ;

also $0 = -t_1 + e^{at_1}$; $\therefore$ $e^{at_1} = t_1$ .....(2), putting this value in (1)

we get, $at_1 = 1 \Rightarrow t_1 = \frac{1}{a}$; now from (1) $ae = 1 \Rightarrow a = \frac{1}{e}$

hence $x_A = t_1 + e^{at_1} = e + e = 2e \Rightarrow A \equiv (2e, 0)$

6. A horse runs along a circle with a speed of 20 km/hr . A lantern is at the centre of the circle. A fence is along the tangent to the circle at the point at which the horse starts . The speed with which the shadow of the horse moves along the fence at the moment when it covers 1/8 of the circle in km/hr is :

एक घोड़ा एक वृत में 20 किमी/घण्टे की दर से दौड़ रहा है। एक लालटेन वृत के केन्द्र पर स्थित है। वृत के एक बिन्दु जहाँ से घोड़ दौड़ना प्रारम्भ करता है, पर स्पर्श रेखा से एक चार दिवारी बनाई जाती है। उस दिवार पर घोड़े की परछाई की गति, जब घोड़ा वृत का 1/8 तय कर चुका है, किमी/घण्टे में होनी चाहिये :

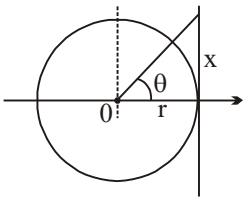
(A) 20 (B) 40 (C) 30 (D) 60

Ans. B

Sol. $\tan\theta = x/r \Rightarrow x = r\tan\theta$

$\Rightarrow$ $dx/dt = r\sec^2\theta\,(d\theta/dt) = r\omega\sec^2\theta = v\sec^2\theta$

where $\theta = 2\pi/8$, $dx/dt = v\sec^2(\pi/4) = 2v = 40$ km/hr ; $\theta = 45^\circ$



7. If $f(x) = 4x^3 - x^2 - 2x + 1$ and $g(x) = \begin{cases} \text{Min}\ \{f(t) : 0 \le t \le x\} & ;\ 0 \le x \le 1 \\ 3 - x & ;\ 1 < x \le 2 \end{cases}$ then $g\left(\frac{1}{4}\right) + g\left(\frac{3}{4}\right) + g\left(\frac{5}{4}\right)$ has the value equal to :

यदि $f(x) = 4x^3 - x^2 - 2x + 1$ तथा $g(x) = \begin{cases} \text{Min}\ \{f(t) : 0 \le t \le x\} & ;\ 0 \le x \le 1 \\ 3 - x & ;\ 1 < x \le 2 \end{cases}$ तब $g\left(\frac{1}{4}\right) + g\left(\frac{3}{4}\right) + g\left(\frac{5}{4}\right)$

का मान है :

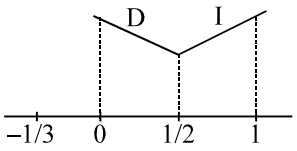
(A) $\frac{7}{4}$ (B) $\frac{9}{4}$ (C) $\frac{13}{4}$ (D) $\frac{5}{2}$

Ans. D

Sol. $f'(x) = 12x^2 - 2x - 2$

$= 2[6x^2 - x - 1] = 2(3x + 1)(2x - 1)$

Hence $g(x) = \begin{cases} f(x) & \text{if } 0 \le x < \frac{1}{2} \\ f\left(\frac{1}{2}\right) & \text{if } \frac{1}{2} \le x \le 1 \\ 3 - x & \text{if } 1 < x \le 2 \end{cases}$



8. Let $f(x) = x^3 - 3x^2 + 2x$. If the equation $f(x) = k$ has exactly one positive and one negative solution then the value of k equals :

माना $f(x) = x^3 - 3x^2 + 2x$, यदि समीकरण $f(x) = k$ का एक ही धनात्मक और एक ही ऋणात्मक हल है तब k =

(A) $-\frac{2\sqrt{3}}{9}$ (B) $-\frac{2}{9}$ (C) $\frac{2}{3\sqrt{3}}$ (D) $\frac{1}{3\sqrt{3}}$

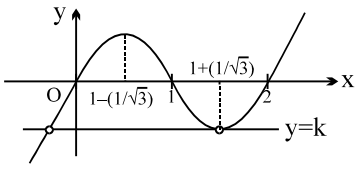
Ans. A

Sol. $f(x) = x(x^2 - 3x + 2)$

$f(x) = x(x - 2)(x - 1)$

graph of $y = f(x)$ is as shown

now $f(x) = k$ to have exactly one positive and negative solution



we have, $k = f\left(1 + \frac{1}{\sqrt{3}}\right)$ (think !) $1 - \frac{1}{\sqrt{3}}$ and $1 + \frac{1}{\sqrt{3}}$ are the roots of $f'(x) = 0$

$\therefore \quad k = \underbrace{\left(1 + \frac{1}{\sqrt{3}}\right)}_{x}\underbrace{\left(\frac{1}{\sqrt{3}} - 1\right)}_{x-2}\underbrace{\left(\frac{1}{\sqrt{3}}\right)}_{x-1} = \left(\frac{1}{3} - 1\right)\left(\frac{1}{\sqrt{3}}\right) = \frac{-2}{3\sqrt{3}}$

9. Let $f(x)$ and $g(x)$ be two continuous functions defined from $R \longrightarrow R$, such that $f(x_1) > f(x_2)$ and $g(x_1) < g(x_2)$, $\forall\ x_1 > x_2$, then solution set of $f\left(g(\alpha^2 - 2\alpha)\right) > f\left(g(3\alpha - 4)\right)$ is :

माना $f(x)$ और $g(x)$, $R \longrightarrow R$ पर परिभाषित दो सतत् फलन है इस प्रकार की $f(x_1) > f(x_2)$ तथा $g(x_1) < g(x_2)$, $\forall\ x_1 > x_2$, तब $f\left(g(\alpha^2 - 2\alpha)\right) > f\left(g(3\alpha - 4)\right)$ का हल है :

(A) R (B) ϕ (C) (1, 4) (D) R – [1, 4]

Ans. C

Sol. obviously f is increasing and g is decreasing in (x_1, x_2)

hence $f\left(g(\alpha^2 - 2\alpha)\right) > f\left(g(3\alpha - 4)\right)$ as f is increasing

$\Rightarrow$ $g(\alpha^2 - 2\alpha) > g(3\alpha - 4)$

$\therefore$ $\alpha^2 - 2\alpha < 3\alpha - 4$ as g is decreasing

$\alpha^2 - 5\alpha + 4 < 0$

$(\alpha - 1)(\alpha - 4) < 0 \quad \Rightarrow \quad \alpha \in (1, 4)$

10. Given $f'(1) = 1$ and $\frac{d}{dx}\left(f(2x)\right) = f'(x) \quad \forall\, x > 0$. If $f'(x)$ is differentiable then there exists a number $c \in (2, 4)$ such that $f''(c)$ equals :

दिया है $f'(1) = 1$ तथा $\frac{d}{dx}\left(f(2x)\right) = f'(x) \quad \forall\, x > 0$. यदि $f'(x)$ अवकलनीय है तब (2, 4) में किसी संख्या c का अस्तित्व इस प्रकार से होगा कि $f''(c) =$

(A) – 1/4 (B) – 1/8 (C) 1/4 (D) 1/8

Ans. B

Sol. $f'(1) = 1; \qquad 2 \cdot f'(2x) = f'(x)$

put $x = 1, \quad f'(2) = \frac{f'(1)}{2} = \frac{1}{2}$

and $f'(4) = \frac{1}{2} f'(2) = \frac{1}{4}$

applying LMVT for $y = f'(x)$ is [2, 4]

$$f''(c) = \frac{f'(4) - f'(2)}{2} = \frac{\frac{1}{4} - \frac{1}{2}}{2} = -\frac{1}{8}$$

11. If $f(x) = \frac{x^2}{2 - 2\cos x}$; $g(x) = \frac{x^2}{6x - 6\sin x}$ where $0 < x < 1$, then :

(A) both 'f' and 'g' are increasing functions (B) 'f' is decreasing and 'g' is increasing function

(C) 'f' is increasing and 'g' is decreasing function (D) both 'f' and 'g' are decreasing function

यदि $f(x) = \frac{x^2}{2 - 2\cos x}$; $g(x) = \frac{x^2}{6x - 6\sin x}$ जहाँ $0 < x < 1$ हो, तो :

(A) 'f' एवं 'g' दोनों वर्द्धमान फलन हैं (B) 'f' ह्रासमान फलन है तथा 'g' वर्द्धमान फलन है

(C) 'f' वर्द्धमान फलन है तथा 'g' ह्रासमान फलन है (D) 'f' एवं 'g' दोनों ह्रासमान फलन हैं

Ans. C

Sol. $f(x) = \frac{x^2}{4}\cos ec^2\frac{x}{2}$

$f'(x) = \frac{x}{2}\cos ec^2\frac{x}{2}\cot\frac{x}{2}$ $\left(\tan\frac{x}{2} - \frac{x}{2}\right)$

For $0 < x < 1$, $\tan\frac{x}{2} > \frac{x}{2}$

$\Rightarrow$ $f'(x) > 0$ $\Rightarrow$ $f(x)$ is increasing.

12. Let $f(x) = \begin{cases} x^{\alpha} l nx, x > 0 \\ 0, \quad x = 0 \end{cases}$. Rolle's theorem is applicable to f for $x \in [0, 1]$, if $\alpha =$

माना $f(x) = \begin{cases} x^{\alpha} l nx, x > 0 \\ 0, \quad x = 0 \end{cases}$ फलन f के लिये $x \in [0, 1]$ में रोले प्रमेय सत्यापित है यदि $\alpha =$

(A) –2 (B) –1 (C) 0 (D) 1/2

Ans. D

Sol. Rolle's theorem is applicable to f for $x \in [0, 1]$.

This means that :

$f(1) = f(0)$ _____(1)

f is continuous in $x \in [0, 1]$ and f is differentiable for $x \in (0, 1)$ _____(2)

From (1) and (2) we get that : $f(1) = 0$

$0 = f(0) = \lim_{x \to 1} x^{\alpha}$ ln x which is true only for positive values of α.

13. If *f* is a strictly increasing function, then $\lim_{x \to 0}\frac{f(x^2) - f(x)}{f(x) - f(0)}$ is equal to :

यदि *f* एक वर्द्धमान फलन है तो $\lim_{x \to 0}\frac{f(x^2) - f(x)}{f(x) - f(0)}$ का मान :

(A) 0 (B) 1 (C) –1 (D) 2

Ans. C

Sol. $\text{Limit} = \lim_{x\to 0}\left(\frac{f(x^2)-f(x)}{f(x)-0}\right)$

$= \lim_{x\to 0}\left(\frac{f'(x^2).2x-f'(x)}{f'(x)}\right)$

$= \lim_{x\to 0}\left(\frac{0-f'(0)}{f'(0)}\right)$

$= -1$

14. Given that $f(x)$ is continuously differentiable on $a \le x \le b$ where $a < b$, $f(a) < 0$ and $f(b) > 0$, which of the following are always true?

(i) $f(x)$ is bounded on $a \le x \le b$.

(ii) The equation $f(x) = 0$ has at least one solution in $a < x < b$.

(iii) The maximum and minimum values of $f(x)$ on $a \le x \le b$ occur at points where $f'(c) = 0$.

(iv) There is at least one point c with $a < c < b$ where $f'(c) > 0$.

(v) There is at least one point d with $a < d < b$ where $f'(c) < 0$.

(A) only (ii) and (iv) are true (B) all but (iii) are true

(C) all but (v) are true (D) only (i), (ii) and (iv) are true

दिया है फलन $f(x)$ अन्तराल [a, b] में सतत् तथा अवकलनीय है जहाँ $a < b$, $f(a) < 0$ तथा $f(b) > 0$, तब निम्न में से कौन–कौन से हमेशा सत्य होगें?

(i) $f(x)$, $a \le x \le b$ में परिबद्ध है।

(ii) $f(x) = 0$ का कम से कम एक मूल $a < x < b$ में होगा।

(iii) $f(x)$ की अधिकतम और न्यूनतम मान $a \le x \le b$ में उन बिन्दुओं पर प्राप्त होगी जहाँ $f'(c) = 0$.

(iv) कम से कम एक बिन्दु c, $a < c < b$ होगा जहाँ $f'(c) > 0$

(v) कम से कम एक बिन्दु d, $a < d < b$ होगा जहाँ $f'(c) < 0$.

(A) केवल (ii) और (iv) सत्य है (B) सभी लेकिन (iii) नहीं

(C) सभी लेकिन (v) नहीं (D) केवल (i), (ii) और (iv) सत्य

Ans. D

Sol. (i) This statement is true, every continuous function is bounded on a closed interval

(ii) True again, by Intermediate Value Theorem

(iii) Not true, because maximum and / or minimum values could also occur at a or b, without the derivatives being 0.

(iv) True. By the Mean Value Theorem there exist a point between a and b where the derivative is exactly $\frac{f(b)-f(a)}{b-a}$, a clearly positive value.

(v) Not always true, for example the function might be strictly increasing guarenteeing the derivative to be always positive.

Thus the true statements are (i), (ii) and (iv) and the correct answer is (D)

15. Give the correct order of initials **T** or **F** for following statements. Use **T** if statement is true and **F** if it is false.

Statement-1: If $f : R \to R$ and $c \in R$ is such that f is increasing in $(c - \delta, c)$ and f is decreasing in $(c, c + \delta)$ then f has a local maximum at c. Where δ is a sufficiently small positive quantity.

Statement-2 : Let $f: (a, b) \to R$, $c \in (a, b)$. Then f can not have both a local maximum and a point of inflection at $x = c$.

Statement-3 : The function $f(x) = x^2 | x |$ is twice differentiable at $x = 0$.

Statement-4 : Let $f : [c - 1, c + 1] \to [a, b]$ be bijective map such that f is differentiable at c then f^{-1} is also differentiable at $f(c)$.

निम्न कथनों के सत्य **T** और असत्य **F** होने का सही क्रम क्या हैं?

कथन-1: यदि $f : R \to R$ और $c \in R$ इस प्रकार से है कि f अन्तराल $(c - \delta, c)$ में वर्धमान और f अन्तराल $(c, c + \delta)$ में ह्यसमान है तब f, c पर स्थानीय उच्चिष्ठ है जहाँ δ छोटी से छोटी धनात्मक संख्या है।

कथन-2 : माना $f : (a, b) \to R$, $c \in (a, b)$ तब f $x = c$ पर स्थानीय उच्चिष्ठ और नति परिवर्तन बिन्दु एक साथ नही देगा।

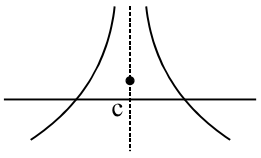
कथन-3 : फलन $f(x) = x^2 | x |$, $x = 0$ पर दो बार अवकलनीय है।

कथन-4 : माना $f : [c - 1, c + 1] \to [a, b]$ एकैकी आच्छादक फलन इसप्रकार है कि f, $x = c$ पर अवकलनीय है तब f^{-1} भी $f(c)$ पर अवकलनीय होगा।

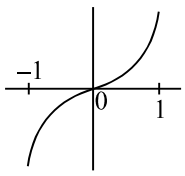
(A) FFTF (B) TTFT (C) FTTF (D) TTTF

Ans. C

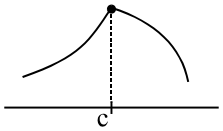
Sol. **S-1 :** f must be continuous $\Rightarrow$ false **S-2:** x=c is local max.but not inflection point $\Rightarrow$ True



S-1



S-4



S-2

S-4 : Consider $f(x) = x^3$ at $x = 0$

$f^{-1}(x) = x^{1/3}$, not differentiable at $x = 0$

16. In a regular triangular prism the distance from the centre of one base to one of the vertices of the other base is l. The altitude of the prism for which the volume is greatest :

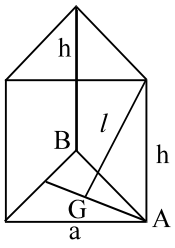
एक समबाहु प्रिज्म में आधार के केन्द्र से दूसरे आधार के शीर्ष की दूरी l है। प्रिज्म की ऊँचाई ज्ञात कीजिए, जिस के लिए आयतन अधिकतम हो

(A) $\frac{l}{2}$ (B) $\frac{l}{\sqrt{3}}$ (C) $\frac{l}{3}$ (D) $\frac{l}{4}$

Ans. B

Sol. $AG = \frac{2}{3}.\frac{a\sqrt{3}}{2} = \frac{a}{\sqrt{3}}$. Now $l^2 = \frac{a^2}{3} + h^2$; area $(A) = \frac{\sqrt{3}}{4}a^2$

$V(h) = \frac{\sqrt{3}}{4}a^2 h = \frac{\sqrt{3}}{4}h(l^2 - h^2)3;$ $\quad V'(h) = 0 \quad \Rightarrow \quad h = \frac{l}{\sqrt{3}}$; $V_{max} = \frac{l^3}{2}$]



17. Let h be a twice continuously differentiable positive function on an open interval J. Let $g(x) = \ln\big(h(x)\big)$ for each $x \in J$

Suppose $\big(h'(x)\big)^2 > h''(x)\, h(x)$ for each $x \in J$. Then :

(A) g is increasing on J (B) g is decreasing on J
(C) g is concave up on J (D) g is concave down on J

यदि h खुले अन्तराल J में दो बार अवकलनीय धनात्मक फलन है

माना $g(x) = \ln\big(h(x)\big)$ $\quad \forall\, x \in J$

एवं $\big(h'(x)\big)^2 > h''(x)\, h(x)$ $\quad \forall\, x \in J$ तब :

(A) g, J में वर्धमान है (B) g , J में ह्यसमान है
(C) g , Jमें ऊपर की और अवतल है (D) g , J में नीचे की ओर अवतल है

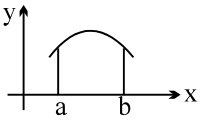
Ans. D

Sol. Given $g(x) = \ln\big(h(x)\big)$

$$g'(x) = \frac{h'(x)}{h(x)}$$

$$g''(x) = \frac{h(x)h''(x) - (h'(x))^2}{h^2(x)} < 0 \qquad \text{(given)}$$

$\therefore$ $\quad g''(x) < 0 \quad \Rightarrow \quad g(x)$ is concave down



18. If the function $f(x) = \frac{t + 3x - x^2}{x - 4}$, where 't' is a parameter has a minimum and a maximum then the range of values of 't' is :

यदि फलन $f(x) = \frac{t + 3x - x^2}{x - 4}$, जहाँ 't' प्राचल है का एक न्यूनतम और अधिकतम मान प्राप्त होता है तब 't' के मानो का परिसर होगा:

(A) $(0, 4)$ (B) $(0, \infty)$ (C) $(-\infty, 4)$ (D) $(4, \infty)$

Ans. C

Sol. $f(x) = \dfrac{t + 3x - x^2}{x - 4}$; $f'(x) = \dfrac{(x-4)(3-2x) - (t + 3x - x^2)}{(x-4)^2}$

for maximum or minimum, $f'(x) = 0$

$-2x^2 + 11x - 12 - t - 3x + x^2 = 0$

$-x^2 + 8x - (12 + t) = 0$

for one M and m,

$D > 0$

$64 - 4(12 + t) > 0$

$16 - 12 - t > 0 \Rightarrow 4 > t$ or $t < 4$

19. P and Q are two points on a circle of centre C and radius α, the angle PCQ being 2θ then the radius of the circle inscribed in the triangle CPQ is maximum when :

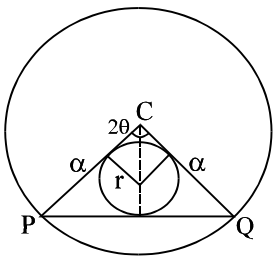
C केन्द्र और α त्रिज्या वाले वृत्त पर P और Q दो बिन्दु है यदि कोण PCQ, 2θ है तब उस त्रिभुज CPQ के अन्तःवृत्त की त्रिज्या अधिकतम होगी यदि :

(A) $\sin\theta = \dfrac{\sqrt{3}-1}{2\sqrt{2}}$ (B) $\sin\theta = \dfrac{\sqrt{5}-1}{2}$ (C) $\sin\theta = \dfrac{\sqrt{5}+1}{2}$ (D) $\sin\theta = \dfrac{\sqrt{5}-1}{4}$

Ans. B

Sol. $r = \dfrac{\Delta}{s}$ where Δ = Area of triangle CPQ and s = semiperimeter of ΔCPQ.

$$r = \frac{\alpha^2 \sin 2\theta}{2s} = \frac{\alpha^2 \sin 2\theta}{2\alpha + 2\alpha \sin\theta} = \frac{\alpha}{2} \cdot \frac{\sin 2\theta}{1 + \sin\theta}$$



Consider $f(\theta) = \dfrac{\sin 2\theta}{1 + \sin\theta}$

$$f'(\theta) = \frac{(1+\sin\theta)\,2\cos 2\theta - \sin 2\theta . \cos\theta}{(1+\sin\theta)^2} = 0$$

$2(1 + \sin\theta)(1 - 2\sin^2\theta) - 2\sin\theta\,(1 - \sin^2\theta) = 0$

$2(1 - 2\sin^2\theta) = 2\sin\theta\,(1 - \sin\theta)$

$1 - 2\sin^2\theta = \sin\theta - \sin^2\theta$

$\sin^2\theta + \sin\theta - 1 = 0$

$\sin\theta = \dfrac{-1 \pm \sqrt{1+4}}{2}$ $\therefore \quad \sin\theta = \dfrac{\sqrt{5}-1}{2}$

20. The range of values of m for which the line $y = mx$ and the curve $y = \frac{x}{x^2+1}$ enclose a region, is :

m के मानों का परिसर क्या होगा जबकि $y = mx$ और वक्र $y = \frac{x}{x^2+1}$ एक परिबद्ध क्षेत्र बनाते है :

(A) $(-1, 1)$ (B) $(0, 1)$ (C) $[0, 1]$ (D) $(1, \infty)$

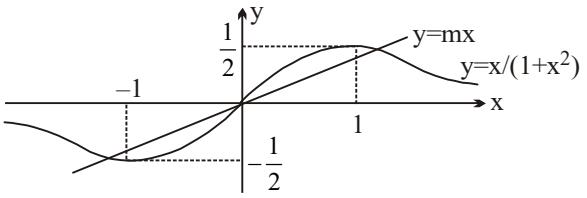
Ans. B

Sol. Solving

$mx = \frac{x}{x^2+1} \Rightarrow x^2 + 1 = \frac{1}{m}$ or $x = 0$

$x^2 = \frac{1}{m} - 1 > 0$ for a region

$\frac{m-1}{m} < 0 \Rightarrow m \in (0, 1)$



Note: for $m = 0$ or 1 the line does not enclose a region

21. For a steamer the consumption of petrol (per hour) varies as the cube of its speed (in km). If the speed of the current is steady at C km/hr then the most economical speed of the steamer going against the current will be :

किसी पानी के जहाज के लिए एक घण्टे में ईधन की खपत उसकी किलोमीटर में चाल के घन के समानुपाती है। यदि धारा की नियत चाल C km/hr है तो जहाज की सबसे कम खर्चीली गति धारा के विपरीत दिशा में क्या होनी चाहिये :

(A) 1.25 C (B) 1.5 C (C) 1.75C (D) 2 C

Ans. B

Sol. Time of journey $= \frac{d}{V-C}$ when V is the speed in still water.

$\therefore$ petrol burnt per hour $= kV^3$

$\therefore$ fuel cost $= \frac{kV^3 d}{V-C} = kd\,\frac{V^3}{V-C} = f(v)$

now proceed

22. Let f, g and h be real-valued functions defined on the interval $[0, 1]$ by $f(x) = e^{x^2} + e^{-x^2}$, $g(x) = x\,e^{x^2} + e^{-x^2}$ and $h(x) = x^2 e^{x^2} + e^{-x^2}$. If a, b and c denote respectively, the absolute maximum of f, g and h on $[0, 1]$, then :

(A) $a = b$ and $c \neq b$ (B) $a = c$ and $a \neq b$ (C) $a \neq b$ and $c \neq b$ (D) $a = b = c$

माना f, g तथा h अन्तराल [0, 1] में $f(x) = e^{x^2} + e^{-x^2}$, $g(x) = x e^{x^2} + e^{-x^2}$ तथा $h(x) = x^2 e^{x^2} + e^{-x^2}$ द्वारा परिभाषित फलन है। यदि a, b तथा c, f, g तथा h के अन्तराल [0, 1] पर परम उच्चिष्ठ मान है, तो :

(A) a = b तथा c ≠ b (B) a = c तथा a ≠ b (C) a ≠ b तथा c ≠ b (D) a = b = c

Ans. D

Sol. If $x \in [0, 1]$

then $x^2 \le x \le 1$

$x^2 e^{x^2} \le x e^{x^2} \le e^{x^2}$

Add e^{-x^2} to all sides

$x^2 e^{x^2} + e^{-x^2} \le x e^{x^2} + e^{-x^2} \le e^{x^2} + e^{-x^2}$

$\Rightarrow$ $h(x) \le g(x) \le f(x)$ ..... (i)

where, $f(x) = e^{x^2} + e^{-x^2}$

$f'(x) = 2x(e^{x^2} - e^{-x^2}) > 0$

$\Rightarrow$ f(x) has a maxima at x = 1

$\Rightarrow$ $a = e + \frac{1}{e}$

$h(x) = x^2 e^{x^2} + e^{-x^2}$

$h'(x) = 2x^3 e^{x^2} + 2x e^{x^2} - 2x e^{-x^2}$

$= 2x^3 e^{x^2} + 2x(e^{x^2} - e^{-x^2}) > 0$

$\Rightarrow$ h(x) has a maxima at x = 1

$\Rightarrow$ $c = e + \frac{1}{e}$

$\because$ $h(x) \le g(x) \le f(x)$

$\Rightarrow$ g(x) also has a maximum value at x = 1

$\Rightarrow$ a = b = c

ONE OR MORE THAN ONE CORRECT TYPE QUESTIONS

23. Consider the curve $f(x) = x^{1/3}$, then :

(A) the equation of tangent at (0, 0) is x = 0 (B) the equation of normal at (0, 0) is y = 0

(C) normal to the curve does not exist at (0, 0) (D) f(x) and its inverse meet at exactly 3 points

माना वक्र $f(x) = x^{1/3}$ है, तो :

(A) (0, 0) पर स्पर्श रेखा का समीकरण x = 0 है (B) (0, 0) पर अभिलम्ब का समीकरण y = 0 है

(C) (0, 0) पर अभिलम्ब अस्तित्व में नहीं है (D) f(x) तथा इसका प्रतिलोम ठीक तीन बिन्दुओं पर मिलते है

Ans. ABD

Sol. $f'(x) = \frac{1}{3x^{2/3}}$

$f'(0) \to \infty$ tangent is vertical at $x = 0$

Equation of tangent at (0, 0) is $x = 0$

Equation of normal is $y = 0$

$f(x) = f^{-1}(x)$

$x^{\frac{1}{3}} = x^3 \quad \Rightarrow \quad x^9 = x$

$\Rightarrow \quad x = 0 \; ; \; 1 \quad ; -1$

24. If the line, $ax + by + c = 0$ is a normal to the curve $xy = 2$, then :

यदि रेखा $ax + by + c = 0$ वक्र $xy = 2$ पर अभिलम्ब हो, तो :

(A) $a < 0, b > 0$ (B) $a > 0, b < 0$ (C) $a > 0, b > 0$ (D) $a < 0, b < 0$

Ans. AB

Sol. Given $xy = 2 \Rightarrow y = \frac{2}{x}$

$\Rightarrow \frac{dy}{dx} = -\frac{2}{x^2}$ ..... slope of tangent

Thus, slope of normal $= \frac{x^2}{2}$ (which is always positive) and it is given that

$ax + by + c = 0$ is normal whose slope $= \frac{-a}{b}$.

$\Rightarrow -\frac{a}{b} > 0$ or $\frac{a}{b} < 0$

$\therefore$ a and b are of opposite sign.

Hence if $a > 0, b < 0$ or $a < 0, b > 0$

25. The normal to the curve $x = a(\cos\theta + \theta \sin\theta)$, $y = a(\sin\theta - \theta \cos\theta)$ at any point 'θ' is such that :

(A) it passes through the origin (B) it makes anlge $\frac{\pi}{2} + \theta$ with the x-axis

(C) it passes through $\left(a\frac{\pi}{2}, -a\right)$ (D) it is at a constant distance from the origin

वक्र $x = a(\cos\theta + \theta\sin\theta)$, $y = a(\sin\theta - \theta\cos\theta)$ के बिन्दु 'θ' पर खींचा गया अभिलम्ब ऐसा है, कि :

(A) वह मूल बिन्दु से होकर जाता है

(B) वह x-अक्ष के साथ $\frac{\pi}{2} + \theta$ का कोण बनाता है

(C) वह बिन्दु $\left(a\frac{\pi}{2}, -a\right)$ से होकर जाता है

(D) वह मूल बिन्दु से अचर दूरी पर है

Ans. BD

Sol. $x = a(\cos\theta + \theta\sin\theta)$ & $y = a(\sin\theta - \theta\cos\theta)$

$\frac{dx}{d\theta} = a(-\sin\theta + \sin\theta + \theta\cos\theta)$,

$\frac{dy}{d\theta} = a(\cos\theta - \cos\theta + \theta\sin\theta)$

$\therefore \frac{dy}{dx} = \frac{\sin\theta}{\cos\theta}$

slope of normal $= -\frac{\cos\theta}{\sin\theta} = -\cot\theta$

it makes angle $\left(\frac{\pi}{2} + \theta\right)$ with the x-axis

eq of normal $y - a\sin\theta + a\,\theta\cos\theta = -\frac{\cos\theta}{\sin\theta}$

$(x - a\cos\theta - a\,\theta\sin\theta)$

$\Rightarrow x\cos\theta + y\sin\theta = a.$

Hence it is at a constant distance 'a' from the origin.

26. Which of the following functions do not satisfy conditions of Rolle's Theorem?

निम्नलिखित में से कौनसे फलन रोल प्रमेय के प्रतिबन्धों को संतुष्ट नहीं करते हैं?

(A) $e^x \sin x, x \in \left[0, \frac{\pi}{2}\right]$

(B) $(x+1)^2(2x-3)^5, x \in \left[-1, \frac{3}{2}\right]$

(C) $\sin|x|, x \in [\pi, 2\pi]$

(D) $\sin\frac{1}{x}, x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

Ans. AD

Sol. (A) $f(0) = 0$

$f\left(\frac{\pi}{2}\right) = e^{\frac{\pi}{2}}$

$f(0) \neq f\left(\frac{\pi}{2}\right)$

(D) $f\left(\frac{-\pi}{2}\right) = -\sin\left(\frac{2}{\pi}\right)$

$f\left(\frac{\pi}{2}\right) = \sin\left(\frac{2}{\pi}\right)$

$f\left(-\frac{\pi}{2}\right) \neq f\left(\frac{\pi}{2}\right)$

27. The function $y = 2x^2 - \ell n\,|x|$ is monotonically increasing in the interval I_1 and montonically decreasing in the interval I_2, $x\ (\neq 0)$, then :

फलन $y = 2x^2 - \ell n\,|x|$ अन्तराल I_1 में एकदिष्ट वर्द्धमान है तथा अन्तराल I_2 में एकदिष्ट ह्रासमान है $x\ (\neq 0)$, तो :

(A) $I_1 = \left(-\frac{1}{2}, 0\right) \cup \left(\frac{1}{2}, \infty\right)$

(B) $I_2 = \left(-\infty, -\frac{1}{2}\right) \cup \left(0, \frac{1}{2}\right)$

(C) $I_1 = \left(-\infty, -\frac{1}{2}\right) \cup \left(0, \frac{1}{2}\right)$

(D) $I_2 = \left(-\frac{1}{2}, 0\right) \cup \left(\frac{1}{2}, \infty\right)$

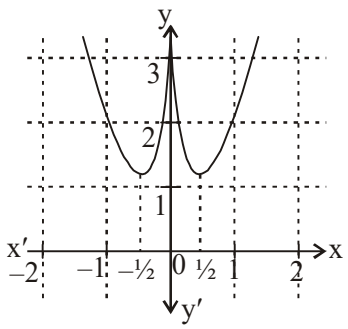
Ans. AB

Sol. $y = 2x^2 - \ln|x|$

$$\frac{dy}{dx} = 4x - \frac{1}{x} = \frac{(2x+1)(2x-1)}{x}$$

Critical points are 0, 1/2, –1/2

Sign scheme of $\frac{dy}{dx}$ is



Clearly f(x) increases in

$\left(-\frac{1}{2}, 0\right) \cup \left(\frac{1}{2}, \infty\right)$ and f(x) decreases

in $\left(-\infty, -\frac{1}{2}\right) \cup \left(0, \frac{1}{2}\right)$.

28. For the function $f(x) = x^4 (12\, l\text{n}x - 7)$:

(A) the point $(1, -7)$ is the point of inflection (B) $x = e^{1/3}$ is the point of minima

(C) the graph is concave downwards in $(0, 1)$ (D) the graph is concave upwards in $(1, \infty)$

फलन $f(x) = x^4 (12\, l\text{n}x - 7)$ के लिए :

(A) बिन्दु $(1, -7)$ नति परिवर्तन बिन्दु है (B) $x = e^{1/3}$ निम्ननिष्ठि का बिन्दु है

(C) आलेख $(0, 1)$ में उत्तल है (D) आरेख $(1, \infty)$ में अवतल है

Ans. ABCD

Sol. $y = f(x) = x^4 (12\log_e x - 7), x > 0$

$$\therefore \frac{dy}{dx} = 16x^3 (3\log_e x - 1) \text{ and } \frac{d^2y}{dx^2} = 16x^2 (9\log_e x)$$

$$\frac{dy}{dx} = 0 \to x = e^{\frac{1}{3}}$$

At $x = -e^{\frac{1}{3}}, \frac{d^2y}{dx^2} > 0$, hence $x = e^{\frac{1}{3}}$ is point of minima

Also for $0 < x < 1, \frac{d^2y}{dx^2} < 0$ and for $x > 1, \frac{d^2y}{dx^2} > 0$

Hence $x = 1$ is point of inflection and for $< x < 1$ graph is concave downward and for $x > 1$ graph is concave upward.

29. Which of the following statements is/are TRUE?

(A) If f has domain $[0, \infty)$ and has no horizontal asymptotes then $\lim\limits_{x\to\infty} f(x) = \infty$ or $\lim\limits_{x\to\infty} f(x) = -\infty$

(B) If f is continuous on $[-1, 1]$ and $f(-1) = 4$ and $f(1) = 3$ then there exist a number r such that $|r| < 1$ and $f(r) = \pi$

(C) $\lim\limits_{x\to\infty} \arcsin\left(\frac{x+1}{x}\right)$ does not exist

(D) For all values of $m \in R$ the line $y - mx + m - 1 = 0$ cuts the circle $x^2 + y^2 - 2x - 2y + 1 = 0$ orthogonally

निम्न कथनों में सत्य कथन है :

(A) यदि f का प्रान्त $[0, \infty)$ है और कोई भी क्षेतिज अनन्तस्पर्शी नहीं तो $\lim\limits_{x\to\infty} f(x) = \infty$ या $\lim\limits_{x\to\infty} f(x) = -\infty$

(B) यदि f,$[-1, 1]$ में सतत् है तथा $f(-1) = 4$ और $f(1) = 3$ तो संख्या r विद्यमान होगी की $|r| < 1$ और $f(r) = \pi$

(C) $\lim\limits_{x\to\infty} \arcsin\left(\frac{x+1}{x}\right)$ विद्यमान नही है

(D) रेखा $y - mx + m - 1 = 0$ वृत्त $x^2 + y^2 - 2x - 2y + 1 = 0$ को सभी $m \in R$ के लिये लम्बवत् काटती है

Ans. BCD

Sol. (A) False, e.g. $f(x) = \sin\sqrt{x}$

(B) True, from IVT

(C) True as $\lim\limits_{x\to\infty} \sin^{-1}\left(1+\frac{1}{x}\right) = \sin^{-1}(\text{a quantity greater than one}) \Rightarrow$ not defined

(D) True, as the line passes through the centre of the circle.

30. A function f is defined by $f(x) = \int_0^{\pi} \cos t \cos(x-t)dt$, $0 \le x \le 2\pi$ then which of the following hold(s) good :

(A) $f(x)$ is continuous but not differentiable in $(0, 2\pi)$

(B) Maximum value of f is π

(C) There exists atleast one $c \in (0, 2\pi)$ s.t. $f'(c) = 0$

(D) Minimum value of f is $-\frac{\pi}{2}$

यदि फलन $f(x) = \int_0^{\pi} \cos t \cos(x-t)dt$, $0 \le x \le 2\pi$ तो निम्न में से सत्य है :

(A) $f(x)$, $(0, 2\pi)$ में सतत् है लेकिन अवकलनीय नहीं

(B) $f(x)$ का अधिकतम मान π है

(C) कम से कम एक ऐसा विद्यमान है कि $c \in (0, 2\pi)$ इस प्रकार कि $f'(c) = 0$.

(D) f का न्यूनतम मान $-\frac{\pi}{2}$ है

Ans. CD

Sol. $f(x) = \int_0^{\pi} \cos t \cos(x-t)dt$ ....(1)

$= \int_0^{\pi} -\cos t \cdot \cos(x-\pi+t)dt$ (using King)

$f(x) = \int_0^{\pi} \cos t \cdot \cos(x+t)dt$ ....(2)

(1) + (2) gives

$$2f(x) = \int_0^{\pi} \cos t\,(2\cos x \cdot \cos t)dt$$

$$\therefore \quad f(x) = \cos x \int_0^{\pi} \cos^2 t\,dt = 2\cos x \int_0^{\pi/2} \cos^2 t\,dt$$

$f(x) = \frac{\pi \cos x}{2}$ Now verify. Only (A) & (B) are correct.

Alternatively: Convert the integrand into sum of two cosine functions.

31. Let $f(x)$ be a polynomial of degree 3 having a local maximum at $x = -1$. If $f(-1) = 2$, $f(3) = 18$, and $f'(x)$ has a local minimum at $x = 0$, then :

(A) distance between $(-1, 2)$ and $(a, f(a))$, which are the points of local maximum and local minimum on the curve $y = f(x)$ is $2\sqrt{5}$

(B) $f(x)$ is a decreasing function for $1 \le x \le 2\sqrt{5}$

(C) $f'(x)$ has a local maximum at $x = 2\sqrt{5}$

(D) $f(x)$ has a local minimum at $x = 1$

दिया है f(x) एक तीन घात का बहुपद फलन इस प्रकार से है कि जो $x = -1$ पर महत्तम है। यदि $f(-1) = 2$, $f(3) = 18$, तथा $f'(x)$ का $x = 0$ पर न्यूनतम है, तो :

(A) बिन्दुओं $(-1, 2)$ तथा $(a, f(a))$ के बीच की दूरी, जो कि वक्र $y = f(x)$ पर महत्तम एवं न्यूनतम बिन्दु है, $2\sqrt{5}$ है

(B) $f(x)$, $1 \le x \le 2\sqrt{5}$ के लिये ह्रासमान फलन है

(C) $f'(x)$, $x = 2\sqrt{5}$ पर उच्चिष्ठ है

(D) $f(x)$, $x = 1$ पर निम्निष्ठ है

Ans. AD

Sol. Let $f(x) = ax^3 + bx^2 + cx + cl$

$f'(x) = 3ax^2 + 2bx + c$

$f'(-1) = 0$ $\qquad$ $f(-1) = 2$

$3a - 2b + c = 0$ $\qquad$ $a + b - c + d = 2$

$f(3) = 18$

$27a + 90 + 3(1-d) = 18$

$f''(x) = 6ax + 2b$

$f''(0) = 0$

$b = 0$

$3a + c = 0$

$-a + b - c + d = 2$ $\qquad$ $2a + d = 2$

$27a + 9b + 3c + d = 18$

$27a + 3c + d = 18$

$16a + 2 = 18$

$a = 1$

$c = -3$

$d = 0$

$f(x) = x^3 - 3x$

32. Consider the function $f(x) = x^2 - x \sin x - \cos x$ then the statements which holds good, are :

(A) $f(x) = k$ has no solution for $k < -1$

(B) f is increasing for $x < 0$ and decreasing for $x > 0$

(C) $\lim\limits_{x \to \pm\infty} f(x) \to \infty$

(D) The zeros of $f(x) = 0$ lie on the same side of the origin

माना फलन $f(x) = x^2 - x \sin x - \cos x$ तो निम्न में से सत्य कथन होगें :

(A) $f(x) = k$ का कोई हल नहीं होगा यदि $k < -1$

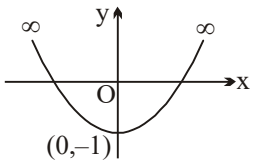
(B) f; $x < 0$ के लिये वर्धमान तथा $x > 0$ के लिये ह्यसमान है

(C) $\lim\limits_{x \to \pm\infty} f(x) \to \infty$

(D) $f(x) = 0$ के मूल बिन्दु के एक ही और होगें

Ans. AC

Sol. Graph of $y = f(x) \Rightarrow$ (A) and (C)



33. For the function $f(x) = \ln(1 - \ln x)$ which of the following do not hold good?

(A) increasing in (0, 1) and decreasing in (1, e) (B) decreasing in (0, 1) and increasing in (1, e)

(C) $x = 1$ is the critical number for $f(x)$. (D) f has two asymptotes

फलन $f(x) = \ln(1 - \ln x)$ के लिये असत्य है :

(A) (0, 1) में वर्धमान और (1, e) में ह्यसमान (B) (0, 1) में ह्यसमान और (1, e) में वर्धमान

(C) $x = 1$, $f(x)$ का क्रान्तिक बिन्दु है (D) f की दो अनन्त स्पर्शीयाँ है

Ans. ABC

Sol. $f(x) = \ln(1 - \ln x)$

domain (0, e)

$$f'(x) = -\frac{1}{(1 - \ln x)} \cdot \frac{1}{x} < 0 \Rightarrow \text{decreasing } \forall \text{ x in its domain} \Rightarrow \text{(A) \& (B) are incorrect}$$

$f'(1) = -1 \quad \Rightarrow \quad$ (C) is also incorrect

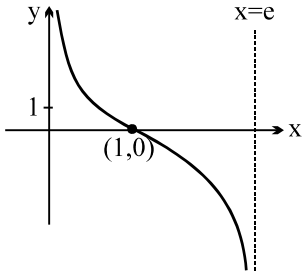
also $f(1) = 0$; $\lim\limits_{x \to e^{-1}} f(x) \to -\infty$; $\lim\limits_{x \to 0^+} f(x) \to \infty$

$$f''(x) = \frac{-\ln x}{x^2(1-\ln x)^2}$$

$f''(1) = 0$ which is a point of inflection

graph is as shown

y axis and x = e are two asynmptotes



34. Which of the following statement(s) is/are True :

(A) If f is a continuous real valued function defined on [0, 1] then $\exists$ positive number c such that $|f(x) - f(y)| \le c$ for all x, y in [0, 1].

(B) If f is continuous on R such that $\lim\limits_{x \to -\infty} f(x) = -\infty$ and $\lim\limits_{x \to \infty} f(x) = \infty$ then the equation $f(x) = k, k \in R$ has at least a root in R

(C) If f and g are derivable and $f(x) > g(x)$ for $a < x < b$ then $f'(x) > g'(x)$ for $a < x < b$

(D) If f and g are continuous and $f(x) > g(x)$ for $a < x < b$ then $\int_a^b f(x)\,dx > \int_a^b g(x)\,dx$

निम्न में से सत्य कथन है :

(A) यदि f वास्तविक मानों का सतत् फलन [0, 1] में परिभाषित है तो धनात्मक c इस प्रकार से होगा की $|f(x) - f(y)| \le c, \ \forall\ x, y \in [0, 1]$.

(B) यदि f, R में सतत् है इस प्रकार की $\lim\limits_{x \to -\infty} f(x) = -\infty$ और $\lim\limits_{x \to \infty} f(x) = \infty$ तो $f(x) = k, k \in R$ का कम से कम एक मूल R में है

(C) यदि f और g अवकलनीय है और $f(x) > g(x)$ for $a < x < b$ के लिये तो $f'(x) > g'(x); a < x < b$

(D) यदि f और g सतत् फलन है और $f(x) > g(x); \ a < x < b$ तो $\int_a^b f(x)\,dx > \int_a^b g(x)\,dx$

Ans. ABD

Sol. (A) as f is continuous, f is bounded on a closed interval $\Rightarrow$ True

(B) Consider $g(x) = f(x) - k$ obviously g is continuous and $\lim\limits_{x \to \infty} g(x) = \infty$ and $\lim\limits_{x \to -\infty} g(x) = -\infty$

hence g (x) must vanish as least once in $(-\infty, \infty)$

$\therefore$ $f(x) - k = 0 \quad \Rightarrow \quad f(x) = k$ (using IVT)

(D) $\left.\begin{array}{l}\text{Consider } f(x) = \cos x \\ \text{and } \quad g(x) = \sin x\end{array}\right]$ on $\left(0, \frac{\pi}{4}\right)$.

Note that $f(x) > g(x)$ on $\left(0, \frac{\pi}{4}\right)$

now $f'(x) = -\sin x$ and $g'(x) = \cos x$

but $f'(x) \not> g'(x)$ on $\left(0, \frac{\pi}{4}\right)$ i.e. $-\sin x \not> \cos x$ in $\left(0, \frac{\pi}{4}\right)$

Exercise C - II

SUBJECTIVE / INTEGER TYPE QUESTIONS

1. If the tangent at the point (x_1, y_1) to the curve $x^3 + y^3 = a^3 (a \neq 0)$ meets the curve again in (x_2, y_2) then show that

$$\frac{x_2}{x_1} + \frac{y_2}{y_1} = -1.$$

यदि वक्र $x^3 + y^3 = a^3 (a \neq 0)$ पर बिन्दु (x_1, y_1) पर स्पर्श रेखा वक्र को (x_2, y_2) पर दुबारा मिलती है तो सिद्ध कीजिये $\frac{x_2}{x_1} + \frac{y_2}{y_1} = -1.$

Ans.

Sol. Given $x^3 + y^3 = a^3$. The derivative is

$$\frac{dy}{dx} = -\frac{x^2}{y^2}$$

Therefore, slope of tangent at (x_1, y_1) is

$$m_T = -\frac{x_1^2}{y_1^2}$$

The tangent passes through (x_2, y_2), therefore, slope of the tangent is also given by

$$m_T = \frac{y_2 - y_1}{x_2 - x_1}$$

Comparing the two slope equations we get,

$$\frac{y_2 - y_1}{x_2 - x_1} = -\frac{x_1^2}{y_1^2}$$

$$\Rightarrow \frac{y_2^3 - y_1^3}{x_2^3 - x_1^3} \times \frac{x_1^2 + x_1x_2 + x_2^2}{y_1^2 + y_1y_2 + y_2^2} = -\frac{x_1^2}{y_1^2}$$

$$\Rightarrow -\frac{x_1^2 + x_1x_2 + x_2^2}{y_1^2 + y_1y_2 + y_2^2} = -\frac{x_1^2}{y_1^2}$$

$$\Rightarrow x_1^2y_1^2 + x_1x_2y_1^2 + x_2^2y_1^2 = x_1^2y_1^2 + x_1^2y_1y_2 + x_1^2y_2^2$$

$$\Rightarrow x_1^2y_2^2 - x_2^2y_1^2 = x_1x_2y_1^2 - x_1^2y_1y_2$$

$$\Rightarrow (x_1y_2 - x_2y_1)(x_1y_2 + x_2y_1) = x_1y_1(x_2y_1 - x_1y_2)$$

$$\Rightarrow x_1y_2 + x_2y_1 = -x_1y_1$$

$$\Rightarrow \frac{x_2}{x_1} + \frac{y_2}{y_1} = -1$$

2. Show that the condition that the curves $x^{2/3} + y^{2/3} = c^{2/3}$ and $(x^2/a^2) + (y^2/b^2) = 1$ may touch if $c = a + b$.

सिद्ध कीजिये की वक्रो $x^{2/3} + y^{2/3} = c^{2/3}$ तथा $(x^2/a^2) + (y^2/b^2) = 1$ के एक दूसरे को स्पर्श करने की शर्त $c = a + b$ होगी।

Ans.

Sol. $C_1 : x^{\frac{2}{3}} + y^{\frac{2}{3}} = c^{\frac{2}{3}}$

$$C_2 : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

Let point of tangency is $P = (c\cos^3\theta, c\sin^3\theta)$

Slope of tangent at C_1

$$\frac{2}{3}x^{-\frac{1}{3}} + \frac{2}{3}y^{-\frac{1}{3}}\frac{dy}{dx} = 0$$

$$\Rightarrow \frac{dy}{dx} = -\left(\frac{y}{x}\right)^{\frac{1}{3}}$$

$$\Rightarrow \left.\frac{dy}{dx}\right|_{\text{at P}} = -\tan\theta \qquad \text{_____(1)}$$

Similarly slope of tangent at C_2

$$\frac{2x}{a^2} + \frac{2y}{b^2}\frac{dy}{dx} = 0$$

$$\Rightarrow \frac{dy}{dx} = -\frac{x}{y}\frac{b^2}{a^2}$$

$$\Rightarrow \left.\frac{dy}{dx}\right|_{\text{at P}} = -\cot^3\theta\frac{b^2}{a^2} \qquad \text{_____(2)}$$

from (1) & (2)

$$-\tan\theta = -\cot^3\theta\frac{b^2}{a^2}$$

$$\Rightarrow \tan\theta = \sqrt{\frac{b}{a}} \Rightarrow \sin\theta = \sqrt{\frac{b}{a+b}} \ \& \ \cos\theta = \sqrt{\frac{a}{a+b}}$$

$$P\left(c\cos^3\theta, c\sin^3\theta\right) = P\left(c\left(\sqrt{\frac{a}{a+b}}\right)^3, c\left(\sqrt{\frac{b}{a+b}}\right)^3\right)$$

$$\text{Put in } C_2 \Rightarrow \frac{c^2a}{(a+b)^3} + \frac{c^2b}{(a+b)^3} = 1$$

$$\Rightarrow c^2 = (a+b)^2$$

$$\Rightarrow c = a + b$$

3. A curve is given by the equations $x = at^2$ & $y = at^3$. A variable pair of perpendicular lines through the origin 'O' meet the curve at P & Q. Show that the locus of the point of intersection of the tangents at P & Q is $4y^2 = 3ax - a^2$.

एक वक्र जिसका समीकरण $x = at^2$ तथा $y = at^3$ है। मूल बिन्दु 'O' से लम्बवत् रेखाओं का चर रेखा युग्म वक्र को P तथा Q पर प्रतिच्छेद करता है। सिद्ध कीजिये P तथा Q पर स्पर्श रेखाओं के प्रतिच्छेद बिन्दु का बिन्दुपथ $4y^2 = 3ax - a^2$ है।

Ans.

Sol. The slope of the line joining any point (at^2, at^3) with the origin $= \frac{at^3}{at^2} = t$

$\Rightarrow$ For any two points t_1, t_2 products of such slopes $= t_1t_2$

Now for perpendicular lines : $t_1t_2 = -1$ ...(1)

We will need the slope of the tangent at a general point lying on the curve

$$\frac{dy}{dx} = \frac{dy}{dt}.\frac{dt}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{3at^2}{2at} = \frac{3}{2}t$$

Using the point-slope form, the equation of the tangent to the curve at

$$P(t_1): y - y(t_1) = \left.\frac{dy}{dx}\right|_{t_1} (x - x(t_1))$$

$$y - at_1^3 = \left.\frac{dy}{dx}\right|_{t_1} \left(x - at_1^2\right)$$

$$y - at_1^3 = \frac{3t_1}{2}\left(x - at_1^2\right) \quad ...(2)$$

And the equation of the tangent to the curve at

$$Q(t_2): y - y(t_2) = \left.\frac{dy}{dx}\right|_{t_2} (x - x(t_2))$$

$$y - at_2^3 = \frac{3t_2}{2}\left(x - at_2^2\right)$$

$$y = \frac{3t_2}{2}\left(x - at_2^2\right) + at_2^3 \quad ...(3)$$

To find the point of intersection of these two lines, equate (3) and (2) and simplify to get

$$x = \frac{a}{3}\left(t_1^2 + t_2^2 + t_1t_2\right)$$

which can be substituted back into (2) to get

$$y = \frac{at_1t_2}{2}(t_1 + t_2)$$

Now, since $t_1t_2 = -1$

Then the point of intersection of the tangents can be written as

$$x = \frac{a}{3}\left(t_1^2 + \frac{1}{t_1^2} - 1\right) = -\frac{a}{2}\left(t_1 - \frac{1}{t_1}\right)$$

Now, to manipulate these equations into the desired form,

$$y = -\frac{a}{2}\left(t_1 - \frac{1}{t_1}\right)$$

$$4y^2 = a^2\left(t_1 - \frac{1}{t_1}\right)^2$$

$$4y^2 = a^2\left(t_1^2 + \frac{1}{t_1^2} - 2\right) \qquad ...(4)$$

$$x = \frac{a}{3}\left(t_1^2 + \frac{1}{t_1^2} - 1\right)$$

$$3x = a\left(t_1^2 + \frac{1}{t_1^2} - 1\right)$$

$$3ax - a^2 = a^2\left(t_1^2 + \frac{1}{t_1^2} - 2\right) \qquad ...(5)$$

Now eliminating t_1 between (4) and (5), we get

$4y^2 = 3ax - a^2$

4. A and B are points of the parabola $y = x^2$. The tangents at A and B meet at C. The median of the triangle ABC from C has length 'm' units. Find the area of the triangle in terms of 'm'.

A तथा B परवलय $y = x^2$ पर स्थित है। A तथा B पर स्पर्श रेखाऐं C पर मिलती है। त्रिभुज ABC में C से माध्यिका की लम्बाई 'm' है। त्रिभुज ABC का क्षेत्रफल 'm' के पदों में निकालिए।

Ans. $\frac{m\sqrt{m}}{\sqrt{2}}$

Sol. Mid point of AB is $\left(\frac{t_1 + t_2}{2}, \frac{t_1^2 + t_2^2}{2}\right)$

Equation of tangent at point A is

$y + t_1^2 = 2xt_1$ ... (i)

Equation of tangent at point B is

$y + t_2^2 = 2xt_2$ ... (ii)

(i) – (ii) $\Rightarrow$ $t_1^2 - t_2^2 = 2x(t_1 - t_2)$

$x = \frac{(t_1 + t_2)}{2}$

$y + t_1^2 = t_1(t_1 + t_2)$

$y = t_1t_2$

Point C is $\left(\frac{(t_1 + t_2)}{2}, t_1t_2\right)$

Length of median is $\left|\frac{t_1^2 + t_2^2}{2} - t_1t_2\right|$

$$= \frac{(t_1 - t_2)^2}{2} = m$$

Area of $\Delta ABC = \frac{1}{2}\begin{vmatrix} t_1 & t_1^2 & 1 \\ t_2 & t_2^2 & 1 \\ \frac{t_1 + t_2}{2} & t_1t_2 & 1 \end{vmatrix}$

$R_2 \to R_2 - R_1$ & $R_3 \to R_3 - R_1$

$$= \frac{1}{2}\begin{vmatrix} t_1 & t_1^2 & 1 \\ (t_2 - t_1) & (t_2^2 - t_1^2) & 0 \\ \frac{(t_2 - t_1)}{2} & t_1(t_2 - t_1) & 0 \end{vmatrix}$$

$$= \frac{(t_2 - t_1)^2}{2}\begin{vmatrix} t_1 & t_1^2 & 1 \\ 1 & (t_2 + t_1) & 0 \\ \frac{1}{2} & t_1 & 0 \end{vmatrix} = \frac{(t_2 - t_1)^3}{4} = \frac{(2m)^{3/2}}{4}$$

5. Show that the angle between the tangent at any point 'A' of the curve $ln(x^2 + y^2) = C\tan^{-1}\frac{y}{x}$ and the line joining A to the origin is independent of the position of A on the curve.

सिद्ध कीजिये कि वक्र $ln(x^2 + y^2) = C\tan^{-1}\frac{y}{x}$ के किसी बिन्दु A पर खीचीं गई स्पर्श रेखा तथा A को मूल बिन्दु से मिलाने वाली रेखा के मध्य कोण वक्र पर A की स्थिति से स्वतंत्र है।

Ans. $\theta = \tan^{-1}\frac{2}{C}$

Sol. Let the point is (x_1, y_1)

Slope of line joining (0, 0) & (x_1, y_1) is

$$m_1 = \frac{y_1}{x_1}$$

$$\frac{(2x+2yy')}{(x^2+y^2)} = \frac{C\left(\frac{y'}{x}-\frac{y}{x^2}\right)}{\left(1+\frac{y^2}{x^2}\right)}$$

$$\frac{2(x_1+y_1y')}{(x_1^2+y_1^2)} = \frac{C(y'x_1-y_1)}{(x_1^2+y_1^2)}$$

$$2x_1 + 2y_1y' = Cx_1y' - Cy_1$$

$$2x_1 + Cy_1 = y'(Cx_1 - 2y_1)$$

$$y' = \frac{(2x_1+Cy_1)}{(Cx_1-2y_1)} = m_2$$

Calculate $\tan\theta = \left|\frac{m_1-m_2}{1+m_1m_2}\right|$

$$\tan\theta = \left|\frac{\frac{y_1}{x_1} - \frac{2x_1+cy_1}{cx_1-2y_1}}{1+\frac{y_1}{x_1\left(\frac{2x_1+cy_1}{cx_1-2y_1}\right)}}\right|$$

$$= \left|\frac{cx_1y_1 - 2y_1^2 - 2x_1^2 - cx_1y_1}{cx_1^2 + cy_1^2 - 2x_1y_1 + 2x_1y_1}\right| = \left|\frac{2}{c}\right|$$

which is independent of (x_1, y_1).

6. If the point P(a, b) lies on the curve $9y^2 = x^3$ such that the normal to the curve at P makes equal intercepts with the axes. Find the value of (a + 3b).

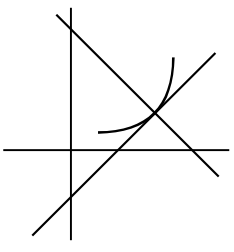
यदि वक्र $9y^2 = x^3$ पर स्थित बिन्दु P(a, b) से अभिलम्ब इस प्रकार डाला जाता है कि अभिलम्ब अक्षो से समान अन्तःखण्ड काटता है तो (a + 3b) का मान ज्ञात कीजिये।

Ans. 12

Sol. Given $9y^2 = x^3$

Let the point on the curve be $x = t^2$ and $y = \frac{t^3}{3}$

$$\frac{dx}{dt} = 2t \quad ; \quad \frac{dy}{dt} = t^2$$



$$\frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx} = \frac{t^2}{2t} = \frac{t}{2} \Rightarrow \text{slope of the normal} = -\frac{2}{t}$$

$\therefore$ normal makes equal intercept

hence $-\frac{2}{t} = -1 \quad \Rightarrow \quad t = 2$

Hence $P = (4, \frac{8}{3}) \quad \Rightarrow \quad a + 3b = 4 + 3 \cdot \frac{8}{3} = 4 + 8 = 12$

7. Water is being poured on to a cylindrical vessel at the rate of 1 m^3/min. If the vessel has a circular base of radius 3 m, find the rate at which the level of water is rising in the vessel.

एक बेलनाकार पात्र में 1 m^3/min की दर से जल भरा जाता है। यदि पात्र के वृत्ताकार आधार की त्रिज्या 3 m है तो जल के सतह के बढ़ने की दर क्या होगी।

Ans. $1/9\pi$ m/min

Sol. $\frac{dv}{dt} = 1\frac{m^3}{min}$

$v = \pi r^2 h$

Differentiating w.r.t. "t"

$\frac{dv}{dt} = \pi r^2 \frac{dh}{dt} \qquad (r = 3m)$

$\Rightarrow \frac{dh}{dt} = \frac{1}{\pi r^2} = \frac{1}{9\pi}\frac{m}{min}$

8. Water is dripping out from a conical funnel of semi vertical angle $\pi/4$, at the uniform rate of 2 cm^3/sec through a tiny hole at the vertex at the bottom. When the slant height of the water is 4 cm, find the rate of decrease of the slant height of the water.

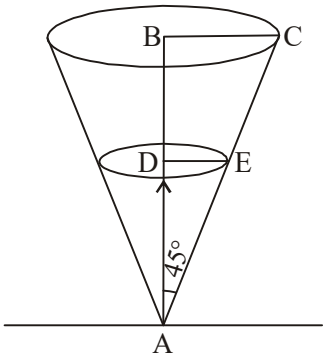
एक शंकु आकार का फनल जिसका अर्धशीर्ष कोण $\pi/4$ है के तल में स्थित एक छोटे छेंद में से पानी 2 cm^3/sec की नियत दर से गिर रहा है। जब पानी की तीर्यक ऊँचाई 4 cm है तो पानी की तीर्यक ऊँचाई के घटने की दर बताइयें।

Ans. $\frac{\sqrt{2}}{4\pi}$ cm/s

Sol. Let the radius, height and the slant height of the funnel are r, h and ℓ respectively.

We know the volume of funnel is $V = \frac{1}{3}\pi r^2 h$

And it's given that semi verticle angle is $\frac{\pi}{4}$



From the figure, $\sin\frac{\pi}{4} = \frac{r}{1} = \frac{1}{\sqrt{2}} \to r = \frac{1}{\sqrt{2}}$

And $\cos\frac{\pi}{4} = \frac{h}{1} = \frac{1}{\sqrt{2}} \Rightarrow h = \frac{1}{\sqrt{2}}$

Let S be the curved surface area of the water cone, then we have

$$S = \pi r\ell = \pi\frac{\ell}{\sqrt{2}}\cdot\ell = \frac{\pi\ell^2}{\sqrt{2}}$$

Differentiating w.r.t. t, we get

$$\frac{dS}{dt} = \frac{\pi}{\sqrt{2}}\cdot 2\ell\frac{d\ell}{dt}$$

Now, $\frac{dS}{dt} = -2cm^2/\sec$ ..... (S is negative, because it is decreasing)

$$\therefore \frac{dS}{dt} = \frac{2\ell\pi d\ell}{\sqrt{2}dt} = -2cm^2/\sec$$

$$\frac{d\ell}{dt} = -\frac{\sqrt{2}}{\pi\ell}$$

Given, $\ell = 4$ cm, then

$$\frac{d\ell}{dt} = -\frac{\sqrt{2}}{4\pi}cm/\sec$$

Hence, the slant height is decreasing at the rate of $\frac{\sqrt{2}}{4\pi}cm/\sec$

9. An air force plane is ascending vertically at the rate of 100 km/h. If the radius of the earth is R km, how fast the area of the earth, visible from the plane increasing at 3min after it started ascending. Take visible area $A = \frac{2\pi R^2 h}{R+h}$ Where h is the height of the plane in kms above the earth.

एक वायुसेना का विमान उर्ध्वाधर में ऊपर की ओर 100 km/h की गति से बढ़ रहा है। यदि पृथ्वी की त्रिज्या R km है तो विमान से दिखाई देने वाला पृथ्वी का क्षेत्रफल किस दर से बढ़ रहा है जबकि 3 मिनट हो चूके हों। मान लिजिये दिखाई देने वाला क्षेत्रफल $A=\frac{2\pi R^2 h}{R+h}$ है जहाँ h विमान की पृथ्वी से kms में ऊँचाई है।

Ans. $200\,\pi\,r^3/(r+5)^2\ km^2/h$

Sol. Let say h = height

and A = area

$$\frac{dh}{dt}=100$$

$$A=\frac{2\pi r^2 h}{r+h}$$

$$\frac{dA}{dt}=\frac{dA}{dh}\frac{dh}{dt}$$

$$=2\pi r^2\frac{d}{dh}\left(\frac{h}{r+h}\right)\cdot 100$$

$$=2\pi r^2\left[\frac{(r+h)-h}{(f+h)^2}\right]\cdot 100$$

$$=\frac{200\pi r^3}{(r+h)^2}$$

$$h=\frac{100}{60}\cdot 3=5km$$

$$\frac{dA}{dt}=\frac{200\pi r^3}{(r+5)^2}km^2/hr.$$

10. A variable ΔABC in the xy plane has its orthocentre at vertex 'B' , a fixed vertex 'A' at the origin and the third vertex 'C' restricted to lie on the parabola $y=1+\frac{7x^2}{36}$. The point B starts at the point (0, 1) at time t = 0 and moves upward along the y axis at a constant velocity of 2 cm/sec. How fast is the area of the triangle increasing when $t=\frac{7}{2}$ sec.

एक चर त्रिभुज ΔABC, xy तल में स्थित है जिसका लम्बकेन्द्र 'B' है शीर्ष 'A' मूल बिन्दु है तथा शीर्ष 'C' परवलय $y = 1 + \frac{7x^2}{36}$ पर रहने के लिये बाध्य है। बिन्दु B का प्रारम्भिक निर्देशांक (0, 1) है, जबt = 0 एवं वह y अक्ष पर 2 cm/sec की दर से y अक्ष पर उपर जा रहा है। $t = \frac{7}{2}$ sec पर त्रिभुज का क्षेत्रफल किस दर से बढ़ रहा होगा ?

Ans. $\frac{66}{7}$

Sol. $y = 1 + \frac{7x^2}{36}$

$\Rightarrow \frac{dy}{dt} = \frac{7x}{18}\frac{dx}{dt}$ $\left(\frac{dy}{dt} = 2cm/sec\right)$

$\Rightarrow \frac{dy}{dt} = \frac{36}{7x}\frac{cm}{sec}$

$\Rightarrow A = \frac{1}{2}xy \Rightarrow \frac{dA}{dt} = \frac{1}{2}\left(\frac{xdy}{dt} + \frac{ydx}{dt}\right)$

at $t = \frac{7}{2}$, y = 8 cm, x = 6 cm, $\frac{dy}{dt} = \frac{6}{7}\frac{cm}{sec}$

$\frac{dA}{dt} = \frac{1}{2}\left[6 \times 2 + 8 \times \frac{6}{7}\right] = \frac{66}{7}\frac{cm^2}{sec}$

11. A circular ink blot grows at the rate of 2 cm^2 per second. Find the rate at which the radius is increasing after $2\frac{6}{11}$ seconds. Use $\pi = \frac{22}{7}$.

एक वृत्ताकार स्याहि का दाग 2 cm^2 प्रति सेकण्ड की दर से बढ़ रहा है। उसकी त्रिज्या की $2\frac{6}{11}$ सेकण्ड पश्चात् वृद्धि दर ज्ञात कीजिये। Use $\pi = \frac{22}{7}$.

Ans. $\frac{1}{4}$ cm/sec.

Sol. Area after $\frac{28}{11}\sec = 2\times\frac{28}{11} = \pi r^2$

Simplifying,

$$r^2 = \frac{56}{11\pi}$$

$$r = \sqrt{\frac{56}{11\pi}}$$

Since $A = \pi r^2$, we have

$$\frac{dA}{dt} = 2\pi r\frac{dr}{dt}$$

$$\Rightarrow 2 = 2\pi\sqrt{\frac{56}{11\pi}}\frac{dr}{dt}$$

$$\Rightarrow 1 = \sqrt{\frac{56\pi}{11}}\frac{dr}{dt}$$

$$\Rightarrow \frac{dr}{dt} = \sqrt{\frac{11}{56\pi}}$$

Using the approximation $\pi \approx \frac{22}{7}$, we get

$$\frac{11}{56\pi} = \frac{11}{56\times\frac{22}{7}} = \frac{11}{56}\times\frac{7}{22} = \frac{1}{16}$$

Hence,

$$\frac{dr}{dt} = \sqrt{\frac{1}{16}} = \frac{1}{4}$$

12. Water is flowing out at the rate of 6 m^3/min from a reservoir shaped like a hemispherical bowl of radius R = 13 m. The volume of water in the hemispherical bowl is given by $V = \frac{\pi}{3}\cdot y^2(3R - y)$ when the water is y meter deep.

Find

(a) At what rate is the water level changing when the water is 8 m deep.

(b) At what rate is the radius of the water surface changing when the water is 8 m deep.

एक 13 cm त्रिज्या के अर्धगोलाकार रिसर्वायर से पानी 6 m^3/min की दर से बाहर निकल रहा है। रिसर्वायर के अन्दर पानी का आयतन $V = \frac{\pi}{3} \cdot y^2 (3R - y)$ से दिया जाता है जहाँ पानी y मीटर गहरा है। ज्ञात कीजिये।

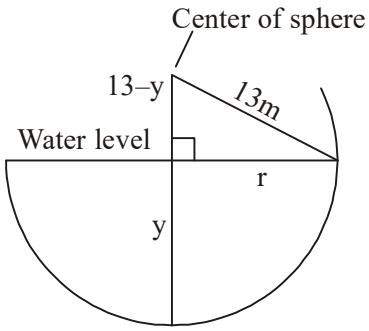
(a) 8m की गहराई पर पानी की सतह किस दर से परिवर्तित हो रही है।

(b) 8 m की गहराई पर पानी की सतह की त्रिज्या किस दर से परिवर्तित हो रही है

Ans. (a) $-\frac{1}{24\pi}$ m/min. (b) $-\frac{5}{288\pi}$ m/min.

Sol. $\frac{dv}{dt} = -6\frac{m^3}{min}$ and $y = 8m$

(a) $V = \frac{\pi}{3} y^2 (39 - y)$



$\Rightarrow V = 13\pi y^2 - \frac{\pi}{3} y^3$

$\Rightarrow \frac{dV}{dt} = \left(26\pi y - \pi y^2\right)\frac{dy}{dt} \Rightarrow -6 = 144\pi \frac{dy}{dt}$

$\Rightarrow \frac{dy}{dt} = -\frac{1}{24\pi}\frac{m}{min}$

(b) $r = \sqrt{(13)^2 - (13 - y)^2}$

$= \sqrt{26y - y^2}$

$\frac{dr}{dt} = \frac{1}{2\sqrt{26y - y^2}} \times (26 - 2y)\frac{dy}{dt}$

$= \frac{13 - y}{\sqrt{26y - y^2}}\frac{dy}{dt} = \frac{13 - 8}{\sqrt{26 \times -8^2}}\left(\frac{1}{-24\pi}\right)$

$\frac{dr}{dt} = -\frac{5}{288\pi}\frac{m}{min}$

13. If in a triangle ABC, the side 'c' and the angle 'C' remain constant, while the remaining elements are changed slightly, show that $\frac{da}{\cos A} + \frac{db}{\cos B} = 0$.

एक त्रिभुज ABC में, भुजा 'c' तथा कोण 'C' सदैव नियत है, जहाँ बाकी अवयवों में थोड़ा बदलाव हो रहा है तो सिद्ध कीजिए

$$\frac{da}{\cos A}+\frac{db}{\cos B}=0$$

Ans.

Sol. $A + B + C = \pi \Rightarrow dA + dB = 0 \Rightarrow dA = - dB$

$$\frac{c}{\sin C} = 2R = \text{constant}$$

$a = 2R\sin A \Rightarrow da = 2R\cos A dA$ .....(i)

similarly $db = 2R\cos B dB$ .....(ii)

Divide (i) by (ii)

$$\frac{da}{db}=\frac{\cos A(dA)}{\cos B(dB)} \Rightarrow \frac{da}{db}=-\frac{\cos A}{\cos B}$$

14. At time $t > 0$, the volume of a sphere is increasing at a rate proportional to the reciprocal of its radius. At $t = 0$, the radius of the sphere is 1 unit and at $t = 15$ the radius is 2 units.
   (a) Find the radius of the sphere as a function of time t.
   (b) At what time t will the volume of the sphere be 27 times its volume at $t = 0$.

   समय $t > 0$ पर, एक गोले के आयतन की वृद्धि दर उसके त्रिज्या के व्युत्क्रमानुपाती है। $t = 0$ पर त्रिज्या 1 इकाई तथा $t = 15$ पर त्रिज्या 2 इकाई है।
   (a) गोले की त्रिज्या को समय t के फलन के रूप में प्रदर्शित करें
   (b) कितने समय t पर आयतन $t = 0$ के आयतन का 27 गुना हो जाएगा

Ans. (a) $r = (1 + t)^{1/4}$ (b) $t = 80$

Sol. $\frac{dv}{dt}=\frac{k}{r} \Rightarrow 4\pi r^2\frac{dr}{dt}=\frac{k}{r}$

$4\pi r^3 dr = k dt$

$\pi r^4 = kt + c$

at $t = 0 ; r = 1 \Rightarrow c = \pi$

$$r^4=\left(\frac{kt}{\pi}+1\right)$$

put $r = 2$ & $t = 15 \Rightarrow k = \pi$

$r^4 = t + 1$

$r = (t + 1)^{1/4}$

at $t = 0$ volume $(v_1) = \frac{4}{3}\pi$

at time t volume $(v_2) = \frac{4}{3}\pi(t+1)^{3/4}$

$v_2 = 27v_1$

$\Rightarrow \quad \frac{4}{3}\pi(t+1)^{3/4} = (27)\frac{4}{3}\pi$

Solve for t

$\Rightarrow (t+1)^{\frac{3}{4}} = 27$

$\Rightarrow t = 80$ sec

15. The chord of the parabola $y = -a^2x^2 + 5ax - 4$ touches the curve $y = \frac{1}{1-x}$ at the point $x = 2$ and is bisected by that point. Find 'a'.

परवलय $y = -a^2x^2 + 5ax - 4$ की एक जीवा $y = \frac{1}{1-x}$ को $x = 2$ पर स्पर्श करती है तथा उसी बिन्दु पर समद्विभाजित होती है। 'a' का मान ज्ञात कीजिये।

Ans. $a = 1$

Sol. $y = -a^2x^2 + 5ax - 4$

$$y = \frac{1}{1-x}$$

Substituting $x = 2$, we have

$$y = \frac{1}{1-2} = -1$$

Hence point of tangency is $(2, -1)$

$$y = \frac{1}{1-x}$$

Differentiating above equation with respect to x, we have

$$\frac{dy}{dx} = -\frac{1}{(1-x)^2} \times (-1) = \frac{1}{(1-x)^2}$$

At point of tangency, i.e., (2, –1)

$$\frac{dy}{dx} = \frac{1}{(1-2)^2} = 1$$

$\therefore$ Equation of tangent is $[(y - y_1) = m(x - x_1)]$

$y - (-1) = 1\,(x - 2)$

$\Rightarrow y - x + 3 = 0$ ...(1)

Now,

$y = -a^2x^2 + 5ax - 4$

$\Rightarrow a^2x^2 - 5ax - y + 4 = 0$

We know that equation of chord having mid point (x_1, y_1) is

$T = S_1$

$$a^2xx_1 - 5a\left(\frac{x + x_1}{2}\right) + \left(\frac{y + y_1}{2}\right) + 4 = a^2x_1^2 - 5ax_1 + y_1 + 4$$

$$a^2xx_1 - 5a\left(\frac{x + x_1}{2}\right) + \left(\frac{y + y_1}{2}\right) = a^2x_1^2 - 5ax_1 + y_1$$

Substituting $(x_1, y_1) = (2, -1)$ as the chord is bisected by the same point,

we have

$$2a^2x - 5a\left(\frac{x + 2}{2}\right) + \left(\frac{y + (-1)}{2}\right) = a^2 2^2 - 10 + (-1)$$

$$2a^2x - \frac{5ax}{2} - 5a + \frac{y}{2} - \frac{1}{2} = 4a^2 - 10a - 1$$

$$x\left(2a^2 - \frac{5a}{2}\right) + \frac{y}{2} = 4a^2 - 5a - \frac{1}{2}$$

$\Rightarrow x(4a^2 - 5a) + y = 8a^2 - 10a - 1$ ...(2)

As equation (1) & (2) represents same line.

Therefore equating the corresponding coefficients, we have coefficient of x in equation (2) = coefficient of x in equation (1)

$4a^2 - 5a = -1$

$\Rightarrow 4a^2 - 5a + 1 = 0$

$4a^2 - 4a - a + 1 = 0$

$\Rightarrow 4a(a-1) - 1(a-1) = 0$

$\Rightarrow (4a-1)(a-1) = 0$

$\Rightarrow a = \frac{1}{4}$ or 1

Hence the required answer is $a = \frac{1}{4}$ or 1.

16. Find the equation of the straight line which is tangent at one point and normal at another point of the curve, $x = 3t^2$, $y = 2t^3$.

वक्र $x = 3t^2$, $y = 2t^3$ के लिये वह रेखा ज्ञात कीजिए जो एक बिन्दु पर स्पर्श रेखा हो तथा अन्य बिन्दु पर अभिलम्ब हो।

Ans. $\sqrt{2}x + y - 2\sqrt{2} = 0$ or $\sqrt{2}x - y - 2\sqrt{2} = 0$

Sol. Given curves are $x = 3t^2$, $y = 2t^3$

Eliminating 't', we get, $27y^2 = 4x^3$

Let the tangent at $P\left(3t_1^2, 2t_1^3\right)$ meets the curve again at $Q\left(3t_2^2, 2t_2^3\right)$

Equation of the tangent at $P\left(3t_1^2, 2t_1^3\right)$ is

$$\frac{y - 2t_1^3}{x - 3t_1^2} = t_1 \left(\text{since} \frac{dy}{dx} = t \right)$$

$\Rightarrow y - t_1 x + t_1^3 = 0$ ....(i)

Equation of normal at $Q(3t_2^2, 2t_2^3)$ is

$$\frac{y - 2t_2^3}{x - 3t_2^2} = -\frac{1}{t_2}$$

$\Rightarrow t_2 y + x - 2t_2^4 - 3t_2^2 = 0$ ....(ii)

Equations (i) and (ii) are identical.

$$t_2 = -\frac{1}{t_1} = -\frac{\left(2t_2^4 + 3t_2^2\right)}{t_1^3}$$

$$\Rightarrow t_1 = -\frac{1}{t_2}, t_1^2 = \left(2t_2^4 + 3t_2^2\right)$$

$$\Rightarrow \left(\frac{1}{t_2}\right)^2 = \left(2t_2^4 + 3t_2^2\right)$$

$$\Rightarrow \left(2t_2^6 + 3t_2^4 - 1\right) = 0$$

$$\Rightarrow 2p^3 + 3p^2 - 1 = 0, p = t_2^2$$

$$\Rightarrow (p+1)\left(2p^2 + p - 1\right) = 0$$

$$\Rightarrow (p+1)^2 (2p-1) = 0$$

$$\Rightarrow p = \frac{1}{2}, p = -1$$

$$\Rightarrow t_2^2 = \frac{1}{2}$$

$$\Rightarrow t_2 = \pm\frac{1}{\sqrt{2}} \text{ and } t_1 = \mp\frac{1}{\sqrt{2}}$$

Hence, the equation of the line PQ is

$$y \pm \sqrt{2}x \pm 2\sqrt{2} = 0$$

17. Let $f(x) = \left(1+\frac{1}{x}\right)^x$ and $g(x) = \left(1+\frac{1}{x}\right)^{x+1}$, both f and g being defined for $x > 0$, then prove that f(x) is increasing and g (x) is decreasing.

माना $f(x) = \left(1+\frac{1}{x}\right)^x$ तथा $g(x) = \left(1+\frac{1}{x}\right)^{x+1}$ दो $x > 0$ के लिये परिभाषित फलन है तो सिद्ध कीजिये f (x) वर्द्धमान है तथा g(x) ह्रासमान है।

Ans.

Sol. $f(x) = \left(1+\frac{1}{x}\right)^x$

$$\ell nf(x) = x\ell n\left(1+\frac{1}{x}\right) = x\left[\ell n(1+x) - \ell nx\right]$$

Differentiating w.r.t. "x"

$$\frac{f'(x)}{f(x)} = \ell n(1+x) - \ell nx + x\left[\frac{1}{1+x} - \frac{1}{x}\right]$$

$$f'(x) = f(x)\left(\ell n\left(\frac{1+x}{x}\right) - \frac{1}{1+x}\right) \qquad \text{_____(1)}$$

Let $h(x) = \ell n\left(\frac{1+x}{x}\right) - \frac{1}{1+x}$

$$\lim_{x\to\infty} h(x) = 0$$

$$h(x) = \ell n(1+x) - \ell nx - \frac{1}{1+x}$$

$$h'(x) = \frac{1}{1+x} - \frac{1}{x} + \frac{1}{(1+x)^2} = \frac{-1}{x(x+1)^2}$$

for $x > 0 \quad \Rightarrow h'(x) < 0$

$\Rightarrow x < \infty$

$\Rightarrow h(x) > \lim_{x\to\infty} h(x)$

$\Rightarrow h(x) > 0$

Now using equation (1)

for $x > 0$, $f(x) > 0$ and $\ell n\left(\frac{1+x}{x}\right) - \frac{1}{1+x} > 0$

$x > 0 \Rightarrow f'(x) > 0$

f(x) is increasing.

$$g(x) = \left(1+\frac{1}{x}\right)^{x+1} \Rightarrow \ell ng(x) = (x+1)\ell n\left(1+\frac{1}{x}\right)$$

$$g'(x) = g(x)\left[\ell n\left(1+\frac{1}{x}\right) - \frac{1}{x}\right] \qquad \text{_____(2)}$$

Let $p(x)=\ell n\left(1+\frac{1}{x}\right)-\frac{1}{x}=\ell n(1+x)-\ell nx-\frac{1}{x}$

$\lim\limits_{x\to\infty} p(x)=0$

$p'(x)=\frac{1}{1+x}-\frac{1}{x}+\frac{1}{x^2}=\frac{x^2-x(1+x)+1+x}{x^2(1+x)}$

$\because x>0\Rightarrow p'(x)>0$

$\Rightarrow x<\infty$

$\Rightarrow p(x)<\lim\limits_{x\to\infty} p(x)$

$\Rightarrow p(x)<0$

If $x>0\Rightarrow g(x)>0$ and $\ell n\left(1+\frac{1}{x}\right)-\frac{1}{x}<0$

from equation (2)

$x>0\Rightarrow g'(x)<0$

g(x) is decreasing.

18. Prove that $\tan^2x+6\,\ell n\sec x+2\cos x+4>6\sec x$ for $x\in\left(\frac{3\pi}{2},2\pi\right)$.

$x\in\left(\frac{3\pi}{2},2\pi\right)$ के लिये सिद्ध कीजिये $\tan^2x+6\,\ell n\sec x+2\cos x+4>6\sec x$.

Ans.

Sol. Let $f(x)=\tan^2x+6\,\ell n\sec x+2\cos x+4-6\sec x$

$f'(x)=2\tan x\sec^2x+6\tan x-2\sin x-6\sec x\tan x$

for $x\in\left(\frac{3\pi}{2},2\pi\right)\Rightarrow f'(x)<0$

for $x\in\left(\frac{3\pi}{2},2\pi\right)\Rightarrow$ f(x) is decreasing.

$\Rightarrow x<2\pi$

$\Rightarrow f(x) > f(2\pi)$

$\Rightarrow \tan^2 x + 6\,\ell n \sec x + 2\cos x + 4 - 6\sec x > 0$

$\Rightarrow \tan^2 x + 6\,\ell n \sec x + 2\cos x + 4 > 6\sec x$

19. If f, ϕ, ψ are continuous in [a, b] and derivable in]a, b[then show that there is a value of c lying between a and b such that, $\begin{vmatrix} f(a) & f(b) & f'(c) \\ \phi(a) & \phi(b) & \phi'(c) \\ \Psi(a) & \Psi(b) & \Psi'(c) \end{vmatrix} = 0$

यदि f, ϕ, ψ अन्तराल [a, b] में सतत् एवं अन्तराल]a, b[में अवकलनीय है तो सिद्ध कीजिये कि c का एक मान a तथा b के मध्य इस प्रकार है कि $\begin{vmatrix} f(a) & f(b) & f'(c) \\ \phi(a) & \phi(b) & \phi'(c) \\ \Psi(a) & \Psi(b) & \Psi'(c) \end{vmatrix} = 0$

Ans. $0 < c < \frac{1}{9}$

Sol. Using LMVT

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

$$\phi'(c) = \frac{\phi(b) - \phi(a)}{b - a}$$

$$\psi'(c) = \frac{\psi(b) - \psi(a)}{b - a}$$

$$\begin{vmatrix} f(a) & f(b) & f'(c) \\ \phi(a) & \phi(b) & \phi'(c) \\ \psi(a) & \psi(b) & \psi'(c) \end{vmatrix}$$

$$C_1 \to \frac{C_2 - C_1}{b - a}$$

$$= (b-a)\begin{vmatrix} \frac{f(b)-f(a)}{b-a} & f(b) & f'(c) \\ \frac{\phi(b)-\phi(a)}{b-a} & \phi(b) & \phi'(c) \\ \frac{\psi(b)-\psi(a)}{b-a} & \psi(b) & \psi'(c) \end{vmatrix}$$

$$=(b-a)\begin{vmatrix} f'(c) & f(b) & f'(c) \\ \phi'(c) & \phi(b) & \phi'(c) \\ \psi'(c) & \psi(b) & \psi'(c) \end{vmatrix}=0$$

20. Using LMVT prove that :

(a) $\tan x > x$ in $\left(0, \frac{\pi}{2}\right)$, (b) $\sin x < x$ for $x > 0$

लेग्रांज प्रमेय का उपयोग करते हुये सिद्ध कीजिये :

(a) $\tan x > x$ in $\left(0, \frac{\pi}{2}\right)$, **(b)** $\sin x < x$ for $x > 0$

Ans.

Sol. (a) let $f(x) = \tan x - x$

Using LMVT for (0, x), where $x \in \left(0, \frac{\pi}{2}\right)$

$$f'(c) = \frac{f(x)-f(0)}{x-0}$$

$$\Rightarrow \left(\sec^2 c - 1\right) = \frac{f(x)}{x}$$

$$\Rightarrow f(x) = x\left(\sec^2 c - 1\right)$$

$$f(x) > 0 \forall x \in \left(0, \frac{\pi}{2}\right)$$

$\Rightarrow \tan x > x$

(b) let $f(x) = \sin x - x$

Using LMVT for (0,x) for $x > 0$

$$f'(c) = \frac{f(x)-f(0)}{x-0}$$

$$\Rightarrow \cos c - 1 = \frac{f(x)}{x}$$

$\Rightarrow f(x) = x(\cos c - 1)$

$\Rightarrow f(x) < 0 \forall x > 0$

$\Rightarrow \sin x - x < 0 \forall x > 0$

$\sin x < x \forall x > 0$

21. Let f defined on [0, 1] be a twice differentiable function such that, $|f''(x)| \leq 1$ for all $x \in [0, 1]$
If $f(0) = f(1)$, then show that, $|f'(x)| < 1$ for all $x \in [0, 1]$

माना f [0, 1] में परिभाषित दो बार अवकलनीय फलन इस प्रकार है कि सभी $x \in [0, 1]$ में $|f''(x)| \leq 1$
यदि $f(0) = f(1)$ है तो सिद्ध कीजिये सभी $x \in [0, 1]$ के लिये $|f'(x)| < 1$ है।

Ans.

Sol. Using Rolle's theorem for f(x) in [0,1]

let $a \in [0,1]$

then $f'(a) = 0$

Now using LMVT for $f'(x)$ in [x, a]

where $x \neq a$ and $x, a \in (0,1)$

Let $c \in (x, a)$

$$f''(c) = \frac{f'(x) - f'(a)}{x - a}$$

$$\Rightarrow f'(x) = f''(c)(x - a)$$

$$\Rightarrow |f'(x)| = |f''(c)||x - a| \qquad (|f''(x)| \leq 1)$$

$$\Rightarrow |f'(x)| \leq |x - a|$$

as $x, a \in (0,1)$ $\Rightarrow |x - a| < 1$

$|f'(x)| < 1$ $\forall x \in [0,1]$

22. Let f be continuous on [a, b] and assume the second derivative f'' exists on (a, b). Suppose that the graph of f and the line segment joining the point $(a, f(a))$ and $(b, f(b))$ intersect at a point $(x_0, f(x_0))$ where $a < x_0 < b$. Show that there exists a point $c \in (a, b)$ such that $f''(c) = 0$.

माना f अन्तराल [a, b] में सतत् फलन है तथा अन्तराल (a, b) में f " अस्तित्व में है। मान लिजिये फलन f का आरेख तथा बिन्दुओं $(a, f(a))$ तथा $(b, f(b))$ को मिलाने वाले रेखा खण्ड का प्रतिच्छेदन बिन्दु $(x_0, f(x_0))$ है जहाँ $a < x_0 < b$ सिद्ध कीजिये। एक बिन्दु $c \in (a, b)$ अस्तित्व में है जिसके लिये $f''(c) = 0$ है।

Ans.

Sol. Using LMVT for f(x) in $x \in (a, x_0)$

Let $p \in (a, x_0)$

$$f'(p) = \frac{f(a) - f(x_0)}{a - x_0} \qquad ...(1)$$

Using LMVT for f(X) in $x \in (x_0, b)$

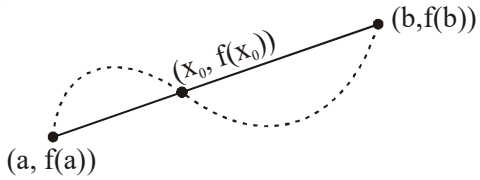
let $q \in (x_0, b)$

$$f'(q) = \frac{f(x_0) - f(b)}{x_0 - b} \qquad ...(2)$$

for some point $p \in (a, x_0)$ & $q \in (x_0, b)$

$f'(p) = f'(q)$

Now for some point $c \in [p, q]$, Using Rolle's Theorem for $f'(x)$



(i) $f'(p) = f'(q)$

(ii) $f'(x)$ is continuous in [p, q]

(iii) $f'(x)$ is differentiable in (p, q)

Hence, $f''(c) = 0$, $c \in [p,q]$ $\quad$ $p, q \in [a, b]$

$f''(c) = 0$, $c \in [a, b]$

23. Let $-1 \le p \le 1$. Show that the equation $4x^3 - 3x - p = 0$ has a unique root in the interval $\left[\frac{1}{2}, 1\right]$ and identify it.

माना $-1 \le p \le 1$ सिद्ध कीजिये समीकरण $4x^3 - 3x - p = 0$ के अन्तराल $\left[\frac{1}{2}, 1\right]$ में अद्वितीय हल है, उसे ज्ञात कीजिये।

Ans. $\cos\left(\frac{1}{3}\cos^{-1} p\right)$

Sol. $f(x) = 4x^3 - 3x - p$

$f\left(\frac{1}{2}\right) = -(p+1)$

$f(1) = (1-p)$

$f(1) \cdot f\left(\frac{1}{2}\right) = -(1-p^2) \le 0 \quad \because p \in [-1, 1]$

$\therefore$ $f(x) = 0$ has atleast one root in $\left[\frac{1}{2}, 1\right]$

$f'(x) = 3(2x-1)(2x+1)$

$\Rightarrow \quad f'(x) > 0 \quad \forall\ x > \frac{1}{2}$

$\Rightarrow$ $f(x) = 0$ has exactly one root in $\left[\frac{1}{2}, 1\right]$

Let the root be $x = \cos\theta$

$\therefore \quad 4\cos^3\theta - 3\cos\theta = p$

$\cos 3\theta = p$

$\Rightarrow \quad \theta = \frac{1}{3}\cos^{-1}(p) \qquad \Rightarrow\ x = \cos\left(\frac{1}{3}\cos^{-1}(p)\right)$

24. (a) Using the relation $2(1-\cos x) < x^2$, $x \neq 0$ or otherwise, prove that $\sin(\tan x) \ge x$, $\forall\ x \in \left[0, \frac{\pi}{4}\right]$.

सम्बन्ध $2(1-\cos x) < x^2$, $x \neq 0$ या अन्यथा, का प्रयोग करते हुये सिद्ध कीजिये $\sin(\tan x) \ge x$, $\forall\ x \in \left[0, \frac{\pi}{4}\right]$

Ans.

Sol. $\cos x - 1 > -\frac{x^2}{2}$ (given) _____(1)

consider $f(x) = \sin(\tan x) - x$

$f'(x) = \cos(\tan x)(1+\tan^2 x) - 1$

$= (\tan^2 x)\cos(\tan x) + \cos(\tan x) - 1$

$\cos(\tan x) - 1 > -\frac{\tan^2 x}{2}$ from (1)

$(\tan^2 x)\cos(\tan x) + \cos(\tan x) - 1$

$> \tan^2 x \left\{\cos(\tan x) - \frac{1}{2}\right\}$

$\Rightarrow \quad f'(x) > \tan^2 x \left\{\cos(\tan x) - \frac{1}{2}\right\}$

$0 \le \tan x \le 1 \quad \left\{\because 0 \le x \le \frac{\pi}{4}\right\}$

$\Rightarrow \quad \cos(\tan x) > \frac{1}{2}$

$\Rightarrow \quad f'(x) > 0 \quad \Rightarrow \quad f(x) \ge f(0) \quad \Rightarrow f(x) \ge 0$

25. Let $f : [0, 4] \to R$ be a differentiable function.
माना $f : [0, 4] \to R$ अवकलनीय फलन है।

(i) Show that there exist $a, b \in [0, 4]$, $(f(4))^2 - (f(0))^2 = 8\, f'(a)\, f(b)$
सिद्ध कीजिये $a, b \in [0, 4]$, इस प्रकार अस्तित्व में है कि $(f(4))^2 - (f(0))^2 = 8\, f'(a)\, f(b)$

(ii) Show that there exist α, β with $\alpha, \beta \in (0, 2)$ such that $\int_0^4 f(t)\, dt = 2\,(\alpha\, f(\alpha^2) + \beta\, f(\beta^2))$

सिद्ध कीजिये α, β जिनके लिये $\alpha, \beta \in (0, 2)$ इस प्रकार अस्तित्व में है कि $\int_0^4 f(t)\, dt = 2\,(\alpha\, f(\alpha^2) + \beta\, f(\beta^2))$

Ans.

Sol. Consider $g(x) = \int_0^{x^2} f(t)dt$ 

$g(1) - g(0)$

$= g'(\alpha), \alpha \in (0, 1)$ {by LMVT in [0,1]}.....(i)

$g(2) - g(1)$

$= g'(\beta), \beta \in (1, 2)$ {by LMVT in [1,2]}.....(ii)

(i) + (ii) $\Rightarrow$ $g(2) - g(0) = g'(\alpha) + g'(\beta)$

$\Rightarrow \quad \int_0^4 f(t)dt = 2\{\alpha f(\alpha^2) + \beta f(\beta^2)\}$

26. Verify Rolles theorem for $f(x) = (x - a)^m (x - b)^n$ on $[a, b]$; m, n being positive integer.

अन्तराल [a, b] में रोले प्रमेय को सत्यापित कीजिये $f(x) = (x-a)^m(x-b)^n$ on [a, b] ; m, n धनात्मक पूर्णांक है।

Ans. $c = \dfrac{mb + na}{m + n}$ which lies between a and b

Sol. $f(x) = (x-a)^m (x-b)^n$

$\because$ 'f' is continuous & derivable in [a, b].

& $f(a) = f(b)$

$\therefore$ according to Rolle's theorem,

there must be atleast one root of the equation

$f'(x) = 0$ in (a, b)

consider $f'(x) = 0$

$m(x-a)^{m-1}(x-b)^n + n(x-b)^{n-1}(x-a)^m = 0$

$(x-a)^{m-1}(x-b)^{n-1}\{mx - mb + nx - na\} = 0$

$\Rightarrow \quad x = \dfrac{mb+na}{m+n} \in (a, b).$

27. A bus contractor agrees to run special buses for the employees of ABC Co. Ltd. He agrees to run the buses if atleast 200 persons travel by his buses. The fare per person is to be Rs. 10 per day if 200 travel and will be decreased for everybody by 2 paise per person over 200 that travels. How many passengers will give the contractor maximum daily revenue?

ABC Co. Ltd. का एक बस ठेकेदार कर्मचारियों के लिये विशेष बसे चलाने के प्रस्ताव पर राजी हो जाता है। वह यह शर्त रखता है कि कम से कम 200 व्यक्ति उसकी बसों से परिगमन करे। यदि 200 व्यक्ति यात्र करते हो, तो प्रत्येक व्यक्ति द्वारा प्रतिदिन का भाड़ा रू. 10 होना चाहिये तथा 200 से अधिक प्रत्येक बढ़ने वाले व्यक्ति के लिये भाड़ा 2 पेसे प्रति व्यक्ति से कम कर दिया जायेगा। कितने व्यक्ति यात्र करने चाहिये कि ठेकेदार को सबसे अधिक मुनाफा हो?

Ans. 350

Sol. Let the number of passengers be $x\ (x \geq 200)$

Fair changed per person $= 10 - (x-200)\dfrac{2}{100}$

Total revenue $= x\left[10 - (x-200)\dfrac{2}{100}\right] = 10x - \dfrac{2x}{100}(x-200) = 10x - \dfrac{2x^2}{100} + 4x$

$f(x) = 14x - \dfrac{2x^2}{100}$

$f'(x) = 14 - \dfrac{4x}{100} = 0 \Rightarrow \quad x = 350$

$f''(x) < 0 \quad \Rightarrow \quad x = 350$ gives maxima

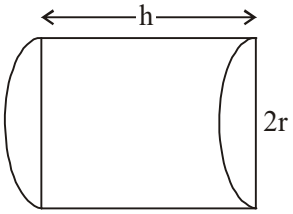
28. A given quantity of metal is to be casted into a half cylinder i.e. with a rectangular base and semicircular ends. Show that in order that total surface area may be minimum , the ratio of the height of the cylinder to the diameter of the semi circular ends is $\pi/(\pi+2)$.

एक दी गई राशि में धातु को आधे बेलन के रूप में बनाया जाता है अर्थात आधार आयताकार हो एवं छोर अर्द्धवृताकार हो । यदि सम्पूर्ण पृष्ठीय क्षेत्रफल न्यूनतम है तो सिद्ध कीजिये कि बेलन की ऊँचाई तथा अर्द्धवृताकार छोरो की त्रिज्या का अनुपात $\pi/(\pi+2)$ है।

Ans.

Sol. Let r be the radius and h be the height of half cylinder

$$\text{Volume} = \frac{1}{2}\pi r^2 h = V \text{ (constant)} \qquad \text{_____(1)}$$



Total surface area of the half cylinder is

$$S = 2\left(\frac{1}{2}\pi r^2\right) + \pi rh + 2rh \qquad \text{_____(2)}$$

From (1) put the value of h in (2)

$$S = \left(\pi r^2\right) + \pi r\left(\frac{2V}{\pi r^2}\right) + 2r\left(\frac{2V}{\pi r^2}\right)$$

$$S = \left(\pi r^2\right) + \left(\frac{1}{r}\right)\left[\frac{4V}{\pi} + 2V\right]$$

$$\frac{dS}{dr} = (2\pi r) + \left(\frac{-1}{r^2}\right)\left[\frac{4V}{\pi} + 2V\right] \qquad \text{...(3)}$$

For maxima/minima ds/dr = 0

$$\Rightarrow (2\pi r) + \left(\frac{-1}{r^2}\right)\left[\frac{4V}{\pi} + 2V\right] = 0$$

$$\Rightarrow (2\pi r) = \left(\frac{1}{r^2}\right)\left[\frac{4V + 2V\pi}{\pi}\right]$$

$$\Rightarrow \pi r^3 = V\left[\frac{2+\pi}{\pi}\right]$$

$$V = \frac{\left(\pi^2 r^3\right)}{(\pi+2)} \qquad ...(4)$$

From (1) and (4)

$$\Rightarrow \frac{1}{2}\pi r^2 h = \frac{\pi^2 r^3}{\pi+2}$$

$$\Rightarrow \frac{h}{2r} = \frac{\pi}{\pi+2}$$

height : diameter = $\pi : \pi + 2$

Differentiating (3) with respect to r

$$\frac{d^2S}{dr^2} = (2\pi) + \left(\frac{2}{r^3}\right)\left[\frac{4V}{\pi} + 2V\right]$$

= positive (as all quantities are +ve)

29. Show that for each $a > 0$ the function $e^{-ax}.\ x^{a^2}$ has a maximum value say F(a), and that F(x) has a minimum value, $e^{-e/2}$.

सिद्ध कीजिये, प्रत्येक $a > 0$ के लिये फलन $e^{-ax}.\ x^{a^2}$ का अधिकतम मान F (a) है तथा F (x) का न्यूनतम मान $e^{-e/2}$ है।

Ans.

Sol. $f(x) = e^{-ax} \cdot x^{a^2}$

$$f'(x) = -ae^{-ax}x^{a^2} + a^2 e^{-ax}x^{a^2-1}$$

$$= ae^{-ax}x^{a^2-1}(-x+a)$$

f′(x) $\quad \overset{+}{\underset{}{\longrightarrow}} \underset{x=a}{\cdot} \overset{-}{\underset{}{\longrightarrow}}$

at x = a, f(x) is maximum

Hence $\quad F(a) = e^{-a^2} \cdot a^{a^2}$

$$F(x) = e^{-x^2} \cdot x^{x^2} = \left(\frac{x}{e}\right)^{x^2}$$

$$\Rightarrow \ell n F(x) = x^2 \ell n\left(\frac{x}{e}\right)$$

$$\Rightarrow \frac{F'(x)}{F(x)} = 2x\ell n\left(\frac{x}{e}\right) + \frac{x^2}{x}$$

$$= x\left[2\ell n\frac{x}{e} + 1\right]$$

$$\underset{\ell n\frac{x}{e} = \frac{1}{2} \text{ or } \left(x = e^{+\frac{1}{2}}\right)}{+ \quad\bullet\quad -}$$

at $x = e^{\frac{1}{2}}$, F(x) has minimum value.

$$F\left(e^{\frac{1}{2}}\right) = \left(\frac{e^{\frac{1}{2}}}{e}\right)^{\left(e^{\frac{1}{2}}\right)^2}$$

$$F\left(e^{\frac{1}{2}}\right) = e^{-\frac{e}{2}}$$

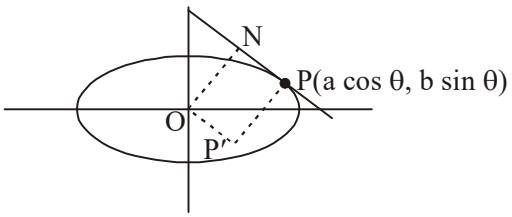
30. A perpendicular is drawn from the centre to a tangent to an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. Find the greatest value of the intercept between the point of contact and the foot of the perpendicular.

दीर्घवृत $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ की एक स्पर्श रेखा पर उसके केन्द्र से लम्ब डाला गया है। स्पर्शी बिन्दु एवं लम्ब के पाद के मध्य रेखा खण्ड की अधिकतम लम्बाई कितनी होगी।

Ans. $|a - b|$

Sol. Equation of PN $\Rightarrow T = 0$

$$\Rightarrow \frac{x\cos\theta}{a} + \frac{y\sin\theta}{b} = 1 \qquad \left[m_T = -\frac{b}{a}\frac{\cos\theta}{\sin\theta}\right]$$



Equation of $PP' \Rightarrow (y - b\sin\theta) = \frac{a\sin\theta}{b\cos\theta}(x - a\cos\theta)$

$$\Rightarrow a\sin\theta x - b\cos\theta y = \left(b^2 - a^2\right)\sin\theta\cos\theta$$

$$\Rightarrow a\sec\theta x - b\,\text{cosec}\theta y = b^2 - a^2$$

$$PN = OP' = \left|\frac{b^2 - a^2}{\sqrt{a^2\sec^2\theta + b^2\,\text{cosec}^2\theta}}\right|$$

$$PN_{max} = \left|\frac{b^2 - a^2}{\sqrt{(a+b)^2}}\right| \left(\left(a^2\sec^2\theta + b^2\,\text{cosec}^2\theta\right)_{min} = (a+b)^2\right)$$

$$PN_{max} = |a - b|$$

31. Suppose f(x) is a function satisfying the following conditions :

(i) $f(0) = 2, f(1) = 1$

(ii) f has a minimum value at $x = \frac{5}{2}$ and

(iii) for all x, $f'(x) = \begin{vmatrix} 2ax & 2ax-1 & 2ax+b+1 \\ b & b+1 & -1 \\ 2(ax+b) & 2ax+2b+1 & 2ax+b \end{vmatrix}$

Where a, b are some constants. Determine the constants a, b and the function f(x).

माना फलन f(x) निम्नलिखित शर्तो को संतुष्ट करता है :

(i) $f(0) = 2, f(1) = 1$ (ii) $x = \frac{5}{2}$ पर f न्यूनतम मान रखता है तथा

(iii) सभी x के लिये, $f'(x) = \begin{vmatrix} 2ax & 2ax-1 & 2ax+b+1 \\ b & b+1 & -1 \\ 2(ax+b) & 2ax+2b+1 & 2ax+b \end{vmatrix}$

जहाँ a, b कोई नियतांक है। नियतांको a, b के मान तथा फलन f(x) ज्ञात कीजिये।

Ans. $a = \frac{1}{4}; b = -\frac{5}{4}; f(x) = \frac{1}{4}(x^2 - 5x + 8)$

Sol. $f(x)=\begin{vmatrix} 2ax & 2ax-1 & 2ax+b+1 \\ b & b+1 & -1 \\ 0 & 0 & 1 \end{vmatrix}\ (R_3 \to R_3-(R_1+2R_2))$

$f'(x)=2ax+b \Rightarrow \quad f(x)=ax^2+bx+c$

$f(x)$ is maximum at $x=\frac{5}{2}$

$f'\left(\frac{5}{2}\right)=0 \qquad \Rightarrow 5a+b=0$

$f(0)=2 \ \Rightarrow \ c=2,\ f(1)=1 \qquad \Rightarrow a+b+c=1$

$\therefore \quad a=\frac{1}{4},\ b=-\frac{5}{4},\ c=2$

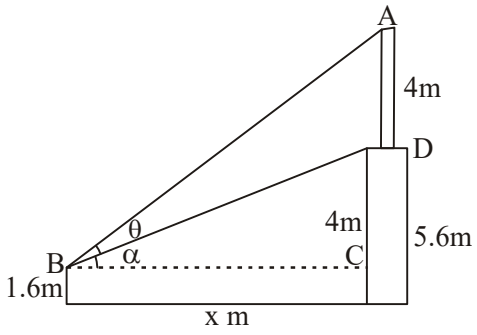
$f(x)=\frac{1}{4}x^2-\frac{5}{4}x+2$

32. A statue 4 metres high sits on a column 5.6 metres high. How far from the column must a man, whose eye level is 1.6 metres from the ground, stand in order to have the most favourable view of statue.

एक 4 मीटर ऊँचाई की प्रतिमा 5.6 मीटर की ऊँचाई के स्तम्भ पर रखी गई है। एक व्यक्ति जिसकी आंखो की ऊँचाई धरातल से 1.6 मीटर है को प्रतिमा के अनुकूल दर्शन के लिये स्तम्भ से कितनी दूर पर खड़ा होना पड़ेगा।

Ans. $4\sqrt{2}$ m

Sol. $\tan\alpha=\frac{4}{x}$



$\tan(\theta+\alpha)=\frac{8}{x}$

$\Rightarrow \frac{\tan\theta+\tan\alpha}{1-\tan\theta\cdot\tan\alpha}=\frac{8}{x}$

$\Rightarrow \frac{\tan\theta+\frac{4}{x}}{1-\frac{4}{x}\cdot\tan\theta}=\frac{8}{x}$

$\Rightarrow \tan\theta+\frac{4}{x}=\frac{8}{x}=\frac{32}{x^2}\tan\theta$

$\Rightarrow \tan\theta+\frac{32}{x^2}\tan\theta=\frac{8}{x}-\frac{4}{x}$

$$\Rightarrow \tan\theta\left(1+\frac{32}{x^2}\right)=\frac{4}{x}$$

$$\Rightarrow \tan\theta=\frac{\frac{4}{x}}{1+\frac{32}{x^2}}$$

$$\Rightarrow \tan\theta=\frac{4x}{x^2+32}=t$$

differentiating w.r.t. x

$$\frac{dt}{dx}=\frac{\left(x^2+32\right)4-4x\left(2x\right)}{\left(x^2+32\right)^2}=0$$

$$\Rightarrow 4x^2+128-8x^2=0$$

$$\Rightarrow 128=4x^2$$

$$\Rightarrow x^2=32$$

$$x=4\sqrt{2}$$

33. A window of fixed perimeter (including the base of the arch) is in the form of a rectangle surmounted by a semicircle. The semicircular portion is fitted with coloured glass while the rectangular part is fitted with clean glass. The clear glass transmits three times as much light per square meter as the coloured glass does. What is the ratio of the sides of the rectangle so that the window transmits the maximum light?

एक नियत परिमाप की खिड़की (आधार का सम्मिलित करके) आयताकार है एवं इसका उपरी भाग अर्द्धवृत्ताकार है। अर्द्धवृत्ताकार में रंगीन कांच लगाया गया है जबकि आयताकार भाग में सफेद कांच लगाया गया है। सफेद कांच से रंगीन कांच की तुलना में तीन गुना प्रति स्वायर मीटर अधिक प्रकाश आता है। आयतकार भाग की भुजाओं की लम्बाई का अनुपात ज्ञात कीजिये जिसके लिये खिड़की से प्रकाश अधिकतम प्राप्त हो ?

Ans. $6/(6+p)$

Sol. Let x and y metres be the length of the sides of rectangle ABCD and let there be a semi-circle on side CD of length x.

Therefore, perimeter of the window (including the base of the arch) = perimeter of the rectangle + perimeter of the semi-circle

$$=2x+2y+\frac{1}{2}\left(\frac{2\pi x}{2}\right)$$

$$= 2x + 2y + \frac{1}{2}\pi x = c\,(\text{constant})$$

$$\therefore y = \frac{1}{2}\left(x - 2x - \frac{1}{2}\pi x\right)$$

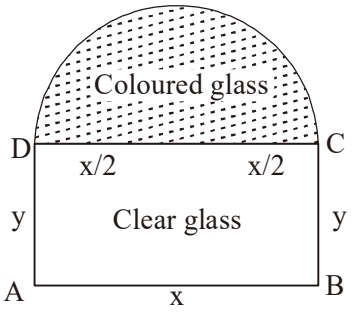
Let k be the light per square metre transmitted by coloured glass so that transmitted by clear glass will be 3k per square metre.

Hence, the total light emitted by the window is given by

A = (Area of coloured glass)k + (Area of clear glass) 3k

$$= \frac{1}{2}\pi\left(\frac{x}{2}\right)^2 k + xy(3k)$$

$$= \frac{1}{8}\pi k x^2 + 3kx\frac{1}{2}\left(x - 2x - \frac{1}{2}\pi x\right)$$



$$= \frac{1}{8}k\left(-5\pi x^2 - 24x^2 + 12cx\right)$$

$$\therefore \frac{dA}{dx} = \frac{1}{8}k\left(-10\pi x - 48x + 12c\right)$$

and $\frac{d^2A}{dx^2} = -\frac{1}{4}(5\pi + 24)k = -ve$

For maximum of minimum of A, $\frac{dA}{dx} = 0$

$$\Rightarrow x = \frac{6c}{(5\pi + 24)}$$

$$\therefore \text{From (1), } y = \frac{(\pi + 6)c}{(5\pi + 24)}$$

Since $\frac{d^2A}{dx^2}$ is –ve, therefore A has maxima

Hence, ration $\frac{x}{y} = 6(\pi + 6)$

34. A trapezium ABCD is inscribed into a semicircle of radius l so that the base AD of the trapezium is diameter and the vertices B & C lie on the circumference. Find the base angle θ of the trapezium ABCD which has the greatest perimeter.

एक समलम्ब चतुर्भुज ABCD जो एक अर्द्धवृत (त्रिज्या l) में स्थित है एवं चतुर्भुज का आधार AD अर्द्धवृत का व्यास है तथा शीर्ष B तथा C परिधि पर स्थित है। अधिकतम परिमाप वाली समलम्ब चतुर्भुज ABCD का आधार का कोण θ ज्ञात कीजिये।

Ans. $q = \mathbf{60^0}$

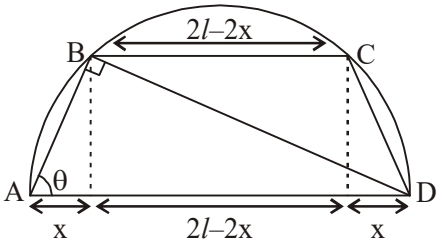
Sol. $AD = 2\ell \cos q$ from ΔABD

$\therefore x = AD\cos\theta = 2\ell\cos^2\theta$

Perimeter $= 2\ell + 2\ell - 2x + 2AB$

$= 4\ell - 2x + 4\ell\cos\theta$

$P(\theta) = 4\ell\left(1 - \cos^2\theta + \cos\theta\right)$



$\therefore P'(\theta) = 4\ell\left(2\cos\theta\sin\theta - \sin\theta\right) = 0$

$\therefore 2\cos\theta\sin\theta = \sin\theta \qquad \theta \neq 0 \text{ or } \pi$

$\therefore \cos\theta = \dfrac{1}{2}$

$\therefore \theta = \dfrac{\pi}{3} = 60^o$

$\therefore$ Perimeter is maximum when $\theta = 60^o$

35. Find the cosine of the angle at the vertex of an isosceles triangle having the greatest area for the given constant length l of the median drawn to its lateral side.

एक समद्विबाहु त्रिभुज के शीर्ष कोण की कोज्या ज्ञात कीजिये जब इसका क्षेत्रफल अधिकतम हो साथ ही शाखाओं की मध्यिकाओं की लम्बाई नियत l है।

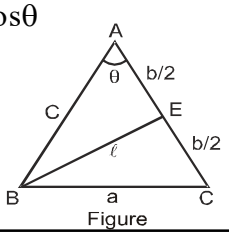
Ans. $\cos A = 0.8$

Sol. $\Delta = \dfrac{1}{2}bc\sin\theta = \dfrac{1}{2}c^2\sin\theta$ ($\because$ b = c)

In Δ ABE, using cosine rule, $\ell^2 = c^2 + \dfrac{b^2}{4} - bc\cos\theta = \dfrac{5c^2}{4} - c^2\cos\theta$

$$c^2 = \frac{4\ell^2}{5 - 4\cos\theta}$$

$\therefore \quad {}_D = 2\ell^2.\ \dfrac{\sin\theta}{5 - 4\cos\theta}$



Figure

$$\frac{d\Delta}{d\theta} = \frac{2\ell^2.((5-4\cos\theta)\cos\theta - \sin\theta\,(4\sin\theta))}{(5-4\sin\theta)^2}$$

$$= \frac{2\ell^2.(5\cos\theta - 4)}{(5-4\sin\theta)^2}$$

$\therefore$ For Δ to be maximum, $\cos\theta = 0.8$

$$\underset{\text{signs of } \frac{d\Delta}{d\theta}}{\overset{+}{\underset{\cos^{-1}0.8}{\overset{-}{\rule{2cm}{0.4pt}}}}}$$

36. From a fixed point A on the circumference of a circle of radius 'a', let the perpendicular AY fall on the tangent at a point P on the circle, prove that the greatest area which the ΔAPY can have is $3\sqrt{3}\,\frac{a^2}{8}$ sq. units.

एक नियत बिन्दु A जो 'a' त्रिज्या के वृत की परिधि पर स्थिति है, माना लम्ब AY, P पर स्पर्श रेखा को Y पर मिलता है सिद्ध कीजिये ΔAPY का अधिकतम क्षेत्रफल का मान $3\sqrt{3}\,\frac{a^2}{8}$ sq. हो सकता है।

Ans.

Sol. Let the circle be $x^2 + y^2 = a^2$. We take point P at (a, 0) and Q at (a cos θ, a sin θ)

Equation of tangent at Q is easily seen to be $x\cos\theta + y\sin\theta = a$ ...(1)

Let PR be perpendicular from P on (1).

Then $PR = \frac{a - a\cos\theta}{\sqrt{\cos^2\theta + \sin^2\theta}} = 2a\sin^2\frac{\theta}{2}$

Also $PQ = \sqrt{(a - a\cos\theta)^2 + a^2\sin^2\theta}$

$$= \sqrt{a^2\left(1 + \cos^2\theta - 2\cos\theta + \sin^2\theta\right)} = a\sqrt{2(1-\cos\theta)} = 2a\sin\frac{\theta}{2}$$

Hence $QR = \sqrt{PQ^2 - PR^2}$

$$= \sqrt{4a^2\sin^2\left(\frac{\theta}{2}\right) - 4a^2\sin^4\left(\frac{\theta}{2}\right)} = 2a\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{\theta}{2}\right)$$

If A denotes the area of ΔPQR, then

$$A = \frac{1}{2}PR\cdot QR = \frac{1}{2}2a\sin^2\left(\frac{\theta}{2}\right)2a\sin\left(\frac{\theta}{2}\right)\cos\left(\frac{\theta}{2}\right) = \frac{1}{2}a^2(1-\cos\theta)\sin\theta = \frac{1}{2}a^2\left(\sin\theta - \frac{1}{2}\sin 2\theta\right)$$

For maxima or minima, we have

$$\frac{dA}{d\theta}=\frac{1}{2}a^2\left(\cos\theta-\cos 2\theta\right)=0$$

Thsi gives $\cos 2\theta=\cos\left(2\pi-2\theta\right)$ or $\theta=2\pi-2\theta$, i.e., $\theta=\frac{2\pi}{3}$

$$\frac{d^2A}{d\theta^2}=\frac{1}{2}a^2\left(-\sin\theta+2\sin 2\theta\right)=\frac{1}{2}a^2\left(\frac{-\sqrt{2}}{2}-\frac{2\sqrt{3}}{2}\right)<0 \text{ at } \theta=\frac{2\pi}{3}$$

Hence A is maximum when $\theta=\frac{2\pi}{3}$

$$\therefore \text{ Maxmimum value of } A=\frac{1}{2}a^2\left[\sin\frac{2\pi}{3}-\frac{1}{2}\sin\frac{4\pi}{3}\right]$$

$$=\frac{1}{2}a^2\left[\frac{\sqrt{3}}{2}+\frac{\sqrt{3}}{4}\right]=\frac{3\sqrt{3}}{8}a^2$$

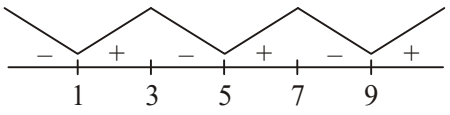
37. Let $f_N(x)$ be a polynomial function which satisfies $f'_N(x)=\prod_{n=1}^{N}(x-n)^n$. If n_1 is sum of all values of x for which $f_{10}(x)$ attains maxima and n_2 denotes the sum of all values of x for which $f_{10}(x)$ attains minima, then find the value of (n_1+n_2).

दिया है $f_N(x)$ एक बहुपद फलन इस प्रकार से है कि $f'_N(x)=\prod_{n=1}^{N}(x-n)^n$. यदि n_1, x के उन सभी मानों का योग है जिसके लिये $f_{10}(x)$ महत्तम है तथा n_2, x के उन सभी मानो का योग है जिसके लिये $f_{10}(x)$ न्यूनतम है तो (n_1+n_2) का मान ज्ञात कीजिए।

Ans. 25

Sol. $f'_{10}(x)=(x-1)(x-2)^2(x-3)^3.....(x-10)^{10}$

$f'_{10}(x)>0\Rightarrow(x-1)(x-3)(x-5)(x-7)(x-9)>0$



$\Rightarrow$ Points of maxima = 3,7 $\quad n_1=10$

Points of minima = 1,5,9 $\quad n_2=15$

$\therefore$ Value of $(n_1+n_2)=10+15=25$

38. Let $x \in \left[-\frac{5\pi}{12}, -\frac{\pi}{3}\right]$, then find the maximum value of $y = \tan\left(x + \frac{2\pi}{3}\right) - \tan\left(x + \frac{\pi}{6}\right) + \cos\left(x + \frac{\pi}{6}\right)$.

दिया है $x \in \left[-\frac{5\pi}{12}, -\frac{\pi}{3}\right]$, तो $y = \tan\left(x + \frac{2\pi}{3}\right) - \tan\left(x + \frac{\pi}{6}\right) + \cos\left(x + \frac{\pi}{6}\right)$ का महत्तम मान ज्ञात करो।

Ans. $\frac{11\sqrt{3}}{6}$

Sol. $f(x) = \tan\left(x + \frac{2\pi}{3}\right) - \tan\left(x + \frac{\pi}{6}\right) + \cos\left(x + \frac{\pi}{6}\right)$

Let $x = -z - \frac{\pi}{6}$, where $z \in \left[\frac{\pi}{6}, \frac{\pi}{4}\right]$

$f(z) = \cot z + \tan z + \cos z$

$= 2 \operatorname{cosec} 2z + \cos z$

Both are monotonically decreasing for $z \in \left[\frac{\pi}{6}, \frac{\pi}{4}\right]$

$f(z)_{max} = f\left(\frac{\pi}{6}\right)$

$= 2\operatorname{cosec}\frac{\pi}{3} + \cos\frac{\pi}{6}$

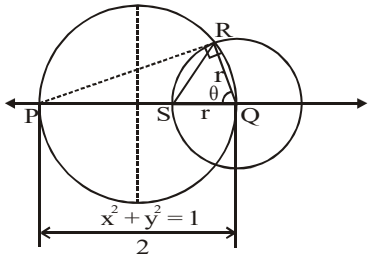
$= \frac{4}{\sqrt{3}} + \frac{\sqrt{3}}{2} = \frac{11\sqrt{3}}{6}$

39. The circle $x^2 + y^2 = 1$ cuts the x-axis at P & Q. Another circle with centre at Q and varable radius intersects the first circle at R above the x-axis & the line segment PQ at S. Find the maximum area of the triangle QSR.

वृत $x^2 + y^2 = 1$, x-अक्ष को P तथा Q पर काटता है। एक अन्य वृत जिसका केन्द्र Q है एवं त्रिज्या चर है प्रथम वृत को x-अक्ष के उपर में R पर प्रतिच्छेद करता है तथा रेखा खण्ड PQ को S पर प्रतिच्छेद करता है। त्रिभुज QSR का अधिकतम क्षेत्रफल ज्ञात कीजिये।

Ans. $\frac{4}{3\sqrt{3}}$

Sol.



$\cos\theta = \frac{r}{2}$

$r = 2\cos\theta$

$A(\Delta SRQ) = f(\theta) = \frac{1}{2}(2\cos\theta)^2.\sin\theta = 2\cos^2\theta\sin\theta$

Now maximize f(θ)

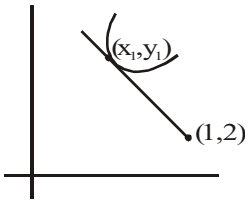
40. Find the equations of the tangents drawn to the curve $y^2 - 2x^3 - 4y + 8 = 0$ from the point (1, 2).

वक्र $y^2 - 2x^3 - 4y + 8 = 0$ पर बिन्दु (1, 2) से स्पर्श रेखाओं का समीकरण ज्ञात कीजिये।

Ans. $2\sqrt{3}\ x - y = 2\left(\sqrt{3} - 1\right)$ or $2\sqrt{3}\,x + y = 2\left(\sqrt{3} + 1\right)$

Sol. $2yy' - 6x^2 - 4y' = 0 \Rightarrow y' = \frac{3x_1^2}{(y_1 - 2)}$

Also slope $= \frac{y_1 - 2}{(x_1 - 1)}$



Equating both terms $\frac{3x_1^2}{(y_1 - 2)} = \frac{(y_1 - 2)}{(x_1 - 1)}$

$\Rightarrow \quad 3x_1^2(x_1 - 1) = (y_1 - 2)^2$

$\Rightarrow \quad 3x_1^3 - 3x_1^2 = y_1^2 - 4y_1 + 4$

$\Rightarrow \quad 3x_1^3 - 3x_1^2 = (2x_1^3 - 8) + 4$

$\Rightarrow \quad x_1^3 - 3x_1^2 + 4 = 0$

$\Rightarrow \quad (x_1 + 1)(x_1^2 - 4x_1 + 4) = 0$

$\Rightarrow \quad (x_1 + 1)(x_1 - 2)^2 = 0$

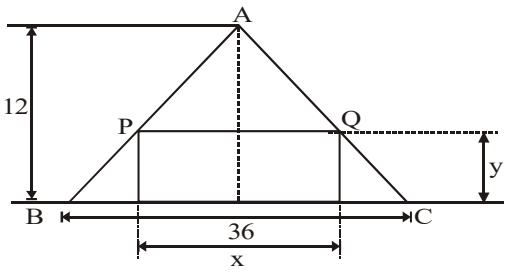
get the equation of tangent at $x_1 = -1, \quad x_1 = 2$

41. What are the dimensions of the rectangular plot of the greatest area which can be laid out within a triangle of base 36 ft. & altitude 12 ft ? Assume that one side of the rectangle lies on the base of the triangle.

त्रिभुज जिसका आधार 36 ft. है तथा लम्ब 12 ft है में बनाये गये अधिकतम क्षेत्रफल वाले आयताकार क्षेत्र की विमाये क्या है ? मान लिजिये आयत की एक भुजा त्रिभुज के आधार पर स्थित है।

Ans. 6' × 18'

Sol.



$\Delta APQ \sim \Delta ABC$

$$\frac{12-y}{12}=\frac{x}{36}$$

$3(12 - y) = x$

$A = xy = 3(12 - y)\, y$

Now maximize the area.

COMPREHENSION TYPE QUESTIONS

Paragraph for Question nos. 42 to 44

a(t) is a function of t such that $\frac{da}{dt} = 2$ for all values of t and a = 0 when t = 0. Further y = m(t) x + c(t) is tangent to the curve $y = x^2 - 2ax + a^2 + a$ at the point whose abscissa is 0. Then

t का एक फलन a(t) इस प्रकार है कि t के सभी मानों के लिए $\frac{da}{dt} = 2$ और जब t = 0 हो, तो a = 0. पुनः y = m(t) x + c(t) वक्र $y = x^2 - 2ax + a^2 + a$ के उस बिन्दु पर स्पर्श रेखा है जिसका भुज 0 है। तो

42. If the rate of change of distance of vertex from the origin with respect to t is k, then k =
यदि शीर्ष की मूलबिन्दु से दूरी में t के सापेक्ष परिवर्तन की दर k है, तो k =

(A) 2 (B) $2\sqrt{2}$ (C) $\sqrt{2}$ (D) $4\sqrt{2}$

Ans. B

Sol. $\frac{da}{dt} = 2 \quad \Rightarrow \quad a = 2t + c$

$\therefore \quad c = 0 \quad \{\because a = 0, \text{ when } t = 0\}$

$\therefore \quad a = 2t$

$\therefore$ the curve $y = x^2 - 2ax + a^2 + a$ becomes

$y = x^2 - 4tx + 4t^2 + 2t$

If $x = 0$, then $y = 4t^2 + 2t$

$\frac{dy}{dx} = 2x - 4t$ $\therefore$ $\left.\frac{dy}{dx}\right|_{\text{at } x=0} = -4t$

$\therefore$ equation of the tangent

$y - (4t^2 + 2t) = -4t\,(x - 0)$

$y = -4t\,x + 4t^2 + 2t$

i.e. $y = -4t\,x + 4t^2 + 2t$ is $(2t, 2t)$

vertex of $y = x^2 - 4t\,x + 4t^2 + 2t$ is $(2t, 2t)$

$\therefore$ distance of vertex from the origin $= 2\sqrt{2}\,t$

$\therefore$ rate of change of distance of vertex from origin with respect to $t = 2\sqrt{2}$

i.e. $k = 2\sqrt{2}$

43. If the rate of change of $c(t)$ with respect to t, when $t = k$, is ℓ, then $\ell =$

यदि $t = k$ पर t के सापेक्ष $c(t)$ में परिवर्तन की दर ℓ है, तो $\ell =$

(A) $16\sqrt{2} - 2$ (B) $8\sqrt{2} + 2$ (C) $10\sqrt{2} + 2$ (D) $16\sqrt{2} + 2$

Ans. D

Sol. $c(t) = 4t^2 + 2t$

$\therefore$ $\frac{dc}{dt} = 8t + 2$ $\therefore$ $\left.\frac{dc}{dt}\right|_{\text{at } t=2\sqrt{2}} = 16\sqrt{2} + 2$ $\therefore$ $\ell = 16\sqrt{2} + 2$

44. The rate of change of $m(t)$, with respect to t, at $t = \ell$ is :

$t = \ell$ पर t के सापेक्ष $m(t)$ में परिवर्तन की दर है :

(A) –2 (B) 2 (C) – 4 (D) 4

Ans. C

Sol. $m(t) = -4t$

$\therefore$ $\frac{dm}{dt} - 4$ $\therefore$ $\left.\frac{dm}{dt}\right|_{\text{at } t=\ell} = -4$

Paragraph for Question nos. 45 to 46

Let C be the curve $f(x) = ln^2x + 2\,ln\,x$ and $A(a, f(a))$, $B(b, f(b))$ where $(a < b)$ are the points of tangency of two tangents drawn from origin to the curve C.

माना C एक वक्र $f(x) = \ln^2 x + 2 \ln x$ है तथा $A(a, f(a))$, $B(b, f(b))$ हैं, जहाँ $(a < b)$, वह बिन्दु है जहाँ पर मूल बिन्दु से वक्र C पर डाली गई स्पर्श रेखायें मिलती है।

45. The value of the product ab is equal to :

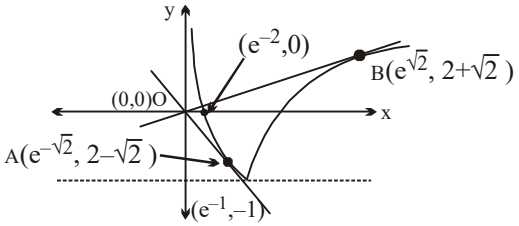
गुणन ab का मान है :

(A) e (B) $\frac{1}{e}$ (C) e^2 (D) 1

Ans. D

Sol. $y = f(x) = \ln^2 x + 2 \ln x$ (Clearly f(x) is defined for $x > 0$)

$$\left.\frac{dy}{dx}\right|_{A(a,f(a))} = \frac{2\ln a}{a} + \frac{2}{a} = \frac{2}{a}(1 + \ln a)$$



Graph of $f(x) = \ln^2 x + 2 \ln x = (\ln x + 1)^2 - 1$

Also $\frac{dy}{dx} = \frac{f(a) - 0}{a - 0}$

$\therefore \quad \frac{f(a)}{a} = \frac{2}{a}(1 + \ln a) \Rightarrow f(a) = 2(1 + \ln a)$

$\therefore \quad \ln^2 a + 2 \ln a = 2 + 2 \ln a$

$\Rightarrow \quad \ln^2 a = 2 \Rightarrow a = e^{\sqrt{2}}$ or $e^{-\sqrt{2}}$

Hence $a = e^{-\sqrt{2}}$ and $b = e^{\sqrt{2}}$

So, $ab = 1$

46. Number of values of x satisfying the equation $5x f'(x) - x \ln 10 - 10 = 0$ is equal to :

समीकरण $5x f'(x) - x \ln 10 - 10 = 0$ को संतुष्ट करने वाले x के मानो की संख्या है :

(A) 0 (B) 1 (C) 2 (D) 3

Ans. C

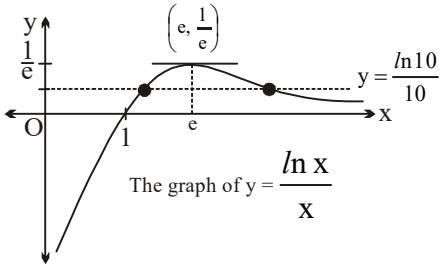
Sol. $5x f'(x) - x \ln 10 - 10 = 0$

$$\Rightarrow \quad 5x\left(\frac{2\ln x}{x} + \frac{2}{x}\right) - x \ln 10 - 10 = 0$$

$\Rightarrow \quad 10 \ln x + 10 - x \ln 10 - 10 = 0$

$\therefore \quad 10 \ln x = x \ln 10 \Rightarrow \frac{\ln x}{x} = \frac{\ln 10}{10}, \; x > 0$

As $y = \frac{\ln x}{x} \Rightarrow y' = \frac{1 - \ln x}{x^2}$



The graph of $y = \frac{\ln x}{x}$

So, two values of x are possible

Paragraph for Question nos. 47 to 50

Consider the function $f(x)=\left(1+\frac{1}{x}\right)^x$ then

माना $f(x)=\left(1+\frac{1}{x}\right)^x$ तो

47. Domain of $f(x)$ is :

$f(x)$ का प्रान्त है :

(A) $(-1, 0)\cup(0, \infty)$ (B) $R - \{0\}$ (C) $(-\infty, -1)\cup(0, \infty)$ (D) $(0, \infty)$

Ans. C

Sol. $f(x)=\left(1+\frac{1}{x}\right)^x$ [For a^x, $a > 0$]

$1+\frac{1}{x}>0 \Rightarrow \frac{x+1}{x}>0$

$\Rightarrow x\in(-\infty,-1)\cup(0,\infty)$

48. Which one of the following limits tends to unity?

निम्न में से कौनसी एक सीमा 1 होगी?

(A) $\lim\limits_{x\to\infty} f(x)$ (B) $\lim\limits_{x\to0^+} f(x)$ (C) $\lim\limits_{x\to-1^-} f(x)$ (D) $\lim\limits_{x\to-\infty} f(x)$

Ans. B

Sol. (A) $\lim\limits_{x\to\infty} f(x)=\lim\limits_{x\to\infty}\left(1+\frac{1}{x}\right)$ (1^∞)

$=e^{\lim\limits_{x\to\infty}\frac{x}{x}}=e$

(B) $\lim\limits_{x\to0^+} f(x)=\lim\limits_{x\to0^+}\left(1+\frac{1}{x}\right)^x$

$\Rightarrow \lim\limits_{x\to0^+} \ell n f(x)=\lim\limits_{x\to0^+} x\ell n\left(1+\frac{1}{x}\right)=0$

$\Rightarrow \lim\limits_{x\to 0^+} f(x) = 1$

(C) $\lim\limits_{x\to 0^-} f(x) = 2$

(D) $\lim\limits_{x\to -\infty} f(x) = \lim\limits_{x\to -\infty}\left(1+\frac{1}{x}\right)^x$ (1^∞)

$= e^{\lim\limits_{x\to -\infty}\frac{x}{x}} = e$

49. The function f(x) :

(A) has a maxima but no minima
(B) has a minima but no maxima
(C) has exactly one maxima and one minima
(D) has neither a maxima nor a minima

फलन f(x) में :

(A) उच्चिष्ठ है लेकिन निम्निष्ठ नहीं
(B) निम्निष्ठ है लेकिन उच्चिष्ठ नहीं
(C) एक ही स्थानिय उच्चिष्ठ और एक ही निनिम्नष्ठ है
(D) न तो उच्चिष्ठ है न ही निम्निष्ठ

Ans. D

Sol. $f(x) = \left(1+\frac{1}{x}\right)^x$

$\ell n f(x) = x\ell n\left(1+\frac{1}{x}\right) = x\left[\ell n(1+x) - \ell n x\right]$

Differentiating w.r.t. "x"

$\frac{f'(x)}{f(x)} = \ell n(1+x) - \ell n x + x\left[\frac{1}{1+x} - \frac{1}{x}\right]$

$f'(x) = f(x)\left(\ell n\left(\frac{1+x}{x}\right) - \frac{1}{1+x}\right)$ ...(1)

Let $h(x) = \ell n\left(\frac{1+x}{x}\right) - \frac{1}{1+x}$

$\lim\limits_{x\to\infty} h(x) = 0$

$h(x) = \ell n(1+x) - \ell n x - \frac{1}{1+x}$

$$h'(x)=\frac{1}{1+x}-\frac{1}{x}+\frac{1}{(1+x)^2}=\frac{-1}{x(x+1)^2}$$

for $x>0$ $\Rightarrow h'(x)<0$

$\Rightarrow x<\infty$

$\Rightarrow h(x)>\lim\limits_{x\to\infty} h(x)$

$\Rightarrow h(x)>0$

Now using equation (1)

for $x>0$, $f(x)>0$ and $\ell n\left(\frac{1+x}{x}\right)-\frac{1}{1+x}>0$

$x>0\Rightarrow f'(x)>0$

f(x) is increasing

Has neither maxima nor a minima.

50. Range of the function f (x) is :

f(x) का परिसर है :

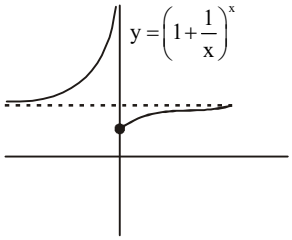
(A) $(0,\infty)$ (B) $(-\infty, e)$ (C) $(1,\infty)$ (D) $(1, e)\cup(e,\infty)$

Ans. D

Sol. $\lim\limits_{x\to-1^-}\left(1+\frac{1}{x}\right)^x\to\infty$

$x\in(-\infty,-1)\cup(0,\infty)$

$f(x)\in(1,e)\cup(e,\infty)$



MATCH THE COLUMN TYPE QUESTION(S)

51. Match the following column :

Column – I	**Column – II**
(A) The Inclination of the tangent to the curve $y=2x^4$ at $x=\frac{1}{2}$.	(P) $\text{cosec}^{-1}\left(\frac{\sqrt{5}}{2}\right)$

(B) The inclination of normal to the curve where the tangent is parallel to line joining (3, 0) and (4,1). (Q) $\frac{3\pi}{4}$

(C) The inclination of tangent to the curve $x = 1 - \cos\theta$, $y = \theta - \sin\theta$ at $\theta = \frac{\pi}{4}$ (R) $\frac{\pi}{4}$

(D) Angle of intersection of the curves $y = [|\sin x| + |\cos x|]$ and $x^2 + y^2 = 5$, where [.] denote the GIF (S) $\frac{\pi}{8}$

निम्न कॉलम को सुमेलित कीजिए :

Column – I	**Column – II**
(A) वक्र $y = 2x^4$ के लिए $x = \frac{1}{2}$ पर स्पर्श रेखा का ढाल होगा	(P) $\text{cosec}^{-1}\left(\frac{\sqrt{5}}{2}\right)$
(B) वक्र के अभिलम्ब का ढाल जहाँ पर स्पर्श रेखा बिन्दुओं (3, 0) तथा (4, 1) को जोड़ने वाली रेखा के समान्तर है	(Q) $\frac{3\pi}{4}$
(C) वक्र $x = 1 - \cos\theta$, $y = \theta - \sin\theta$ के लिए $\theta = \frac{\pi}{4}$ पर स्पर्श रेखा का ढाल होगा	(R) $\frac{\pi}{4}$
(D) वक्र $y = [\|\sin x\| + \|\cos x\|]$ तथा $x^2 + y^2 = 5$ का प्रतिछेदन कोण है जहाँ [.] महतम पूर्णांक फलन दर्शाता है	(S) $\frac{\pi}{8}$

Ans. (A) R, (B) Q , (C) S, (D) P

Sol. (A) $y = 2x^4$ at $x = \frac{1}{2}$

$$\frac{dy}{dx} = 8x^3 = M_T$$

$$\tan\theta = 8 \times \left(\frac{1}{2}\right)^3 = 1$$

$$\theta = \frac{\pi}{4}$$

(B) A(3, 0) B(4, 1)

Slope of tangent = $M_{AB} = \frac{1-0}{4-3} = 1$

Slope of normal = –1

$\tan\theta = -1$

$\theta = \frac{3\pi}{4}$

(C) $x = 1 - \cos\theta \quad y = \theta - \sin\theta$

$\frac{dx}{d\theta} = \sin\theta \qquad \frac{dy}{d\theta} = 1 - \cos\theta$

$\frac{dy}{dx} = \frac{1-\cos\theta}{\sin\theta}$

$\theta = \frac{\pi}{4}$

$M_T = \frac{1-\frac{1}{\sqrt{2}}}{\frac{1}{\sqrt{2}}} = \sqrt{2-1}$

$\tan\theta = \sqrt{2} - 1$

$\theta = \frac{\pi}{8}$

(D) $x^2 + y^2 = 5$

$y = [|\sin x| + |\cos x|]$ put y = 1

$1 \le |\sin x| + |\cos x| \le \sqrt{2}$ $X^2 = 4 \;\; X = \pm 2$

point (2, 1) and (–2, 1)

So, y = 1

Now $x^2 + y^2 = 5$

y = 1 $2x + 2y\frac{dy}{dx} = 0$

$m_1 = 0$ $\frac{dy}{dx} = \frac{-x}{y}$

or

$$m_2 = \frac{-2}{1} = -2$$

$$\tan\theta = \left|\frac{m_1 - m_2}{1 + m_1 m_2}\right| = \left|\frac{0+2}{1+0}\right| = 2$$

$$\tan\theta = \frac{2}{1}$$

$$\cos ec = \frac{\sqrt{5}}{2}$$

$$\theta = \cos ec^{-1}\left(\frac{\sqrt{5}}{2}\right)$$

52. Match the following column :

Column – I	**Column – II**
(A) $f(x) = 2x^3 + 3x^2 - 12x + 7$ is strictly increasing in the interval	(P) $[6, +\infty)$
(B) $f(x) = 3x^4 + 8x^3 - 6x^2 - 24x + 19$ is strictly increasing in the interval	(Q) $[7, +\infty)$
(C) $f(x) = x^3 - 6x^2 - 36x + 7$ is strictly increasing in the interval	(R) $[1, +\infty)$
(D) $f(x) = 2x^3 - 15x^2 - 84x + 7$ is strictly increasing	(S) $(1, +\infty)$

निम्न कॉलम को सुमेलित कीजिए :

Column – I	**Column – II**
(A) फलन $f(x) = 2x^3 + 3x^2 - 12x + 7$ किस अन्तराल में वर्धमान है	(P) $[6, +\infty)$
(B) फलन $f(x) = 3x^4 + 8x^3 - 6x^2 - 24x + 19$ किस अन्तराल में वर्धमान है	(Q) $[7, +\infty)$
(C) फलन $f(x) = x^3 - 6x^2 - 36x + 7$ किस अन्तराल में वर्धमान है	(R) $[1, +\infty)$
(D) फलन $f(x) = 2x^3 - 15x^2 - 84x + 7$ किस अन्तराल में वर्धमान है	(S) $(1, +\infty)$

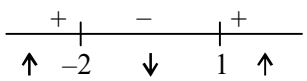
Ans. (A) P,Q,R,S (B) P,Q, R, S (C) P, Q (D) Q

Sol. (A) $f(x) = 2x^3 + 3x^2 - 12x + 7$

$f(x) = 6x^2 + 6x - 12$

$= 6(x^2 + x - 2)$

$= 6(x + 2)(x - 1)$



Increasing in $(-\infty, -2) \cup (1, \infty)$

(B) $f(x)\ 3x^4 + 8x^3 - 6x^2 - 24x + 19$

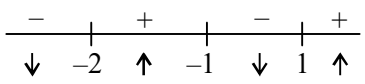
$f'(x) = 12x^3 + 24x^2 - 12x - 24$

$= 12(x^3 + 2x^2 - x - 2)$

$\Rightarrow 12(x^2(x + 2) - (x + 2))$

$= 12(x + 2)(x^2 - 1)$

$= 12(x + 2)(x - 1)(x + 1)$

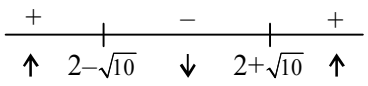


increasing in $(-2, -1) \cup (1, \infty)$

(C) $f(x) = x^3 - 6x^2 - 36x + 7$

$f'(x) = 3x^2 - 12x - 36$

$= 3(x^2 - 4x - 6)$



(D) $f(x) = 2x^3 - 15x^2 - 84x + 7$

$f'(x) = 6x^2 - 30x - 84$

$= 6(x^2 - 5x - 14)$

$= 6(x - 7)(x + 2)$

